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Optimal Market Making in Cryptocurrency Perpetual Futures: A Stochastic Control Extension for Discrete Funding Settlements and Nonlinear Alpha Signals

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The views stated in this paper are those of the authors and not necessarily those of the supervisor, Erasmus School of Economics or Erasmus University Rotterdam.

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Abstract

Market makers on cryptocurrency perpetual futures face a risk classical inventory-control models omit. At fixed times a discrete funding payment taxes, or pays, held positions on a clock the whole market can see. This thesis extends the Guéant-Lehalle-Tapia framework with an analytical correction for this payment, smoothing it into a continuous drain concentrated in a short pre-settlement window the model sizes itself, and pairs it with a LightGBM drift signal learned from Level-2 microstructure, expected to supply the correction with situational awareness. Both go through a six-strategy ablation on 64 out-of-sample days of self-recorded Binance Bitcoin data in a fee-exact, last-in-queue replay engine. The measurement environment is itself a finding: at the retail fee tier, fee and adverse selection outweigh the captured spread by six cents per fill, almost identically across strategies. Raw profit therefore ranks strategies by how often they trade, which is why we instead focus on risk-adjusted performance and mechanism behind the trades. The learned model, that was trained to forecast price drift, turns out to work as a risk clock, anticipating when resting quotes are about to be hit. Used defensively, widening only the threatened quote, it cuts mean loss by 26% and tail loss by over a fifth on 27% fewer fills. Shuffling those forecasts in time removes most of the gain, so the timing is genuine. The funding correction activates at the predicted minutes-scale horizon and rotates inventory onto the side that receives the payment, 2.1 lots in the final minute against GLT's near-zero. On BTC's mild carry it earns little, as its magnitude bound predicts, but its advantage grows monotonically with carry, at rates thinner perpetuals see. We initially expected the two extensions to reinforce each other, but they do not seem to interact at all. BTC's funding carry is simply too small to offset the transaction fees.

1 Introduction

Anyone who has exchanged money at an airport kiosk has traded against a market maker. The kiosk posts two prices at once, one at which it buys euros and a higher one at which it sells them, and lives off the gap. It does not care whether the euro rises or falls, it cares that customers keep arriving at both of its service windows, some selling euros to it, some buying. Its two real fears are narrower. The first is being stuck holding too many euros when the exchange rate moves sharply, and the second is the customer who exchanges a large sum thirty seconds before a news headline moves the rate, a customer who, in retrospect, knew something. Modern electronic markets are the industrial-scale version of this kiosk. On a modern exchange, a market maker, continuously posts *limit orders* on both sides of the *limit order book*: a standing offer to buy at the bid, and a standing offer to sell at the ask. Every pair of trades it wins, one on each side, earns it the *bid-ask spread*, the difference between those two prices. But every trade it wins against a better-informed counterparty costs it more than the spread it earned: the maker has bought something that is about to be worth less, or sold something that is about to be worth more. This is the kiosk's second fear, industrialised, and that is: *adverse selection*, the central antagonist of this thesis.

The first fear, inventory, is the one mathematics conquered first. A market maker filled on its bid accumulates a long position, and that position's value now rises and falls with a price the maker holds no view on: an unhedged bet it never meant to place. The seminal work of Avellaneda & Stoikov (2008) formalised the trade-off between earning spread and carrying

inventory as a problem in stochastic optimal control, and Guéant et al. (2013) (hereafter referred to as GLT, both the paper and the model) made it operational by reducing the governing Hamilton-Jacobi-Bellman (HJB) equation, the equation that dictates the optimal continuous-time quoting policy, to a solvable system of ordinary differential equations (ODEs). The GLT solution has an elegant behavioural core: it skews both quotes away from whichever side would increase the maker's current position, long or short, making position-reducing fills more likely than position-growing ones. The maker accepts a little less to get flat, and demands a little more to take on more.

This thesis uses that framework for an asset class it was never designed for. Cryptocurrency *perpetual futures* are the dominant instrument of crypto speculation, and their defining quirk matters here: unlike a conventional future, the contract never expires, so nothing ever forces its price to converge to the spot price it tracks. Exchanges anchor the two through the *funding rate* (or referred to interchangeably as carry). At fixed settlement times, every eight hours on Binance's Bitcoin perpetual futures (BTCUSDT[1] or BTC) contract, every open position pays or receives a cash transfer proportional to its size. The sign and magnitude are set by how far the perpetual has been trading above or below spot: when it trades above spot, longs pay shorts, and below spot, the payment reverses. For a directional trader this is a modest carry cost. For a market maker it is something sharper: a deterministic clock event at which held inventory triggers a cash transfer, which the whole market can see coming, and around which the incentive to hold inventory changes. The theoretical models of Avellaneda-Stoikov and GLT assume an asset with zero cost-of-carry and no such clock. The discrete, position-proportional funding settlement creates a recurring boundary that classical continuous-time formulations are mathematically ill-equipped to handle.

Quoting through these settlements demands two capabilities at once: rigorous control of inventory risk, which now generates a deterministic cash flow of its own, and real-time awareness of the book conditions the quotes rest in. Analytical stochastic models in the GLT line carry no representation of order book state, so their quotes answer to inventory and to the terminal horizon but never to the warning signs, leaving orders resting where they are most likely to be picked off. Data-driven strategies read those warning signs well, but manage inventory without the closed-form quotes and optimality guarantees that make the analytical approach worth having. The studies that do place a predictive signal inside an HJB framework work almost entirely on traditional asset classes. What remains open is the combination, on an instrument where the trader's wealth process is subject to discrete funding settlements

Addressing this intersection matters to both the literature and practice. Scientifically, it contributes to the stochastic optimal control literature by formalising a discrete cost-of-carry mechanism inside a continuous-time market-making model. Practically, it offers an alternative to simply halting algorithms through deterministic liquidity shocks: a maker that prices the settlement boundary correctly can remain active through it with quantified risk.

To that end, this thesis proposes a hybrid market-making architecture that explicitly decouples endogenous inventory risk from exogenous microstructural dynamics. The GLT framework is mathematically extended by converting the discrete funding payment into a continuous

[1] All price values are in USDT (Tether), trading 1:1 with USD.

drain inside the wealth process, concentrated in a short window before each settlement so the quotes anticipate it. An exact discrete jump would break the ODE reduction, and the smoothed drain preserves it. The length of this anticipation window is set by the model itself: by how quickly the effect of the upcoming settlement fades as one looks further back from it in time. Concurrently, a LightGBM machine learning (ML) price drift signal, driven by high-frequency microstructure features such as order flow and order book imbalance, is used to capture the non-linear interactions a linear forecast cannot. Whether that complexity is needed is itself part of the question, so one arm of the ablation prices a drift estimated from price alone by a rolling autoregression.

This motivates the primary research question of the thesis: *to what extent can the risk-adjusted performance of a GLT market-making strategy on cryptocurrency perpetual futures be enhanced by integrating a non-linear ML drift signal and an HJB-derived analytical correction for discrete funding rates?* While each component is evaluated in isolation, the central thesis is that the two components are structurally complementary: the HJB correction provides analytical timing around the funding settlement, while the ML signal supplies real-time microstructural awareness that the analytical correction cannot capture. Neither component can do the other's job. This form of complementarity carries a testable footprint: running on different clocks and on different information, the two contributions could reinforce one another at the settlement, collapse into redundancy, or merely add. The ablation is designed to measure which of the three, and whether the tested contract's funding effect permits an answer at all.

To answer this question empirically, a six-strategy ablation study is conducted on 94 days of self-recorded, message-level Level-2 (L2) Bitcoin order book data from Binance, using a daily walk-forward design where each 1-day evaluation window is trained on the preceding 30 days. The strategies are replayed through a fee-exact, queue-conservative execution engine. One result must be stated up front, because it conditions how everything else in this thesis should be read. At the retail fee tier this study pays, passive market making on this venue loses money in every configuration tested, ours and the baselines alike, because the average fill costs more in adverse selection and fees than it captures in spread. The evaluation also stacks its execution assumptions against itself (quotes rest at the back of the queue and re-price on a discrete decision grid), so this toll is measured under pessimistic rather than flattering conditions.

This loss is itself a finding, and its mechanics dictate how strategies must be scored: raw profit merely counts fills, so the questions that carry the thesis are risk-adjusted and mechanistic. What does inventory control buy? When does the learned drift signal act, and what exactly does it know? Is the funding boundary priced correctly when it arrives? That last question the sample itself cannot fully answer: its funding rate never approached the levels routinely reached elsewhere. A Monte Carlo stress suite therefore drives the same execution engine with synthetic funding paths, asking what the correction is worth where the transfer is large enough to matter.

The answers run as follows. Inventory control buys what profit and loss (P&L) cannot show: at matched participation the controlled book carries half the position risk of a naive strategy quoting at a matched width, and touches its inventory cap 0.03% of the time against 9.5%. The drift signal carries real information, but not the direction it was trained on. It improves trade selection by trading less, not the outcome of individual fills, so its effective use is defensive,

widening the quote it expects to be hit. Used that way it cuts GLT's mean daily loss by 26% and its tail loss by over a fifth on 27% fewer fills. Even the symmetric coupling, forced to lean into its forecast rather than away from it, loses less than GLT, through the same channel: the quote it widens sheds more fills than the quote it tightens attracts. The same controller reading a simple price-only drift shows no timing content at all, so the complexity of the learned signal is buying something a simple price-based rule cannot.

The funding correction behaves as the mathematics prescribes, engaging a median 815 seconds ahead, inside the predicted window, and ending on the side that receives the payment at 68% of settlements against GLT's 48%, which turns the 0.6 USDT GLT pays over the 64 days into 6.6 collected. On BTC's mild carry it earns little, but its advantage scales monotonically with carry across the Monte Carlo regimes, collecting 59.6 USDT per path against GLT's 0.3 where the rate is pinned at the suite's ± 75 basis-point cap, above BTC's exchange-enforced cap, but inside the tier thinner perpetuals demonstrably reach. Whether the two combine is the question this sample cannot close: at BTC's carry, and in a loss regime where totals track how much a strategy trades, no test available here resolves an interaction, and no evidence of one is found.

The thesis proceeds as follows. After reviewing the stochastic control, microstructure, and perpetual-futures literatures (Section 2) and detailing dataset integrity (Section 3), Section 4 develops the extended HJB framework, with proofs deferred to Appendix D. Section 5 outlines calibration, the ML pipeline, and the replay engine, setting up the ablation, mechanism analyses, and stress tests in Section 6, which answers the research question, before Section 7 concludes.

2 Literature Review

To contextualise the hybrid architecture, this review begins with the two charges that determine liquidity provision, adverse selection and the venue's fee, and then synthesises three strands of literature. The first traces the mathematical formalisation of inventory risk, from foundational models to the stochastic control frameworks that form the baseline of our execution engine. The second examines short-term alpha generation within limit order books, and the shift from linear, parametric signal models to non-linear machine learning. The third turns to cryptocurrency perpetual futures, where model-free learning approaches and the derivatives pricing literature on funding mechanisms leave the same microstructural gap, the one motivating the analytical HJB extension of Section 4.

Adverse selection, named in Section 1 as the central antagonist, is not an artifact of high-frequency markets but the founding problem of market microstructure theory. Kyle (1985) shows that an informed trader optimally conceals their information within the order flow, so that price impact scales with the very flow a liquidity provider absorbs. Glosten & Milgrom (1985) put a price on that exposure: a competitive market maker facing informed and uninformed counterparties must post a strictly positive spread simply to break even, taxing the uninformed to fund the losses it takes to the informed. The spread therefore acts, in part, as insurance against the informed, and it persists even where liquidity provision is otherwise costless. Whether it covers what it insures against is the question this thesis evaluates for every fill.

The exchange's fee sits in the same balance. Under maker-taker pricing, where the exchange

charges the order that takes liquidity and charges less, or pays a rebate, to the order that supplies it, a maker's compensation is the spread it captures minus the fee it pays. The fee therefore shapes liquidity supply rather than merely taxing it. Colliard & Foucault (2012) model that channel: fees move the prices at which limit orders are posted, and hence how often they fill, so that even a fee cut can leave traders worse off. Malinova & Park (2015) supply the empirical counterpart: when the Toronto Stock Exchange began paying liquidity suppliers a rebate, having previously charged its whole fee to the demanders, competition among makers passed the subsidy into narrower quotes. The demanders then paid a higher fee on a tighter spread, so their all-in cost did not move. These tighter spreads attracted higher taker volume, which diluted informed trades and lowered measured adverse selection. Both results work through quote adjustment, and this thesis reaches the point where that adjustment runs out: the exponential fill intensity bounds the half-spread a maker can post profitably, so a maker fee above that bound cannot be quoted around.

The stochastic control literature responded not to this informational risk but to a second factor, the risk of holding inventory while the price moves. Avellaneda & Stoikov (2008) (hereafter AS) gave that risk its standard formalisation, carrying the inventory-dependent 'reservation price' of Ho & Stoll (1981) out of the continuous dealer market and into a high-frequency setting. The construction has two steps: first an indifference valuation price that moves with the maker's own inventory, then the bid and ask quoted around it. They model liquidity-consuming market orders as independent Poisson processes whose intensities decay in the quoted distance from the reference price (or anchor). Maximising Constant Absolute Risk Aversion (CARA) utility of terminal wealth then sets immediate spread capture against the variance the inventory carries. The result is the canonical baseline of quantitative market making.

The AS framework, while seminal, reaches its quotes only by approximation, expanding its value function in inventory. Guéant et al. (2013) (GLT) solve it exactly: a change of variables, exploiting the CARA utility structure and the exponential form of the arrival intensity, turns the HJB system into a system of linear ODEs. GLT further impose an inventory bound $q \in \{-Q, \dots, Q\}$, in their own reading both a realistic risk limit and what allows the problem to be solved rigorously: it makes the ODE system finite. They obtain both exact solutions under inventory constraints and closed-form asymptotic approximations via spectral analysis.

Guéant (2017) later extends the reduction itself to general intensity functions, at the cost of the system's linearity, and unifies the CARA and risk-neutral criteria of this literature in one ODE system. We adopt the CARA utility case, where adverse price movements and inventory accumulation are penalised natively by the utility itself, with no need for an exogenous inventory penalty. The same paper also extends the framework to multiple assets. We nonetheless build on the 2013 single-asset formulation, because neither generalisation helps here: the funding extension lives entirely in one asset's wealth process, and the linearity the general-intensity case surrenders is what the settlement-boundary regime switch of Section 4 rests on.

Building directly on GLT's ODE reduction, which circulated as a preprint from 2011, Fodra & Labadie (2012) extended the framework to non-martingale mid-price dynamics. While GLT briefly address a constant drift, Fodra and Labadie generalise this to a broader class of processes, including mean-reverting Ornstein-Uhlenbeck dynamics, and numerically demonstrate how dir-

ectional bets affect the optimal quotes and P&L distribution. They achieve this by skewing the indifference price toward the anticipated future mid-price, embedding the maker's directional view directly into her quotes. While this thesis builds on that directional extension to integrate predictive drift signals, we diverge from their treatment of utility and inventory risk on two counts. First, like much of the directional literature, their formulation appends an exogenous inventory-risk-aversion penalty to the objective regardless of utility choice: even their exponential (CARA) case layers this penalty into the terminal utility. As noted, we reject the exogenous inventory penalty. In this asset class, inventory risk is not an abstract preference parameter but a tangible, deterministic cost-of-carry shock at funding settlement, which we capture endogenously within the cash flow dynamics of the wealth process. Retaining an exogenous penalty on top would double-count the same risk.

The second divergence concerns the drift itself. Where they allow it to depend on the price level, as in their Ornstein-Uhlenbeck specification, we restrict to drifts exogenous to the price: a price-dependent drift breaks the change of variables behind the ODE reduction, and preserving that tractability is the architectural premise of this thesis.

With the drift in place, we turn to the predictive signal by which a market maker anticipates adverse selection rather than merely pricing it. The existing literature typically embeds the microstructure directly into the stochastic control problem, which induces mathematical trade-offs. For instance, Á. Cartea et al. (2014) model market order flow as a multi-factor mutually exciting Hawkes process, so that influential orders cluster and carry short-term drift, at the cost of carrying the excitation intensities as additional state variables. Their headline conclusion, that a market maker who ignores short-term alpha is adversely selected by better-informed traders, is a hypothesis this thesis tests directly rather than a premise it adopts. Similarly, A. Cartea et al. (2018) demonstrate that order book imbalance predicts both the sign (buy or sell) of the next market order and the price innovation that follows it, and carry the imbalance as a state that modulates the two sides' arrival rates asymmetrically. Closest to the present work, Á. Cartea & Wang (2020) make the drift signal explicit, modelling a mean-reverting momentum process driven by market-order arrivals that enters the value function as its own state variable. Each of these embeddings exacts the same structural price: the exact ODE solutions available under CARA utility must be abandoned, and one falls back on numerical schemes or asymptotic approximations within risk-neutral frameworks employing exogenous inventory penalties.

This thesis avoids that price through a decoupled architecture that keeps the two sides' fill intensities symmetric to preserve the tractability of the GLT mathematics. Rather than expanding the HJB state space, the detection of toxic flow is outsourced to an exogenous, non-linear ML layer that aggregates microstructural footprints, such as book imbalance, order flow imbalance (Cont et al., 2014) and VPIN (Easley et al., 2012), into a single deterministic drift parameter, whose shift of the reservation price moves the maker's two quotes in opposite directions, one tightening as the other widens: the asymmetric execution profile advocated by Cartea and co-authors.

Outsourcing that signal also frees its functional form: a drift carried as a state variable must stay linear or low-dimensional, an exogenous one does not. As Sirignano & Cont (2019) demonstrate, high-frequency price formation relies on non-linear feature interactions that render

classical linear approximations sub-optimal. We test that premise in a pre-analysis, as a forecasting comparison rather than a strategy: LightGBM (Ke et al., 2017) against a classical linear (Ridge) baseline, to establish whether capturing high-dimensional interactions, such as between order flow imbalance and extreme trade intensities, yields a predictive advantage. Gradient boosted trees offer a decisive practical advantage over deep neural networks here: inference is branchwise deterministic, needs no GPU, and costs a fixed number of branch comparisons per prediction, bounded by tree count times depth, unlike the matrix multiplications of deep networks, which are latency-prohibitive for a live quoting loop.

Some ML approaches abandon the HJB framework altogether, as Sadighian (2020) does for cryptocurrency market making with model-free deep reinforcement learning. Such approaches give up the closed-form quotes and the optimality guarantee of a verification theorem, and with them the auditable link between a calibrated parameter and the quote it moves. Where reinforcement learning is instead used inside the framework, it serves as a numerical solver for dimensions the classical schemes cannot reach, as in the corporate-bond problem of Guéant & Manziuk (2019), a difficulty the single-asset formulation here does not have.

The gap that matters lies between two literatures. The stochastic control literature has largely overlooked the mechanics of cryptocurrency perpetual futures. The pricing literature on these contracts, by contrast, has characterised their economic structure and funding mechanisms, theoretically and empirically, confirming that periodic funding payments are a first-order feature (Ackerer et al., 2025; Angeris et al., 2023; He et al., 2022). That perspective ignores the execution problem: it does not address how a market maker carrying inventory should adjust their quotes in anticipation of discrete funding settlements. These settlements arrive on a fixed clock, carrying a cost of inventory that is closely forecastable ahead of time. Standard GLT frameworks cannot see that structure at all: they assume zero cost-of-carry. Bridging this gap between continuous limit order book execution and discrete funding settlement motivates the analytical HJB extension developed in this thesis.

3 Data

This study uses a high-frequency dataset of Binance USDT-margined perpetual futures on Bitcoin, self-recorded message by message from the exchange WebSocket API. The study window spans 94 contiguous *funding days*, each running from 04:00 Coordinated Universal Time (UTC) to the following 04:00 UTC, starting 19 January and ending 23 April 2026. Every boundary falls midway between the settlements at 00:00, 08:00 and 16:00 UTC, so no day is split by one. Recording began four hours before the window opens, so every rolling feature is fully initialised at the first row. The window is partitioned into a 30-day pre-analysis and initial training period, and a 64-day out-of-sample period for the daily walk-forward evaluation of Section 5.

The resulting object is large. Across the full window the recorder captured approximately 2.06×10^8 aggregate trades, each one market order's fills at a single price level, so a sweep through several levels arrives as several records. Alongside them, on a nominal 100 ms cadence, came 8.0×10^7 incremental L2 diff messages, each batching every book change in the preceding interval, and 7.5×10^7 partial snapshots of the twenty best price levels per side. These streams align into a single reconstructed market state that feeds the ML features, calibrates the HJB

framework's parameters, and serves as the limit order book for execution replay.

Standard aggregated feeds, for instance one-minute price bars, are structurally insufficient for calibrating a continuous-time control model or for extracting sub-second microstructure feature signals, so message-level L2 capture is a hard requirement. The instrument is deliberately restricted to Bitcoin. Its funding rates are less volatile and its premium deviations narrower than those of lower-capitalisation altcoins, which dampens the funding correction. The trade-off is accepted: a trustworthy replay needs a deep book and uninterrupted trade flow. That depth keeps out the artificial spread widenings and optimistic fill assumptions that contaminate backtests on thin instruments, so the control engine is validated where fills can be trusted. The stress suite of Section 6.6 then applies the funding magnitudes of thinner contracts to a controlled synthetic environment.

3.1 Acquisition and Order Book Reconstruction

Two independent servers in different locations recorded the same streams in parallel, since otherwise a single dropped connection would cause gaps in the dataset. Their recordings were merged by each stream's own identifier, so an outage at one machine is invisible unless both lost the connection at once, which is what keeps total feed downtime negligible. Four primary streams came off the feed: incremental L2 diffs, each listing, for every price level that changed, the new *absolute* quantity resting there, a total and not a change; partial snapshots of the top 20 levels per side; aggregate trade executions; and the 1-second mark price stream, which delivers the exchange's mark price together with the live funding rate estimate.

The local order book is reconstructed on a fixed 100 ms grid: incremental diffs are applied in sequence to advance the book state, and after a gap the book is re-anchored on a partial snapshot. The reconstruction keeps the top 20 price levels per side, deep enough for the high-frequency predictive features while staying computationally light for the replay engine. From this state come the reference price series and the model inputs of Section 5: the book advances every 100 ms, but features and targets are sampled onto a uniform one-second grid, which absorbs the sub-100 ms randomness in message arrival that would otherwise leak timing noise into them.

Quotes are decided on the same 100 ms grid the book is reconstructed on, on a timer rather than on message arrivals. The cadence is not ours: the depth feed is published in 100 ms batches. Deciding on a timer is the standard choice for a study that compares control models rather than chases profit: all six strategies then face the same decision instants with the same information, and what separates them is what they do with it. The cost is staleness, because the update actual cadence runs at a median of 102 ms and drifts against the grid, so order decisions observe the book as of the most recent batch, a median of 51 ms old. By high-frequency standards that is a long delay and the most pessimistic feature of the replay. Appendix C.1 therefore tests it directly, re-running the ablation with each quote submitted the moment the exchange publishes, on the book that publication carries. That arm stays a side experiment rather than the headline because a fresh book does not come with fresh inputs: the drift signal and the calibrated parameters can only be those of the previous grid point, taking them from the next one being leakage, and this residual lag does not necessarily affect all strategies equally.

3.2 Data Integrity and Gap Handling

Self-recorded WebSocket data inevitably suffers network jitter and server disconnections. Forward-filling the book across those gaps would be actively harmful, since it produces phantom fills, freezes the last-in-queue position tracking, and flatlines the order flow features during the volatility that most likely caused the disconnection. The gap protocol therefore treats missing data the way a live system treats a network partition.

Gaps in the order book are detected by update-identifier discontinuity: the reconstruction flags an interval as invalid the moment a diff's previous-update identifier proves that messages were lost. Within an invalid interval the book features are null and the row is marked invalid, any row whose feature or target window overlaps the interval is dropped from the learning set. The replay engine hard-pauses: resting quotes are cancelled, inventory is held frozen and unhedged, no P&L accrues, and quoting resumes only after a warm-up window has rebuilt the rolling features the gap invalidated. Book gaps are unrecoverable, but missing aggregate trade data is retroactively fetched from the Binance REST API, keeping trade-driven features such as trade intensity and VPIN exact across reconnection boundaries.

Exogenous slow-moving variables, namely the funding rate and the mark price, arrive on a 1 s grid and are forward-filled across gaps, since these are step functions or valuation indices rather than tradeable book state. A row is flagged as *stale* when the mark price has not been refreshed within the preceding 5 seconds, and on such a row the funding-related feature is null so that no stale funding feature signal enters the learning set, though the row itself stays valid for the book features.

How much data the protocol discards is worth settling, and at both layers the answer is minimal. At the raw feed layer, the aggregate-trade record is complete on every day, at 100% update-identifier integrity once the refetched messages are in. The L2 diff stream lost 345 seconds across the 94-day window, 0.0042% of wall-clock time. Fifteen days carry an interruption, with one out-of-sample day, 26 March, accounting for 222 of those seconds and no other for more than 35 (Appendix A, Table 5). At the feature layer, a row is invalid whenever the book state at its second boundary falls inside a gap: this discards 0.0028% of pre-analysis rows and 0.0049% of out-of-sample rows, while mark price staleness separately nulls the funding-related feature on 0.0019% and 0.0008%. Both effects are of the same order in each period, so the protocol does not silently thin one study period more than the other.

3.3 Market Conditions and Comparability

Beyond data integrity, the credibility of the walk-forward evaluation rests on the two study periods spanning comparable yet non-trivial market conditions. Figure 1 summarises the raw market across the 94-day window along the three dimensions that drive the market-making problem: price level, realised volatility, and the funding rate, with the corresponding period statistics collected in Appendix A, Tables 6 and 7.

The window opens at its price peak of roughly 93,400 USDT, then enters a sustained decline, with realised volatility peaking in a high-volume sell-off on 5 February (7.8% on the day, and almost eleven million trades), before the price stabilises in the mid 60,000s and recovers to near 78,000 USDT at the close. The pre-analysis window accordingly averages a higher daily realised

volatility (2.91%, 55.5% annualised) than the out-of-sample window (2.26%, 43.1% annualised), though the two volatility ranges overlap substantially. What matters most for the replay is that top-of-book conditions are near-identical across the split: the mean spread is close to one tick (the BTC price increment, 0.10 USDT per BTC), around 0.015 basis points of the asset price, and the mean displayed size at the two best quotes combined is 10 to 11 BTC in both periods (Table 6). That stability, with the overlapping volatility ranges, is what licenses transporting a model and its calibrated parameters from the pre-analysis window onto the out-of-sample period.

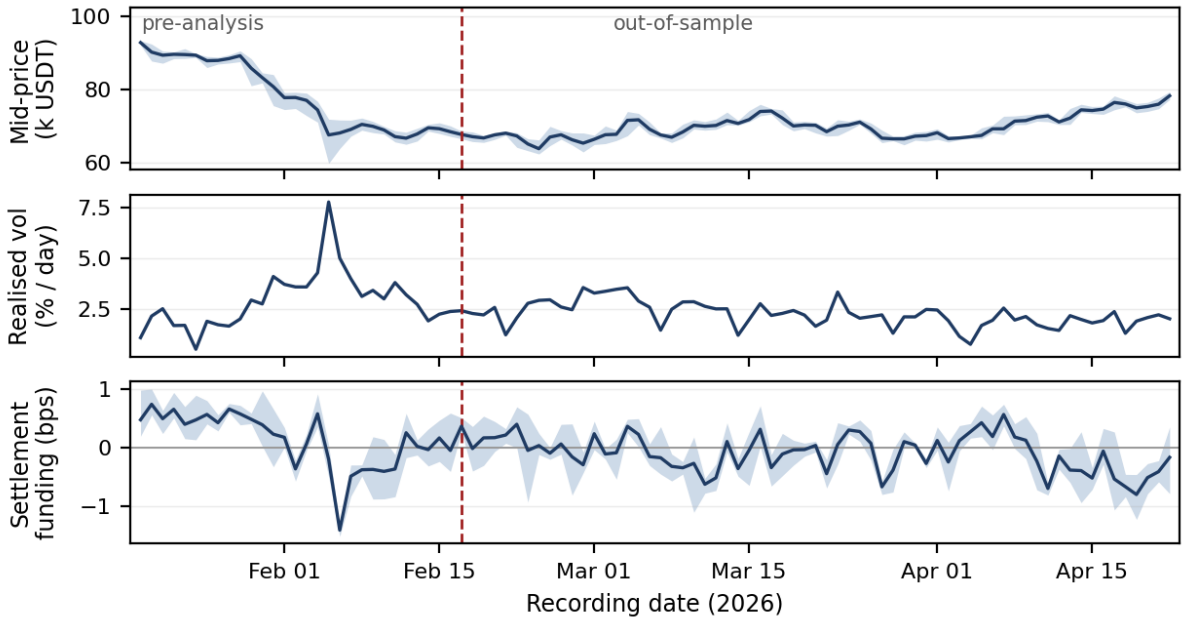


Figure 1: Raw market conditions across the 94-day study window: daily mean mid-price in thousands of USDT with its daily low–high band (top), daily realised volatility (middle), and daily mean settlement funding rate in basis points (bottom), with the shaded band spanning the day's lowest to highest settlement rate. The dashed line marks the boundary between the 30-day pre-analysis and 64-day out-of-sample windows.

The funding environment, summarised in Table 7, is equally informative. The 8-hour rate is small: period means fall inside ± 0.15 basis points per settlement and the extremes span roughly -1.5 to $+1.0$, the expected profile of a deeply liquid instrument. Its sign, however, is regime-dependent: funding is positive at 67% of pre-analysis settlements and, as the market declines, at only 43% of out-of-sample ones. The correction of Section 4 is driven by the rate live ahead of each individual settlement, never by a period average, so what matters for the evaluation is that the out-of-sample window exercises both directions in quantity. A recognised limitation of this strict chronological split is that the out-of-sample period is structurally calmer and lower-priced than the training set. However, this regime shift serves as a natural test for out-of-sample generalisation, while the algorithm's robustness to under-represented funding carry extremes is systematically validated via the Monte Carlo stress tests in Section 6.6.

4 Mathematical Foundation

This section develops the mathematical framework underpinning the proposed market-making strategy. Its overall structure is inherited from GLT: the ansatz (a trial form posited for the value function and confirmed afterwards), the reduction to a system of ODEs, the verification argument, and the asymptotic analysis. The proofs in the appendix proceed along analogous lines, adapted to two structural extensions. The first is a drift in the reference price, whose resulting quote formulas GLT state without proving and our proofs carry through every result. The second, new to this thesis, is the continuous funding penalty.

The drift is treated throughout as a generic input: any measurable forecast of the reference-price increment can play the role, however it is produced. One structural restriction applies. It must enter the control problem as a scalar, not as a function of the price itself. A signal built from the price history qualifies, since each solve freezes it at its current value. A drift that tracks the running price, such as the mean-reverting dynamics of Fodra & Labadie (2012), does not: it keeps the price inside the transformed problem and leaves partial differential equations (PDEs) where ODEs should be. The two estimators used in the ablation, a linear trend and the non-linear ML signal, are specified in Section 5.

We begin with the probability space and the state dynamics, then extend the wealth process to carry the funding rate. The HJB equation follows, characterising the value function of a market maker who maximises expected utility under inventory risk, and the work of the section is to reduce it to a tractable ODE system, read the optimal quotes off that system, and verify that the candidate behind them really is the value function. What remains is behaviour: how those quotes look far from settlement, how the strategy is stitched across successive funding epochs (the 8-hour intervals between settlements), and how it switches near settlement between two regimes that are computational rather than market ones. Proofs, the three approximations that preserve ODE-tractability, and a table of the notation are collected in Appendix D.

4.1 Probability Space and Controlled State Dynamics

We consider a high-frequency market maker operating on a single perpetual contract, without the market power to influence the price process, so that the reference-price dynamics below are exogenous. Let the market be defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$, with Ω the set of possible market outcomes, $\mathcal{F}$ the events to which a probability can be assigned, $\mathbb{P}$ the physical measure, and the filtration $\{\mathcal{F}_t\}$ recording what is known at time t. The filtration satisfies the usual conditions of right-continuity and completeness: anything knowable an instant after t is already known at t, and every event of probability zero counts as known from the start.

Let W_t be a standard one-dimensional Brownian motion under $\mathbb{P}$, and let N_t^a, N_t^b be point processes counting the market orders that execute against the maker's ask and bid quotes, with $\mathcal{F}_t$-predictable intensities λ_t^a, λ_t^b (fixed by information available strictly before t) specified below. The two are assumed, as in GLT, to be mutually independent and independent of W_t, so a trade against one of the maker's quotes carries no information about the subsequent move in the reference price: the model contains no adverse selection. The filtration is generated by these three processes, so the only information in the model is the price path and the record of trades.

The market maker continuously proposes a bid price S_t^b (at which it stands ready to buy) and an ask price S_t^a (at which it stands ready to sell), trading whenever an arriving market order consumes one of these quotes.[2] The control variables are the bid and ask half-spreads, $\delta_t^b = S_t - S_t^b$ and $\delta_t^a = S_t^a - S_t$, measured from a reference price S_t, the fair-value anchor formalised below.

No maker can absorb inventory without limit: finite capital and risk limits confine it to $q_t \in \{-Q, \dots, Q\}$, where $Q \in \mathbb{N}$ is the maximum absolute inventory. At maximum long inventory ($q_t = Q$) the maker stops quoting on the bid, and symmetrically on the ask at $q_t = -Q$. This bound is both an institutional reality and what makes the state space finite, a prerequisite for solving the control problem.

4.1.1 Reference Price Process

The standard GLT formulation models the reference price as an arithmetic Brownian motion, $dS_t = \sigma dW_t$ in the driftless baseline, with $\sigma > 0$ its volatility, and GLT note that the anchor does not have to be the raw mid-price. The derivation that follows is likewise independent of which order book observable plays the role, requiring only that the chosen anchor be well described by the dynamics below. The concrete choice of reference price is settled in Section 5.2.

Directional information enters through a drift term, giving the generalised reference-price dynamics

$$dS_t = \mu\, dt + \sigma\, dW_t, \tag{1}$$

where μ is a forecast of its drift, exogenous to the control problem like σ. Both may be constants, or may be estimated as time-varying quantities. In either case they are held at their current values within a single solve: the control problem is posed over a micro-horizon Δt short enough for that to hold, then re-solved continuously with refreshed values. This is the certainty-equivalence approximation, whose variance error is bounded in Appendix D.4.1.

4.1.2 Inventory Process

The inventory changes only when a market order executes against one of the market maker's resting quotes. Recall that N_t^b and N_t^a count those executions, or fills, so each bid fill is a purchase that raises the inventory by one unit and each ask fill a sale that lowers it by one:

$$dq_t = dN_t^b - dN_t^a. \tag{2}$$

Baseline liquidity is assumed symmetric across the two sides, with arrival intensities following the Avellaneda-Stoikov (AS) exponential specification

$$\lambda^b(\delta^b) = A\exp(-k\delta^b), \qquad \lambda^a(\delta^a) = A\exp(-k\delta^a), \tag{3}$$

so that $A^b = A^a = A$ and $k^b = k^a = k$. Here A sets the overall arrival rate of marketable flow reaching the book, and k the rate at which fills become rarer as a quote sits further from the

[2] All transactions are of constant size, normalised to 1; the framework carries through unchanged for any constant size V, with risk aversion then measured per unit traded.

reference price. The symmetry costs little in terms of expressiveness, since its quotes still skew, through inventory and through the drift, and a common k is in any case what the change of variables of Proposition 1 requires. How A and k are estimated is deferred to Section 5; like μ and σ, they may be constants or time-varying estimates, and are held at their current values within a single solve.

4.1.3 Cash Process and the Funding Extension

In the standard GLT framework, the cash process tracks only accumulated trading revenue:

$$dX_t^{\text{standard}} = (S_t + \delta_t^a)dN_t^a - (S_t - \delta_t^b)dN_t^b.$$

Perpetual futures, however, exchange a funding payment between long and short positions at discrete intervals. Let $f_{T_{\text{fund}}}$ denote the funding rate applied at the deterministic settlement time T_{fund}, under the convention that $f_{T_{\text{fund}}} > 0$ implies long positions pay short positions. The true, unapproximated cash process incorporates a discrete jump:[3]

$$X_{T_{\text{fund}}^+} = X_{T_{\text{fund}}^-} - q_{T_{\text{fund}}^-} S_{T_{\text{fund}}} f_{T_{\text{fund}}}. \tag{4}$$

The derivation that follows uses four main properties of this settlement: its timing is deterministic and known in advance, settlements recur at a known interval, the payment is proportional to the notional value of the position held at the settlement instant, and an estimate of the rate is available before it is applied.[4] Any contract with a periodic funding settlement has them, so the extension is not tied to a particular instrument.

Modelled directly, the jump imposes a matching condition on the value function at each settlement that ties the reduced system of Section 4.3 to the price level, so the ODE reduction is lost. The two approximations that follow restore it, both formally justified in Appendix D.4.

Continuous Decay Approximation. The discrete settlement jump is approximated as a continuous cash drain over the horizon leading up to T_{fund}. Let f_t denote the observable pre-settlement estimate of the rate, converging to $f_{T_{\text{fund}}}$ by the settlement and treated as locally constant over the micro-horizon. The drain decays backwards from the settlement at a rate $\rho > 0$, a timescale identified in Section 5.6 from the model's own relaxation speed, and is scaled by the constant $\tilde{\rho} := \rho / \left(1 - e^{-\rho\, \Delta T_{\text{epoch}}}\right)$, where ΔT_{epoch} is the interval between settlements. That scaling is what makes the drain and the funding transfer agree in magnitude: for a static position and funding rate, what the drain accumulates over a full epoch equals exactly the discrete transfer. The cash differential becomes:

$$dX_t = (S_t + \delta_t^a)dN_t^a - (S_t - \delta_t^b)dN_t^b - q_t S_t f_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))dt. \tag{5}$$

[3] Exchanges settle the payment on mark price notional. The derivation and the continuous drain use the reference price instead, since it updates with the book rather than on the exchange's one-second grid and differs from the mark price by far less than the approximation errors of Appendix D.4. The replay books the settlement itself to P&L at the exact mark price.

[4] The last property holds because the published rate is built from a time-weighted average of the contract's premium over the index across the settlement interval, so most of it is already fixed by the time the boundary layer of Section 4.8 begins (Binance Futures, 2026).

The settlement now enters the wealth process continuously, so the optimal quotes depend on the time remaining to T_{fund}.

Affine Constraint via Localised Reference Price. The drain introduces a cross-term $q_t S_t$. Dependence on q is harmless, since the reduced system is indexed by inventory, but S_t would survive the change of variables into the coefficients, breaking the affine structure the reduction requires. The structure is restored by a localised reference price $S_{\text{ref}} = S_{t_0}$, the value of S_t observed at the start of each solve. Substituting $S_t \approx S_{\text{ref}}$ *exclusively* within the funding term collapses it to a deterministic scalar, $F_t = S_{\text{ref}} f_t$, and gives the execution-ready cash dynamics:

$$dX_t = (S_t + \delta_t^a)dN_t^a - (S_t - \delta_t^b)dN_t^b - q_t F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))dt. \tag{6}$$

Both substitutions carry a cost, and both are forced. The affine one is cheap by construction: the problem is re-solved over rolling micro-horizons Δt with S_{ref} refreshed at each solve, so the pathwise error $|S_t - S_{\text{ref}}|$ is of order $\mathcal{O}(\sigma\sqrt{\Delta t})$. Appendices D.4.2 and D.4.3 bound the error each one induces, and show why the exact alternative fails to close in each case.

The dynamics are evaluated throughout over a localised intra-epoch horizon satisfying $t \leq T \leq T_{\text{fund}}$, so that no integration crosses a settlement. The global infinite-horizon strategy is obtained by treating the problem as a sequence of continuous control epochs, joined at successive settlements by the continuation condition of Section 4.7.

4.2 The Optimisation Problem and HJB Equation

The control problem is posed over a finite horizon $[0, T]$, which must assign a value to whatever position the market maker still holds when it ends. Inventory is not cash: closing it requires trading, and trading a position out in one go concedes price. Terminal wealth is therefore the accumulated cash X_T, plus the mark-to-market value $q_T S_T$ of the remaining position, less a linear cost of unwinding it, $\nu|q_T|$ with $\nu \geq 0$, identical for long and short positions. The unwinding cost is a modelling term: it prices the terminal position inside the objective, so that the strategy is discouraged from carrying inventory into T that it would have to dump there. It could be given an empirical value, but nothing in the derivation depends on which one, and ν is never itself charged as a cash flow.

Preferences are Constant Absolute Risk Aversion (CARA), $U(x) = -\exp(-\gamma x)$ with absolute risk aversion coefficient $\gamma > 0$, and the market maker's problem is to choose the quoting strategy that maximises the expected utility of terminal wealth. A quoting strategy is a pair of $\mathcal{F}_t$-predictable processes $\delta = (\delta_t^a, \delta_t^b)_{t\in[0,T]}$, the two half-spreads set at each instant on the information available strictly before it. It is admissible, written $\delta \in \mathcal{A}$, when it also respects a finite lower bound $\delta_t^a, \delta_t^b \geq \delta_{\min}$ for all t, almost surely.[5] The value function, the best expected utility attainable from a given state, is

$$v(t, X, S, q) = \sup_{\delta\in\mathcal{A}} \mathbb{E}_t \left[-\exp\left(-\gamma(X_T + q_T S_T - \nu|q_T|)\right)\right], \tag{7}$$

where $\mathbb{E}_t[\cdot]$ is shorthand for the conditional expectation $\mathbb{E}[\,\cdot \mid X_t = X,\, S_t = S,\, q_t = q]$ under $\mathbb{P}$,

[5] Any finite bound serves this purpose, ensuring mathematical well-posedness by preventing infinite fill intensities. Formal treatment and boundary clipping are deferred to Appendices D.2.1 and D.2.5.

the average over all paths the market can take from the current state. At time T there is nothing left to choose, and the value function reduces to the utility itself, which gives the terminal condition $v(T, X, S, q) = -\exp(-\gamma(X + qS - \nu|q|))$ for every inventory level.

Applying Itô's Lemma for semi-martingales with jump processes gives the infinitesimal change of the value function along the controlled state. Taking conditional expectations then makes the Brownian term vanish and replaces the jump increments by their intensities, leaving the expected instantaneous change of the value function. The Dynamic Programming Principle, the observation that a strategy optimal over the whole horizon must remain optimal when re-started from any intermediate state, requires this expected change to vanish at every state under the best admissible controls, formally $\sup_{\delta\in\mathcal{A}} \mathbb{E}_t[dv]/dt = 0$. This is the HJB equation. Because the inventory moves in discrete units, it is not one equation but a system of $2Q + 1$, one per inventory level. Each is a PDE in (t, X, S), and each is linked to its two neighbours through the fill terms, since a fill carries the state to an adjacent inventory level.

For interior inventory states ($|q| < Q$), on the state domain $(t, X, S) \in [0, T) \times \mathbb{R}^2$, the equation reads

$$\begin{aligned} 0 = {} & \partial_t v + \mu \partial_S v + \tfrac{1}{2}\sigma^2 \partial_{SS} v - \left(q F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))\right) \partial_X v \\ & + \sup_{\delta^a \geq \delta_{\min}} \left\{ \lambda^a(\delta^a) \left[v(t, X + S + \delta^a, S, q - 1) - v(t, X, S, q) \right] \right\} \\ & + \sup_{\delta^b \geq \delta_{\min}} \left\{ \lambda^b(\delta^b) \left[v(t, X - S + \delta^b, S, q + 1) - v(t, X, S, q) \right] \right\}. \end{aligned} \tag{8}$$

The first line collects the passive evolution: time, the drift and diffusion of the reference price, and the funding drain. Each supremum prices one quote. The bracketed difference is the jump in value when that quote is filled On the ask side, for instance, cash rises by the sale proceeds $S + \delta^a$ while the inventory drops from q to $q - 1$, and this jump is weighted by the arrival intensity $\lambda^a(\delta^a)$ of a quote resting at that distance. At the upper inventory bound $q = Q$ a bid-side fill would breach the capacity constraint, so the bid-side supremum is dropped, and symmetric logic removes the ask-side supremum at $q = -Q$. The full Itô expansion and the explicit boundary-state equations are provided in Appendix D.2.2.

4.3 ODE Reduction via the Extended Ansatz

Equation (8) is a coupled system of $2Q+1$ non-linear PDEs, one per inventory state, in the three continuous arguments t, X and S. Solved as it stands, it would demand a numerical scheme on a three-dimensional grid, replicated across every one of those coupled states. A change of variables removes two of those dimensions. The route is the one the AS literature established: posit a trial form for the value function, the ansatz, and verify that it solves the equation. CARA utility allows the mark-to-market value $X + qS$ to be factored out, and together with the exponential decay of the AS intensity functions this leaves a system of coupled linear ODEs in time alone. The change of variables is due to GLT. What is new is the funding penalty, which varies with both time and inventory, and Proposition 1 establishes that the reduction still goes through once it is present.

4.3.1 Change of Variables for the Extended HJB

Proposition 1. *Let $(\omega_q(t))_{|q|\le Q}$ be a family of strictly positive functions satisfying the following system of linear ODEs on $t \in [0,T]$, where the dot denotes differentiation with respect to time:*

$$\dot{\omega}_q(t) = \left[\alpha q^2 - \beta q + \Phi(t,q)\right]\omega_q(t) - \eta\left(\omega_{q-1}(t) + \omega_{q+1}(t)\right), \quad |q| < Q, \tag{9}$$

$$\dot{\omega}_Q(t) = \left[\alpha Q^2 - \beta Q + \Phi(t,Q)\right]\omega_Q(t) - \eta\omega_{Q-1}(t), \tag{10}$$

$$\dot{\omega}_{-Q}(t) = \left[\alpha Q^2 + \beta Q + \Phi(t,-Q)\right]\omega_{-Q}(t) - \eta\omega_{-Q+1}(t), \tag{11}$$

with the terminal condition $\omega_q(T) = \exp(-k\nu|q|)$ for all $q \in \{-Q,\dots,Q\}$,[6] *and with parameters defined as:*

$$\alpha = \frac{k}{2}\gamma\sigma^2, \quad \beta = k\mu, \quad \Phi(t,q) = kqF_t\tilde{\rho}\exp(-\rho(T_{\text{fund}} - t)), \quad \eta = A\left(1 + \frac{\gamma}{k}\right)^{-\left(1+\frac{k}{\gamma}\right)}. \tag{12}$$

Then the extended ansatz

$$v(t,X,S,q) = -\exp\left(-\gamma(X + qS)\right)\cdot\omega_q(t)^{-\frac{\gamma}{k}} \tag{13}$$

is a classical solution of the partitioned HJB equation (8).

The four parameters carry distinct roles: α penalises inventory quadratically through variance, β tilts the system in the direction of the drift, $\Phi(t,q)$ carries the funding drain, and η, built from the intensity scale A, sets the strength of the coupling between neighbouring inventory levels, since only a fill moves the system from one to the next. The proof substitutes the ansatz into the HJB equation, solves the two suprema through their First-Order Conditions, and factors out the mark-to-market component, which the CARA exponential makes possible. Appendix D.2.3 gives it in full.

That the funding penalty survives this reduction is structural rather than fortunate. Differentiating the ansatz in X returns the value function itself, so the drain contributes a term proportional to it, with a coefficient linear in q. That coefficient is free of X because CARA makes it so, and free of S because the localisation of Section 4.1.3 had already reduced the funding term to the deterministic scalar F_t. In the system of Proposition 1 that coefficient multiplies $\omega_q(t)$ alone, exactly as the drift term βq does, and leaves the coupling to $\omega_{q\pm1}$ untouched. Had S been retained there instead, ω_q would have to carry a price argument of its own, the diffusion term would return derivatives in S, and the system would have stayed a PDE. The same argument shows why the exact settlement cannot be kept. As a jump, it would enter the HJB equation as a matching condition, $v(T^-_{\text{fund}},X,S,q) = v(T^+_{\text{fund}},X - qSf_{T_{\text{fund}}},S,q)$, and substituting the ansatz gives $\omega_q(T^-_{\text{fund}}) = \omega_q(T^+_{\text{fund}})\,e^{-kqSf_{T_{\text{fund}}}}$: the price would again enter ω_q. Freezing S inside that condition and refreshing it at every solve preserves the ODE system, but the frozen value then stands in for the price at the settlement itself, an error that grows with the time to settlement and is shared across consecutive solves; Appendix D.4.3 shows it to be substantially larger than that of the drain.

[6] Setting $\nu = 0$ recovers the terminal condition $\omega_q(T) = 1$ of GLT.

4.3.2 Matrix Formulation

The ODE system of Proposition 1 admits a compact matrix formulation. Let the state vector $\omega(t) \in \mathbb{R}^{2Q+1}$ be ordered by inventory level, $\omega(t) = (\omega_Q(t), \omega_{Q-1}(t), \ldots, \omega_{-Q}(t))'$, and define the time-dependent diagonal entries $D_q(t) = \alpha q^2 - \beta q + \Phi(t, q)$.

Proposition 2. *Let $M(t)$ be the symmetric tridiagonal matrix with diagonal entries $D_q(t)$ and constant off-diagonal entries $-\eta$:*

$$M(t) = \begin{pmatrix} D_Q(t) & -\eta & 0 & \cdots & 0 \\ -\eta & D_{Q-1}(t) & -\eta & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & -\eta & D_{-Q+1}(t) & -\eta \\ 0 & \cdots & 0 & -\eta & D_{-Q}(t) \end{pmatrix}. \tag{14}$$

The system is governed by $\dot{\omega}(t) = M(t)\omega(t)$, and the unique classical solution defines a family of strictly positive functions $(\omega_q(t))_{|q|\leq Q}$ satisfying the equations of Proposition 1.

The two claims of Proposition 2 rest on different arguments. Existence and uniqueness of a classical solution on $[0, T]$ follow from the global Picard-Lindelöf theorem, whose hypothesis is met here: the system is linear, and its coefficients are continuous and bounded in time. The one coefficient that could have failed this is $\Phi(t, q)$, the only term carrying an explicit time dependence, and it carries it as a decaying exponential, bounded on $[0, T]$ and smooth in t. Strict positivity, which Proposition 1 assumes and which the ansatz needs because it raises $\omega_q(t)$ to a negative power, is established separately in Appendix D.2.4 by a comparison argument of maximum-principle type.

The funding penalty's dependence on time, harmless for existence and uniqueness, is what denies a closed form. It makes $M(t)$ non-autonomous, and its values at different instants do not commute, their product depending on the order in which they are taken. The evolution over an interval therefore cannot be collapsed into a single matrix exponential, and the solution $\omega(t) = \exp(-M(T-t))\,\omega(T)$ of the autonomous GLT setting no longer solves the system. Section 4.8 returns to this and builds the two-regime treatment on it.

4.4 Optimal Quotes

The First-Order Conditions of the Hamiltonian maximisation, derived in Appendix D.2.3, yield the optimal controls:

$$\begin{aligned} \delta_t^{b*}(q) &= \frac{1}{k} \ln\left(\frac{\omega_q(t)}{\omega_{q+1}(t)}\right) + \frac{1}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right), \quad q \neq Q, \\ \delta_t^{a*}(q) &= \frac{1}{k} \ln\left(\frac{\omega_q(t)}{\omega_{q-1}(t)}\right) + \frac{1}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right), \quad q \neq -Q. \end{aligned} \tag{15}$$

At $q = Q$ the bid is withdrawn, and symmetrically for the ask at $q = -Q$.

Their sum is the total optimal spread $\psi^*(t, q) = \delta_t^{a*}(q) + \delta_t^{b*}(q)$, defined for $|q| < Q$, and it separates into two parts. The first, $\frac{2}{\gamma}\ln(1 + \gamma/k)$, is strictly positive and independent of both time and inventory. It is where risk aversion settles against the intensity decay k, since quoting

further out earns more per fill and fills more rarely. The second, $-\frac{1}{k}\ln\left(\omega_{q+1}(t)\,\omega_{q-1}(t)/\omega_q(t)^2\right)$, weighs the two neighbouring inventory states against the current one, measuring how sharply $\ln\omega_q(t)$ curves in inventory, and it carries the whole of the spread's dependence on the state.

4.5 Verification Theorem

Proposition 1 produces a smooth solution of the HJB equation, which is a candidate and not yet an answer: it becomes the value function only once it is shown to dominate every admissible strategy and to be attained by one.

Theorem 1. *Let $\omega_q(t)$ be the unique strictly positive classical solution of the ODE system of Proposition 2, with terminal condition $\omega_q(T) = \exp(-k\nu|q|)$, and let v be the candidate of Equation (13). Then v is the value function of the control problem with the cash dynamics of Equation (6), and the controls of Equation (15) attain the supremum: no admissible quoting strategy achieves higher expected utility.*

The proof, given in Appendix D.2.5, follows an arbitrary admissible quoting strategy along the path it generates and shows that no such strategy can end up ahead of the candidate, with equality exactly when the quotes are those of Equation (15). Neither the drift nor the funding drain complicates that argument: inventory is capped and the horizon is finite, so both contribute at most a fixed, finite amount to terminal wealth.

4.6 Autonomous Asymptotic Limit

Recall that the funding penalty $\Phi(t,q)$ is the only entry of $M(t)$ carrying an explicit time dependence, the remaining coefficients being held at their current values across each solve by certainty equivalence. Far from a settlement, the exponential decay drives the funding penalty to zero ($\Phi(t,q) \to 0$), leaving an autonomous system whose diagonal entries $D_q = \alpha q^2 - \beta q$ retain only the quadratic variance penalty and the drift. Two results follow, and they are of different kinds. Theorem 2 is exact: it shows that the optimal quotes converge, as the remaining horizon grows, to a time-invariant limit expressed through a single eigenvector of M. That limit is explicit in the eigenvector, but the eigenvector is itself defined implicitly, as the solution of a minimisation. Proposition 3 then approximates it by a Gaussian, and only there do the quotes become a formula in the model parameters alone.

Because M is symmetric its eigenvectors form a basis, so the solution of the autonomous system can be written as a combination of them. Integrating backward from the horizon multiplies the eigenvector belonging to eigenvalue λ_i by $\exp(-\lambda_i(T-t))$, and the eigenvector together with that factor is an eigenmode. As the remaining horizon $(T-t)$ widens, the mode with the smallest eigenvalue outgrows every other, and the quotes settle onto a time-invariant plateau (Guéant et al., 2013). How quickly they reach it is set by the gap between the two smallest eigenvalues, the leftover from the next mode shrinking like $\exp(-(\lambda_1-\lambda_0)(T-t))$.

4.6.1 Asymptotic Limit of the Optimal Quotes

Theorem 2. *Consider the autonomous system obtained by setting $\Phi \equiv 0$, with any strictly positive terminal condition $\omega(T)$. Then the optimal quotes of Equation (15) admit time-invariant*

limits as the remaining horizon expands:

$$\begin{aligned}\lim_{(T-t)\to\infty}\delta_t^{b*}(q) &= \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right)+\frac{1}{k}\ln\left(\frac{f_q^0}{f_{q+1}^0}\right), \quad q\neq Q,\\ \lim_{(T-t)\to\infty}\delta_t^{a*}(q) &= \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right)+\frac{1}{k}\ln\left(\frac{f_q^0}{f_{q-1}^0}\right), \quad q\neq -Q,\end{aligned} \tag{16}$$

where $f^0\in\mathbb{R}^{2Q+1}$ is the strictly positive eigenvector associated with the smallest eigenvalue of the symmetric matrix M, characterised, once normalised to unit length, by

$$f^0\in\underset{f\in\mathbb{R}^{2Q+1},\|f\|_2=1}{\operatorname{argmin}}\left[\sum_{q=-Q}^{Q}(\alpha q^2-\beta q)f_q^2+\eta\sum_{q=-Q}^{Q-1}(f_{q+1}-f_q)^2+\eta f_Q^2+\eta f_{-Q}^2\right]. \tag{17}$$

The bracket is the quadratic form of $M+2\eta I$ rather than of M. The two differ only by the term $2\eta\|f\|_2^2$, which the normalisation fixes at the constant 2η. Minimising either therefore yields the exact same f^0. The quadratic form of M alone contains the products $-2\eta f_q f_{q+1}$, which carry no immediate intuition. The shift supplies the missing $\eta f_q^2+\eta f_{q+1}^2$, and the three combine into $\eta(f_{q+1}-f_q)^2$. The criterion then reads directly: the first sum evaluates the weights f_q placed on each inventory level, quadratically through the variance and linearly through the drift, while the second penalises variation between neighbouring levels. The proof, in Appendix D.2.6, establishes that the smallest eigenvalue is simple and that its eigenvector is strictly positive in every component, by a Perron-Frobenius-style argument resting only on the sign of the off-diagonal entries $-\eta$. Both are needed for Equation (16): simplicity makes the eigenvector unique, so that a single f^0 can be spoken of, and positivity keeps the logarithms finite. Summing the two half-spreads gives the total asymptotic spread $\psi_\infty^*(q)$, the analogue of $\psi^*(t,q)$ with f^0 in place of $\omega(t)$.

4.6.2 Continuous Gaussian Approximation

The discrete criterion of Equation (17) has no closed-form minimiser. Following GLT, it is relaxed to a continuous counterpart. The inventory index becomes a real variable x and the sums become integrals. The squared differences $(f_{q+1}-f_q)^2$ between neighbouring inventory levels become the squared derivative $(\tilde{f}'(x))^2$, and the two boundary terms go with them, because the relaxation pushes the inventory cap out to infinity. That last step is the substantive one. In exchange, the problem can be solved exactly.

Proposition 3. *Let*

$$\tilde{f}^0\in\underset{\|\tilde{f}\|_{L^2(\mathbb{R})}=1}{\operatorname{argmin}}\int_{\mathbb{R}}\left[(\alpha x^2-\beta x)\tilde{f}(x)^2+\eta\left(\tilde{f}'(x)\right)^2\right]dx. \tag{18}$$

Then, taking the positive branch (unique up to sign, fixed as in Theorem 2),

$$\tilde{f}^0(x)=\pi^{-\frac{1}{4}}\left(\frac{\alpha}{\eta}\right)^{\frac{1}{8}}\exp\left(-\tfrac{1}{2}\sqrt{\tfrac{\alpha}{\eta}}\left(x-\tfrac{\mu}{\gamma\sigma^2}\right)^2\right). \tag{19}$$

The proof, in Appendix D.2.7, uses the calculus of variations. Completing the square turns the criterion into a harmonic well, which is why the minimiser is a Gaussian, and the appendix establishes that it exists and is unique.

Returning to the discrete inventory grid, Proposition 3 suggests approximating the components of the discrete eigenvector, up to the multiplicative constant that cancels in every quote, by the sampled Gaussian

$$f_q^0 \approx \exp\left(-\tfrac{1}{2}\sqrt{\tfrac{\alpha}{\eta}}\left(q - \tfrac{\mu}{\gamma\sigma^2}\right)^2\right). \tag{20}$$

Substituting this Gaussian into the limits of Equation (16) and expanding α, β and η yields the closed-form approximations

$$\begin{aligned}
\delta_\infty^{b*}(q) &\approx \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right) + \left(-\frac{\mu}{\gamma\sigma^2} + \frac{2q+1}{2}\right)\sqrt{\frac{\sigma^2\gamma}{2kA}\left(1+\frac{\gamma}{k}\right)^{1+\frac{k}{\gamma}}}, \\
\delta_\infty^{a*}(q) &\approx \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right) + \left(\frac{\mu}{\gamma\sigma^2} - \frac{2q-1}{2}\right)\sqrt{\frac{\sigma^2\gamma}{2kA}\left(1+\frac{\gamma}{k}\right)^{1+\frac{k}{\gamma}}}.
\end{aligned} \tag{21}$$

The approximation is at its best near flat inventory but, as GLT note, worsens toward the capacity limits $\pm Q$, because quotes rely on ratios of adjacent components that magnify small errors. Section 4.8 gives the reason in full and draws the operational conclusion. Summing the two half-spreads gives the corresponding approximation to the total asymptotic spread:

$$\psi_\infty^*(q) \approx \frac{2}{\gamma}\ln\left(1+\frac{\gamma}{k}\right) + \sqrt{\frac{\sigma^2\gamma}{2kA}\left(1+\frac{\gamma}{k}\right)^{1+\frac{k}{\gamma}}}. \tag{22}$$

Adding the two half-spreads cancels the drift, at any drift signal strength, leaving the spread to risk aversion, volatility and the liquidity parameters alone. The inventory level cancels too, because the logarithm of a Gaussian curves by the same amount at every level and the spread is built from that curvature, which is why the expression above carries no q; both cancellations are exact in the relaxed form alone. The drift changes not how wide the maker quotes but where it wants to be: $\mu/(\gamma\sigma^2)$ is the inventory it aims to hold, long on a positive signal, short on a negative one, in proportion to the signal. Guéant et al. (2013) work the remaining comparative statics out of the same closed form, parameter by parameter.

4.7 Epoch Stitching and Continuous Operation

Everything so far ended at a fixed horizon T under the liquidation terminal condition $\omega_q(T) = \exp(-k\nu|q|)$. That deadline is an artifact of posing the problem on a finite horizon, not something a market maker necessarily faces. The only deadline that genuinely recurs is the settlement itself. Replacing the liquidation condition by the value of entering the next epoch is what lets the funding extension run indefinitely rather than once, and it is the last piece the model needs before it can be solved.

Let $\{T_{\text{fund}}^{(1)}, T_{\text{fund}}^{(2)}, \dots\}$ be the schedule of settlements. In the instant after settlement $T_{\text{fund}}^{(k)}$, the funding cash flow has been booked, the penalty Φ has fallen back to a negligible level, and the next settlement $T_{\text{fund}}^{(k+1)}$ lies a full ΔT_{epoch} away. Because Section 4.1.3 replaced the discrete

settlement by a continuous drain, the state does not jump at a settlement and the value function passes through it continuously, so Bellman's Principle of Optimality reads plainly: the value of ending epoch k is the value of beginning epoch $k+1$, and one epoch's terminal condition is the next one's continuation value.

That looks circular, since the next epoch's value is itself unknown. It is not, because Theorem 2 asks nothing of a terminal condition beyond strict positivity, which Proposition 2 already guarantees. Except in the short window running up to its own settlement, epoch $k+1$ is governed by the autonomous matrix M, so Theorem 2 applies to it directly, with a full epoch as the remaining horizon. Section 4.6 showed that what fades is the transient part of the continuation value, at a speed set by the gap between the two smallest eigenvalues of M, and an epoch is far longer than that fade requires.[7] Neglecting the thin non-autonomous layer at the far end of epoch $k+1$, the principal mode alone survives, $\omega((T^{(k)}_{\text{fund}})^{+}) \approx c\, f^0$, where the single positive scalar c captures the continuation value's projection onto f^0 and its decay over the epoch. The system is linear, so rescaling a terminal condition rescales the entire solution, and the quotes read only ratios of adjacent components, $\ln\big(cf^0_q/(cf^0_{q\pm1})\big) = \ln\big(f^0_q/f^0_{q\pm1}\big)$. Setting $c=1$ therefore costs nothing, and the condition imposed at every settlement is

$$\omega_q(T^{(k)}_{\text{fund}}) = f^0_q, \qquad q \in \{-Q, \dots, Q\}. \tag{23}$$

What this costs is an assumption already made everywhere else. Through α, β and η, the vector f^0 depends on the current values of μ, σ, A and k, where the liquidation condition depended only on k and ν. The next epoch's parameters cannot be known at the settlement, so holding them at their current values is the same certainty equivalence already made within each epoch. The liquidation penalty ν then survives only to close the bounded-horizon problem Theorem 1 is stated on. Nothing established earlier is lost: strict positivity still holds, the integrability estimate behind the Verification Theorem still holds, and the spectral representation is in fact sharpened by the new condition, all three checked in Appendix D.2.8.

The epochs then join at each settlement without a seam: integrate up to the settlement, hand over the value the next epoch begins from, and repeat. Carrying the reduced system through a settlement instead is not an option, since matching across one would put the price back into ω_q, which is what the change of variables removed. The handover is therefore not a convenience but what the reduction requires in order to run at all.

4.8 Non-Autonomous Regime Switching

Section 4.6 took the autonomous limit of $M(t)$ by sending the funding penalty $\Phi(t,q)$ to zero, which describes most of an epoch. As a settlement nears, the penalty instead grows until it is comparable to the inventory terms on the diagonal, and the argument behind Theorem 2 fails: no single matrix governs the system long enough for the solution to settle onto that matrix's principal eigenvector, so the quotes of Equation (16) cease to describe the optimum.

An epoch therefore divides in two. Theorem 2 stated its limits for a horizon that grows without bound, which leaves open whether a maker with a single epoch to run ever reaches them.

[7] As an illustration at the reference calibration of Section 5: the spectral gap is $\lambda_1 - \lambda_0 = 6.46 \times 10^{-3}\,\text{s}^{-1}$, so after an eight-hour epoch the relative weight of the first transient mode is of order e^{-186}.

It does, and quickly. The funding penalty distorts ω near a settlement, and that distortion decays as the backward solve moves away from the settlement. Its decay rate is the one Section 4.6 identified, the gap between the two smallest eigenvalues of M, and Section 4.7 showed an epoch to be far longer than that decay needs. Theorem 2 asks only that ω be strictly positive where the distortion ends, which Proposition 2 guarantees. For all but the last stretch of an epoch the asymptotic quotes are therefore right, up to a residual exponentially small in the epoch's length. In that stretch they are not: no time-invariant description survives there, and the system must be integrated numerically backwards from the settlement. Regime I and Regime II are these two treatments, switched between at the point where they stop agreeing; they divide the computation, not the market.

4.8.1 Regime I: Away from Settlement ($t \ll T_{\text{fund}}$)

Here the quotes do not depend on time. Theorem 2 gives them as a fixed function of inventory, so the whole surface of $2Q + 1$ half-spread pairs follows from one evaluation of f^0 and changes only when the parameters change.

Two routes to f^0 are available. Proposition 3 relaxed the inventory cap to infinity, so its Gaussian is defined over unbounded inventory. The Gaussian is accurate only when it is already negligible at $\pm Q$, it can then be used directly. Otherwise f^0 comes from the discrete criterion of Equation (17), a tridiagonal eigenproblem of dimension $2Q + 1$, cheap at any realistic capacity. The discrete route is the safer default, because how quickly the Gaussian decays is fixed by the model parameters and can leave it still substantial at $\pm Q$, where it is describing inventory the maker is not permitted to hold. The error is then largest near the cap, where the quotes are most extreme. Appendix E.1 measures the error, and Section 5 states which route this thesis takes.

4.8.2 Regime II: The Boundary Layer ($t \to T_{\text{fund}}$)

Inside the boundary layer the quotes come from the system itself. Proposition 2's $\dot{\omega}(t) = M(t)\omega(t)$ is integrated numerically backwards from the settlement, starting at $\omega(T_{\text{fund}}) = f^0$ by Equation (23), and Equation (15) turns the resulting $\omega(t)$ into quotes at each instant. Those quotes now depend on time, and they skew to rotate inventory off the side the settlement will charge and onto the side it will pay. A maker that kept quoting the Regime I asymptote through the layer would make no such attempt, and would meet the settlement with whatever inventory the market had left it.

Which side is penalised depends only on the sign of qF_t, because the funding penalty enters the diagonal linearly in q where the variance penalty αq^2 enters quadratically and cannot tell the two signs apart. Consider, for instance, positive inventory into a settlement where longs pay shorts. The penalty $\Phi(t, q) = kqF_t\tilde{\rho}\exp(-\rho(T_{\text{fund}} - t))$ grows with inventory, so the backward induction assigns lower $\omega_q(t)$ the more is held, and lower is worse, since the ansatz raises ω to a negative power. What the two ratios then do is opposite: $\ln(\omega_q/\omega_{q-1})$ falls, pulling the ask δ^{a*} in toward $\delta_{\min}$ to shed the position, and $\ln(\omega_q/\omega_{q+1})$ rises, pushing the bid δ^{b*} out against taking on more. Both quoted prices move down together.

4.8.3 Hybrid Architecture

Because the boundary layer solution depends dynamically on only two state variables, backward time and inventory, it can be solved once in advance and stored as a lookup table (LUT) of $\ln \omega$. Parameters enter when this table is built rather than when it is read, and the stochastic price path never enters at all. The table only needs to be rebuilt when those parameters move. Quoting is then reduced to a fast lookup and two subtractions, since Equation (15) reads differences of $\ln \omega$.

What remains is where to switch. Regime I governs while $t < t^*$, and t^* sits where the two descriptions stop agreeing: working back from the settlement, it is the last point at which the funding penalty still moves the solved quotes by less than one tick relative to the autonomous ones. The lead $u^* = T_{\text{fund}} - t^*$ is therefore set by the carry and the volatility prevailing at the time, not fixed in advance.

This completes the model: a time-invariant quote surface for almost all of an epoch, a numerically solved one in the short window before each settlement, and a value handed from each epoch to the next, so that the strategy runs indefinitely rather than to a horizon. Appendix D.3 gives the stopping rule that fixes t^*, and Appendix E the solver and the memory layout behind it.

5 Methodology and Experimental Design

The mathematical foundation of Section 4 specifies the structural form of the optimal quotes and the regime-switching architecture that implements them, taking the drift and the market parameters as given. This section supplies everything the mathematics left unspecified, in three parts: the predictive layer that generates the ML drift signal α_{ML} and the coupling that maps its raw output onto the drift the controller actually consumes (Sections 5.1 to 5.5); the causal calibration of every remaining control parameter and the replay engine that executes the quotes (Sections 5.6 and 5.7); and the experimental design under which the six strategies are compared (Section 5.8), along the Monte Carlo suite designed to evaluate the impact of higher funding rates that the quiet historical sample cannot capture (Section 5.9).

5.1 Microstructure Feature Engineering

All features are computed on the uniform one-second grid of the processed dataset (Section 3), 86,400 rows per day. The features fall into six families, each described here in plain terms, with the exact formula for every one collected in Table 8 of Appendix B.1. Every feature is causal, using only information available at or before its timestamp: every window in that table is backward-looking. Throughout, P_i^b, Q_i^b and P_i^a, Q_i^a denote the price and resting size at the i-th best bid and ask level, indexed from $i = 0$ at the top of book. Two derived prices recur: the mid-price $S_t = \frac{1}{2}(P_0^a + P_0^b)$, which Section 5.2 settles on as the reference the quotes are anchored to, and the volume-weighted micro-price $P_t^{\text{micro}} = \frac{Q_0^b P_0^a + Q_0^a P_0^b}{Q_0^b + Q_0^a}$. By shifting closer to the quote with *less* resting size (the side more susceptible to depletion), the micro-price effectively embeds the standing order book imbalance into a single metric.

Book-shape features. These describe the static state of the resting book: how the liquidity is distributed across levels, and how lopsided it is. The central static feature is order book imbalance: the normalised difference between the size resting on the bid side and on the ask side. A book heavy on bids has more buying interest waiting than selling interest and leans the next move upward; a book heavy on asks leans it downward. It is computed over four book bands $B \in \{\text{L1}, \text{L2-5}, \text{L6-10}, \text{L11-20}\}$, where L1 is the top price level and the other bands aggregate levels 2 to 5, 6 to 10 and 11 to 20, each normalised onto $[-1, 1]$. The family also includes the bid-ask spread, per-side liquidity slopes, the absolute depths as ratios to their own trailing 300 s mean, and the age of the most recent book update. The slopes measure how quickly resting size accumulates away from the best quotes, the depth ratios keep each band comparable as the price and volume regime drifts, and the book age is a freshness check.

Order flow imbalance. Two limit order books with the exact same static shape can arrive at that state from opposite directions. Because the static book-shape features cannot capture this historical trajectory, order flow imbalance (Cont et al., 2014) instead measures the active pressure reshaping the book. It does so by accumulating every signed change in resting volumes (orders added, orders cancelled and price levels stepping), over the same four bands B. Writing a prime for the previous book update (P', Q'), the signed event at book level i is

$$e_i = \underbrace{Q_i^b \mathbf{1}_{\{P_i^b \geq P_i^{b\prime}\}} - Q_i^{b\prime} \mathbf{1}_{\{P_i^b \leq P_i^{b\prime}\}}}_{\text{bid side}} + \underbrace{Q_i^{a\prime} \mathbf{1}_{\{P_i^a \geq P_i^{a\prime}\}} - Q_i^a \mathbf{1}_{\{P_i^a \leq P_i^{a\prime}\}}}_{\text{ask side}}, \qquad \text{OFI}_B = \sum_{i \in B} e_i, \tag{24}$$

the signs arranged so that a growing bid depth and a shrinking ask depth both count as positive: both are expressions of buying pressure. The feature reported at second t sums OFI_B over the ten 100 ms book updates in $(t - 1\,\text{s}, t]$.

Trade-flow features. The book displays quoted liquidity; trades show what was actually paid. Over rolling 1, 5 and 30 second windows we record the trade count; buy- and sell-initiated volume, with their signed difference and normalised imbalance; the largest single trade; and the log change in the volume-weighted average traded price from one window to the next. Together they say how heavy recent market-order flow is and which side is paying the spread.

VPIN. The volume-synchronised probability of informed trading (Easley et al., 2012) estimates what fraction of recent volume is one-sided, a proxy for the chance that the flow is informed and will adversely select a resting quote. It is measured on a volume clock: trades fill equal-volume buckets, and VPIN averages the absolute buy-minus-sell imbalance, as a fraction of bucket volume, over the last $n = 50$ buckets in our configuration. Which side initiated a trade is taken from the exchange's own record of which side crossed the spread, rather than inferred by the bulk-volume classification of the original specification, removing that estimator's main source of error. It is emitted at three bucket sizes spaced in factor-five steps, so the model sees toxicity accumulating over fast, medium and slow horizons, with the wall-clock fill time of the buckets recorded alongside as a measure of trading speed.

Price dynamics. Micro-price log-returns over 1, 5 and 30 second lookbacks give the model recent price momentum, and they stay informative even in seconds with no trade, because the micro-price moves with the resting book. Additionally, the realised volatility of these one-second returns over a trailing 60-second window is the model's proxy for the prevailing market noise.

Funding context. The final feature family situates the quote within the funding cycle. It includes the time remaining to the next settlement, the basis (the gap between the perpetual's mark price and the book mid price), and the staleness of the mark price. The explicit funding rate is excluded from the feature set to preserve the integrity of our ablation study. Because the rate acts as the direct analytical input to the control correction $\Phi(t, q)$, allowing the ML model to ingest it would cause the learned channel to overlap with the structural channel, obstructing our ability to measure their separate marginal contributions. However, by retaining the basis, a live proxy for the premium index, and the time to settlement, the model remains capable of capturing the microstructure effects of carry pressure, leaving only the deterministic pricing of the rate to the stochastic controller.

5.2 Prediction Target

Section 4.1.1 required only that the reference price be well described by the reference diffusion, and the mid price named in Section 5.1 is the observable that instantiates it, because the price anchor and the prediction target must be the same object. The mid carries no queue-imbalance skew, and that imbalance is forecast information which must reach the controller through the features and the drift channel, never through the price itself, where it would set the reservation price chasing moves inside the spread. It also sits better with that requirement than the alternative: the mid moves only when a best quote moves, whereas the micro-price moves whenever the sizes behind those quotes change too, adding to its increments a bounded term the arithmetic Brownian reference does not have. The choice of target naturally follows from the choice of price. Predicting the micro-price drift instead would add a mechanical bid-ask bounce that carries no mid-price information, and would buy nothing, since the two candidates correlate above 0.9996 at every horizon mentioned below, their accuracy is indistinguishable fold to fold, and the bounce is worth at most 0.7% of the achievable R^2 (Appendix B.2). The micro-price therefore enters only as a feature.

The drift signal α_{ML} is trained to predict the drift of the mid-price over a horizon h,

$$\text{drift}_h^{\text{mid}}(t) \;=\; \frac{S_{t+h} - S_t}{h}, \tag{25}$$

expressed in price units per second, the units of the drift μ in the control problem. It is not fed to the quotes in that raw form: how the prediction is mapped onto the drift the stochastic controller consumes is described in Section 5.5.

Labels are computed at the four candidate horizons of 1, 5, 10 and 30 seconds, and dropped whenever the forward window $[t, t + h]$ crosses a data gap, since S_{t+h} cannot be formed across one ($\sim$ 0.01% of rows). The operational horizon is fixed at $h = 10\,\text{s}$, matched to how long liquidity-taking flow takes to reach a quote at the distance we post, so that the forecast spans the interval that decides whether a fill is a good one. Under the last-in-queue assumption of Section 5.7, our order joins the back of the queue displayed at its price and is advanced only by trades that consume the size ahead of it. This wait is driven by the time the queue ahead takes to clear: on the pre-analysis days alone, the typical top-of-book queue clears in about 8 seconds, and $h = 10\,\text{s}$ is the closest candidate in log-space (Appendix B.3). Quoting deeper than the top

of book only lengthens the wait, so 8 seconds is a floor rather than a point estimate.[8]

An individual quote lives nowhere near that long. The ideal quote follows the reference price, which at the calibrated volatility moves by more than a tick inside a single 100 ms step, so a resting order is usually cancelled and re-posted almost at once. The horizon is unaffected by that, because what re-posting resets is our place in the queue, not the rate at which liquidity-taking flow arrives at the distance we are quoting, and it is that arrival the forecast has to span.

5.3 Statistical Characterisation

Two modelling choices depend on properties of the data rather than on assumption: which features must be transformed to transfer across walk-forward folds, and whether the target warrants a non-linear predictor over a linear one. A per-day diagnostic battery settles both, with the full tables in Appendix B.4.

Stationarity is assessed using the Augmented Dickey-Fuller (ADF) test, which identifies only two required feature transformations.[9] The book depths enter as ratios to their trailing mean, rendering them stationary by construction. The four non-stationary VPIN series, namely the three bucket-fill times and the macro-scale toxicity estimate, are first-differenced.

The target is stationary on every day at every horizon, so a learned mapping can transfer across folds, but it also carries structure a fixed linear predictor cannot represent. It is conditionally heteroskedastic, with the Engle ARCH-LM test rejecting homoskedasticity on every day, and heavy-tailed, with excess kurtosis of 22 at the ten-second horizon and even 98 at one second. While heteroskedasticity strictly describes the second moment, it dictates the signal-to-noise ratio of our features. For example, order flow imbalance is a much noisier predictor of price drift in high-volatility regimes than in calm ones. Because the optimal response depends on the prevailing market state, this relationship is inherently non-linear. A globally fixed linear coefficient cannot capture this state-dependency, whereas a tree model naturally captures it by splitting on regime-identifying features. That is a hypothesis rather than a demonstration, and Section 5.4 tests it head to head against a linear baseline. The drift signal must in any case be conditional rather than a level: the target's unconditional mean is a negligible -0.0018 price units per second, identical at all four horizons (Table 13).

5.4 Predictive Microstructure Modelling

Sections 5.1 and 5.2 fix what the model must map, features in and drift out, and Section 5.3 has handed over the question of whether that mapping needs to be non-linear at all. What remains is the function class, and the protocol that trains it.

Deep networks are excluded on latency grounds (Section 2): the one-second drift signal cadence is a concession of the replay, and a live engine refreshing on the book update instead

[8] A natural theoretical alternative would be the mean wait $1/(A\,e^{-k\delta})$ implied by the fill-intensity model itself. However, this is unsuitable for defining the machine learning target because it yields a dynamic wait time that varies with the quote distance δ, whereas the predictive horizon h must be a single, fixed constant.

[9] We cross-check the ADF results against the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, leveraging their opposite null hypotheses for robustness. However, the decision to first-difference relies on the ADF test, as it explicitly tests for the unit root that differencing removes.

would find a matrix multiplication per prediction the wrong cost structure. LightGBM (Ke et al., 2017) is the non-linear contender that does fit it, since a prediction reduces to a fixed number of branch comparisons.

That justifies the choice of non-linear model, but not the necessity of non-linearity itself. The primary theoretical advantage of a tree-based model is its capacity to capture feature interactions. We test this hypothesis directly against a Ridge regression baseline using the 30 pre-analysis days. To ensure a realistic evaluation, this preliminary test employs a rolling walk-forward out-of-sample structure that closely mirrors the main experimental design (which later evaluates 64 out-of-sample days using trailing 30-day training windows). The Ridge model receives the same feature set, less the exactly-collinear columns it must drop (Appendix B.5). The loss is squared error throughout, in this comparison and in the production, and its minimiser is the conditional mean, which is how the drift μ of Section 4 is defined.[10]

The verdict of this pre-analysis depends on the drift horizon. LightGBM's advantage is largest at one second, where its out-of-sample squared-error loss is significantly below Ridge's (Diebold-Mariano statistic +34.5); it narrows to +11.4 at five seconds and +4.45 at the operational ten horizon, still significant and winning 15 of the 16 folds; and it reverses at thirty seconds, where Ridge is significantly better (-2.58, $p = 0.010$). The direction is what one would expect. The predictable part of the target shrinks as the horizon lengthens, so there is less non-linear structure left to find, and a flexible model with nothing to fit pays for its flexibility in variance. Non-linearity pays where the signal-to-noise ratio remains high, and that includes the horizon the strategy actually quotes on (Appendix B.5). The ablation proceeds with LightGBM throughout.

5.4.1 Training Protocol

The production model is retrained daily on a 30-day rolling window. The first window is the 30 pre-analysis days and supplies the drift prediction model for the first out-of-sample day. Every later window slides forward by one day and drops its oldest, giving 64 windows, one per out-of-sample day. Hyperparameters and the boosting round count come from an expanding cross-validation inside the training window itself: three inner folds, validating on days 21 to 23, 24 to 26 and 27 to 30 of the window. Early stopping is applied independently to each validation split, yielding three optimal boosting round counts. For the full-window fit, the final model is trained using the maximum of these three optima. Because the inner folds train on fewer days (20 to 26) than the full 30-day window, they are more likely to trigger early stopping prematurely compared to a model trained on the complete dataset. Selecting the largest of the three optima therefore provides the least downward-biased estimate for the final iteration count.

The search itself runs weekly rather than daily. One window in seven re-runs the inner cross-validation with Optuna in full, producing both a new hyperparameter configuration and a new boosting round count. The next six windows hold those values fixed and train an entirely new model on their own 30 days: only the hyperparameter search is weekly, the actual model fit is daily. Even the last window in a cycle still shares 24 of its 30 days with the window the

[10] Retraining the whole pre-analysis pipeline under a Huber objective changes nothing a Diebold-Mariano test can detect ($p = 0.57$, Appendix B.5).

configuration was tuned on, so a reused configuration is being applied to almost the data it was tuned on. A transfer experiment confirms that even the worst-case performance degradation is negligible. On the pre-analysis folds (following the same simplified walk-forward evaluation described earlier), training a fold's model under hyperparameters tuned on a window that shares no days with it at all moves out-of-sample R^2 by less than a tenth of its fold-to-fold standard deviation (Appendix B.6). The tuned configurations are not themselves stable, swinging across much of the search space from window to window, but the performance they produce is. Seeds and the exact search space are fixed in the code (Appendix F).

To prevent data information leakage, we purge any training row within τ seconds of a transition to a validation or out-of-sample set (De Prado, 2018). The purge window τ is re-estimated on each rolling window as the time the slowest feature's autocorrelation takes to decay (a median of 15 minutes across all 64 windows). This process discards under 0.06% of training rows (Appendix B.7).

5.4.2 Model Interpretability

An ensemble of several hundred trees is not readable by inspection, so the evaluation carries its own interpretability layer. SHapley Additive exPlanations (SHAP; Lundberg & Lee, 2017) split each prediction of α_{ML} into per-feature contributions, and we ask two questions of them: which features carry the signal, and how the attribution shifts as a settlement approaches. The model is given the funding clock but never the funding rate, so whatever its attribution does toward the settlement is inferred, not given. The attributions are reported in Section 6.4.

5.5 Instantiating the Drift

The mathematics of Section 4 consumes a single scalar drift μ. Two concrete estimators instantiate it in the ablation. The non-ML benchmark is a first-order autoregressive forecast, AR(1), of the same quantity the ML model predicts: the mid-price change over the coming $h = 10\,\text{s}$, regressed through the origin on the change over the preceding 10 s, $\hat{\mu}(t) = \beta\,(S_t - S_{t-h})/h$ in price units per second. The price-only directional input is fitted exactly as the ML model is, on the same rolling window, the same purge and the same test days, so target, horizon, and the two estimators differ only in their feature set and their function class. Because the earlier comparison against a Ridge baseline already validated the purely predictive advantage of the non-linear function class, this AR(1) benchmark serves a pure operational purpose: it isolates whether the complex, multi-feature drift signal actually improves the trading policy when integrated into the stochastic controller.

The ML drift signal is the LightGBM prediction α_{ML} of Section 5.4, over that same horizon. Both pass through the identical coupling described next, so the machinery is not what separates them. Their natural scales differ, so at a common horizon they drive the quote by different amounts, and the matched-participation sweep of Section 6.2 is what compares them at equal turnover rather than at whatever turnover their dispersions happen to produce.

A drift signal cannot be fed to the quotes raw, because of a horizon mismatch. The asymptotic quotes of Theorem 2 price a *permanent* drift: feed the controller a drift μ and its reservation price shifts by μ times an implied holding horizon h_{eff}, the controller's own timescale. That ho-

rizon is not a free parameter. It is measured by rebuilding the exact discrete eigenvector f^0 of Section 4.6 with a small test drift already in it, and reading off how far the resulting quotes shift per unit of drift. It is remeasured whenever the market constants roll, since it depends on σ, A and k. At the pre-analysis calibration of Section 5.6 it is $h_{\text{eff}} \approx 168\,\text{s}$. The prediction, however, is a forecast over $h = 10\,\text{s}$, refreshed every second, so it never describes a permanent move at all. Treating it as one overstates the warranted response more than sixteenfold. The production coupling instead prices the signal at its own horizon: the reservation price moves by the expected price change over the signal's lifetime, where $\hat{\mu}$ is whichever estimator the strategy carries,

$$\text{shift} \;=\; \text{clip}(\hat{\mu}\, h,\; \pm c_1)\,, \qquad c_1 = \frac{1}{\gamma}\ln\!\Big(1+\frac{\gamma}{k}\Big)\,. \tag{26}$$

The cap is c_1, the inventory-independent term of the half-spread, so that the drift signal on its own can move a quote no further than the base, a limit that binds on 0.5% of the ML strategies' out-of-sample quotes. To reach the solver the shift is translated back into the drift that produces it, $\mu = \text{shift}/h_{\text{eff}}$. The mathematics stays untouched, only its input is now horizon-consistent. We refer to this symmetric form as the *horizon* coupling: a positive signal tightens the bid and widens the ask in equal measure, exactly as Theorem 2 prescribes for a drift.[11]

The second form, the *defensive* coupling, keeps the widening and discards the tightening. A drift already widens the quote on the side the signal marks as threatened, and defensive leaves that alone. It changes the other quote instead, flooring it at the half-spread it would have had with no signal rather than letting the drift tighten it. Nothing in Theorem 2 prescribes asymmetric quoting: the theoretical drift naturally widens the threatened side, but it does not mandate a floor on the favoured side. Defensive is therefore the horizon coupling with the chasing half taken out, and comparing the two measures what that half is worth: whether a weak directional signal pays a passive maker more by taking the fills it favours or by refusing the ones it warns about. The horizon coupling is the headline throughout, and defensive is run beside it to settle that (Section 6.4).

Defensive's asymmetry does not come out of the control problem. The horizon coupling is the exact solution to the HJB objective function, and flooring one of its two quotes leaves the defensive variant the optimum of none, with no theoretical basis to calibrate the floor. What the floor encodes is a belief the model does not hold: that the drift signal describes the flow about to fill us rather than the price itself, so that a fill on the marked side should be refused rather than a fill on the other side chased. It is therefore evaluated purely as an ad-hoc heuristic applied over the theoretical quotes.

5.6 Calibration of the Control Parameters

The controller reads two kinds of parameter. Volatility and the fill intensities are estimated on a rolling window that uses only data from before the quote it prices, so they move continuously

[11] One approximation is involved: h_{eff} is a small-signal slope, so the response it summarises is linear only for small signals, flattening as the eigenvector presses against the inventory cap. Measured against a full finite-horizon solve in which the drift genuinely decays on the signal's own timescale, at the median operating signal the deployed quote sits within a tick of that optimum at zero inventory and within four ticks at the inventories the strategy typically holds, the residual reaching seven ticks only in the outermost inventory states. At the strongest drift signals the deployed quote under-responds, the conservative direction.

through the evaluation days. The second kind are the control problem's own hyperparameters, distinct from the ML model's: the boundary-layer rate ρ, the drift signal horizon and coupling, the risk aversion, the inventory cap and the quote bounds. These are set once on the 30 pre-analysis days and never move again, so no evaluation day informs any of them.

Lot convention. The mathematical framework of Section 4 is written for a generic indivisible trade unit, one lot of size V. In the replay we fix $V = 0.005$ BTC, one size increment above the median aggressing trade of 0.004 BTC on the 30 pre-analysis funding days and five times the exchange minimum, so that the price-impact-free fill assumption is credible at the scale of a single quote. Inventory is the integer lot count $q_t \in \{-Q, \dots, Q\}$ with $Q = 10$, capping the position at ± 0.05 BTC, a deliberately tight risk limit that also keeps the backward-induction dimension $N_q = 2Q + 1 = 21$ small.

Prices are quoted in USDT per BTC throughout: the reference S, the half-spreads $\delta^{a,b}$ and the funding scalar F_t directly, the volatility σ per square-root second, and the drift α_{ML} per second. The decay rate k is measured in inverse USDT per BTC, written $(\text{USDT/BTC})^{-1}$, so that the product $k\delta$ is a pure number. Inventory, by contrast, is counted in lots, which is what makes $\gamma = 2 \times 10^{-5}$ a risk aversion per lot rather than per unit of cash.[12] Realised P&L, exchange maker and taker fees, and the discrete funding transfer at T_{fund} are all booked on the true notional $q_t V S_t$, never on a unit-lot proxy.

Volatility σ. The HJB consumes the realised volatility of one-second mid-price increments over a trailing 60 s window, matching the arithmetic Brownian reference dynamics. The estimator is gap-aware: increments spanning a data gap are dropped and σ freezes at its last value rather than extrapolating through a feed outage. Three alternatives were checked before settling on it. A sparse-grid variant and an exponentially weighted variant agree with ours within 9% at the median, so the level is not an artefact of the estimator. A jump-robust bipower variant understates volatility by about 30%, and it was rejected once that deficit had been traced to tick discreteness rather than to jumps (Appendix B.8). This 60 s feed prices quotes and is a different object both from the daily realised volatility of Section 3, which describes market regimes, and from the realised-volatility feature of Section 5.1, which the model reads off micro-price log-returns.

Fill intensities (A, k). The exponential intensity $\lambda(\delta) = Ae^{-k\delta}$ is the load-bearing empirical object of the framework, and the standard way of estimating it is subtly wrong for a queue-based exchange. The obvious estimator counts every trade print that reaches a quote resting δ from the mid, but as Section 3 notes, one market order sweeping several price levels is published as several records, so a single sweep is counted many times over, while a resting order it swept through can fill only once. The resulting overstatement is more than thirtyfold for the tightest quotes and still sevenfold at the distances the controller usually quotes, and no single rescaling repairs it, since it shrinks as the quote moves out. We instead measure the object the formulas actually assume, the fill rate of a freshly posted last-in-queue order. *Virtual* orders, orders posted only on paper and never sent to the market, are placed every 60 s on a grid of distances reaching 30 USDT per BTC, by which point the fill rate has fallen tenfold, and each is followed

[12] A conventional CARA on wealth in USDT is recovered by $\gamma_{\text{USDT}} = \gamma/V = 4 \times 10^{-3}$. The rescaling is exact rather than approximate: the quotes are scale-homogeneous, so the per-BTC solve with integer lots reproduces the fully per-lot solve to machine precision.

on the trade clock until it fills or expires under the same queue mechanics the replay engine applies to real quotes. These per-distance fill rates are then fitted by $\ln \lambda = \ln A - k\delta$, an order that expired unfilled contributing the time it was exposed rather than being dropped, giving $A = 0.274$ per side and $k = 0.090\,(\text{USDT/BTC})^{-1}$ on the pre-analysis window.

Because the whole framework rests on this one functional form, both of its assumptions are tested rather than asserted (Appendix B.9). The exponential shape holds: across 60 distance bins and five million tracked orders the log-rate fit returns $R^2 = 0.998$, and although a quadratic term is statistically detectable on a sample that size, it moves the fitted rate by under 4% over the distances the controller usually quotes. Fitting the bid side and the ask side separately, as a check on the symmetry $A^b = A^a$, $k^b = k^a$ the model imposes, gives differences of 3.7% in k and 1.1% in A, so treating the two sides as one costs the fit almost nothing. The replay engine consumes a rolling version of the same estimator, refitted through each day on a trailing window[13] On the pre-analysis window itself that rolling fit reads a slightly steeper $k = 0.101\,(\text{USDT/BTC})^{-1}$ against the pooled 0.090, so the decay the engine quotes on is the tighter of the two. The pooled fit is the reference calibration, the fixed parameter point at which later solver validations and latency measurements are evaluated. The rolling window size, the refit stride, and the lifetime of a virtual order are all set on the pre-analysis days and derived in Appendix B.9.

Boundary-layer rate ρ. The funding drain needs one timing input: how far before a settlement the controller should start pricing the transfer that is coming. The rate is a modelling choice rather than an estimate, and we take it from the model itself. Looking backward from the settlement, the terminal condition's influence on the quotes fades as the system of Section 4.6 settles onto the eigenvector of M that dominates far from any settlement, and the rate at which it fades is the gap between the two smallest eigenvalues of M. That gap is the natural choice for ρ, because it is the window within which the controller can still respond to what is coming: a faster rate would compress the response into a layer shorter than the quotes take to move, and a slower one would start pricing the settlement long before a strategy that can see it coming would have any reason to act. At the pre-analysis calibration it gives $\rho = 6.46 \times 10^{-3}\,\text{s}^{-1}$: the drain's weight falls to about a third of itself every 155 seconds, so nearly all of it lands in the last few minutes before a settlement. It shapes only *when* anticipation concentrates, never how much changes hands, and it is held fixed across the funding-aware strategies and across the 64 days: re-estimating it as σ, A and k roll would stretch and compress the layer on transient inputs. The selection argument is given in full in Appendix D.4.2, and the census verifying ex post that the resulting layer is thin against the epoch in Table 22.

Quote bounds and the liquidation penalty. The admissibility bound $\delta_{\min}$ is set to zero, so a quote may reach the anchor but never cross it. Under extreme inventory, a strong predictive signal can activate this bound and push the computed half-spread below zero, pinning the quote at the anchor.[14] The terminal liquidation penalty ν of the finite-horizon formulation is not used. Setting it would apply to every strategy alike, so the comparison is fair either way and leaving it out is the simpler choice. The replay books the actual taker fee on any forced

[13] Windows too sparse to support a fit use the last valid estimate; under 4% of out-of-sample time.

[14] Crossing the opposite side of the market is a separate constraint, enforced by the replay engine's post-only clamp (Section 5.7): a quote that would execute on arrival takes liquidity rather than providing it, and the engine holds it one tick inside the opposite best quote.

end-of-day liquidation. The two inputs held fixed rather than re-estimated as the sample runs, γ and ρ, are swept in the sensitivity analysis of Section 6.8.

5.7 The Execution Engine

All strategy results are generated offline in Python. However, high-frequency backtests are only credible if their computational footprint fits a live market. To prove deployability, the logic is mirrored fill-for-fill in a hybrid C++/Python execution engine (Figure 2). This C++ layer acts as a strict verification tool, confirming that the offline results, whether on recorded data or synthetic price paths (Section 5.9), are mathematically and computationally achievable in production. The Python engine and its C++ twin agree to 10^{-6} per fill event at three levels: 1,000 randomised fill-core scenarios, a synthetic day run end to end, and a full real recorded day on which all six strategies together reproduce every one of its roughly 117,000 fills.

The architecture explicitly divides the workload. Python handles the heavy research math (predictive signals, calibrations, and boundary-layer lookup tables (LUT)), while C++ manages the latency-critical hot path (book state, HJB quotes, and fill matching). In live trading, they communicate asynchronously via ZeroMQ, using atomic pointer swaps to update LUTs in locked memory without blocking the quoting path (Appendix E.3).

For this thesis, the engine runs in offline replay, consuming identical pre-computed artifacts through the same in-memory interfaces.[15] Consequently, the exercised quoting path is exactly what a live system would execute. It differs from a live environment in only two off-path respects: data transport is simulated, and order book reconstruction is processed offline in Python rather than incrementally in C++.

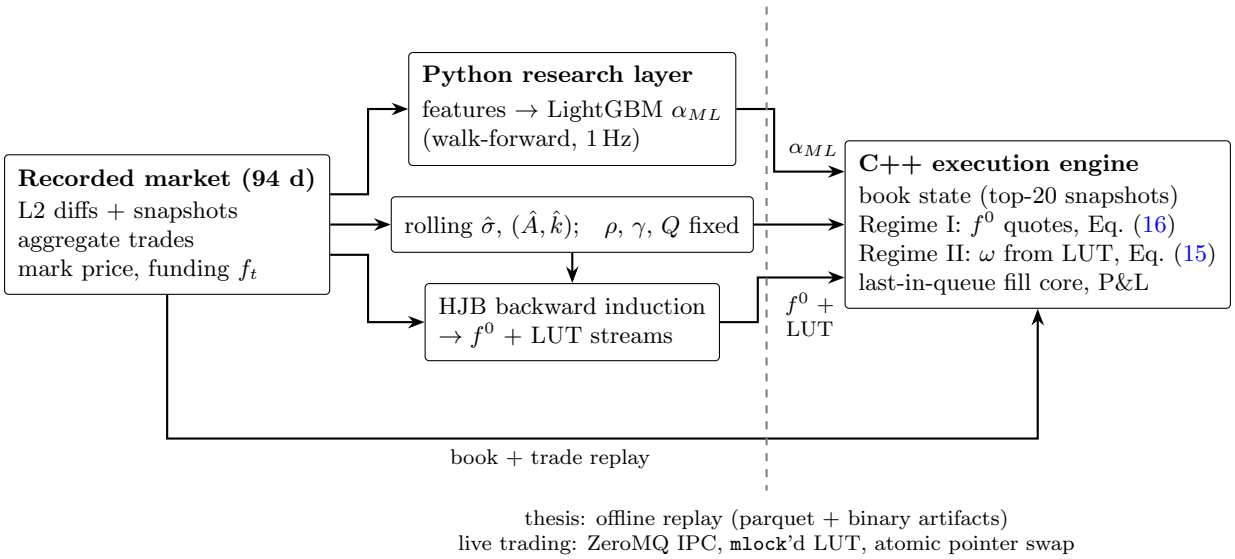


Figure 2: The hybrid C++/Python architecture. The Python layer owns everything trained, calibrated, or solved (signal, calibrations, boundary-layer LUTs); the C++ engine owns the per-quote hot path. The dashed boundary marks the data transport: replay consumes the identical artifacts from disk that the live system would receive over ZeroMQ.

[15] Both the far-from-settlement eigenvector and the near-settlement LUT are solved in Python and streamed to the C++ engine, rebuilt only when inputs move; see Appendix E.3.

Table 34 collects the latency measurements that justify our core architectural decisions. A quote has to be ready before the book changes again, and the C++ engine produces one in 3 to 4 ns against 0.3 to 1.3 μs for the same two lookups in the Python reference, a factor of 80 to 400. That difference is the whole case for writing the quoting path in C++, and the whole of what C++ is needed for. Everything else is heavy numerical work of the kind Python's scientific libraries already perform at compiled speed, and none of it runs more than once a second: the eigenvector far from a settlement solves in 27 μs, and the near-settlement problem, where the ODE system of Section 4 is integrated backward from the settlement, and stored as a LUT indexed by inventory and by time remaining, takes 64 ms to rebuild in full (Appendix E.2). There is nothing left for C++ to win there, so the drift signal, the calibrations and the backward induction stay in Python in the live trading design too. That rebuild fits only because the problem was reduced first: the ODE system solves in 38 ms (without the LUT storing), where the un-reduced PDE it replaces takes 0.60 s on a fixed grid of 512 price points, sixteen times longer than the ODE solve.

Our replay engine is built to under-promise. The recording says which trades happened, but it can never say whether one of them would have reached our order had it been resting there, and nothing recovers that. Every such gap is therefore closed against us, so on everything the recording can settle, queue position, fills, fees and reaction speed, the results are a lower bound on what this quoting logic would have achieved live. Quotes reprice on the 100 ms grid of Section 3. Fills occur on the trade clock: an order fills only when a recorded trade printed at or through its price, never on a probabilistic draw. The drift signal updates at an even lower frequency: it is produced once a second, so one prediction prices ten consecutive quote updates. Order latency itself is set to zero, so a quote decided on one grid step is live at that step. The reaction cost is already carried by the grid rather than by the wire: a decision cadence of 100 ms dominates any plausible message latency, so the grid, not the transport, is what makes the strategy slow. Every resting order joins the back of the displayed queue (*last-in-queue*) and its queue position advances only as size ahead of it is traded, never as that size is cancelled; when we move a quote, the resting order is cancelled and a new one joins the back, exactly as a cancel and replace does live. A sweep, aggressive volume trading through the quoted price in one stroke, fills the order by at most the volume that actually traded through it.

Two variants of the queue model, each run for the standard GLT strategy on three out-of-sample days against a baseline that reproduces the headline run exactly, price the two choices separately (Appendix E.4). Giving every order priority at the front the moment it is placed, the most generous assumption available, leaves net profit per fill unmoved, so last-in-queue costs us nothing we would otherwise have earned. Suspending the cancel and replace rule instead, so that a resting order keeps its queue position until the ideal quote has moved 5, 20 or 50 ticks away from it, makes matters worse at every step. The asymmetry carries the message: on this venue queue priority is nearly worthless while reaction speed is expensive. The binding execution constraint is therefore the 100 ms decision grid itself. A faster live system would do better, not worse, so the grid is a conservative choice. The strategy's own market impact is assumed zero. At a 0.005 BTC lot against the 10 to 11 BTC displayed at the two best quotes (Section 3) this is a small assumption, but it is an assumption: a real order would sit in the

book and change what others do. The reference dynamics of Section 4 assume the same thing, in common with the market-making literature this work builds on.

P&L is booked exactly, on Binance's retail (VIP 0) schedule: a 2 basis-point maker fee on every fill, a 5 basis-point taker fee on any forced end-of-day liquidation, and the discrete funding transfer $-q\,V S_{\text{mark}}\,f_{T_{\text{fund}}}$ (position in lots times the lot size V times the mark price, times the settled rate) applied at the exact settlement millisecond. The continuous drain of Section 4 exists only inside the HJB that shapes quotes, it never touches the booked cash.

In the thesis, we deploy the exact discrete eigenvector in Regime I rather than the Gaussian continuum form of Section 4.6, and the reason is measurable. The Gaussian approximation assumes that the inventory envelope it produces dies away well inside the position limits. At this calibration it does not, and the quotes it returns are wrong by up to 62 ticks at the largest inventories the strategy is allowed to hold (Appendix E.1).

Furthermore, baking α_{ML} into the LUT would demand a full rebuild at every signal update, so it enters as a first-order correction instead: the solver stores how the quotes respond to a small drift, and the live signal scales that stored response. Against an exact solve with the drift built in, the shortcut is off by at most 2.4 ticks at the median operating signal, and re-running full replay days with the drift baked exactly into every LUT rebuild moves terminal P&L by 0.4% , so what it costs the quotes reaches no reported result.

5.8 Experimental Design and Evaluation Metrics

The core empirical design is a six-strategy ablation, evaluated over the 64-day out-of-sample window and ordered so that each mechanism can be isolated by a pairwise contrast within the model family.

Strategy 1, naive constant spread, is a purely static quoting baseline, symmetric at all times and bounded only by the position cap. It is run at a fixed half-spread of the GLT family's own median quoted half-spread of 8.6 USDT per BTC, so that the comparison of quoting logic is less confounded by quoting width. **Strategy 2, standard GLT**, is the classical GLT framework on the mid-price reference (S_t), with zero continuous drift ($\mu = 0$) and no funding correction or settlement boundary knowledge. **Strategy 3, GLT plus a fitted price-only drift**, instantiates μ with the AR(1) forecast of Section 5.5, which reads no order book. With target, horizon and fitting protocol held fixed, the 3-versus-4 contrast asks whether the richer feature set and the more flexible model, which do predict better, are worth more once the forecast has passed through the stochastic controller. **Strategy 4, GLT plus LightGBM drift**, instantiates μ with α_{ML}. **Strategy 5, GLT plus the endogenous funding correction**, is the structural HJB extension for discrete settlement in isolation ($\mu = 0$). **Strategy 6, the full hybrid**, combines the LightGBM drift with the funding correction; the 6-versus-5 and 6-versus-4 contrasts measure each component's marginal value on top of the other, and their interaction term measures complementarity.

A *placebo* variant re-runs each drift strategy with its signal's values permuted in time within each day (fixed per-day seed, timestamps unchanged), preserving that signal's distribution while destroying its timing, so the real-minus-placebo gap isolates its timing content from the mechanical effect of sometimes quoting wider. A *matched-participation* sweep traces each family's

loss against its fill count by varying quote width (the naive family) or the signal-pricing horizon h (the ML and AR families), so strategies can be compared at equal turnover rather than at whatever turnover their parameters happen to produce (Section 6.2). Strategies 3, 4, and 6 are additionally run under both couplings of Section 5.5 (symmetric horizon and pre-registered defensive).

The research question is posed on the risk-adjusted axis, and the metric set is chosen to resist two specific failure modes of naive market-making evaluation. The first primary metric is the **post-trade markout**: for a fill at price p with direction s (+1 buy, −1 sell), $\mathrm{mo}_\tau = s\,(S_{t+\tau} - p)/p \times 10^4$ basis points at $\tau = 5\,\mathrm{s}$.[16] Measured from the fill price, it nets the captured half-spread against the subsequent adverse move, so markout is spread capture plus adverse selection fill by fill, the two being components of one measurement rather than competing names for it. Adverse selection is measured from the mid price instead of the fill price. They will therefore both be used in the analysis. Being per-fill it is free of the volume confound by construction. The metric is pre-registered with a one-sided alternative, that a better strategy shows better per-fill markouts, and the two mechanisms by which a drift signal could deliver that leave opposite signatures. If the signal grades individual fills, per-fill markout improves. If instead it times them, warning that a resting quote is about to be run over, per-fill markout does not improve and the effect appears in fill counts instead. Section 6.4 reports which signature the data carries.

The second metric is the **Sharpe ratio of daily P&L**[17], annualised from the 64 daily observations at $\sqrt{365}$. One caveat is stated here and respected throughout: for strategies with negative mean P&L, a higher volatility mechanically improves the Sharpe ratio, so Sharpe differences are never interpreted alone; mean loss, daily volatility and the 5% expected shortfall are always reported alongside. These are complemented by the **inventory diagnostics**: mean and mean-square inventory, and the fraction of time at the position cap. These are the objects the CARA controller actually manages.

Because per-fill expected value turns out to be negative for every feasible quoting configuration on this venue, total P&L is monotone in participation and a strategy can "win" simply by trading less. The placebo control and the matched-participation frontier exist to separate information from turnover, and every uncertainty statement is a paired difference across identical market days rather than a level. Realised P&L is additionally decomposed per day into spread capture, adverse selection, funding, exchange fees, and an inventory residual. The residual is what the inventory itself then earns or loses, the mark-to-market of a position the maker is still carrying once that five-second window (like adverse selection) has passed, and with it the five components close exactly against the booked terminal P&L.

5.9 Monte Carlo Stress Testing

The 64 observed days are, by the standards of crypto markets, quiet: realised carry runs an order of magnitude below documented extremes. The sample can price the tail it contains, but not one it never saw, so claims about extreme carry are assigned instead to a Monte Carlo

[16] $\tau = 1$ and $\tau = 30$ were computed but left out of the analysis since they provide us with the same conclusions.

[17] Sharpe intervals come from the stationary block bootstrap of Politis & Romano (1994), with the mean block length selected automatically by Politis & White (2004) as corrected by Patton et al. (2009).

suite that drives the same execution engine with synthetic funding paths, paired by seed across scenarios so every contrast sees the same underlying draws.

Each scenario is run over two hundred paths, balancing statistical precision with computational runtime. A path is one compressed funding day, two settlements an hour apart rather than three eight hours apart, which is what makes running that many paths affordable. The compression is not neutral, and the one thing it changes that matters is the share of each epoch the boundary layer occupies, so the suite is read as a test of the mechanism's direction and scaling rather than of its magnitude on a real day. The intensity inputs are idealised by design: every strategy is given the intensity parameters the generator itself uses. Volatility, however, is estimated exactly as in the replay, from a trailing 60 s window on the (synthetic) mid price, so the strategies run on an estimate that lags rather than on the truth. The suite tests the machinery rather than the inputs. Appendix E.5 specifies the generator, the scenario grid, and what else the compression does and does not change.

What the suite can isolate is the funding correction, whose input is a rate rather than a forecast. It replaces the exchange's published pre-settlement rate, which the replay reads directly, with Ornstein-Uhlenbeck paths under six separate carry regimes: a quiet baseline, documented carry at 5.5 basis points per settlement, persistent one-sided carry (both positive and negative), carry pinned at the suite's ±75 basis-point cap, and rapid sign flips. Of these, only the quiet baseline and documented carry are within BTC's own range: it clears the 5.5 basis points on about one in two hundred settlements, with the largest in the settlement history sampled for Table 36 is 8.8.

Every scenario runs the naive constant-spread strategy as the uncontrolled-inventory reference, standard GLT, and the funding correction, all three drift-free. The suite settles one pre-stated point: whether the funding correction's value grows with carry as the mechanism predicts.

6 Out-of-Sample Results

This chapter reports the six-strategy ablation over the 64 walk-forward out-of-sample days. One fact organises it: a passive fill on this venue costs more than it pays, and no quote width we test reverses that, so raw P&L counts tolls rather than ranking intelligence. What the research question actually poses therefore lives elsewhere, in how much a strategy trades, in inventory risk, in the timing of fills, and in the pricing of the funding boundary. The chapter takes those in order on the recorded days (Sections 6.1 to 6.5) and then into carry regimes the sample never produced (Section 6.6). It then analyses the interaction effect between the two extensions (Section 6.7), before closing with a robustness assessment across key parameter choices (Section 6.8). The funding-aware strategies act only in the short settlement boundary layer, and inside it this sample's funding rate is too small to move a daily total, so nothing separates them from their funding-blind twins until Section 6.5 examines the mechanism at the settlement itself.

6.1 Headline Ablation and the Anatomy of the Loss

Table 1 is the chapter's reference table: six strategies, one tape, one fee schedule, differing only in how a quote is chosen, with the horizon coupling for the drift strategies.

Table 1: Headline ablation over the 64 out-of-sample days (USDT, 0.005 BTC lots), horizon coupling for the drift strategies. Sharpe is annualised from daily P&L net of all fees, with a stationary-block-bootstrap 95% CI (automatic block length). Mean/d and SD/d are the mean and standard deviation of daily P&L in USDT, as is $ES_{5\%}$, while Net/fill is that P&L per fill, in USDT cents. $ES_{5\%}$ is the mean of the three worst days, the tail of the daily P&L distribution. $\overline{|q|}$ is the mean absolute inventory in lots over live time.

Strategy	Sharpe	95% CI	Mean/d	SD/d	Net/fill (¢)	$ES_{5\%}$	Fills/d	$\overline{\lvert q\rvert}$
1 Naive (matched)	-32.7	[-42.9, -28.4]	-670	391	-5.72	-1,741	11,705	5.67
2 GLT	-43.9	[-53.7, -39.3]	-717	312	-5.80	-1,404	12,377	2.78
3 GLT + AR	-44.4	[-54.5, -39.8]	-725	312	-5.80	-1,412	12,493	2.70
4 GLT + ML	-40.0	[-49.9, -35.8]	-688	329	-5.84	-1,455	11,784	2.67
5 GLT + funding	-43.9	[-53.8, -39.4]	-717	312	-5.80	-1,403	12,378	2.78
6 Hybrid	-40.0	[-49.9, -35.9]	-689	329	-5.84	-1,458	11,805	2.69

Read as a scoreboard, the table says only that every strategy loses money. The attribution says why. For the full hybrid strategy the day-average decomposition is +278 USDT of spread capture against −487 USDT of adverse selection and −489 USDT of exchange fees (Figure 3; the full attribution is in Table 20). Per fill that is $+0.024 - 0.041 - 0.041 \approx -0.058$ USDT: a filled quote is run over by more than it captured, and the maker fee costs as much again. Do the same division for the other five strategies and the toll barely moves. All strategies sit within 0.12 cents of each other, between −5.72 and −5.84 cents per fill.

The fee half is structural, decidable from the calibration before a single day is replayed. The optimal quote is a base $c_1 = \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right)$ plus a state-dependent skew. That base is bounded above by $1/k$ whatever the risk aversion: 8.4 USDT per BTC at the median of the rolling fit the engine quotes on and 11.8 in even its flattest tenth of windows, while the 2 basis-point maker fee at the sample's mean fill price is 14.0 USDT per BTC.[18] In practice the state-dependent skew adds little: GLT's median quoted half-spread is 8.6 USDT per BTC, and the naive baseline is matched to exactly that width, so neither quoting family comes close to the fee. No parameterisation of the fee-blind objective closes that gap. The natural next question, why not simply make the model fee-aware and quote wider than the fee, deserves a mathematical answer rather than a heuristic one. The maker fee enters the fill payoff as an affine correction that the optimisation absorbs exactly, requiring no calibration at all, and Appendix D.2.9 derives this repair while Section 6.2 tests it.

The adverse selection component is not fixed prior to calibration; it is determined exogenously rather than chosen. We quote uninformed against flow that is not: 84 to 89% of fills arrive as sweeps, and with the touch spread at about a tick (Section 3) there is no width to hide behind. Adverse selection cannot say how much of the move that follows was private information and how much was the market simply repricing, and for a passive quote the distinction changes little, since the price moved away either way. Uninformed quotes also lack the warning that informed

[18] This fee is tier-specific. Professional market makers on major exchanges pay reduced or negative maker fees; Table 27 re-prices the recorded fill streams across that range, from a 2 basis-point charge to a 1 basis-point rebate.

flow is arriving, and Section 6.4 is where the ML signal is found to supply that warning.

The same engine on the informationless synthetic tape of Section 6.6 books *positive* per-fill markouts: the sign is thus a property of the flow, not a defect of the model or its accounting. This settles, in the negative, the question Section 2 attached to every fill: the spread does not cover what it insures against, the capture paying for a little over half of the adverse move. Passive quoting on this venue at the retail tier amounts to selling free options to faster traders while paying a fee for the privilege.

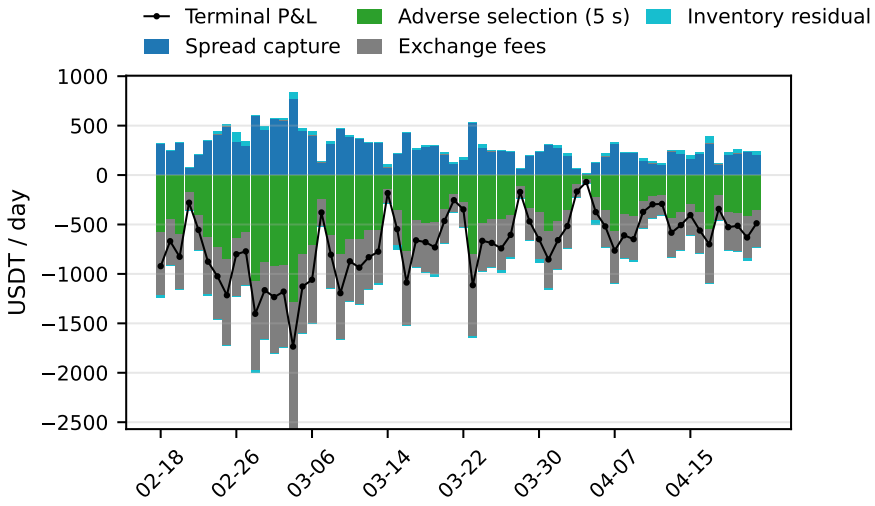


Figure 3: Daily P&L attribution of the full hybrid strategy across the 64 out-of-sample days: spread capture, adverse selection (the 5 s mid move after each fill), and exchange fees. The funding component, never reaching half a USDT on any day at this sample's carry, is too small to see at this scale and is omitted from the bars (Table 20 gives the full attribution, funding and inventory residual included).

The caveat of Section 5.8 now binds. With a negative mean it is *additional* volatility that raises the Sharpe ratio, so the column rewards what a real trader would penalise, and the size of that effect is measurable rather than rhetorical. A difference of Sharpes splits exactly into a volatility term $m_A(1/s_A - 1/s_B)$ and a mean term $(m_A - m_B)/s_B$. Of the naive baseline's 11-point advantage over GLT, 8.3 is the volatility term alone, bought by running about 25% more daily volatility. The intervals close off the other route: 14 Sharpe points wide, they overlap heavily across all five risk-managed strategies, so no level separates any pair. Sharpe levels are therefore never used for ranking, but instead to break down the underlying components.

Mean P&L, Volatility and the expected shortfall (ES), here the mean of the three worst days, are reported beside every Sharpe for that reason, and together the three locate where a strategy's risk actually comes from. The naive baseline's extra volatility reappears in its tail, whose −1,741 USDT ES is roughly a fifth worse than any of the risk-managed strategies, and that gap is what quoting a fixed width regardless of the market costs.[19] Behind part of it sits the exposure the risk-managed strategies is built to control, the inventory standing behind each quote, which Section 6.3 measures. The drift term is the other mechanism. A signal large enough to move its quotes leans the book directionally, and microstructure forecasts are noisy by nature, so some of that lean is taken in the wrong direction. This exposure widens the daily

[19] Expected shortfall significance is not tested: at 64 days, its 5% tail contains just three observations.

distribution and thickens the tail. Balancing the predictive advantage against its induced tail risk remains a central trade-off of an ML model trained on noisy data.

One feature of the engine could manufacture that loss regime rather than measure it: every number in this chapter is produced against a book a median 51 ms old. Re-running the whole ablation on the exchange's publication instants, anchored to the book and inventory of that instant, leaves the sign of everything intact. A fresh state improves every fill, per-fill markout by 0.14 to 0.17 basis points on every strategy, and cuts GLT's daily loss by 41%, from −717 to −424 USDT, yet the best markout anywhere in the ablation is still −0.70 basis points against a 2.00 basis-point fee. The improvement arrives through the channel every other lever in this chapter uses: posting on the publication clock costs each strategy 30 to 37% of its fills, and pricing those at the toll they paid before accounts for 84 to 89% of the gain. Book staleness is a fourth participation dial, not the origin of the loss, and no conclusion in this chapter changes when it is removed (Appendix C.1).

6.2 The Matched-Participation Frontier

Section 6.1 found about the same per-fill toll on all six strategies, each at its own participation level. If that toll belongs to the tape rather than to any one quoting rule, pushing a family away from its natural level should not move that toll: a strategy's loss would then follow from how much it trades. Figure 4 tests this by sweeping the three families that have a participation dial against mean daily P&L, with GLT and the fee-aware controller of Appendix D.2.9 plotted as the single points they are. For the naive family the dial is the fixed half-spread, swept from 5 to 40 USDT per BTC; for the ML and AR families it is the signal-pricing horizon h, swept from 5 to 40 s. A longer h prices the same drift forecast as a larger expected move, so by Equation (26) it scales the shift response by 0.5 to 4 times its deployed value. Because the fill intensity $Ae^{-k\delta}$ is convex, the tightened quote then gains more fills than the widened one gives up, and participation rises with h until the c_1 clip caps the skew. The two dials therefore run in opposite directions along the axis: the naive family trades less as the half-spread rises, the drift signal families more as h rises. Each point is a full 64-day replay.

In the left panel every configuration of every family, from a fixed spread to the full drift signal-driven controller, falls on essentially the same line. Widening the naive quote from 5 to 16 USDT per BTC cuts its fill count fifteenfold and leaves its toll between −5.5 and −5.9 cents per fill, a band that contains all six strategies of Section 6.1, and the ML horizon adjustments sweep along the same line. Daily loss is that near-constant toll times fill count, so what separates the rows of Table 1 is how much they trade rather than how they decide to trade, and the only way to lose less is to trade less. The toll does thin further out, to −1.8 cents at a half-spread of 40 USDT per BTC, but only because a strategy quoting that deep barely trades at all: 218 fills a day, and mean daily P&L of −4 USDT with a 95% interval from −16 to +8. At no width we test, then, does quoting carry positive per-fill value.

The line has a mechanism behind it. A quote resting deeper is reached only by a larger sweep, so it captures more spread and is run over by more, and the two move almost together: mean capture per fill rises from 1.2 to 5.7 basis points across the naive widths while adverse selection deepens from −2.1 to −5.6, a near-exact linear trade in which each extra basis point

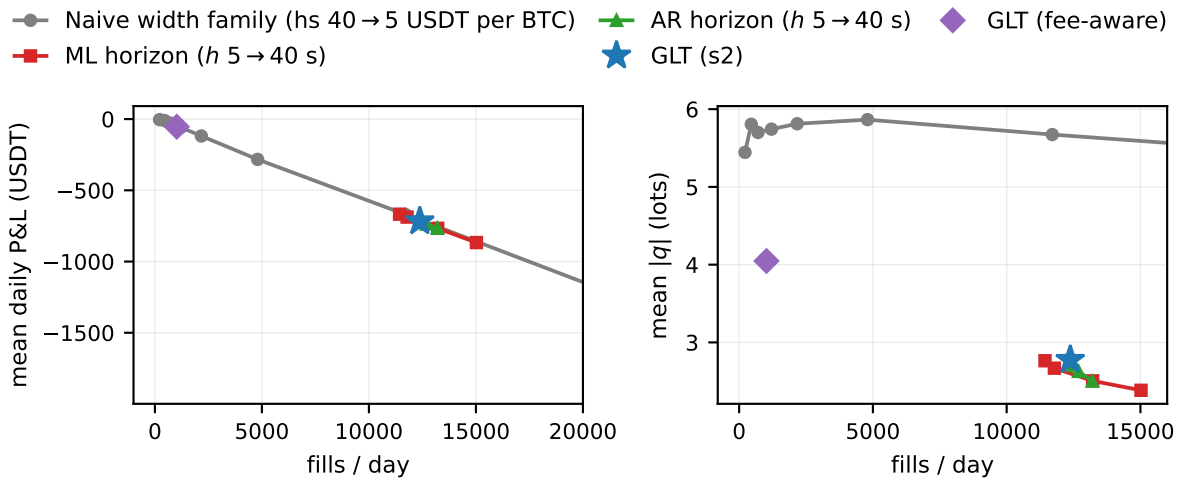


Figure 4: Matched-participation frontier, 64 out-of-sample days. Left: mean daily P&L against fills per day; right: mean absolute inventory against the same axis. Both panels are clipped, leaving the narrowest naive width, a half-spread of 5 USDT per BTC at 33,261 fills per day, off-scale on the same line.

of capture costs 0.78 of a basis point in adverse selection ($R^2 = 0.99$). The fee-aware controller of Appendix D.2.9 shows what that does to a repair aimed at the fee. The optimisation absorbs the correction exactly: quotes shift outward by the fee, and every closed form survives with a rescaled fill-intensity level. This is the principled version of "just quote wider than the fee", and over the 64 days it cuts the daily loss by an order of magnitude, −54 USDT against GLT's −717. All of it is participation. The controller makes 12 times fewer fills and still books −5.3 cents on each, just below the naive line: at the same half-spread, about 23 USDT per BTC, a fixed spread does marginally better.

The right panel shows what the control machinery buys. At any participation level the naive family carries roughly twice the inventory of the controlled families (5.4 to 5.9 lots against 2.5 to 2.9), because it buys its lower fill count with width alone while they buy it with width *and* a restoring skew. The fee-aware controller shows the cost from the other side: it has the skew but not the fills to express it, so its book carries 4.0 lots against GLT's 2.8. Covering the fee and controlling inventory pull the quote width in opposite directions, and on this tape no width satisfies both. What the optimal-control machinery purchases is therefore not P&L, which the toll fixes, but the right to operate at matched participation with half the inventory risk. The volume confound this frontier exposes is not specific to our design: any market-making backtest that ranks strategies on raw P&L in a negative-toll regime is, to first order, ranking them on fill count.

6.3 What Inventory Control Buys

Section 6.2 showed the naive family carrying twice the inventory of the controlled families at every participation level; Figure 5 shows what that looks like as a distribution. A fixed-width symmetric quoter's inventory is a random walk, buying and selling with equal willingness whatever it holds, so its position wanders until the hard position cap stops it, a limit imposed on the strategy rather than by it: the matched-spread naive book spends 9.5% of live time pressed

against that cap, and its distribution piles up at both limits. The GLT book concentrates near zero and touches the cap only on 0.03% of live time, because the reservation-price skew acts like a spring: a long position pushes both quotes down and a short position pushes them up, always paying the strategy something to return to flat. In mean-square inventory, the quantity the CARA objective actually penalises, the naive book runs 41 against GLT's 12. The ML strategy's distribution lies on top of GLT's, 2.67 lots of mean inventory against 2.78 and the same 0.03% at the cap, so the drift overlay does not loosen the control the skew provides.

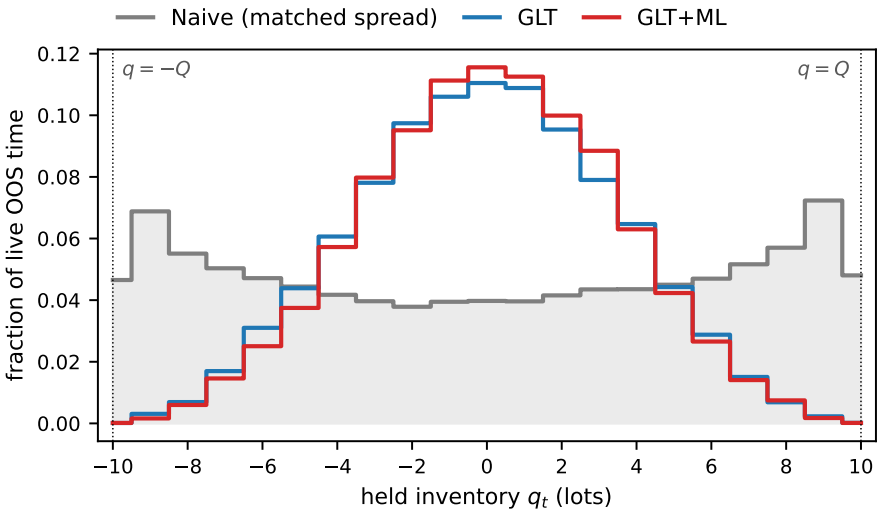


Figure 5: Distribution of held inventory over all live out-of-sample seconds, for the matched-spread naive strategy, GLT, and GLT + ML.

Section 6.2 left P&L fixed by fill count and a common toll, so the extra position risk leaves no clear mark on the realised losses. Because the only risk functional available is the CARA objective optimised by the controlled family, using it to rank them would be circular. Instead, the histogram confirms something simpler: constant-spread quoting operates against its risk limit, whereas the optimal-control models manage inventory endogenously.

6.4 The Mechanism of the ML Signal: Timing, Not Selection

A drift forecast can be spent two ways: chase the fills it favours, or refuse the ones it warns about (Section 5.5). Table 2 prices both. The verdict is one-sided, and it belongs to the ML signal alone. Under the horizon coupling the ML strategy improves on GLT modestly, −688 against −717 USDT per day, while under the defensive coupling it takes a quarter off the mean loss and over a fifth off both daily volatility and the tail. The gain is not in the fills themselves: per-fill markout barely moves, a span of 0.020 against a level of about 0.86 basis points, while the fill count falls by more than 25%. Whatever the defensive coupling buys, it thus buys the way every lever in Section 6.2 did, by *declining* trades.

The two couplings split the risk columns as sharply as the P&L one: of the five rows the ML horizon strategy carries the widest daily distribution and the worst tail, 329 USDT and −1,455, while the ML defensive strategy carries the narrowest and the best, 244 and −1,076. Because horizon coupling shifts quotes toward the predicted drift, an incorrect signal leaves the

Table 2: Horizon versus defensive coupling for each drift signal strategy, with GLT as the reference row. $\text{mo}_{5\text{s}}$ is the mean per-fill markout at 5 s over maker fills (Section 5.8). Δ placebo is the daily P&L gain over the same strategy run on its time-shuffled signal, with intervals and t-statistics in Table 21. Mean/d, SD/d, $\text{ES}_{5\%}$, Fills/d and $\overline{|q|}$ carry the definitions and units of Table 1.

Strategy / coupling	Mean/d	SD/d	$\text{ES}_{5\%}$	Fills/d	$\overline{\lvert q \rvert}$	$\text{mo}_{5\text{s}}$ (bp)	Δ placebo
2 GLT (reference)	-717	312	-1,404	12,377	2.78	-0.852	—
3 GLT + AR (horizon)	-725	312	-1,412	12,493	2.70	-0.856	-6
3 GLT + AR (defensive)	-696	308	-1,375	11,998	2.76	-0.849	+8
4 GLT + ML (horizon)	-688	329	-1,455	11,784	2.67	-0.869	+159
4 GLT + ML (defensive)	-527	244	-1,076	8,981	2.86	-0.860	+75

strategy standing closer to a trade it wanted to avoid. Even so it loses less than GLT on fewer fills, 11,784 against GLT's 12,377, and not by quoting much narrower: its median half-spread roughly matches GLT's, 8.62 against 8.60 USDT per BTC. A risk clock is already running underneath, thinning fills where they are most dangerous, but the chasing half then spends that advantage back. Defensive only keeps the widening and refuses the tightening into the wrong side, at the cost of a little inventory, 2.86 lots against 2.67, since the chasing it discards was also pulling the book toward flat. The defensive strategy retains the benefit of the risk clock without spending it back on signal-chasing.

At first glance the AR benchmark repeats the pattern at a tenth of the scale, which is also the ratio of the two amplitudes: one standard deviation of it moves a quote by 0.28 USDT per BTC against the ML signal's 2.73, and under the defensive coupling it improves on GLT by 21 USDT per day against 190. A tenth of the response buying a tenth of the effect is what a quieter version of the same information would look like, and it is not what this is. On the 64 out-of-sample days the ML signal correctly predicts the direction of the coming ten-second move on 60.1% of rows against the AR's 52.7%,[20] and they separate further still when the drift signal is strong. Sorted by its own signal strength the ML model climbs from 51% in its weakest tenth to 67% in its strongest, clearing a coin flip on all 64 days, while the AR holds near 53% across every tenth and is worst in its strongest, 51%, above a coin flip there on only 24 days of 64.

Whether either drift signal survives inside the controller is a separate question, since a strategy that widens at random also trades less and can look better for it. The placebo settles it by shuffling each signal in time, preserving its distribution and every quote adjustment it makes, destroying only when they happen. The two answers are opposite. For the ML signal under the defensive coupling the shuffle costs 75 USDT per day ($t = 19.9$): of the 190 USDT by which defensive beats GLT, roughly 60% is the mechanical value of sometimes quoting wider and 40% is knowing when to do it. Under the horizon coupling there is no mechanical fallback, a symmetric skew at a random moment being pure noise, and the placebo lands 159 USDT per day ($t = 21.7$) below the real signal and below GLT itself, randomly timed skews tightening into sweeps as often as away from them. The AR reverses it, beating its own shuffle by 8 USDT per day under the defensive coupling and losing to it by 6 under the horizon, under one percent of the daily loss and of opposite sign, so there is no timing content to detect in either direction. A price-only drift does not do inside the controller what the ML signal does, and the quality of

[20] Rows with no direction to call, in the realised move or in the forecast, are excluded from both numbers.

its timing is what it lacks.

We now know the ML drift signal is real. Table 3 shows the mechanism by which it works as a risk clock. The table splits live time into thirds by signal strength and classes every fill of both books (GLT and ML) as *aligned* when it goes the way the signal points, buying into a predicted rise or selling into a predicted fall, and *counter* when it goes the other way. GLT, which never sees the drift signal, marks where the danger sits: in the strongest third it is hit 777 times an hour against 349 in the weakest, and 468 of those land on the counter side against 309 aligned. The ML book, quoting the same tape, takes that counter side at 0.45× GLT's rate and the aligned side at 1.48×, and both halves survive on the exchange's publication clock (Appendix C.1). Yet GLT's markout is the same in every cell, −0.84 to −0.87 basis points, large signal or small, with it or against it. The drift signal thus provides anticipation rather than evaluation: it identifies when and on which side toxicity arrives.

Table 3: Fill rates against the ML drift signal, which runs the horizon coupling. Live time is split into three equal-duration tiers (T1 to T3, T3 the strongest) by signal strength $|\alpha_{ML}|$, so each rate is per hour spent in that tier. Fills are bucketed by the signal at quote placement, the value the strategy could act on. A fill is *aligned* when it goes the way the signal points, buying into a predicted rise or selling into a predicted fall, and *counter* when it goes the other way.

$\|\alpha\|$ tier	side	fills/h (GLT)	fills/h (ML)	ratio	GLT 5 s markout (bps)
T1	aligned	168	201	1.20	-0.86
T1	counter	181	180	0.99	-0.87
T2	aligned	184	243	1.32	-0.85
T2	counter	236	178	0.75	-0.86
T3	aligned	309	458	1.48	-0.85
T3	counter	468	213	0.45	-0.84

Funding settlement does not act as a toxicity event in this data. The appropriate benchmark is therefore the model without a drift signal, whose quotes cannot react to the settlement clock by construction. Splitting GLT's fills by time to settlement, adverse selection inside the final 15 minutes is −1.93 basis points against −1.99 more than 2 hours out, a paired difference over the 64 days indistinguishable from zero ($t = 1.30$, $p = 0.20$). An effect of any size would have shown: the same two fill samples that give the null on adverse selection give a post-fill price movement 0.45 basis points *smaller* near the settlement ($p = 0.0002$), nearly four times the smallest difference this test could resolve at 80% power, so the null bounds the window rather than failing to look at it. Settlements fall at fixed hours, so neither result separates the settlement from the three 15-minute windows it occupies. Instead, the contrast evaluates the exact window over which a strategy could intervene, which is the relevant empirical question. Although the drift signal includes the settlement countdown and mark basis in its feature set, it assigns them no real predictive weight. Against the signal-blind control its fills land on the funding-collecting side no more often in any window before the settlement, the largest deviation being under one percentage point and none exceeding 1.5 standard errors. The drift signal is uncorrelated with the funding rate across the out-of-sample fills, and its magnitude neither grows into the settlement nor scales with the rate. There is no settlement-specific mode to find.

The model's internal feature attributions confirm this same pattern. Pooled over all 64 folds, SHAP values concentrate heavily on the standing book rather than on flow: top-of-book

imbalance first at a 16.8% share, the relative level-one depths and the book-slope imbalance next at 10 to 12% each, with order flow (2.7%), trade intensity (2.0%) and VPIN (0.8%) far behind (Table 28). A model trained to forecast drift that ends up reading queue state rather than the flow crossing it is the interpretability counterpart of the timing result above. Those shares barely move across the epoch, holding to within 4% from two hours out to the final minute (Figure 6a). The one feature that does move is the countdown, `seconds_to_funding`, rising to 1.48 times its bulk share inside the final thirty minutes pooled across folds, but a countdown growing informative as it runs out is close to mechanical. The remaining features standing above their bulk share show no increase into the settlement at all. They reach their level about a hundred minutes out, long before a market pricing the settlement would, and hold it flat to the last minute (Figure 6b). No toxicity feature gains weight as the settlement approaches, and `basis_bps`, the one price feature carrying funding-adjacent information, does not either, sitting at 0.94 of its bulk share in the final five minutes, in line with the book features rather than above them. The funding rate itself is deliberately absent from the feature set, since a model given the rate settled at each boundary could learn the transfer rather than the book, while `basis_bps` carries the same information at finer resolution as the premium that rate is computed from. Handed the countdown and the basis and nothing else, the model finds nothing at the settlement worth reorganising for.

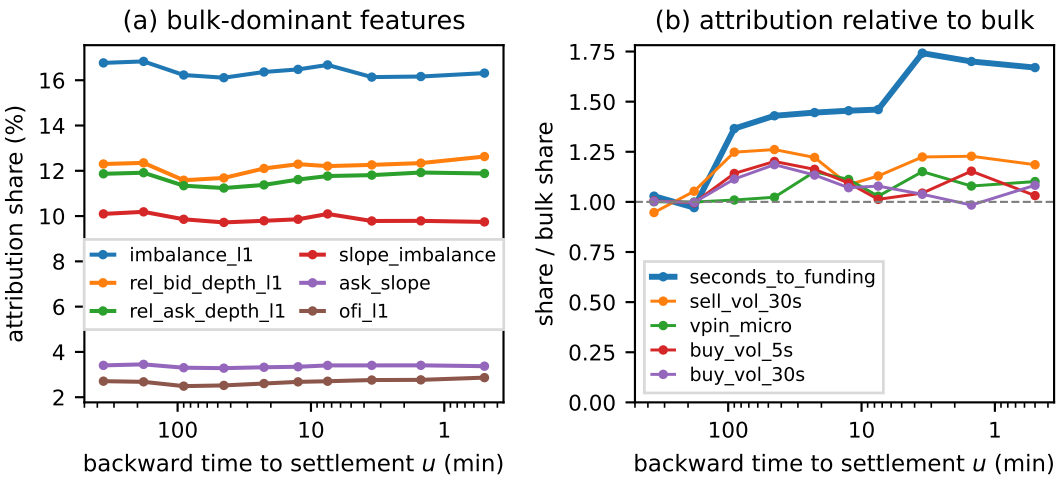


Figure 6: SHAP attribution into the funding settlement. Left: attribution share against backward time to settlement for the six features carrying the largest share in the bulk of the epoch. They hold their shares flat across the whole clock, to within 4% of the bulk level in the final minutes. Right: share relative to bulk share, for the five largest risers carrying non-negligible attribution. Only the funding clock climbs into the settlement (1.48× pooled over the final thirty minutes, Table 28); the others reach their level about a hundred minutes out and hold it flat thereafter.

6.5 The Funding Correction at the Boundary

The final mechanism in this chapter is the funding correction, which at this sample's carry level has a negligible impact on total P&L: the inventory cap is 0.05 BTC, so a book pinned against it through a 90th-percentile settlement is charged only about 0.3 USDT. Everything interesting about the correction on this sample is therefore in the mechanism rather than in the money,

and the mechanism appears where the mathematics predicted.

The boundary layer forms when and where predicted. The funding-aware strategies latch into Regime II at 92.7% of the 192 out-of-sample settlements, a median 815 seconds before the settlement lands and half of them between 9 and 18 minutes before it, thin against the eight-hour epoch. The layer widens with the carry it is anticipating, and the 14 settlements at which nothing activates have almost nothing to anticipate: the largest $|F_t|$ among them is 0.26 USDT per BTC, against a median of 2.1 among those that do latch (census in Table 22, per-settlement detail in Figure 8). Figure 7 shows what happens inside the layer.

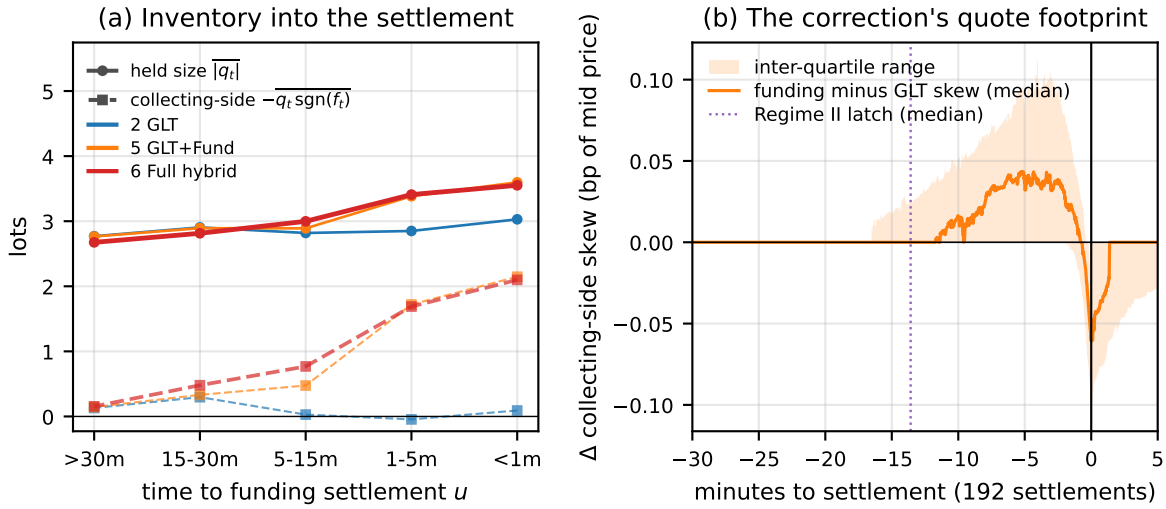


Figure 7: The funding correction at the settlement boundary. (a) Inventory into the settlement, averaged over every live second in each time-to-settlement bucket: held size $\overline{|q_t|}$ (solid) and the collecting-side component $-\overline{q_t \operatorname{sgn}(f_t)}$ (dashed), the part of the book that will be paid rather than charged. (b) The funding strategy's quote skew minus GLT's, in basis points of the mid price toward the collecting side, pooled across all 192 out-of-sample settlements as a median with its inter-quartile band; the dotted line marks the median Regime II latch.

Inside the layer the correction does what a penalty linear in q says it should, which is subtler than shedding inventory. Because $\Phi(t, q) \propto qF_t$ charges only the paying side and *rewards* the collecting side, the response to an approaching settlement is not to flatten but to rotate. Panel (a) catches that rotation: averaged across the 192 settlements, the collecting-side component of the funding book climbs from 0.14 lots outside half an hour to 2.1 in the final minute, against a flat near-zero for GLT. It arrives where it was aimed. GLT, blind to the funding clock by construction, ends a settlement collecting 48% of the time and pays 0.6 USDT in total over the 64 days, while the funding-aware strategies collect at close to 70% of settlements and turn that total into +6.6 USDT. Rotating is not flattening, and the held size proves it: the correction leaves *more* unsigned size on the book at the settlement instant, 0.43 lots more than GLT on average ($t = 2.9$)[21], because collecting requires holding a position.

Panel (b) is the same event in quote space, and it answers how a transfer worth cents moves a quote at all. A quote is a price, and the funding scalar $F_t = S_t f_t$ is written in the units a price is written in: the median settlement's 2.1 USDT per BTC is 21 ticks, and the correction spends

[21] Also, $t = 2.9$ when each day's three settlements are collapsed to a daily mean

a few of them. GLT and its funding-aware twin quote identically until the switch. After it, the funding quotes lean toward the collecting side, at their widest a median 0.044 basis points of the mid price across the 192 settlements, about three ticks, some five minutes out. In the last half-minute the difference turns negative, and panel (a) says why: by then the funding book holds around two lots of collecting-aligned inventory against GLT's zero, and GLT's own inventory skew leans against further accumulation at that size, so the pooled difference is comparing books in different states rather than just two quoting rules. The event itself settles the reading: once it lands the funding term leaves the quotes entirely, the rotated book unwinds, and the difference is back at zero within two minutes.

6.6 Carry Stress in a Controlled Market

Section 6.5 could show the correction firing where the mathematics puts it, but at a carry worth about 0.3 USDT to a pinned book it could not show what the term is worth once it is large. That needs an environment where the carry is set rather than observed and where nothing else changes when it is, and the value of the Monte Carlo suite here is its controlled environment. Its flow is generated without information, so no informed trader competes for the fills. The generator's true mid is known, so nothing is measured against a price the strategy helped to make, and paired seeds hold the tape fixed across strategies, so the only difference between two rows is the funding term itself. Table 4 displays the six carry regimes, 200 paths each (definitions in Appendix E.5, full columns and the paired inference in Appendix C).

Table 4: Funding stress, 200 paired paths per scenario. Cells are mean terminal P&L per path with funding collected in parentheses, both USDT; the last line is the paired mean gap with its standard error. Columns are the scenarios of Table 35, at a mean per-settlement rate of base (1 bp), doc. (5.5 bp), pers.− (−45 bp), pers.+ (+45 bp), pinned (75 bp) and flip (0 bp, sign-reversing). The naive baseline and the remaining columns are in Table 23.

	base	doc.	pers.−	pers.+	pinned	flip
GLT	−729 (0.0)	−729 (0.0)	−730 (−0.2)	−729 (0.2)	−729 (0.3)	−730 (−0.9)
+ funding	−729 (0.1)	−726 (2.4)	−690 (33.7)	−690 (33.7)	−661 (59.6)	−693 (31.4)
gap ± s.e.	+0.3 (0.05)	+3.1 (0.14)	+39.6 (0.88)	+38.9 (0.88)	+67.6 (1.38)	+37.4 (1.55)

Magnitude first, and in funding rather than in profit, since funding capture is what the correction optimises and the only quantity on which the two books differ by construction. Funding collected per path rises monotonically with the carry supplied: 0.1 USDT at the constant baseline rate, 2.4 at the carry documented on thinner perpetuals, 33.7 under sustained carry of either sign (positive or negative rate), and 59.6 with the rate pinned at the suite's ±75 basis-point cap. GLT, blind to the clock, takes between −0.9 and +0.3 USDT in every one of them, inheriting whatever sign the rate happened to carry. The baseline scenario serves as a control: with almost zero carry to capture, the funding term vanishes and both models perform identically.

Two properties of that ladder matter more than its levels. The gain is sign-symmetric: the book collects the same 33.7 USDT whether persistent carry runs positive or negative, while GLT's funding merely inherits the sign of the rate. Section 6.5 saw that rotation on the recorded tape as an inventory statistic, here it is put under an exact controlled check, and the payoff is

symmetric because the rotation is. The rotation also survives a rate that reverses underneath it. The sign-flip scenario is built to punish a strategy that commits early to a side, and the correction still collects 31.4 USDT there, within a couple of dollars of the persistent funding scenarios, because the boundary layer re-solves as the forecast moves instead of betting on it once. What it spends for that is inventory risk. With the rate pinned at the suite's cap (75 bp), the correction stands at the position limit for 1.55% of the path against GLT's 0.28%, because collecting a payment means standing on one side of the book as the settlement arrives.

At high carry levels, inventory capacity plays a different role. With the rate pinned at that same cap, quadrupling Q scales funding collected from 59.6 to 281.2 USDT per path, while the funding-blind control's own expected shortfall deteriorates from -774 to -876. Capacity that is pure risk for a funding-blind book is a return dial for a funding-aware one, at the price of a larger directional bet into each settlement (Table 24).

None of this is a claim about live profitability. The epochs are compressed, the flow is synthetic, and the price path does not respond to the carry imposed on it, so the collecting side banks the payment without the basis reversion that would accompany it in a real high-carry regime. The suite shows the mechanism scales, it does not show the money is there. Scaling the recorded tape instead, which keeps real price dynamics and the true eight-hour epoch, reaches the same place: multiplying the live funding estimate and the rate finally settled by 15 and by 124, to mean magnitudes of 5.5 and 45 basis points, the correction collects 4.29 and 40.91 USDT per day and ends on the receiving side at every one of the 64 days, where the funding-blind book pays 0.13 and 1.07. Carry is exogenous there too, so those are magnitudes rather than forecasts. What the two instruments agree on is the mechanism, not the money.

6.7 The Two Extensions Together

The research question posits that the two extensions are structurally complementary: that the ML drift signal supplies the correction with situational awareness, so it does not chase a scheduled payment into flow about to run it over, and that the settlement supplies the signal with a predictable toxicity window, the one moment in the day when the whole market knows a cash flow is coming. The second half of that expectation fails at its premise. Section 6.4 finds no concentration of adverse selection at the settlement, and no toxicity feature that gains weight into it. There is nothing scheduled there for a drift signal to price.

The first half is not resolved here, because both components act on the same quantity. The correction rotates inventory onto the collecting side at roughly two settlements in three (Section 6.5). Crossing the drift signal with it gives a two-by-two design on collecting-side inventory at the 192 out-of-sample settlements, and the signal's contribution differs across the two halves: $+0.37$ lots when it acts alone, $+0.07$ lots when the correction is already present. The interaction is -0.31 lots ($t = -1.85$, $p = 0.066$). If the drift overlay were additive it would still contribute $+0.37$ lots with the correction present; if it were completely absorbed it would contribute nothing. It contributes $+0.07$, much the nearer to the second, though the standard error of 0.17 rules out neither. That contribution does not reflect the signal responding to settlement as discussed earlier. The drift signal is uncorrelated with the funding rate across the out-of-sample fills, points toward the collecting side in 50.2% of them, and neither grows nor

scales with the rate as the settlement approaches.

Settling the question economically is beyond what this venue allows. Realised profit per fill is identical across the strategies to within a spread an order of magnitude below their day-to-day variation (Section 6.1), so no per-fill test separates them, and daily totals confound participation (Section 6.2). Complementarity would become measurable on a contract whose carry is large against its fee toll, which BTC is not over this period. The pairing is therefore neither demonstrated nor excluded by this sample, and the conditions under which it could be decided lie outside this contract and these 64 days.

6.8 Sensitivity

The risk aversion γ is a choice: nothing in the data sets it, so the question is whether the chapter's conclusions depend on where it was put. The decay rate ρ is this thesis's own addition, read once from the model's spectral gap and then held constant across all 64 days, so the question is what it controls. Each is swept over a factor of sixteen on a deterministic sixteen-day subset of the out-of-sample window fixed before any cell was run, γ on the full hybrid and ρ on the two funding strategies. γ enters the quotes mainly through the inventory penalty $\alpha = \frac{1}{2}k\gamma\sigma^2$, which prices the book the trader is carrying, so raising it makes the trader lean harder against its own inventory. This effect holds empirically: mean $|q|$ falls monotonically from 2.79 to 2.34 lots. The lean, however, is not free, because the side that would reduce inventory sits closer to the market, so the trader trades more, 10,978 to 11,252 fills a day, and pays the per-fill toll of Section 6.1 more often. Both the daily loss and its dispersion therefore rise with γ rather than falling with it. Every column moves within two to three per cent across the whole factor of sixteen, less inventory risk against slightly more of every other kind (Table 25).

The structural parameter ρ sets when the boundary layer opens, not what it is worth: across the same factor of sixteen the Regime II time share thins from 4.8% to 2.5%, exactly as the mathematics predicts. The funding collected rises with ρ, a shorter layer concentrating the rotation nearer the settlement, and flattens above the calibrated value: ten cents more at four times ρ, against eighty cents less at a quarter of it (Table 26). Given this dataset, neither hyperparameter has much of an impact on the conclusions established throughout the paper.

7 Conclusion

Perpetual futures charge a position-proportional cash transfer every eight hours, at a moment the whole market can see coming, a feature classical inventory models are not built to address. This thesis asked what a GLT controller gains from pricing that settlement analytically, from a learned microstructure drift signal, and from the two together. On this contract at retail fees, neither extension changes what a fill is worth. Per-fill quality is flat across every strategy tested, the funding leg is worth almost nothing at this contract's carry, and the daily totals that appear to separate the strategies are counting fills rather than ranking decisions. Instead, each extension acts on a narrower set of execution mechanics: which fills the strategy accepts, and which inventory side it holds into settlement.

The discrete settlement enters the wealth process as a normalised drain concentrated before

each settlement, which preserves the ODE reduction an exact jump would destroy and lets the model size its own anticipation window. Additionally, a LightGBM signal trained on L2 features supplies the drift. Six strategies, from a fixed-width quoter, through GLT, to the full hybrid, were then evaluated over a 64-day daily walk-forward period using self-recorded BTC data. The model is retrained on a rolling window so that no day is ever priced by a signal that has seen it. The Python implemented engine is fee-exact and queue-conservative, with a C++ version agreeing on it event by event. The evidence of the results is read through a matched-participation frontier, placebo controls, and a Monte Carlo suite to simulate synthetic carry scenarios that were not in the recorded data.

The measurement environment is the first finding, and it constrains every other. A passive fill on this venue at the retail tier costs between 5.5 and 5.9 cents more than it earns, and that toll barely moves however wide the quotes are placed or however the drift signal is used. Daily profit is therefore a fill counter, and with a negative mean the Sharpe ratio rewards the volatility a real trader would penalise, so neither quantity ranks the six strategies. At retail fees passive quoting on this contract sells options to faster traders and pays a fee for the privilege. Even the fee-aware objective, an exact and calibration-free transformation of the same closed forms, cuts the loss by an order of magnitude without flipping its sign, because a deeper quote is reached only by a larger trading sweep, where the additional spread capture is almost entirely offset by increased adverse selection. Comparing strategies at equal turnover is therefore not a refinement but a precondition, and it is the methodological contribution this thesis would defend first. Judged there, inventory control does its advertised job: mean-square inventory of 12 against the fixed-width quoter's 41 at the same spread, 0.03% of live time at the position cap against 9.5%, and a daily mean that differs by less than a sixth of either strategy's own daily dispersion.

The learned signal turns out to work as a risk clock instead of a straightforward drift forecast. Shuffling it in time costs 159 USDT a day ($t = 21.7$), demonstrating that its value lies in its timing rather than its signal frequency. Although per-fill quality remains uniform across all metrics, fills taken against its own forecast drop to 0.45 times GLT's baseline rate, while fills aligned with the forecast increase to 1.48 times. It anticipates when a resting quote is about to be run over instead of where the price will go, and its best expression is to stand aside. A price-only drift put through the same controller beats its own shuffle by under one per cent of the daily loss, so what the LightGBM signal adds is not available from the simpler alternative.

The funding correction fires where the mathematics predicts, activating within the boundary layer at 92.7% of settlements and rotating the book onto the side that receives the payment. At BTC's own carry it turns GLT's -0.6 USDT over the 64 days into $+6.6$, as its magnitude bound predicts, so this contract can validate the mechanism but cannot price it. What it collects rises monotonically with carry across the Monte Carlo stress regimes, from 0.1 USDT per path at the sample's own rate to 33.7 under sustained carry and 59.6 with the rate pinned at the suite's cap. Whether the two extensions combine therefore remains unresolved: crossing the two on collecting-side inventory leaves an interaction of -0.31 lots ($t = -1.85$), an additive quote cannot express the coupling the question posited, and the carry that would make that coupling measurable never occurred in the sample.

The theoretical contribution stands independently of all these results. A discrete, position-proportional settlement can be carried inside a continuous-time control problem without giving up the ODE reduction that makes the controller solvable, and the construction translates to any market with such a clock.

The primary limitation of these empirical results is the sample itself. It is one contract over 94 contiguous days, chosen liquid so that a queue-conservative replay could be relied upon. That choice inevitably shaped the observable effects: a touch spread of roughly one tick, a fee schedule under which no configuration achieves profitability, and a carry so mild that the funding leg remained economically inert. Every mechanism, however, stayed measurable, only its economics did not, and a contract whose carry is large against its fee toll would put the same instruments to a harder test. That is the natural starting point for future work: testing higher-carry contracts, evaluating fee tiers where net P&L can flip positive, and eventually deploying to a live feed, for which the system architecture and measured latencies present no barrier. The second continuation follows from the failure to pair the two extensions. An additive quote cannot express the coupling the research question posited, and letting the signal modulate the funding term directly instead, scaling the drain or its decay rate by predicted toxicity, is the form in which that question deserves to be asked again. Third, the defensive behaviour that drives most of the drift signal's value remains an operational floor imposed on an optimal policy, rather than an optimum derived directly. Formulating the underlying objective function that naturally yields this asymmetry remains an open problem.

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A Data Quality and Comparability

This appendix contains the integrity and descriptive tables referenced in Section 3. Table 5 summarises raw WebSocket feed quality at the recording layer: L2 diff continuity, total gap time per day, book-snapshot capture rate, and mark price coverage. Tables 6 and 7 characterise the raw market and its funding environment per study period, substantiating the regime comparison of Section 3.3. The per-feature and per-target statistical diagnostics, which justify the modelling choices of Section 5 rather than the integrity of the recorded data, are deferred to Appendix B.4.

Table 5: Raw WebSocket feed quality over the 94-day recording window (00:00 UTC on 19 January 2026 to 04:00 UTC on 23 April 2026). Individual rows are shown for every day with a non-zero L2 diff gap, every day with a book-snapshot capture rate below 85 %, and one representative clean day per study period (Pre = pre-analysis days 1–30; OOS = out-of-sample days 31–94). Aggregate rows span all days in each period; percentage columns aggregate as the unweighted mean of daily values, Gap (s) as the period sum (†). *Diffs (%):* fraction of L2 update identifiers received without sequence gaps. *Gap (s):* total L2 feed downtime in seconds (individual days) or †period total (aggregate rows); the two-period sum of 345 s equals the figure cited in Section 3.2. *Book cap. (%):* fraction of the theoretical 100 ms book snapshots delivered by Binance; shortfalls reflect the exchange's periodic snapshot scheduling, not irrecoverable data loss. *Mark cov. (%):* mark-price rows as a fraction of expected seconds in the day (86 400 for full days). Trade update-identifier integrity was 100.00 % on all 94 days and is omitted from the table.

Date	Per.	Diffs (%)	Gap (s)	Book cap. (%)	Mark cov. (%)
Jan 20	Pre	99.9997	0.7	92.57	100.00
Jan 24	Pre	100.0000	0.0	82.21	100.00
Jan 25	Pre	99.9991	29.6	89.24	99.97
Jan 29	Pre	99.9997	4.2	90.89	100.00
Feb 1	Pre	99.9513	31.7	92.97	99.96
Feb 3	Pre	99.9998	4.0	93.93	100.00
Feb 11	Pre	99.9987	1.1	94.72	100.00
Feb 13	Pre	99.9998	0.7	93.70	100.00
Feb 14	Pre	99.9996	0.9	90.10	100.00
Feb 15	Pre	100.0000	0.0	91.22	100.00
Feb 21	OOS	100.0000	0.0	84.84	100.00
Feb 26	OOS	99.9997	4.5	93.12	100.00
Mar 13	OOS	99.9986	0.9	94.01	100.00
Mar 26	OOS	99.8548	222.1	93.45	100.00
Mar 31	OOS	99.9996	4.4	95.22	100.00
Apr 3	OOS	99.9995	0.8	87.94	100.00
Apr 4	OOS	100.0000	0.0	82.34	100.00
Apr 7	OOS	100.0000	0.0	94.21	100.00
Apr 16	OOS	99.9996	5.2	94.00	100.00
Apr 19	OOS	99.9393	34.3	92.07	99.96
All 30 days	Pre	99.9983	72.8†	92.44	100.00
All 64 days	OOS	99.9967	272.1†	92.61	100.00

Table 6: Descriptive market characteristics of the BTCUSDT perpetual over the study window (04:00 UTC 19 January to 04:00 UTC 23 April 2026), by period (Pre = pre-analysis days 1–30; OOS = out-of-sample days 31–94; All = full 94-day window). All quantities are computed on the 1 s grid over valid book ticks; the mid-price range is the daily low–high envelope, all other entries are period means of the per-day statistic. Realised volatility is the daily root of summed squared 1 s mid-price log-returns, also reported annualised, i.e. multiplied by $\sqrt{365}$. Spread in basis points is $10^4 \times \text{spread}/S_t$ (relative to mid-price); spread in ticks is spread/\$0.10 (absolute tick units). Depth is the resting quantity displayed at the best bid and best ask combined; traded notional is price times quantity summed over the day's trades.

Characteristic	Pre	OOS	All
Mid-price range (thousand USDT)	59.9–93.4	62.4–79.4	59.9–93.4
Realised volatility (% / day)	2.91	2.26	2.46
Realised volatility (% ann.)	55.5	43.1	47.1
Mean spread (bps)	0.015	0.015	0.015
Mean spread (ticks)	1.15	1.04	1.08
Mean top-of-book depth (BTC)	9.96	11.21	10.81
Mean daily trades (millions)	2.85	1.88	2.19
Mean daily traded notional (billion USDT)	15.81	12.85	13.80

Table 7: Funding and carry environment of the BTCUSDT perpetual over the study window (04:00 UTC 19 January to 04:00 UTC 23 April 2026), by period (Pre / OOS / All as in Table 6). Statistics are computed over the funding rate *as realised at each 8-hour settlement* (the rate actually charged on position notional, recovered as the last published estimate immediately before settlement), not the time-average of the continuously-published estimate, which mean-reverts within the funding window and would understate the realised carry; all entries are in basis points (1 bp = 0.01%). The mean pools over individual settlements across the period. The premium basis is (mark − mid)/mid in basis points.

Characteristic	Pre	OOS	All
Mean settlement funding (bps)	+0.137	-0.102	-0.026
Mean abs. settlement funding (bps)	0.457	0.335	0.374
Min settlement funding (bps)	-1.518	-1.228	-1.518
Max settlement funding (bps)	+1.000	+0.739	+1.000
Settlements with funding > 0 (%)	66.7	42.7	50.4
Number of settlements	90	192	282
Mean abs. premium basis (bps)	0.797	0.602	0.665

B Pre-Analysis and Validation Studies

This appendix collects everything that was decided or validated on the 30 pre-analysis days before the out-of-sample period was touched: the exact feature definitions, the choice of learning target, the operational label horizon, the statistical characterisation of features and target, the linear-baseline comparison behind the model-class choice, and the validation studies for the leakage controls and for the volatility and intensity estimators the controller consumes. Every choice made here is decided on the 30 pre-analysis days alone; two exceptions touch data from the out-of-sample period without being decisions: a post-hoc stability recheck in Appendix B.3, computed after the horizon was already fixed and playing no part in fixing it, and the purely descriptive target-distribution table of Appendix B.4 (Table 13), which characterises the label rather than choosing anything about it.

B.1 Feature Definitions

This appendix gives the exact definition of every feature in the model input set, complementing the plain descriptions in Section 5. Throughout, book bands are $B \in \{\text{L1}, \text{L2-5}, \text{L6-10}, \text{L11-20}\}$ (disjoint level ranges), trade windows are $w \in \{1, 5, 30\}$ seconds, and a prime denotes the previous 100 ms book update. The VPIN base bucket size is $V_{\text{base}} \approx 70.33$ BTC, calibrated once on the first 30-day fold as the median hourly traded volume scaled so that the $n = 50$ base-scale buckets span roughly 37.5 minute. The short and long scales use $0.2\,V_{\text{base}}$ and $5\,V_{\text{base}}$, which at the same typical volume span about 7.5 minutes and 3.1 hours respectively.

Realised volatility overlaps a quantity the control problem also uses, and the overlap is deliberate rather than double-counting. The feature is the standard deviation of one-second micro-price log-returns, a pure number that reads the current noise level; it is close in spirit to the volatility feed σ that prices the quotes, but it is not the same object, and σ is defined in Section 5.6. The two also act on different moments: there, σ sizes the variance penalty common to every strategy in the ablation; here, the feature lets the learned drift α_{ML} condition its first-moment forecast on the volatility regime.

Table 8: Exact definitions of the model features, by family. $S_t = \frac{1}{2}(P_0^a + P_0^b)$ is the mid-price and $P_t^{\text{micro}} = (Q_0^b P_0^a + Q_0^a P_0^b)/(Q_0^b + Q_0^a)$ the volume-weighted micro-price; $P_i^{b/a}, Q_i^{b/a}$ are the price and resting size at the i-th best level; $V^{\text{buy}}, V^{\text{sell}}$ are buy- and sell-initiated traded volume.

Feature	Definition
Book shape	
`spread`, `spread_bps`	$P_0^a - P_0^b$; and $(P_0^a - P_0^b)/S_t \times 10^4$
`imbalance_`B (OBI)	$\big(\sum_{i\in B} Q_i^b - \sum_{i\in B} Q_i^a\big)/\big(\sum_{i\in B} Q_i^b + \sum_{i\in B} Q_i^a\big)$, per band B
`bid_slope`, `ask_slope`	$\sum_{i<20} Q_i^b/(P_0^b - P_{19}^b)$; and $\sum_{i<20} Q_i^a/(P_{19}^a - P_0^a)$
`slope_imbalance`	(`bid_slope` − `ask_slope`)/(`bid_slope` + `ask_slope`)
`rel_bid_depth_`B, `rel_ask_depth_`B	band depth divided by its own trailing 300 s mean, per side and band
`book_age_ms`	milliseconds since the last book update
Order flow imbalance	
`ofi_`B	$\sum_{i\in B} e_i$ accumulated over $(t-1\text{s}, t]$, with $e_i = Q_i^b \mathbf{1}_{\{P_i^b \geq P_i^{b\prime}\}} - Q_i^{b\prime}\mathbf{1}_{\{P_i^b \leq P_i^{b\prime}\}} + Q_i^{a\prime}\mathbf{1}_{\{P_i^a \geq P_i^{a\prime}\}} - Q_i^a \mathbf{1}_{\{P_i^a \leq P_i^{a\prime}\}}$
Trade flow (each over window w)	
`n_trades_`w`s`	number of trades in $[t-w, t)$
`buy_vol_`w`s`, `sell_vol_`w`s`	buy- and sell-initiated volume in the window
`signed_vol_`w`s`	$V^{\text{buy}} - V^{\text{sell}}$
`trade_imbalance_`w`s`	$(V^{\text{buy}} - V^{\text{sell}})/(V^{\text{buy}} + V^{\text{sell}})$
`largest_trade_`w`s`	largest single trade size in the window
`return_`w`s`	$\log\big(\text{vwap}_w(t)/\text{vwap}_w(t-w)\big)$, where $\text{vwap}_w(t) = \sum_j p_j q_j / \sum_j q_j$ is the volume-weighted average traded price over the window $[t-w, t)$
VPIN	
`vpin_{micro,base,macro}`	$\frac{1}{n}\sum_{j=1}^{n} \lvert V_j^{\text{buy}} - V_j^{\text{sell}}\rvert / V$ over the last $n = 50$ equal-volume buckets, at the three bucket sizes $V = \{0.2, 1, 5\}\, V_{\text{base}}$ (micro, base, macro)
`vpin_window_{micro,base,macro}_s`	wall-clock seconds the n buckets took to fill at that scale (small when trading is fast)
Price dynamics	
`micro_return_`w`s`	$\log\big(P_t^{\text{micro}}/P_{t-w}^{\text{micro}}\big)$
`realized_vol_60s`	rolling standard deviation of 1 s micro-price log-returns over 60 s
Funding context	
`seconds_to_funding`	seconds to the next settlement T_{fund}
`basis_bps`	$(\text{mark} - S_t)/S_t \times 10^4$
`mark_stale_age_ms`	milliseconds since the last mark price update

B.2 Mid-Price versus Micro-Price as the Learning Target

This appendix validates the claim from Section 5.2 that using the adopted mid-price drift, rather than the micro-price drift, sacrifices no predictive power.

Writing the micro-price as the mid-price plus a bounded skew, $P_{\text{micro}}(t) = P_{\text{mid}}(t) + s(t)$

with $|s(t)| \le \text{spread}(t)/2$, the two targets decompose exactly as

$$\text{drift}_h^{\text{micro}}(t) = \text{drift}_h^{\text{mid}}(t) + \frac{s(t+h) - s(t)}{h}. \tag{27}$$

The second term is the bid-ask bounce. Since $s(t)$ is known at time t, a model could predict its $-s(t)/h$ component at no cost, so this term sets the only mechanical advantage a micro-price target could confer. We bound it with a no-model diagnostic computed directly on the labelled pre-analysis rows, assuming nothing about whether the skew reverts: treating the mechanical channel as additive with unit coefficient and neglecting its covariance with the mid-price drift, the maximum coefficient of determination attainable from the current skew alone, expressed against the adopted mid-price target, is

$$R^2_{\text{mech}} \;\le\; \frac{\text{Var}\big(s(t)/h\big)}{\text{Var}\big(\,\text{drift}_h^{\text{mid}}\,\big)}. \tag{28}$$

The bound is conservative in both directions. The denominator uses the mid-price target, whose variance is smaller than the micro-price target's by exactly the bounce increment in (27), inflating the reported ceiling; and the full-reversion predictor $\hat{y} = -s(t)/h$ attains a negative out-of-sample R^2 at every horizon, so the skew does not in fact reliably revert and the true mechanical contribution lies below the ceiling.

Table 9 evaluates this ceiling on the pre-analysis rows: it stays under 0.7% of the mid-price R^2 at every horizon. Table 10 confirms the equivalence directly, training the production LightGBM pipeline on each target: R^2 is indistinguishable, every gap falling well inside the per-fold R^2 standard deviation, and so is directional accuracy once both targets are scored on the same rows.

Table 9: Equivalence of mid-price and micro-price as the LightGBM learning target, computed on the pre-analysis split ($n \approx 2.59 \times 10^6$ rows). The mechanical ceiling (28) is the maximum R^2 any model could extract purely from the current skew, expressed as a fraction of the adopted mid-price target R^2.

h (s)	corr(mid, micro)	R^2_{mech} ceiling	mid-target R^2	ceiling / R^2
1	0.99966	3.6×10^{-4}	0.0653	0.56%
5	0.99994	6.3×10^{-5}	0.0199	0.32%
10	0.99997	3.0×10^{-5}	0.0109	0.28%
30	0.99999	1.0×10^{-5}	0.0014	0.70%

Table 10: Head-to-head pre-analysis comparison of the LightGBM pipeline trained on the mid-price versus the micro-price drift target, pooled over the 16 walk-forward folds ($n \approx 1.38 \times 10^6$ out-of-sample rows per horizon). Directional accuracy is in per cent, scored for both targets on the rows where the mid-price label is non-zero.

	pooled OOS R^2		directional accuracy (%)	
h (s)	mid	micro	mid	micro
1	0.0653	0.0633	70.5	70.4
5	0.0199	0.0194	59.7	59.7
10	0.0109	0.0105	56.5	56.5
30	0.0014	0.0011	52.9	52.9

B.3 Operational Horizon Selection

The label horizon is set to the expected fill time of a passive order quoted at the best bid or ask, computed on the pre-analysis window so that no out-of-sample data enters the choice. Because the replay fills orders last-in-queue, that time is how long the top-of-book queue ahead of the order takes to be consumed. Summing top-of-book depth over the valid 100 ms book snapshots and the market-order volume that actually depletes the top-of-book queue gives the lower bound

$$\tau_{\mathrm{LB}} = \Delta t_{\mathrm{book}} \frac{\sum_t \left(Q_0^b(t) + Q_0^a(t)\right)}{\sum_{j \in \mathcal{L}_1} V_j} = \frac{\bar{V}}{v_{\mathcal{L}_1}}, \tag{29}$$

where the numerator sums the top-of-book depth on both sides over the valid book snapshots t, $\mathcal{L}_1$ is the set of trades that consume top-of-book volume, $\bar{V}$ is the mean two-sided top-of-book depth, and $v_{\mathcal{L}_1}$ the level-one volume-depletion rate. Quoting at the top of book is the most aggressive operating point a GLT strategy uses, so any inventory-aware depth $\delta^* > 0$ only lengthens the wait and τ_{LB} is a lower bound on the operational fill time.[22] On the pre-analysis window, $\bar{V} = 9.96\,\mathrm{BTC}$ and $v_{\mathcal{L}_1} = 1.25\,\mathrm{BTC/s}$ give $\tau_{\mathrm{LB}} = 7.96\,\mathrm{s}$, and that fixes the horizon. Recomputed after the fact on each of the 64 walk-forward training windows, τ_{LB} has median 7.15 s and range $[5.46, 11.74]$ s, snapping to $h = 10\,\mathrm{s}$ in log-space throughout; those windows contain out-of-sample days, so this is a stability diagnostic reported after the choice was made and played no part in it.

B.4 Statistical Characterisation of Features and Target

This appendix collects the per-day diagnostics behind the modelling choices of Section 5.3, in two blocks. The feature block comes first: the joint ADF and KPSS stationarity classification of Table 11, which marks the VPIN series that are first-differenced. Where ADF rejects a unit root and KPSS still rejects stationarity, the feature is kept raw: that disagreement is the known finite-sample over-rejection of KPSS under high persistence, structural breaks or heavy tails, none of which differencing would address. The feature block closes with the variance

[22] The bound replaces $\mathbb{E}[V/v]$ with $\mathbb{E}[V]/\mathbb{E}[v]$, so by Jensen's inequality the true mean fill time is longer; estimating it directly, by averaging the per-bin ratio over bins of width 30 to 600 s, gives a bin-width-dependent mean of 21 to 45 s but a bin-stable median of 11 to 13 s, which still snaps to $h = 10\,\mathrm{s}$ at every width.

inflation table of Table 12 that fixes the Ridge-only collinearity drop. The target block is a single table, the distribution of the adopted mid-price drift target (Table 13); its stationarity and heteroskedasticity are identical at all four horizons and are stated in that table's caption rather than tabulated, the other metric families classify identically, and the mid-versus-micro choice is settled in Appendix B.2. Diagnostics that could change the model use the 30 pre-analysis days only, to keep model design clear of the out-of-sample period; the purely descriptive target distribution uses all 94 days.

Table 11: Joint ADF + KPSS stationarity classification of every model feature of Table 8, per funding day, on the 30 pre-analysis days. Every series was tested on all $n = 30$ days. ADF H_0: unit root; KPSS H_0: stationary. ADF rej. % and KPSS rej. % are the fraction of days each test rejects its null. The S/D/N column counts the day-by-day joint classification: S, both tests indicate stationarity; D (Mixed), the tests disagree, ADF finding no unit root while KPSS rejects stationarity; N (Non.), both indicate a unit root. Modal is the most common class across the 30 days. The four VPIN scale and timing series (`vpin_window_{micro,base,macro}_s` and `vpin_macro`) are first-differenced, three falling in class N (a unit root by both tests) and the fast-bucket fill time borderline; all other series, where ADF rejects the unit root, are left untransformed (see Section 5.3).

Group	Series	ADF rej. %	KPSS rej. %	S/D/N	Modal
Book shape	spread	100%	80%	6/24/0	Mixed
	spread_bps	100%	83%	5/25/0	Mixed
	imbalance_l1	100%	47%	16/14/0	Stat.
	imbalance_l2_5	100%	93%	2/28/0	Mixed
	imbalance_l6_10	100%	57%	13/17/0	Mixed
	imbalance_l11_20	100%	77%	7/23/0	Mixed
	bid_slope	100%	80%	6/24/0	Mixed
	ask_slope	100%	73%	8/22/0	Mixed
	slope_imbalance	100%	43%	17/13/0	Stat.
	rel_bid_depth_l1	100%	0%	30/0/0	Stat.
	rel_bid_depth_l2_5	100%	0%	30/0/0	Stat.
	rel_bid_depth_l6_10	100%	0%	30/0/0	Stat.
	rel_bid_depth_l11_20	100%	0%	30/0/0	Stat.
	rel_ask_depth_l1	100%	0%	30/0/0	Stat.
	rel_ask_depth_l2_5	100%	0%	30/0/0	Stat.
	rel_ask_depth_l6_10	100%	0%	30/0/0	Stat.
	rel_ask_depth_l11_20	100%	0%	30/0/0	Stat.
	book_age_ms	100%	20%	24/6/0	Stat.
OFI	ofi_l1	100%	40%	18/12/0	Stat.
	ofi_l2_5	100%	30%	21/9/0	Stat.
	ofi_l6_10	100%	53%	14/16/0	Mixed
	ofi_l11_20	100%	40%	18/12/0	Stat.
Trade flow	n_trades_1s	100%	100%	0/30/0	Mixed
	buy_vol_1s	100%	100%	0/30/0	Mixed
	sell_vol_1s	100%	93%	2/28/0	Mixed
	signed_vol_1s	100%	3%	29/1/0	Stat.
	trade_imbalance_1s	100%	80%	6/24/0	Mixed
	largest_trade_1s	100%	100%	0/30/0	Mixed
	return_1s	100%	0%	30/0/0	Stat.
	n_trades_5s	100%	100%	0/30/0	Mixed

Group	Series	ADF rej. %	KPSS rej. %	S/D/N	Modal
	buy_vol_5s	100%	97%	1/29/0	Mixed
	sell_vol_5s	100%	93%	2/28/0	Mixed
	signed_vol_5s	100%	3%	29/1/0	Stat.
	trade_imbalance_5s	100%	70%	9/21/0	Mixed
	largest_trade_5s	100%	100%	0/30/0	Mixed
	return_5s	100%	0%	30/0/0	Stat.
	n_trades_30s	100%	100%	0/30/0	Mixed
	buy_vol_30s	100%	97%	1/29/0	Mixed
	sell_vol_30s	100%	90%	3/27/0	Mixed
	signed_vol_30s	100%	0%	30/0/0	Stat.
	trade_imbalance_30s	100%	30%	21/9/0	Stat.
	largest_trade_30s	100%	100%	0/30/0	Mixed
	return_30s	100%	0%	30/0/0	Stat.
VPIN	vpin_micro	100%	87%	4/26/0	Mixed
	vpin_window_micro_s	77%	100%	0/23/7	Mixed
	vpin_base	100%	97%	1/29/0	Mixed
	vpin_window_base_s	7%	100%	0/2/28	Non.
	vpin_macro	10%	100%	0/3/27	Non.
	vpin_window_macro_s	3%	100%	0/1/29	Non.
Price dynamics	micro_return_1s	100%	0%	30/0/0	Stat.
	micro_return_5s	100%	0%	30/0/0	Stat.
	micro_return_30s	100%	0%	30/0/0	Stat.
	realized_vol_60s	100%	100%	0/30/0	Mixed
Funding context	seconds_to_funding	100%	100%	0/30/0	Mixed
	basis_bps	100%	97%	1/29/0	Mixed
	mark_stale_age_ms	100%	13%	26/4/0	Stat.

Table 12: Features with a Variance Inflation Factor above a threshold of 10, computed on the full feature matrix on 2026-02-15. Collinearity is localised: the nine {buy,sell,signed}_vol columns are exactly collinear by construction (signed_vol = buy_vol − sell_vol, so each trade-window triple is rank 2) and have infinite VIF, the two spread terms are large but finite, and every remaining feature is below about 22. The gradient-boosted model is invariant to this; the Ridge baseline drops only the three signed_vol_* columns, which suffices to break the exact dependency, and retains the rest under its L_2 penalty.

Feature	VIF
spread_bps	7683.2
spread	7677.6
slope_imbalance	21.7
n_trades_30s	18.2
imbalance_l1	16.8
n_trades_5s	10.5
9 exactly-collinear flow columns ({buy,sell,signed}_vol_{1,5,30}s)	∞

B.5 Linear Baseline Comparison and Loss Robustness

The model-class choice is tested head to head on the pre-analysis window: the production LightGBM pipeline against a Ridge regression trained under the identical walk-forward pro-

Table 13: Distribution of the adopted learning target, the mid-price drift $\text{drift}_h^{\text{mid}}(t) = (S_{t+h} - S_t)/h$ in price units per second, the direct α_{ML} target, at each label horizon. Each statistic is the mean of the 94 per-funding-day values (the Jarque-Bera column is the daily median). The excess kurtosis far exceeds the Gaussian value of zero at every horizon and the Jarque-Bera test rejects normality on all 94 days, confirming a heavy-tailed target. The target is also stationary (joint ADF + KPSS class S, i.e. ADF rejects the unit root and KPSS does not reject stationarity, on every day at every horizon) and conditionally heteroskedastic (Engle ARCH-LM rejects homoskedasticity on every day); both verdicts are identical across horizons and so are stated here rather than tabulated. It is the conditional heteroskedasticity, not the marginal non-Gaussianity, that motivates the non-linear function class for α_{ML}.

h (s)	Mean (USDT/s)	Std (USDT/s)	Skewness	Excess kurtosis	JB stat (med.)
1	-0.0018	6.018	0.219	98.1	7.4×10^6
5	-0.0018	2.898	0.127	29.5	1.0×10^6
10	-0.0018	2.087	0.162	22.2	5.6×10^5
30	-0.0018	1.217	0.151	15.8	3.1×10^5

tocol, identical features (less the collinear set dropped by the VIF screen of Table 12), and identical purge discipline. Table 14 reports the Diebold-Mariano test at all four label horizons. LightGBM's squared-error loss is significantly lower at 1, 5 and 10 s, the advantage shrinking monotonically with the horizon until it reverses at 30 s, where the pooled R^2 has decayed to 0.0014 and Ridge is marginally better. Non-linear interactions carry value exactly at the short horizons where predictive signal exists. The result is also robust to the training loss: retraining the full pipeline under a Huber objective moves pooled R^2 by +0.00014 and the Diebold-Mariano test cannot separate the two ($p = 0.57$), so the quadratic loss is not doing hidden work.

Table 14: Ridge versus LightGBM on the pre-analysis pooled out-of-sample predictions, target `drift_mid`, at all four label horizons. The Diebold-Mariano statistic is on squared-error loss with $d_t = (y - \hat{y}_{\text{ridge}})^2 - (y - \hat{y}_{\text{lgb}})^2$, so a positive value means LightGBM has the lower loss; standard errors are HAC-robust at 33 lags, on $n \approx 1.38 \times 10^6$ pooled rows per horizon. Directional accuracy is in per cent, over rows with a non-zero label. The last column counts the folds, of 16, on which LightGBM has the lower fold mean squared error.

			pooled OOS R^2		dir. acc. (%)		
h (s)	DM	p	LGB	Ridge	LGB	Ridge	LGB wins
1	+34.48	$< 10^{-5}$	0.0653	0.0472	70.5	69.7	16 / 16
5	+11.37	$< 10^{-5}$	0.0199	0.0173	59.7	59.4	16 / 16
10	+4.45	$< 10^{-5}$	0.0109	0.0096	56.5	56.1	15 / 16
30	−2.58	0.010	0.0014	0.0027	52.9	52.9	9 / 16

B.6 Hyperparameter Transfer Across Folds

The weekly cadence of the hyperparameter search rests on the flat-basin claim of Section 5.4.1: performance is insensitive to which near-optimal configuration is used, even though the tuned hyperparameters themselves swing across much of the search space from fold to fold. The transfer experiment makes this precise. Every pre-analysis fold is refit under the hyperparameter vector tuned on a different donor fold, under three pairings of increasing hostility: donors eight folds away (fully time-disjoint training windows), adjacent donors, and the fold order reversed. Table 15 reports the outcome at the operational horizon: the mean own-minus-foreign R^2 difference is below 0.08 fold-to-fold standard deviations in every pairing, and the same ratio

restricted to leak-free donors, strictly earlier than the test fold so no hyperparameter look-ahead is possible, does not change the verdict.

Table 15: Cross-fold hyperparameter transfer at the operational horizon ($h = 10\,\text{s}$), 16 pre-analysis folds. Each fold is refit under the hyperparameter vector tuned on a different (donor) fold. The three rows are the pairings, in increasing hostility: Offset +8 takes the donor eight folds away, so its training window is time-disjoint from the recipient's, Offset +2 takes an adjacent donor with overlapping data, and Reversed order pairs fold j with fold $n-1-j$, sending late-sample hyperparameters back to early folds and the reverse. ΔR^2/sd is the mean own-minus-foreign R^2 difference as a fraction of the fold-to-fold standard deviation of own R^2. The leak-free column is the same ratio restricted to donor folds strictly earlier than the test fold, over the same fold-to-fold standard deviation.

			ΔR^2/sd	
Pairing	$\bar{R}^2$ own	$\bar{R}^2$ foreign	all donors	leak-free
Offset +8 (time-disjoint)	0.0144	0.0139	+0.075	−0.019 (n=8)
Offset +2 (adjacent)	0.0144	0.0147	−0.053	−0.226 (n=2)
Reversed order	0.0144	0.0140	+0.060	−0.046 (n=8)

B.7 Purge-Window Sizing

The purge gap of Section 5.4.1 is sized from the data rather than assumed. For every event-indicator feature, the daily autocorrelation functions on the 30 pre-analysis days are stacked and their median taken, which is robust to anomalous days; the feature's decay time is the smallest lag at which the median ACF stays inside the $z/\sqrt{n}$ noise band for five consecutive lags. The purge must cover the slowest feature, the deep-book imbalance `imbalance_l11_20` at 990 s, which rounded up to the nearest minute gives $\tau_{\text{purge}} = 1020\,\text{s}$ on the pre-analysis window. Table 16 lists the per-feature decay times behind that figure. Regime-indicator features, realised volatility and the VPIN scales, are deliberately excluded from the maximum: their slow autocorrelation reflects volatility clustering, a property of the market that persists across any train/test seam, not feature memory that a purge could remove. That pre-analysis number is worked here as the example; in production the same procedure is rerun on every rolling training window rather than fixed, which under the stationarity classification of Table 11 is statistically equivalent but keeps every model-relevant parameter on the walk-forward discipline. Across the 64 production windows it returns a median of 900 s and a range of 720 to 1500 s.

Table 16: Per-feature decay timescales from the median ACF across the 30 pre-analysis days. For each microstructure feature the daily ACFs are stacked and the median curve is taken across days; τ is the smallest lag at which $|\widetilde{\mathrm{ACF}}(k)|$ stays inside the noise band $z/\sqrt{n} = 0.007$ for five consecutive lags. The last two columns give the median ACF at two reference lags, so entries above 0.007 are still autocorrelated there. The slowest feature sets the purge: its τ, rounded up to the nearest minute, is the operational τ_{purge}. Regime-indicator features, realised volatility and the VPIN scales, are excluded because their slow ACF reflects regime persistence, not seam-specific contamination.

Feature	τ (s)	median ACF (60 s)	median ACF (300 s)
ofi_l1	11	0.006	0.003
ofi_l2_5	6	0.005	0.003
ofi_l6_10	7	0.003	0.000
ofi_l11_20	10	0.004	0.002
imbalance_l1	100	0.012	0.004
imbalance_l2_5	856	0.031	0.014
imbalance_l6_10	738	0.022	0.013
imbalance_l11_20	990	0.031	0.013
trade_imbalance_30s	666	0.075	0.018
signed_vol_30s	381	0.060	0.015
micro_return_30s	58	-0.003	-0.011

B.8 Volatility Estimator Validation

Table 17 collects the estimator diagnostics behind the σ feed of Section 5.6. Three results matter downstream. First, the bipower rejection: on the one-second grid, where 59% of mid increments are exactly zero, the bipower-to-realised ratio is 0.689; recomputed on a five-second grid it recovers to 0.880, identifying tick discreteness rather than jumps as the cause and disqualifying bipower as the level feed. Second, level robustness: the sparse-grid and EWMA variants agree with the production estimator within 9% at the median. Third, window robustness: halving or doubling the 60 s trailing window moves the median estimate by about 1%, and even a 300 s window moves it by 7.5%. Volatility reaches the quotes only through the inventory penalty $\alpha = \frac{1}{2}k\gamma\sigma^2$, so the γ sweep of Table 25, run over a factor of sixteen, spans a factor of four in σ at the same time; no plausible window choice comes close to that span.

Table 17: Realised-volatility estimator validation on the 30 pre-analysis days. Each row is the across-day distribution of the stated ratio, against the production 60 s estimator σ; subscript sp denotes the sparse five-second grid. σ_{BP} is the bipower variation, jump-insensitive by construction, so a ratio far below one indicates either jumps or, as here, returns too often exactly zero to pair; σ_{sparse} and σ_{EWMA} are the same window subsampled and exponentially weighted, testing microstructure-noise bias and responsiveness; the window rows compare shorter and longer trailing windows against the 60 s production choice. A ratio near one is agreement, and the three verdicts are read in Appendix B.8.

Analysis	Metric	Median	P5	P95
Jump diagnostic, 1 s grid	σ_{BP}/σ	0.689	0.000	0.981
Jump diagnostic, 5 s grid	$\sigma_{BP,sp}/\sigma_{sp}$	0.880	0.734	0.955
Noise bias	σ_{sparse}/σ	1.087	0.733	1.447
EWMA responsiveness	σ_{EWMA}/σ	1.043	0.800	1.723
Window, $w = 30$ s	σ_{30}/σ_{60}	0.994	0.989	1.014
Window, $w = 120$ s	σ_{120}/σ_{60}	1.009	1.005	1.014
Window, $w = 300$ s	σ_{300}/σ_{60}	1.075	1.046	1.104

B.9 Intensity Estimation Window and Stride

The rolling (A, k) estimator has three free operational constants, all selected on pre-analysis and validated here. The window length is chosen by leave-future-out predictive deviance: each candidate window fits the queue-aware model and is scored on the virtual-order outcomes of the following, out-of-window evaluation block. The 45-minute window attains the minimum raw deviance per block at every evaluation-block length tested, from two minutes to one hour, beating shorter windows (noisier fits, per-block deviance 167 at 15 minutes against 114 at 45) and longer ones (staler fits). The stride is validated by measuring how fast the fitted parameters actually move: Table 18 shows the median relative change in k between fits 60 s apart is 2.2%, comparable to the fit's own noise floor of 0.6%, so refitting more often than every minute would track estimation noise rather than the market.

The third constant is the lifetime $H = 2.5$ s over which a virtual order is tracked before it is treated as expired. It is set at the time the mid needs to diffuse across one characteristic distance, $(1/(k\sigma))^2$, so a virtual order lives exactly as long as the market state that priced it rather than for a tuned interval; the diffusion scale here is the jump-insensitive one-second mid volatility of 6.6 USDT per BTC per square-root second, not the 60 s level feed of Section 5.6, since a lifetime should be robust to the outliers the level feed exists to see. The recomputed timescale at the final calibration came out at 2.85 s, and H was kept at the 2.5 s committed on pre-analysis rather than refitted.

Table 19 tests the two assumptions the exponential specification makes. Over 5.18 million virtual orders and the 0.99 million of them that filled, the log-rate fit returns $R^2 = 0.998$, and the alternative power-law shape is rejected outright ($\Delta\text{AIC} = +274$); a quadratic term is detectable at this sample size ($p = 1.5 \times 10^{-12}$) but changes the fitted rate by under 4% across the distances the controller occupies. Fitting the bid and ask sides separately, the check on the symmetry $A^b = A^a$, $k^b = k^a$ imposed throughout Section 4, gives differences of 3.7% in k and 1.1% in A: statistically resolvable on a sample this size, and economically negligible against the 10.8% week-to-week variation in k itself.

Table 18: Stride validation for the rolling queue-aware (A, k) estimator on the pre-analysis days: relative parameter change $|\Delta k/k|$ and $|\Delta A/A|$ between fits separated by the stated lag, against the estimator's own resampling noise floor.

metric & lag (s)	n_{pairs}	median (%)	p95 (%)	p99 (%)	floor (%)
k, 60	41668	2.232	9.602	21.348	0.579
k, 300	41456	5.521	22.203	44.341	1.294
k, 900	40983	9.787	38.142	71.882	2.241
A, 60	41668	2.054	7.455	10.609	0.486
A, 300	41456	5.716	18.422	27.446	1.087
A, 900	40983	11.122	38.392	64.783	1.883

Per-lag noise floor $= (\sqrt{2} \times \text{rel-SE}) \times \sqrt{\text{lag}/L}$, $L = 2700$ s.
Bound at lag$\geq L$: $k = 3.88\%$, $A = 3.26\%$. Default stride $S = 60$ s.

Table 19: Functional-form and symmetry validation of the queue-aware intensity estimator on the 30 pre-analysis days. $\hat{A}$ and $\hat{k}$ are the pooled fit of $\lambda(\delta) = Ae^{-k\delta}$ over all resting virtual orders, each tracked for $H = 2.5\,\text{s}$. A positive ΔAIC favours the exponential over the stated alternative, so the power law is rejected and the quadratic term is a detectable but economically negligible refinement. The symmetry block fits the bid and ask sides separately: the Wald test resolves a difference in k at this sample size, while the size of that difference stays well inside the week-to-week variation of k itself, which is the comparison that licenses the symmetric specification of Section 4.

Quantity	Value
Sample	
Resting virtual orders	5,183,880
of which filled ($H = 2.5\,\text{s}$)	985,142
Fitted intensity	
$\hat{A}$ (per second per side)	0.274
$\hat{k}$ (USDT^{-1})	0.0900
Functional form	
log-rate R^2, exponential	0.998
log-rate R^2, with quadratic term	0.999
ΔAIC, power law − exponential	+274
ΔAIC, quadratic − exponential	−51
quadratic coefficient (p)	3.98×10^{-4} (1.5×10^{-12})
Side symmetry	
$\lvert k^b - k^a \rvert / k$	3.7%
$\lvert A^b - A^a \rvert / A$	1.1%
Wald p, $k^b = k^a$	1.6×10^{-5}
weekly variation in k (CV)	10.8%

C Additional Results

This appendix holds the supplementary tables and figures behind results the main chapter states in one line each (Section 6).

P&L attribution. Table 20 gives the five-component daily decomposition of Section 5.8 for the full hybrid, behind Figure 3.

Table 20: Daily P&L attribution for the full hybrid over the 64 out-of-sample days (USDT). Every component is a signed contribution to the daily total, so negative entries subtract from P&L; funding is a net receipt over the window. The inventory residual is the mark-to-market of the position carried beyond the five-second markout window, computed as the balancing term of the identity.

Component	Mean / day	SD / day	Total
Spread capture	278.10	145.29	17,798
Adverse selection (5 s)	-487.03	245.74	-31,170
Funding received	0.10	0.12	7
Exchange fees	-489.25	231.59	-31,312
Inventory residual	9.15	26.21	585
Terminal P&L	-688.94	329.08	-44,092

Placebo control. Table 21 reports the real-versus-placebo contrast of Section 6.4: the time-shuffled drift signal with identical marginal distribution, under both couplings, with the paired per-day differences.

Table 21: Real (r) versus placebo (p) for the drift signal strategies under both couplings; the placebo is the same signal shuffled in time, preserving its marginal distribution and destroying only when its moves happen. Δ is the paired per-day difference, with a stationary-bootstrap 95% CI and paired t.

	mean/day (USDT)		fills/day		Δ (r − p)		
Coupling, strategy	r	p	r	p	mean/day	95% CI	t
Horizon, AR	-725	-719	12,493	12,411	−6	$[-8, -4]$	−5.2
Horizon, ML	-688	-847	11,784	14,727	+159	$[150, 167]$	21.7
Horizon, Hybrid	-689	-850	11,805	14,790	+161	$[152, 170]$	21.9
Defensive, AR	-696	-705	11,998	12,161	+8	$[6, 11]$	7.2
Defensive, ML	-527	-602	8,981	10,297	+75	$[69, 82]$	19.9
Defensive, Hybrid	-527	-603	8,982	10,299	+76	$[69, 83]$	20.1

Funding boundary census. Table 22 is the per-settlement census of the Regime-II boundary layer on the out-of-sample days behind the activation and switch-lead numbers of Section 6.5.

Table 22: Census of the funding boundary layer for strategy 5 over the 192 out-of-sample settlements: fraction of settlements at which the strategy latches into Regime II, and the distribution of the switch lead u^*, in seconds before settlement, over the settlements that switch. The last block gives the funding scalar $|F_t| = S_t|f_t|$ at settlement, in USDT per BTC, the magnitude that drives the layer and so sets how far ahead of settlement the switch occurs.

	Out-of-sample
Settlements	192
Layer switches on (%)	92.7
Regime-II switch lead u^* (s), switching settlements	
min	60
median	815
mean	923
max	4070
Funding scalar at settlement (USDT per BTC)	
median $\lvert F_t \rvert$	2.000
mean $\lvert F_t \rvert$	2.371

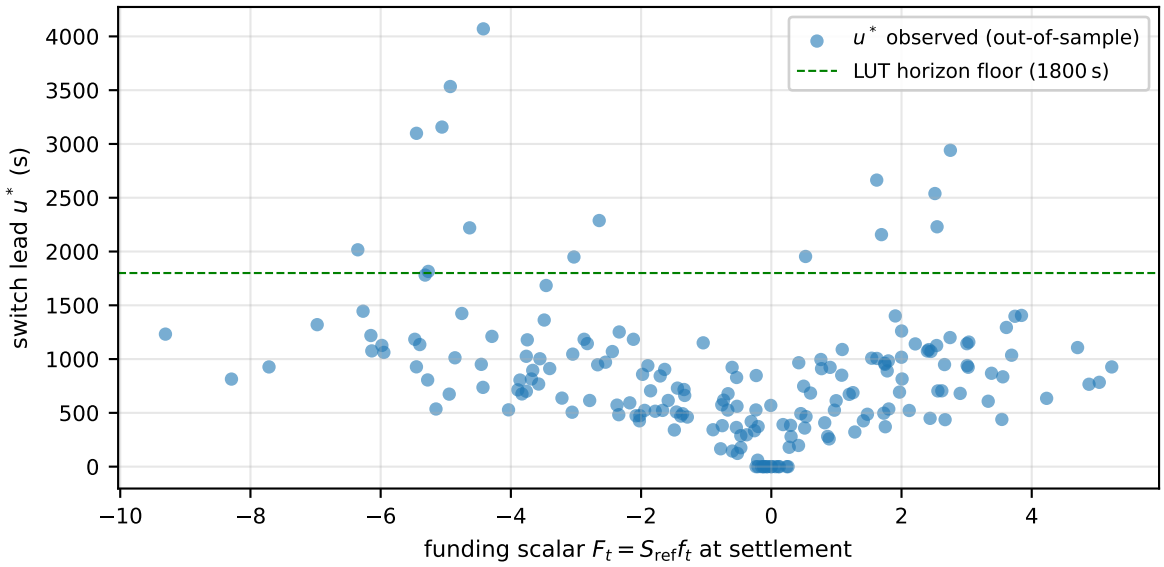


Figure 8: The boundary-layer census behind Table 22, per settlement. Observed Regime-II switch leads u^* widen with the funding scalar $|F_t|$, in USDT per BTC, and vanish in the near-zero-funding cluster where no layer forms. The dashed line marks the lookup table's minimum horizon of 1800 s: the rebuilt table always covers at least this window, so switch leads below the line are set purely by the one-tick trigger, while leads above it occur at high carry, when the rebuild extends the table's horizon beyond its floor.

Stress-test table. Table 23 gives the full Monte Carlo results of Section 6.6 across the five funding scenarios, whose generator and grid are specified in Appendix E.5. It carries the columns the main text omits: the uncontrolled naive baseline, quoting at GLT's own median half-spread so the comparison is not one of participation; fills and per-fill P&L, which separate volume from per-fill economics; time at the inventory cap; and the footer line giving the paired mean gap with its standard error. The absolute P&L levels inherit the retail-fee toll of Section 6.1 by construction, since the suite books the same fees as the replay; the informative objects are the paired same-seed contrasts between rows. Cap (%) is the mean fraction of path time spent at the inventory cap; Fills and P&L/fill separate participation from per-fill economics, mirroring the decomposition of Section 6.2. The Markout column is the per-fill 5 s markout against the generator's true mid: *positive* for every strategy on the synthetic tape, at roughly the quoted depth, because the synthetic flow carries no information; this is the no-informed-flow benchmark cited in Section 6.1, and it shows the losses here are carried by the fee alone.

Table 23: Ornstein-Uhlenbeck funding-rate stress: 200 paired synthetic paths per scenario, funding cap ±75 bp. The naive arm (strategy 1) quotes at a width-matched fixed half-spread of 8.6 USDT. Every entry is a mean over the 200 paths of that scenario: Mean P&L, $ES_{5\%}$ (the mean of the worst 5% of paths) and Funding collected are USDT per path, Cap (%) is the share of path time spent at the inventory limit $|q| = Q$, Fills is the path fill count, and P&L/fill and Markout are per fill, in USDT and in basis points of the fill price respectively. Scenario definitions are in Appendix E.5.

Scenario	Strat.	Mean P&L	$ES_{5\%}$	Cap (%)	Funding	Fills	P&L/fill	Markout (bp)
baseline const	s1	-992	-1,013	7.98	0.0	17,388	-0.057	+0.86
baseline const	s2	-729	-744	0.28	0.0	15,240	-0.048	+1.04
baseline const	s5	-729	-744	0.28	0.1	15,237	-0.048	+1.04
cap pinned	s1	-991	-1,054	7.98	1.2	17,388	-0.057	+0.86
cap pinned	s2	-729	-774	0.28	0.3	15,240	-0.048	+1.04
cap pinned	s5	-661	-686	1.55	59.6	15,048	-0.044	+1.04
documented carry	s1	-992	-1,013	7.98	0.1	17,388	-0.057	+0.86
documented carry	s2	-729	-745	0.28	0.0	15,240	-0.048	+1.04
documented carry	s5	-726	-741	0.34	2.4	15,221	-0.048	+1.04
persistent neg	s1	-993	-1,038	7.98	-0.7	17,388	-0.057	+0.86
persistent neg	s2	-730	-757	0.28	-0.2	15,240	-0.048	+1.04
persistent neg	s5	-690	-710	1.06	33.7	15,114	-0.046	+1.04
persistent pos	s1	-992	-1,033	7.98	0.7	17,388	-0.057	+0.86
persistent pos	s2	-729	-759	0.28	0.2	15,240	-0.048	+1.04
persistent pos	s5	-690	-712	1.06	33.7	15,110	-0.046	+1.04
sign flip	s1	-994	-1,046	7.98	-1.4	17,388	-0.057	+0.86
sign flip	s2	-730	-765	0.28	-0.9	15,240	-0.048	+1.04
sign flip	s5	-693	-728	0.96	31.4	15,129	-0.046	+1.04

Paired mean P&L gap, s5 − s2, USDT/path ± s.e.: baseline const $+0.3 \pm 0.05$; cap pinned $+67.6 \pm 1.38$; documented carry $+3.1 \pm 0.14$; persistent neg $+39.6 \pm 0.88$; persistent pos $+38.9 \pm 0.88$; sign flip $+37.4 \pm 1.55$.

Inventory-capacity sweep under carry stress. Table 24 substantiates the capacity-lever discussion of Section 6.6: the funding-aware strategy and its funding-blind control re-run at doubled and quadrupled inventory capacity on identical stress paths.

Table 24: Inventory-capacity sweep under the two carry scenarios (200 paired paths each, horizon coupling; seeds are shared across Q legs and strategies, and the $Q = 10$ rows are the main funding suite's own). Mean terminal P&L, expected shortfall (mean of the worst 5% of paths), and funding collected, all in USDT per path. The final column is the paired per-path mean P&L gap, strategy 5 minus strategy 2, with its standard error. The hybrid (strategy 6) tracks strategy 5 within 2 USDT in every cell and is omitted.

		GLT (s2)		Funding (s5)			
Scenario	Q	P&L	$ES_{5\%}$	P&L	Funding	$ES_{5\%}$	Δ(s5−s2)
Cap-pinned	10	-729	-774	-661	59.6	-686	+67.6 ± 1.4
Cap-pinned	20	-734	-821	-595	133.3	-626	+138.5 ± 2.8
Cap-pinned	40	-733	-876	-453	281.2	-511	+279.5 ± 4.9
Doc. carry	10	-729	-745	-726	2.4	-741	+3.1 ± 0.1
Doc. carry	20	-735	-755	-729	7.6	-752	+6.5 ± 0.4
Doc. carry	40	-737	-768	-725	18.4	-769	+12.3 ± 1.1

Parameter sensitivity. Tables 25 and 26 report the sweeps behind Section 6.8; Table 27 re-prices the recorded fill streams across the maker-fee tier range. The two replay sweeps re-run the full engine on a deterministic sixteen-day subset of the out-of-sample window, every fourth day starting 18 February 2026, a rule fixed before any cell was run. Each table carries the diagnostic column its parameter acts through directly, inventory for γ and the Regime II time share for ρ.

Table 25: Sensitivity of the full hybrid strategy to miscalibration of the risk-aversion hyperparameter γ over a factor of sixteen, on the sixteen-day sweep subset. $\gamma \times 1$ is the calibrated configuration. Mean $|q|$ is in lots of 0.005 BTC.

	γ	P&L/day (USDT)	daily s.d. (USDT)	mean $\lvert q \rvert$ (lots)	fills/day
$\gamma \times 0.25$	5.00e-06	-631	305	2.79	10978
$\gamma \times 0.5$	1.00e-05	-633	307	2.76	11017
$\gamma \times 0.75$	1.50e-05	-635	307	2.71	11043
$\gamma \times 1$	2.00e-05	-638	309	2.68	11082
$\gamma \times 1.5$	3.00e-05	-640	310	2.63	11128
$\gamma \times 2$	4.00e-05	-643	312	2.57	11180
$\gamma \times 4$	8.00e-05	-646	314	2.34	11252

Table 26: Sensitivity of the funding strategies to misspecification of the boundary-layer decay rate ρ (ρ-scale 1.0 is the calibrated value). Only quantities ρ directly acts on are shown: the discrete funding collected and the share of live time spent in Regime II.

ρ scale	Strat.	Funding P&L (USDT)	Regime II frac.
0.25	s5	1.3	0.048
0.25	s6	1.5	0.048
0.5	s5	1.7	0.037
0.5	s6	1.8	0.037
0.75	s5	1.8	0.033
0.75	s6	2.1	0.033
1	s5	2.1	0.031
1	s6	2.3	0.031
1.5	s5	2.2	0.028
1.5	s6	2.4	0.028
2	s5	2.3	0.027
2	s6	2.4	0.027
4	s5	2.4	0.025
4	s6	2.4	0.025

Table 27: Fee-tier sensitivity (post hoc re-pricing of the recorded fills). A positive maker fee is charged to the maker, a negative one is paid to it. Mean daily P&L rather than Sharpe, since a Sharpe comparison across strategies is unsafe in a negative-mean regime; m^* is the maker fee at which the full hybrid's P&L crosses zero. Each row re-prices the same recorded fill streams at one maker and taker fee pair, so only the fee columns change between rows; the last column records whether the full hybrid beats the GLT baseline on mean daily P&L at that pair.

Maker (bp)	Taker (bp)	s6 P&L/day (USDT)	s2 P&L/day (USDT)	s6 > s2
-1.00	3.0	44	55	no
-1.00	4.0	44	55	no
-1.00	5.0	44	55	no
-1.00	6.0	44	55	no
-0.50	3.0	-78	-74	no
-0.50	4.0	-78	-74	no
-0.50	5.0	-78	-74	no
-0.50	6.0	-78	-74	no
+0.00	3.0	-200	-202	yes
+0.00	4.0	-200	-202	yes
+0.00	5.0	-200	-202	yes
+0.00	6.0	-200	-203	yes
+1.00	3.0	-444	-460	yes
+1.00	4.0	-444	-460	yes
+1.00	5.0	-445	-460	yes
+1.00	6.0	-445	-460	yes
+2.00	3.0	-689	-717	yes
+2.00	4.0	-689	-717	yes
+2.00	5.0	-689	-717	yes
+2.00	6.0	-689	-717	yes
taker 3.0 bp: m^* = -0.82 bp; s6 and s2 cross at -0.17 bp				
taker 4.0 bp: m^* = -0.82 bp; s6 and s2 cross at -0.17 bp				
taker 5.0 bp: m^* = -0.82 bp; s6 and s2 cross at -0.17 bp				
taker 6.0 bp: m^* = -0.82 bp; s6 and s2 cross at -0.17 bp				

SHAP attribution shares. Table 28 lists the per-feature attribution shares in the bulk of the funding epoch versus the pre-settlement boundary window, behind Figure 6.

Table 28: SHAP attribution shares (feature mean |SHAP| as a fraction of the total) in the bulk of the epoch versus the final pre-settlement window, and their ratio. Arrows mark the features whose share rises in the boundary layer.

feature	bulk share	boundary share	ratio
imbalance_l1	16.8%	16.4%	0.98
rel_bid_depth_l1	12.3%	12.2%	0.99
rel_ask_depth_l1	11.9%	11.6%	0.97
slope_imbalance	10.1%	9.8%	0.97
ask_slope	3.4%	3.4%	0.98
ofi_l1	2.7%	2.7%	0.99
spread_bps	2.7%	2.6%	0.97
bid_slope	2.7%	2.6%	0.96
micro_return_1s	2.3%	2.2%	0.98
imbalance_l11_20	2.2%	2.3%	1.01
micro_return_30s	2.1%	2.0%	0.97
imbalance_l6_10	2.0%	2.0%	0.99
n_trades_30s	2.0%	2.0%	1.04
micro_return_5s	1.4%	1.4%	1.01
seconds_to_funding$^{\uparrow}$	0.9%	1.4%	1.48
sell_vol_30s$^{\uparrow}$	1.1%	1.4%	1.20
vpin_micro$^{\uparrow}$	0.8%	0.9%	1.12

boundary = $u \leq 30$ min (344,853 rows), bulk = $u > 2$ h (4,146,839 rows).
Above the rule: largest features by bulk share. Below it: the smaller features whose share rises most toward the settlement; $^{\uparrow}$ = boundary share $\geq 1.1\times$ bulk share.

C.1 Quoting on the Publication Clock

Every number so far was produced against a book a median 51 ms old, the phase cost Section 3 described. That staleness belongs to the replay rather than to the strategy, and it is the one feature of the engine that could manufacture the loss regime rather than measure it, so this subsection removes it: the same decisions, submitted at the exchange's publication instants and anchored to the book and inventory of that instant. Table 29 reports every strategy and coupling on both clocks.

What the clock changes is what the per-fill columns can resolve: the five markouts now spread twice as far apart, 0.041 bps against 0.020, with less noise around each. Markout is adverse selection, the price move against a fill in the five seconds after it, plus the spread that fill captured on the way in, so both track how far from the mid a strategy rests. The horizon coupling's fills look less adverse than GLT's, -1.818 against -1.930 bps, only because they capture least, 1.076 against 1.217; the defensive coupling, resting furthest out, is the same trade reversed, worst adverse selection in the table and best markout. Across the five strategies markout rises with captured spread at $+0.148$ bps per bps ($t = 4.7$), one line explaining 88% of the difference between them, where the deployed clock left no slope at all ($t = 0.1$). Matched for quoted width the five span 0.018 bps. Neither column ranks a strategy, which is why every per-fill comparison across strategies returns a null, now on a fresh book as well.

This is where the question fixed before these cells were run gets its answer: was the drift signal only compensating for a book we quoted against too late, which a faster clock should erase, or a real risk clock, which it should leave standing? It stands. Re-running Section 6.4's

Table 29: The ablation on both clocks, 64 out-of-sample days. Each strategy appears twice, on the deployed clock of Table 1 and on the exchange's publication clock; P&L/day, SD, Sharpe, $ES_{5\%}$, Fills/day and $\overline{|q|}$ carry the definitions and units of that table. Capture and adverse selection are per-fill means in basis points of the fill price, and sum fill by fill to the markout. The naive row is the width-matched baseline; the "(def.)" rows carry the defensive coupling of Table 2.

Strategy	Clock	P&L/day (USDT)	SD (USDT)	Sharpe	$ES_{5\%}$ (USDT)	Fills/day	$\overline{\lvert q \rvert}$ (lots)	Capture (bps)	Adv. sel. (bps)
1 Naive (matched)	deployed	-670	391	-32.7	-1,741	11,705	5.67	1.238	-2.098
	publication	-390	232	-32.1	-1,021	7,388	5.72	1.262	-1.991
2 GLT	deployed	-717	312	-43.9	-1,404	12,377	2.78	1.202	-2.054
	publication	-424	186	-43.5	-834	7,861	2.65	1.217	-1.930
3 GLT + AR	deployed	-725	312	-44.4	-1,412	12,493	2.70	1.192	-2.048
	publication	-428	185	-44.3	-837	7,947	2.56	1.206	-1.928
4 GLT + ML	deployed	-688	329	-40.0	-1,455	11,784	2.67	1.120	-1.989
	publication	-438	211	-39.7	-933	8,180	2.42	1.080	-1.821
5 GLT + funding	deployed	-717	312	-43.9	-1,403	12,378	2.78	1.202	-2.055
	publication	-423	186	-43.6	-834	7,859	2.64	1.216	-1.930
6 Full hybrid	deployed	-689	329	-40.0	-1,458	11,805	2.69	1.117	-1.986
	publication	-439	211	-39.8	-935	8,205	2.45	1.076	-1.818
3 GLT + AR (def.)	deployed	-696	308	-43.2	-1,375	11,998	2.76	1.220	-2.070
	publication	-409	183	-42.6	-812	7,583	2.63	1.237	-1.942
4 GLT + ML (def.)	deployed	-527	244	-41.3	-1,076	8,981	2.86	1.344	-2.204
	publication	-312	147	-40.6	-656	5,793	2.73	1.339	-2.041
6 Full hybrid (def.)	deployed	-527	244	-41.3	-1,075	8,982	2.87	1.344	-2.205
	publication	-313	147	-40.6	-657	5,800	2.75	1.339	-2.042

tercile panel on the publication clock, the ML book takes the counter side of strong-signal fills at 0.44× GLT's rate and the aligned side at 1.78×, against 0.45× and 1.48× on the deployed clock (Table 3). Both halves of the coupling sharpen: it declines more of what the signal warns about, and the tightening it applies on the other side now converts into fills instead of being blunted by a stale state. Raw P&L shows the second half. The ML strategy trades 319 fills a day more than GLT where it used to trade 593 fewer, so its 29 USDT advantage becomes a 15 USDT deficit, every extra fill paying the toll. Once again the quality of the fills returns a null: the whole of the drift signal's value sits in when to decline fills.

Two patterns run through the full record. Sharpe moves by less than a point on either clock, since daily mean and daily volatility fall together, which is the split of Section 6.1 operating in reverse. And the inventory ordering is untouched: the naive baseline carries 5.7 lots on both clocks against the risk-managed family's 2.4 to 2.9, so refreshing the state changes how much a strategy trades, not how much risk it chooses to hold.

One asymmetry runs against the drift signal strategies on this clock. With no fixed grid, each quote carries the inputs of the preceding grid point, and the drift forecast, produced once a second on a computational-cost argument, is more stale still (Section 3). The comparison is therefore mildly conservative for the ML arms, since they are the only strategies that carry this additional stale parameter.

D Mathematical Appendix

The mathematical material is collected here in four parts: a table of the notation used throughout, the formal proofs and extended derivations behind Section 4, the boundary that separates the two operating regimes, and the justification of the three structural approximations underpinning the scalar ansatz.

D.1 Notation

Symbols used throughout Section 4 and this appendix are collected below. Symbols introduced only within a single appendix are defined where they appear.

Symbol	Meaning
Time and horizons	
t	Running time
T	Terminal horizon of a single control problem
Δt	Micro-horizon between successive re-solves
T_{fund}	Next discrete funding settlement time
ΔT_{epoch}	Interval between successive settlements
$u = T_{\text{fund}} - t$	Backward time remaining to settlement
u^*, t^*	Regime-transition boundary, in backward and forward time
Market and maker state	
S_t	Reference price
S_{ref}	Reference price frozen at the start of the current solve
μ	Drift of the reference price, held constant within a solve
σ	Volatility of the reference price, held constant within a solve
W_t	Standard one-dimensional Brownian motion
q_t	Inventory, in lots
Q	Inventory capacity, $q_t \in \{-Q, \dots, Q\}$
$N_q = 2Q + 1$	Dimension of the reduced ODE system
V	Lot size, the constant trade size; q_t and Q are counted in lots
X_t	Cash
Quoting and fills	
S_t^b, S_t^a	Bid and ask prices posted by the maker
δ_t^b, δ_t^a	Bid and ask half-spreads, $S_t - S_t^b$ and $S_t^a - S_t$, the control variables
$\delta_{\min}$	Lower bound on the half-spreads
$c_1 = \frac{1}{\gamma}\ln(1 + \gamma/k)$	Inventory-independent term of the half-spread
ψ^*	Total optimal spread, $\delta^{a*} + \delta^{b*}$
N_t^b, N_t^a	Point processes counting fills on the bid and the ask
λ^b, λ^a	Fill intensities, $A\exp(-k\delta)$
A	Intensity scale, the arrival rate of marketable flow
k	Intensity decay, the rate at which fills thin with distance

Symbol	Meaning
$\mathcal{A}$	Set of admissible quoting strategies
Preferences and value	
γ	Absolute risk aversion coefficient
$U(x) = -e^{-\gamma x}$	CARA utility, the argument being wealth
$v(t, X, S, q)$	Value function
$\mathbb{E}_t[\cdot]$	Expectation conditional on the state at t
ν	Terminal unwinding cost coefficient
Funding	
$f_{T_{\text{fund}}}$	Funding rate applied at settlement, the terminal value of f_t
f_t	Pre-settlement estimate of that rate
$F_t = S_{\text{ref}} f_t$	Deterministic funding scalar
ρ	Decay rate of the continuous funding drain, Equation (6)
$\tilde{\rho}$	Epoch-normalising constant, $\rho/(1 - e^{-\rho \Delta T_{\text{epoch}}})$
$\Phi(t, q)$	Continuous funding penalty, $kqF_t\tilde{\rho}e^{-\rho(T_{\text{fund}}-t)}$
Reduced system	
$\omega_q(t)$	Risk-penalty state vector, one component per inventory level
$\alpha = \frac{1}{2}k\gamma\sigma^2$	Quadratic inventory penalty coefficient
$\beta = k\mu$	Drift coefficient
η	Fill-coupling constant, $A(1 + \gamma/k)^{-(1+k/\gamma)}$
$M(t)$	Tridiagonal state matrix, $\dot{\omega} = M(t)\omega$
$D_q(t)$	Diagonal entry of $M(t)$, $\alpha q^2 - \beta q + \Phi(t, q)$
f, f_q	Trial vector of Equation (17), whose minimiser is f^0
f^0	Principal eigenvector of the autonomous M, belonging to its smallest eigenvalue
x	Continuous inventory variable of the relaxed criterion, Proposition 3 and Appendix D.2.7
λ_0, λ_1	Smallest and next eigenvalue of M; their difference is the spectral gap
Signals	
α_{ML}	Machine learning drift signal, in price units per second

D.2 Proofs and Extended Derivations

This appendix contains the formal proofs of the propositions and theorems stated in Section 4, alongside the complete derivation of the partitioned HJB equation and technical details regarding boundary conditions and admissibility. Notation follows Section 4 throughout and is collected in Appendix D.1. Appendices D.2.3 to D.2.7 are structured along the lines of GLT, and depart from it in two places: the funding penalty $\Phi(t, q)$, which has no counterpart there, and the drift μ, whose resulting system that paper states without proof. Both are carried through every result below. The derivations are set out at greater length than a research article would allow them,

so that every step can be checked where it stands rather than reconstructed by the reader.

D.2.1 Admissibility of the Control Set

Let $\mathcal{A}$, the set of admissible quoting strategies, consist of all $\mathcal{F}_t$-predictable processes $\delta = (\delta_t^a, \delta_t^b)_{t\in[0,T]}$ satisfying $\delta_t^a, \delta_t^b \geq \delta_{\min}$ almost surely, where $\delta_{\min}$ is a finite scalar lower bound. Any finite value serves the mathematical purpose set out below, whatever its sign. Throughout we take $\delta_{\min} \leq 0$. The sign is what permits an admissible strategy to concede price in order to shed inventory at all, which a strictly positive bound would forbid; the finiteness of the bound, not its sign, is what caps how far that concession may go. Section 5.6 states the value used.

Economically, a negative bound is the largest such concession, crossed half-spread plus exchange taker fee, that the market maker is willing to make in order to shed inventory quickly, and a bound of zero forbids crossing altogether, holding the bid at or below the reference price and the ask at or above it.

Mathematically, the floor keeps the problem well posed by bounding the fill intensity. The exponential intensity $A\exp(-k\delta)$ is unbounded as a quote moves inward, so without a lower bound on δ an admissible strategy could drive the arrival rate arbitrarily high, and N^a and N^b could then explode, recording infinitely many fills in a finite interval. With the floor in place the intensity never exceeds $A\exp(-k\delta_{\min})$, the uniform bound on the jump compensators used in Appendix D.2.5.

Conversely, no upper bound is imposed on the controls. A quoting distance $\delta^{a/b} \to \infty$ economically corresponds to the market maker completely withdrawing their quotes from that side of the limit order book. Mathematically, this simply drives the exponential arrival intensities to zero ($\lambda^{a/b} \to 0$), shutting off the corresponding counting process without introducing any ill-posedness to the HJB formulation.

The bound is therefore a well-posedness device rather than an execution rule, and the analysis that follows works throughout with the unconstrained interior maximisers of the Hamiltonian in Proposition 1. Where the constraint is slack those maximisers are the optimal controls and the linear ODE system of Proposition 1 is exact. Whether it is slack depends on where the floor is set. The higher $\delta_{\min}$, the more states in which a quote is pushed past it; there the supremum is attained at $\delta_{\min}$, the corresponding row of the system acquires the clipped Hamiltonian in place of the interior one, and the solved ω bounds the attainable value from above. This is a consequence of the operational choice of $\delta_{\min}$, not of the reduction itself.

Finally, admissibility is defined broadly on the full space of predictable controls. The Markovian feedback structure of the optimal controls δ^* is not imposed as an initial constraint, but rather emerges naturally as a property of the analytical solution. No separate integrability condition is needed: the floor, the inventory capacity and the finite horizon together make the expectation in Equation (7) well defined for every admissible strategy, as Appendix D.2.5 verifies.

D.2.2 Full Derivation of the Partitioned HJB Equation

To derive the HJB equation introduced in Section 4.2, we must first define the stochastic dynamics of the market maker's state variables over an infinitesimal time step dt. The reference price

evolves as a diffusion process, the inventory transitions via discrete trade executions modelled as point processes, and the cash balance is driven by both trade cash flows and continuous funding charges:

$$dS_t = \mu dt + \sigma dW_t, \tag{30}$$

$$dq_t = dN_t^b - dN_t^a, \tag{31}$$

$$dX_t = (S_t + \delta_t^a) dN_t^a - (S_t - \delta_t^b) dN_t^b - q_t F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t)) dt. \tag{32}$$

With the state dynamics established, we apply Itô's Lemma for semi-martingales with jump processes to the value function $v(t, X, S, q)$ of Equation (7). For notational brevity throughout the differential expansion, we suppress the state arguments and denote the value function at the current state (t, X_t, S_t, q_t) simply as v, writing the arguments out only where the function is evaluated at the displaced state a fill produces. This is the heuristic derivation, working with differentials over an infinitesimal dt, so it does not distinguish a state from its left limit. That distinction does real work in the Verification Theorem, where predictability of the integrands is what makes the compensated jump integrals true martingales, and Appendix D.2.5 carries the left-limit notation $\tau-$ throughout for that reason.

Substituting the state dynamics into the multidimensional Itô formula, the infinitesimal change in the value function is given by:

$$\begin{aligned} dv = &\left(\partial_t v + \mu \partial_S v + \tfrac{1}{2}\sigma^2 \partial_{SS} v - (q_t F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))) \, \partial_X v\right) dt \\ &+ \sigma \partial_S v \, dW_t \\ &+ \left[v(t, X_t + S_t + \delta_t^a, S_t, q_t - 1) - v\right] dN_t^a \\ &+ \Big[v(t, X_t - S_t + \delta_t^b, S_t, q_t + 1) - v\Big] \, dN_t^b. \end{aligned} \tag{33}$$

By taking the conditional expectation $\mathbb{E}_t[\cdot]$ of both sides over the infinitesimal interval $[t, t + dt]$, two simplifications occur:

1. The expectation of the Brownian increment is zero ($\mathbb{E}_t[dW_t] = 0$), causing the stochastic diffusion term to vanish entirely.

2. The expectations of the point-process increments are given by their respective predictable compensators: $\mathbb{E}_t[dN_t^a] = \lambda^a(\delta_t^a) dt$ and $\mathbb{E}_t[dN_t^b] = \lambda^b(\delta_t^b) dt$.

Substituting these expected jump rates into the differential yields an expression factored by the deterministic time step dt:

$$\begin{aligned} \mathbb{E}_t[dv] = \Bigg(&\partial_t v + \mu \partial_S v + \tfrac{1}{2}\sigma^2 \partial_{SS} v - (q_t F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))) \, \partial_X v \\ &+ \lambda^a(\delta_t^a) \left[v(t, X_t + S_t + \delta_t^a, S_t, q_t - 1) - v\right] \\ &+ \lambda^b(\delta_t^b) \Big[v(t, X_t - S_t + \delta_t^b, S_t, q_t + 1) - v\Big] \Bigg) dt. \end{aligned} \tag{34}$$

Under the Dynamic Programming Principle the value process is a supermartingale along any admissible control, so its expected increment is non-positive, and a martingale along an optimal

one, so the increment is zero there. Together these say that the supremum over admissible controls of the expected increment vanishes. Dividing by dt gives the continuous-time HJB equation:

$$\sup_{\delta^a,\delta^b\geq\delta_{\min}} \frac{\mathbb{E}_t[dv]}{dt} = 0. \tag{35}$$

To enforce the inventory capacity limits $q \in \{-Q,\dots,Q\}$, this continuous-time formulation is partitioned. The interior states ($|q| < Q$) yield the primary equation detailed in Section 4.2. At the inventory bounds, fills that would breach the capacity are excluded by dropping the respective suprema, yielding the following boundary-state equations:

For the upper inventory bound ($q = Q$):

$$\begin{aligned} 0 = {} & \partial_t v + \mu\partial_S v + \tfrac{1}{2}\sigma^2\partial_{SS} v - \left(QF_t\tilde{\rho}\exp(-\rho(T_{\text{fund}}-t))\right)\partial_X v \\ & + \sup_{\delta^a\geq\delta_{\min}} \left\{\lambda^a(\delta^a)\left[v(t,X+S+\delta^a,S,Q-1) - v(t,X,S,Q)\right]\right\}. \end{aligned} \tag{36}$$

For the lower inventory bound ($q = -Q$):

$$\begin{aligned} 0 = {} & \partial_t v + \mu\partial_S v + \tfrac{1}{2}\sigma^2\partial_{SS} v - \left(-QF_t\tilde{\rho}\exp(-\rho(T_{\text{fund}}-t))\right)\partial_X v \\ & + \sup_{\delta^b\geq\delta_{\min}} \left\{\lambda^b(\delta^b)\left[v(t,X-S+\delta^b,S,-Q+1) - v(t,X,S,-Q)\right]\right\}. \end{aligned} \tag{37}$$

As established in Section 4.2, the terminal condition $v(T,X,S,q) = -\exp(-\gamma(X+qS-\nu|q|))$ closes the partitioned system, pricing the forced unwind of the remaining inventory.

D.2.3 Proof of Proposition 1: Change of Variables for the Extended HJB

Let $v(t,X,S,q)$ follow the extended ansatz defined in Equation (13). At this stage v is a candidate function rather than the value function; Theorem 1 identifies the two. We first compute its partial derivatives:

$$\begin{aligned} \partial_t v &= -\frac{\gamma}{k} v(t,X,S,q)\frac{\dot{\omega}_q(t)}{\omega_q(t)} \\ \partial_X v &= -\gamma v(t,X,S,q) \\ \partial_S v &= -\gamma q v(t,X,S,q) \\ \partial_{SS} v &= \gamma^2 q^2 v(t,X,S,q). \end{aligned}$$

Next, we evaluate the jump components of the Hamiltonian. For the bid side (valid for $q < Q$), the supremum term of Equation (8) is:

$$\sup_{\delta^b\geq\delta_{\min}} \left\{\lambda^b(\delta^b)\left[v(t,X-S+\delta^b,S,q+1) - v(t,X,S,q)\right]\right\}.$$

Substituting the explicit Avellaneda-Stoikov intensity function $\lambda^b(\delta^b) = A\exp(-k\delta^b)$ and

applying our ansatz to the jump state yields:

$$\begin{aligned} v(t, X - S + \delta^b, S, q+1) &= -\exp\Big(-\gamma(X - S + \delta^b + (q+1)S)\Big) \cdot \omega_{q+1}(t)^{-\frac{\gamma}{k}} \\ &= -\exp\Big(-\gamma(X + qS + \delta^b)\Big) \cdot \omega_{q+1}(t)^{-\frac{\gamma}{k}} \qquad (38) \\ &= v(t, X, S, q)\exp(-\gamma\delta^b)\left(\frac{\omega_{q+1}(t)}{\omega_q(t)}\right)^{-\frac{\gamma}{k}}. \qquad (39) \end{aligned}$$

The maximisation problem thus becomes:

$$\sup_{\delta^b \geq \delta_{\min}} \left\{ A\exp(-k\delta^b) v(t, X, S, q) \left[\exp(-\gamma\delta^b)\left(\frac{\omega_{q+1}(t)}{\omega_q(t)}\right)^{-\frac{\gamma}{k}} - 1 \right] \right\}.$$

Differentiating the objective with respect to δ^b and factoring out the non-zero $A\exp(-k\delta^b) \cdot v(t, X, S, q)$ leaves the bracket $k - (k+\gamma)\exp(-\gamma\delta^b)\left(\omega_{q+1}(t)/\omega_q(t)\right)^{-\gamma/k}$, which is strictly increasing in δ^b and crosses zero exactly once. Because $v < 0$, the derivative of the objective carries the opposite sign to the bracket: the objective rises to the left of that crossing and falls to the right of it. The crossing is therefore the unique global maximum, not merely a stationary point, and the First-Order Condition (FOC) alone locates it. Setting the bracket to zero,

$$k - (k+\gamma)\exp(-\gamma\delta^{b*})\left(\frac{\omega_{q+1}(t)}{\omega_q(t)}\right)^{-\frac{\gamma}{k}} = 0,$$

and rearranging for δ^{b*} provides the optimal bid half-spread:

$$\delta_t^{b*} = \frac{1}{k}\ln\left(\frac{\omega_q(t)}{\omega_{q+1}(t)}\right) + \frac{1}{\gamma}\ln\left(1 + \frac{\gamma}{k}\right). \qquad (40)$$

Before fully evaluating the supremum, we evaluate the two components of the jump term at the optimal control. First, substituting δ_t^{b*} into the exponential jump bracket yields a deterministic scalar:

$$\exp(-\gamma\delta_t^{b*})\left(\frac{\omega_{q+1}(t)}{\omega_q(t)}\right)^{-\frac{\gamma}{k}} - 1 = -\frac{\gamma}{k+\gamma} = -\frac{\gamma}{k}\left(1 + \frac{\gamma}{k}\right)^{-1}.$$

Second, substituting δ_t^{b*} into the intensity function yields:

$$\begin{aligned} \lambda^b(\delta_t^{b*}) &= A\exp(-k\delta_t^{b*}) \\ &= A\exp\left(-\ln\left(\frac{\omega_q(t)}{\omega_{q+1}(t)}\right) - \frac{k}{\gamma}\ln\left(1 + \frac{\gamma}{k}\right)\right) \\ &= A\left(\frac{\omega_{q+1}(t)}{\omega_q(t)}\right)\left(1 + \frac{\gamma}{k}\right)^{-\frac{k}{\gamma}}. \end{aligned}$$

Multiplying the two components by the value function gives the bid-side jump term in closed form:

$$\sup_{\delta^b \geq \delta_{\min}} \{\dots\} = -v(t, X, S, q)\left[A\left(1 + \frac{\gamma}{k}\right)^{-\left(1+\frac{k}{\gamma}\right)}\right]\frac{\gamma}{k}\left(\frac{\omega_{q+1}(t)}{\omega_q(t)}\right).$$

Substituting the constant $\eta = A\left(1+\frac{\gamma}{k}\right)^{-\left(1+\frac{k}{\gamma}\right)}$, we achieve the final simplified term:

$$\sup_{\delta^b \geq \delta_{\min}} \left\{ \lambda^b(\delta^b) \left[v(t, X - S + \delta^b, S, q+1) - v(t, X, S, q) \right] \right\} = -v(t, X, S, q) \cdot \eta \frac{\gamma}{k} \frac{\omega_{q+1}(t)}{\omega_q(t)}. \quad (41)$$

By symmetric derivation, the optimal ask half-spread δ_t^{a*} (for $q > -Q$) and the evaluation of its corresponding supremum are:

$$\delta_t^{a*} = \frac{1}{k} \ln\left(\frac{\omega_q(t)}{\omega_{q-1}(t)}\right) + \frac{1}{\gamma} \ln\left(1+\frac{\gamma}{k}\right). \quad (42)$$

Together the two half-spreads are Equation (15) of Section 4.4.

Similarly, substituting δ_t^{a*} into the ask-side exponential jump bracket yields the identical deterministic scalar:

$$\exp(-\gamma \delta_t^{a*}) \left(\frac{\omega_{q-1}(t)}{\omega_q(t)}\right)^{-\frac{\gamma}{k}} - 1 = -\frac{\gamma}{k+\gamma}.$$

Plugging this scalar and the optimal intensity $\lambda^a(\delta_t^{a*})$ back into the HJB yields the final simplified ask term:

$$\sup_{\delta^a \geq \delta_{\min}} \{ \lambda^a(\delta^a) \left[v(t, X + S + \delta^a, S, q-1) - v(t, X, S, q) \right] \} = -v(t, X, S, q) \cdot \eta \frac{\gamma}{k} \frac{\omega_{q-1}(t)}{\omega_q(t)}. \quad (43)$$

Finally, putting all the terms together for the interior states ($|q| < Q$), the full HJB equation and its substituted partial derivatives yield:

$$\begin{aligned} 0 &= \partial_t v + \mu \partial_S v + \tfrac{1}{2}\sigma^2 \partial_{SS} v - \left(q F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))\right) \partial_X v \\ &\quad + \sup_{\delta^a \geq \delta_{\min}} \{\dots\} + \sup_{\delta^b \geq \delta_{\min}} \{\dots\} \\ &= -\frac{\gamma}{k} v \frac{\dot{\omega}_q(t)}{\omega_q(t)} - \mu\gamma q v + \tfrac{1}{2}\sigma^2\gamma^2 q^2 v + \left(q F_t \tilde{\rho} \exp(-\rho(T_{\text{fund}} - t))\right) \gamma v \\ &\quad - v \cdot \eta \frac{\gamma}{k} \frac{\omega_{q-1}(t)}{\omega_q(t)} - v \cdot \eta \frac{\gamma}{k} \frac{\omega_{q+1}(t)}{\omega_q(t)}. \end{aligned}$$

Dividing the entire equation by the strictly positive term $\left(-\frac{\gamma}{k}\frac{v(t,X,S,q)}{\omega_q(t)}\right)$ removes v, and with it every occurrence of X and S. This is the step that reduces the coupled PDEs to ODEs in time alone, and Section 4.3 explains why the funding term survives it:

$$0 = \dot{\omega}_q(t) + (k\mu)\, q\omega_q(t) - \left(\frac{k}{2}\gamma\sigma^2\right) q^2 \omega_q(t) - \left(kqF_t\tilde{\rho}\exp(-\rho(T_{\text{fund}} - t))\right)\omega_q(t) + \eta\left(\omega_{q-1}(t) + \omega_{q+1}(t)\right).$$

Before isolating the time derivative, we group the coefficients of $\omega_q(t)$ and substitute the system parameters defined in Proposition 1 ($\alpha = \frac{k}{2}\gamma\sigma^2$, $\beta = k\mu$, and $\Phi(t,q) = kqF_t\tilde{\rho}\exp(-\rho(T_{\text{fund}} - t))$):

$$0 = \dot{\omega}_q(t) - \left(\alpha q^2 - \beta q + \Phi(t,q)\right)\omega_q(t) + \eta\left(\omega_{q-1}(t) + \omega_{q+1}(t)\right).$$

Rearranging this equation to isolate $\dot{\omega}_q(t)$ provides the final ODE for the interior inventory

states ($|q| < Q$):

$$\dot{\omega}_q(t) = \left(\alpha q^2 - \beta q + \Phi(t,q)\right)\omega_q(t) - \eta\left(\omega_{q-1}(t) + \omega_{q+1}(t)\right).$$

For the boundary states, the market maker stops quoting on the side that would breach the capacity, which is what Equations (36) and (37) of Appendix D.2.2 express.

At the upper inventory limit ($q = Q$), the market maker ceases quoting on the bid side. Consequently the bid-side supremum is absent, so no coupling to ω_{Q+1} appears, yielding:

$$\dot{\omega}_Q(t) = \left(\alpha Q^2 - \beta Q + \Phi(t,Q)\right)\omega_Q(t) - \eta\omega_{Q-1}(t).$$

At the lower inventory limit ($q = -Q$), the market maker ceases quoting on the ask side. The ask-side supremum is absent, so no coupling to ω_{-Q-1} appears. Substituting $q = -Q$ into the drift term flips the sign of the β component, yielding:

$$\dot{\omega}_{-Q}(t) = \left(\alpha Q^2 + \beta Q + \Phi(t,-Q)\right)\omega_{-Q}(t) - \eta\omega_{-Q+1}(t).$$

Finally, to verify the terminal condition, we evaluate the ansatz at the liquidation horizon $t = T$:

$$v(T,X,S,q) = -\exp\left(-\gamma(X+qS)\right)\cdot\omega_q(T)^{-\frac{\gamma}{k}}.$$

Substituting the terminal condition of the ODE system, $\omega_q(T) = \exp(-k\nu|q|)$, yields:

$$\begin{aligned} v(T,X,S,q) &= -\exp\left(-\gamma(X+qS)\right)\cdot\left(\exp(-k\nu|q|)\right)^{-\frac{\gamma}{k}} \\ &= -\exp\left(-\gamma(X+qS)\right)\cdot\exp(\gamma\nu|q|) \\ &= -\exp\left(-\gamma(X+qS-\nu|q|)\right). \end{aligned}$$

This exactly recovers the terminal boundary condition of the HJB equation established in Section 4.2. Because the derived ODEs match Equations (9), (10) and (11), and the terminal boundary condition is explicitly satisfied, the extended ansatz is a classical solution. The proposition is proven. □

D.2.4 Proof of Proposition 2: Strict Positivity

We now prove Proposition 2, establishing the strict positivity ($\omega_q(t) > 0$) of the functions $(\omega_q(t))_{|q|\le Q}$. This condition is a prerequisite for the definition of the value function $v(t,X,S,q)$. Because $\omega_q(t)$ is raised to the strictly negative fractional power $-\frac{\gamma}{k}$ (where risk aversion $\gamma > 0$ and intensity decay $k > 0$), any violation of this positivity breaks the control problem in two ways:

1. **The Zero Bound ($\omega_q = 0$):** A negative exponent means $\omega_q^{-\gamma/k} = \frac{1}{\omega_q^{\gamma/k}}$, so $\omega_q(t) = 0$ is a division by zero. The value function diverges to $-\infty$ and the ansatz is undefined there.

2. **The Negative Bound ($\omega_q < 0$):** Because $\frac{\gamma}{k}$ is not an integer, a negative base raised to this power is undefined over the reals. Utility maximisation is posed on the real line, so an expected utility that is not real-valued has no meaning within this framework.

Conversely, the other state variables, cash (X) and the reference price (S), are permitted to cross zero. Because they enter only through the exponential factor $-\exp(-\gamma(X+qS))$, negative values map to lower utilities, which is well defined. Only $\omega_q(t)$ carries the domain restriction.

Neither the drift coefficient β nor the funding penalty $\Phi(t,q)$ is sign-restricted, so the diagonal entries $D_q(t)$ are not sign-definite and may vary in time. We prove that the solutions $\omega_q(t)$ are nonetheless bounded below by a strictly positive function and never reach zero.

By the Extreme Value Theorem, a continuous real-valued function on a compact set attains an absolute maximum and minimum. The solutions exist and are continuous on the closed interval $[0,T]$ by Proposition 2, and the inventory space $\{-Q,\dots,Q\}$ is finite, so their product is compact.

Recall from the main text the diagonal entries $D_q(t) = \alpha q^2 - \beta q + \Phi(t,q)$. Section 4.3 establishes that these coefficients are continuous and bounded on $[0,T]$, which is what the existence argument of Proposition 2 requires: α, β and η are each either constant or, if estimated as time-varying quantities, held at their current values across a solve under certainty equivalence; q ranges over a finite set, and $\Phi(t,q)$ carries its time dependence as a decaying exponential. A continuous function on a compact domain is bounded, so we may define the global bounds

$$C_{\max} = \sup_{t\in[0,T],\,|q|\le Q} D_q(t), \quad C_{\min} = \inf_{t\in[0,T],\,|q|\le Q} D_q(t). \tag{44}$$

Furthermore, let $m > 0$ be the lower bound of the terminal condition evaluated at the horizon T. Because the terminal liquidation boundary is defined as $\omega_q(T) = \exp(-k\nu|q|)$, and the absolute inventory capacity is finite ($|q| \le Q$), the global minimum of this boundary condition is bounded strictly away from zero. For any $\nu \ge 0$ this minimum is attained at the maximum absolute inventory capacity, symmetrically at $q = Q$ and $q = -Q$. We therefore define $m = \exp(-k\nu Q) > 0$.

We claim that the solutions remain strictly positive and satisfy the following lower bound, obtained by integrating the worst-case scalar comparison equation $\dot{y} = (C_{\max} - \eta)y$ backward from $y(T) = m$, in which every diagonal entry sits at its supremum and the coupling contributes least:

$$\forall t \in [0,T], \quad \forall q \in \{-Q,\dots,Q\}, \quad \omega_q(t) \ge m\exp(-(C_{\max}-\eta)(T-t)). \tag{45}$$

The constant is $C_{\max} - \eta$ rather than the sharper $C_{\max} - 2\eta$ because the bound must hold uniformly over all inventory states, and at $q = \pm Q$ only one neighbouring state contributes to the coupling.

The argument is a comparison of maximum-principle type: we assume the bound fails, locate the worst point, and show that the differential inequality there cannot hold. Suppose then that the bound is violated somewhere, so that $\exp(-\zeta(T-t))\,(\omega_q(t) - m\exp(-(C_{\max}-\eta)(T-t)))$ is strictly negative at some point of the domain. Fix $\epsilon > 0$ small enough that adding $\epsilon(T-t)$, which is at most ϵT, leaves that value negative. The role of ϵ is to convert a weak inequality into a strict one. Without it, the first-order condition at the minimum would only guarantee the derivative is non-negative; the added term $\epsilon(T-t)$ instead guarantees it is at least ϵ, a strictly positive number, which is what the contradiction below needs.

The weight $\exp(-\zeta(T-t))$ is strictly positive, so it changes no sign, and the constant

$\zeta = 2\eta - C_{\min}$ is chosen so that the coefficient $\zeta + D_q - 2\eta$ collapses to $D_q - C_{\min}$, which is non-negative by the definition of $C_{\min}$. Under this hypothesis the composite function attains a strictly negative global minimum:

$$\min_{t\in[0,T],\,|q|\le Q} \left[\exp(-\zeta(T-t))\left(\omega_q(t) - m\exp(-(C_{\max}-\eta)(T-t))\right) + \epsilon(T-t)\right] < 0. \tag{46}$$

Because the term evaluated at $t = T$ is non-negative ($\omega_q(T) \ge m$), this minimum must be achieved at some coordinate (t^*, q^*) with $t^* < T$, leaving room to perturb t forward without leaving the domain. Since the function value cannot decrease below its minimum under such a perturbation, the time derivative of the composite test function is non-negative at this minimum; since the $\epsilon(T-t)$ term contributes $-\epsilon$ to that derivative, the remaining part satisfies:

$$\frac{d}{dt}\left[\exp(-\zeta(T-t))\left(\omega_{q^*}(t) - m\exp(-(C_{\max}-\eta)(T-t))\right)\right]\Big|_{t=t^*} \ge \epsilon. \tag{47}$$

Evaluating this derivative using the product and chain rules, and factoring out the common $\exp(-\zeta(T-t^*))$ term, yields:

$$\begin{aligned}\exp(-\zeta(T-t^*))\Big[&\zeta\left(\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*))\right)\\ &+ \dot\omega_{q^*}(t^*) - m(C_{\max}-\eta)\exp(-(C_{\max}-\eta)(T-t^*))\Big] \ge \epsilon.\end{aligned} \tag{48}$$

Multiplying through by the strictly positive term $\exp(\zeta(T-t^*))$ gives:

$$\begin{aligned}&\zeta\left(\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*))\right) + \dot\omega_{q^*}(t^*)\\ &\quad - m(C_{\max}-\eta)\exp(-(C_{\max}-\eta)(T-t^*)) \ge \epsilon\exp(\zeta(T-t^*)).\end{aligned} \tag{49}$$

We now substitute the governing ODE. If $|q^*| < Q$, we have $\dot\omega_{q^*}(t^*) = D_{q^*}(t^*)\omega_{q^*}(t^*) - \eta(\omega_{q^*-1}(t^*) + \omega_{q^*+1}(t^*))$. Substituting this into Equation (49) yields:

$$\begin{aligned}&\zeta\left(\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*))\right) + D_{q^*}(t^*)\omega_{q^*}(t^*) - \eta(\omega_{q^*-1}(t^*) + \omega_{q^*+1}(t^*))\\ &\quad - m(C_{\max}-\eta)\exp(-(C_{\max}-\eta)(T-t^*)) \ge \epsilon\exp(\zeta(T-t^*)).\end{aligned}$$

Define the composite test function $F(t,q)$:

$$F(t,q) = \exp(-\zeta(T-t))\left[\omega_q(t) - m\exp(-(C_{\max}-\eta)(T-t))\right] + \epsilon(T-t).$$

The global minimum gives $F(t^*,q^*) \le F(t,q)$ for all t and q. Fixing the time at t^* leaves $F(t^*,q^*) \le F(t^*,q)$ for all $q \in \{-Q,\dots,Q\}$, so q^* minimises F across the inventory grid at that instant.

Evaluating F at the adjacent inventory states $q^* \pm 1$ gives:

$$\begin{aligned}&\exp(-\zeta(T-t^*))\left[\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*))\right] + \epsilon(T-t^*)\\ &\le \exp(-\zeta(T-t^*))\left[\omega_{q^*\pm1}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*))\right] + \epsilon(T-t^*).\end{aligned}$$

Every term other than ω depends on t alone and is identical on both sides, so subtracting $\epsilon(T-t^*)$, dividing by the positive weight $\exp(-\zeta(T-t^*))$ and cancelling the $m\exp(\dots)$ term leaves:

$$\omega_{q^*}(t^*) \le \omega_{q^*\pm1}(t^*). \tag{50}$$

Consequently we can bound the coupling terms in the governing ODE,

$$-\eta\,(\omega_{q^*-1}(t^*)+\omega_{q^*+1}(t^*)) \le -2\eta\,\omega_{q^*}(t^*),$$

and applying this inequality to the ODE-substituted expression yields:

$$\begin{aligned}&\zeta\,(\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*))) + D_{q^*}(t^*)\omega_{q^*}(t^*) - 2\eta\omega_{q^*}(t^*)\\ &\quad - m(C_{\max}-\eta)\exp(-(C_{\max}-\eta)(T-t^*)) \ge \epsilon\exp(\zeta(T-t^*)).\end{aligned}$$

To use the contradiction hypothesis we need the quantity assumed negative, $(\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*)))$, to appear explicitly. We therefore add and subtract $(D_{q^*}(t^*) - 2\eta)m\exp(-(C_{\max}-\eta)(T-t^*))$, then simplify the coefficients:

$$\begin{aligned}&(\zeta + D_{q^*}(t^*) - 2\eta)\,(\omega_{q^*}(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*)))\\ &\quad + m(D_{q^*}(t^*) - C_{\max} - \eta)\exp(-(C_{\max}-\eta)(T-t^*)) \ge \epsilon\exp(\zeta(T-t^*)).\end{aligned} \tag{51}$$

We can now evaluate the signs of the left-hand side of Equation (51) at the global minimum coordinate:

1. By the choice $\zeta = 2\eta - C_{\min}$, the first coefficient $(\zeta + D_{q^*}(t^*) - 2\eta)$ simplifies to $(D_{q^*}(t^*) - C_{\min})$. Note that (t^*, q^*) minimises the composite test function, not the diagonal entry $D_q(t)$, so this term need not be zero; but $C_{\min}$ is the global infimum of $D_q(t)$ over the whole domain, so $D_{q^*}(t^*) \ge C_{\min}$ and the coefficient is non-negative (≥ 0). Under the contradiction hypothesis the bound is breached at (t^*, q^*), so $(\omega_{q^*}(t^*) - m\exp(\dots))$ is strictly negative. The product of a non-negative and a strictly negative term ensures the entire first component is non-positive (≤ 0).

2. By the definition of the global supremum, the term $(D_{q^*}(t^*) - C_{\max})$ is non-positive (≤ 0). Because the fill-coupling constant η is strictly positive, subtracting it guarantees the coefficient $(D_{q^*}(t^*) - C_{\max} - \eta)$ is strictly negative. Since the lower bound m and the exponential function are strictly positive, their product forms a strictly negative second component (< 0).

Consequently, the sum of all terms on the left-hand side is strictly negative. This contradicts the inequality, as the right-hand side $\epsilon\exp(\zeta(T-t^*))$ is strictly positive.

The same contradiction occurs at the boundaries. If $q^* = Q$, the governing ODE drops the bid-side coupling, yielding $\dot\omega_Q(t^*) = D_Q(t^*)\omega_Q(t^*) - \eta\omega_{Q-1}(t^*)$. Applying the minimum property $-\eta\omega_{Q-1}(t^*) \le -\eta\omega_Q(t^*)$ and substituting into the inequality gives:

$$\begin{aligned}&(\zeta + D_Q(t^*) - \eta)\,(\omega_Q(t^*) - m\exp(-(C_{\max}-\eta)(T-t^*)))\\ &\quad + m(D_Q(t^*) - C_{\max})\exp(-(C_{\max}-\eta)(T-t^*)) \ge \epsilon\exp(\zeta(T-t^*)).\end{aligned}$$

Because $(\zeta + D_Q(t^*) - \eta) = (\eta + D_Q(t^*) - C_{\min}) > 0$, the first term is strictly negative, forcing the contradiction once again. A symmetric argument holds for $q^* = -Q$.

Every case yields a contradiction, so the initial assumption is false. Therefore:

$$\omega_q(t) \geq m \exp(-(C_{\max} - \eta)(T - t)) > 0. \tag{52}$$

This concludes the proof that the family of functions $(\omega_q(t))_{|q| \leq Q}$ remains strictly positive, so the ansatz of Equation (13) is well defined throughout. □

D.2.5 Proof of Theorem 1: Verification Theorem

To verify that the candidate solution coincides with the true value function, we evaluate the system dynamics under an arbitrary admissible control policy. The proof is stated on the fixed horizon $[t, T]$ of a single control problem; in the implementation T is the next settlement T_{fund}, and the receding-horizon engine re-solves the system repeatedly inside that horizon as new initial conditions arrive (Section 4.8). Throughout this proof, the parameters μ, σ, A, and k, together with the funding scalar F_t, are treated as deterministic continuous functions of time on this horizon, whether held constant or, if estimated as time-varying quantities, refreshed periodically, consistent with the certainty-equivalence approximation established in Appendix D.4.1. The convention removes their stochasticity within each individual re-solve while preserving the slower time-variation captured by the continuous refresh of initial conditions. The generic admissible control is written ϑ throughout this proof, reserving δ for the optimal controls of Equation (15), so that the two are never confused in the inequalities below. Let $\vartheta = (\vartheta^a_t, \vartheta^b_t) \in \mathcal{A}$ be an arbitrary pair of admissible controls. Let $t \in [0, T)$, and consider the controlled state processes for $\tau \in [t, T]$:

$$\begin{aligned} dS^{t,s}_\tau &= \mu \, d\tau + \sigma \, dW_\tau, \quad S^{t,s}_t = s \\ dX^{t,x,\vartheta}_\tau &= (S_\tau + \vartheta^a_\tau) \, dN^a_\tau - (S_\tau - \vartheta^b_\tau) \, dN^b_\tau \\ &\qquad - q_{\tau-} F_\tau \tilde{\rho} \exp(-\rho(T_{\text{fund}} - \tau)) \, d\tau, \quad X^{t,x,\vartheta}_t = x \\ dq^{t,q,\vartheta}_\tau &= dN^b_\tau - dN^a_\tau, \quad q^{t,q,\vartheta}_t = q. \end{aligned} \tag{53}$$

To enforce the market maker's finite risk capacity within the stochastic model, the counting processes N^b and N^a, which almost surely have no common jump times, carry $\mathcal{F}_\tau$-predictable intensities λ^b_τ and λ^a_τ truncated by indicator functions:

$$\lambda^b_\tau = A \exp(-k\vartheta^b_\tau) \mathbf{1}_{\{q_{\tau-} < Q\}}, \quad \lambda^a_\tau = A \exp(-k\vartheta^a_\tau) \mathbf{1}_{\{q_{\tau-} > -Q\}}. \tag{54}$$

These are not Poisson processes. Their intensities depend on the control and on the current inventory, so they are point processes with predictable intensities, and it is the compensated differences $dM^{a/b}_\tau = dN^{a/b}_\tau - \lambda^{a/b}_\tau \, d\tau$ that are martingales.

The indicator functions guarantee that the inventory satisfies $q_\tau \in \{-Q, \ldots, Q\}$ almost surely under any control policy. This gating is the stochastic counterpart of the treatment in Appendix D.2.2, where the capacity is enforced by dropping the blocked side's supremum from Equations (36) and (37). The two are the same operation: at $q = Q$ the bid intensity is zero, so the bid contributes nothing to the generator whatever ϑ^b is, and the value of the blocked quote

is immaterial.

With the state dynamics and capacity constraints established, we can evaluate the evolution of our candidate value function v along an arbitrary controlled trajectory. Because v is smooth, we apply Itô's formula for semi-martingales with jumps between t and a localising stopping time τ_n. We define $\tau_n = T \wedge \inf\{\tau > t : |S_\tau - s| \geq n \text{ or } |N^a_\tau - N^a_t| \geq n \text{ or } |N^b_\tau - N^b_t| \geq n\}$ for $n \in \mathbb{N}$.

Notation: To preserve visual clarity within the extended Itô expansion, we suppress the explicit initial condition and control superscripts on the state variables. Specifically, we write $X_{\tau-}$, S_τ, and $q_{\tau-}$ instead of their fully expanded forms $X^{t,x,\vartheta}_{\tau-}$, $S^{t,s}_\tau$, and $q^{t,q,\vartheta}_{\tau-}$. All processes remain bound to the initial state (t, x, s, q) and to the control policy ϑ. The subscript $\tau-$ denotes the left limit of the jump processes, the state of the system at the instant immediately prior to an order fill, which is what makes the integrands predictable, a requirement for jump-diffusion integration.

Expanding the infinitesimal generator, the evolution of the value function along the trajectory is given by:

$$
\begin{aligned}
v(\tau_n, X_{\tau_n-}, S_{\tau_n}, q_{\tau_n-}) = v(t,x,s,q) \\
&+ \int_t^{\tau_n} \left(\partial_\tau v + \mu \partial_S v + \frac{1}{2}\sigma^2 \partial_{SS} v - (q_{\tau-} F_\tau \tilde{\rho} \exp(-\rho(T_{\text{fund}} - \tau))) \, \partial_X v \right) d\tau \\
&+ \int_t^{\tau_n} \left[v(\tau, X_{\tau-} + S_\tau + \vartheta^a_\tau, S_\tau, q_{\tau-} - 1) - v(\tau, X_{\tau-}, S_\tau, q_{\tau-}) \right] \lambda^a_\tau d\tau \\
&+ \int_t^{\tau_n} \left[v(\tau, X_{\tau-} - S_\tau + \vartheta^b_\tau, S_\tau, q_{\tau-} + 1) - v(\tau, X_{\tau-}, S_\tau, q_{\tau-}) \right] \lambda^b_\tau d\tau \\
&+ \int_t^{\tau_n} \sigma \partial_S v(\tau, X_{\tau-}, S_\tau, q_{\tau-}) dW_\tau \\
&+ \int_t^{\tau_n} \left[v(\tau, X_{\tau-} + S_\tau + \vartheta^a_\tau, S_\tau, q_{\tau-} - 1) - v(\tau, X_{\tau-}, S_\tau, q_{\tau-}) \right] dM^a_\tau \\
&+ \int_t^{\tau_n} \left[v(\tau, X_{\tau-} - S_\tau + \vartheta^b_\tau, S_\tau, q_{\tau-} + 1) - v(\tau, X_{\tau-}, S_\tau, q_{\tau-}) \right] dM^b_\tau,
\end{aligned}
\tag{55}
$$

where M^a and M^b are the previously defined compensated martingales.

To prove the stochastic integrals in Equation (55) are true martingales on $[t, \tau_n]$, we systematically bound the state variables and integrands. First, because the functions $\omega_q(\tau)$ are strictly positive and continuous on the compact set $[0, T]$, they attain a strictly positive lower bound, ensuring the fractional component $\omega_q(\tau)^{-\frac{\gamma}{k}}$ is bounded above. Second, the admissible controls are bounded below ($\vartheta^{a/b}_\tau \geq \delta_{\min}$), which imposes a strict upper bound on the jump intensities: $\lambda^{a/b}_\tau \leq A \exp(-k\delta_{\min})$.

Third, the localising stopping time τ_n explicitly bounds the price process and jump counts, while inventory is constrained by capacity Q. Consequently, the mark-to-market wealth $X_{\tau-} + q_{\tau-} S_\tau$ attains a finite lower bound. This wealth floor holds even across jumps: an ask fill shifts wealth to $(X_{\tau-} + q_{\tau-} S_\tau) + \vartheta^a_\tau$, which remains bounded below because the control floor $\vartheta^a_\tau \geq \delta_{\min}$ prevents infinite losses on executed quotes. The bid side is symmetric.

Because the wealth floor and bounded temporal components prevent the negative exponential utility function from diverging, the value function v and its derivatives are bounded on $[t, \tau_n]$. This boundedness is sufficient for the stochastic integrals: the bounded integrand against the

Brownian motion ($\sigma\partial_S v = -\gamma q\sigma v$) forms a square-integrable true martingale. Similarly, because the differences in the value function across jumps are bounded, and the intensities satisfy $\lambda_\tau^{a/b} \leq A\exp(-k\delta_{\min})$, the expected variation of the jump integrals is finite. This guarantees that the integrals with respect to the compensated processes $dM_\tau^{a/b}$ are true martingales rather than merely local ones.

Taking the conditional expectation $\mathbb{E}_t[\cdot]$, the dW_τ, dM_τ^a, and dM_τ^b integrals vanish:

$$\begin{aligned}\mathbb{E}_t\left[v(\tau_n, X_{\tau_n-}, S_{\tau_n}, q_{\tau_n-})\right] &= v(t,x,s,q)\\ &+ \mathbb{E}_t\left[\int_t^{\tau_n}\left(\partial_\tau v + \mu\partial_S v + \frac{1}{2}\sigma^2\partial_{SS}v - \left(q_{\tau-}F_\tau\tilde{\rho}\exp(-\rho(T_{\text{fund}}-\tau))\right)\partial_X v\right)d\tau\right]\\ &+ \mathbb{E}_t\left[\int_t^{\tau_n}\left[v(\tau, X_{\tau-}+S_\tau+\vartheta_\tau^a, S_\tau, q_{\tau-}-1) - v\right]\lambda_\tau^a d\tau\right]\\ &+ \mathbb{E}_t\left[\int_t^{\tau_n}\left[v(\tau, X_{\tau-}-S_\tau+\vartheta_\tau^b, S_\tau, q_{\tau-}+1) - v\right]\lambda_\tau^b d\tau\right].\end{aligned}$$

The sum of the expected continuous drift and the expected jump integrals is non-positive. This is the supermartingale half of the Dynamic Programming Principle, which Appendix D.2.2 sets out in full. The integrand is the generator of v evaluated under the particular control ϑ, and Equation (8) sets the supremum of that expression over all admissible controls to zero, so any single admissible control gives at most zero. At the inventory bounds $q = \pm Q$ the blocked side's intensity vanishes, and the governing equations are the boundary pair (36) and (37).

Therefore, we obtain:

$$\mathbb{E}_t\left[v(\tau_n, X_{\tau_n-}, S_{\tau_n}, q_{\tau_n-})\right] \leq v(t,x,s,q), \tag{56}$$

with equality when the controls are taken equal to the maximisers of the Hamiltonian, $\vartheta^a = \delta^{a*}$ and $\vartheta^b = \delta^{b*}$ of Equation (15). These optimal Markovian controls are globally well-defined because $\omega_q(\tau)$ is strictly positive on $[0,T]$, ensuring the log-ratios remain finite. They are admissible wherever they clear the floor, $\delta^{a*}, \delta^{b*} \geq \delta_{\min}$, which is exactly the regime in which the linear reduction of Proposition 1 is exact (Appendix D.2.1). In states where the floor binds, the constrained supremum is attained at $\delta_{\min}$ instead, Equation (56) continues to hold for every admissible control, and v bounds the value function from above rather than equalling it. The identification below is stated under the former case.

To pass to the limit as $n \to \infty$, we observe that the localising sequence $\tau_n \to T$ is eventually exact. Because the price process S has continuous paths on the compact interval $[t,T]$ and the jump intensities are bounded by $A\exp(-k\delta_{\min})$, the state processes are almost surely non-explosive. Consequently, for almost every path, there exists a finite index N such that for all $n \geq N$, $\tau_n = T$. The sequence $v(\tau_n, X_{\tau_n-}, S_{\tau_n}, q_{\tau_n-})$ is therefore almost surely eventually constant. To pass this limit inside the expectation, we establish uniform integrability by bounding the sequence in L^2.

A sufficient condition for uniform integrability is boundedness in L^2. This requires the second

moment of the value function sequence to be uniformly bounded over n, yielding the condition:

$$\sup_n \mathbb{E}_t \left[\exp\left(-2\gamma(X_{\tau_n-} + q_{\tau_n-} S_{\tau_n})\right) \omega_{q_{\tau_n-}}(\tau_n)^{-\frac{2\gamma}{k}} \right] < \infty.$$

The terminal risk profile at T, whether defined by a liquidation penalty or an epoch continuation vector (Appendix D.2.8), serves exclusively as the boundary condition for the ODE system defining $\omega_q(T)$. Because the Itô expansion evaluates the process at the interior localising stopping time $\tau_n \leq T$, the terminal condition enters the expectation only implicitly through the continuous evolution of $\omega_q(\tau_n)$.

The continuity and strict positivity of $\omega_q(\tau)$ on the compact interval $[0, T]$ guarantee that the inventory-dependent component $\omega_{q_{\tau_n-}}(\tau_n)^{-\frac{2\gamma}{k}}$ is uniformly bounded above by a finite deterministic constant. By replacing this component with its supremum and factoring the bound outside the expectation, it suffices to bound the expected square of the unpenalised mark-to-market wealth process, $\mathbb{E}_t\left[\exp(-2\gamma(X_{\tau_n-} + q_{\tau_n-} S_{\tau_n}))\right]$. Integrating the wealth and inventory dynamics yields:

$$\begin{aligned} X_{\tau_n-} + q_{\tau_n-} S_{\tau_n} = x + qs + \int_t^{\tau_n} \vartheta_\tau^a dN_\tau^a + \int_t^{\tau_n} \vartheta_\tau^b dN_\tau^b + \int_t^{\tau_n} \mu q_{\tau-} d\tau \\ - \int_t^{\tau_n} F_\tau \tilde{\rho} \exp(-\rho(T_{\text{fund}} - \tau)) q_{\tau-} d\tau + \int_t^{\tau_n} \sigma q_{\tau-} dW_\tau. \end{aligned} \tag{57}$$

Here, the integrals run over the half-open interval $[t, \tau_n)$, matching the left limits X_{τ_n-} and q_{τ_n-} on the left-hand side, ensuring any jump occurring exactly at τ_n is excluded from both sides.

As established at the beginning of the proof, the structural parameters (μ, σ, F, A, k) are treated as deterministic continuous functions on the compact horizon $[0, T]$. While this continuity previously sufficed to ensure the state processes and local martingales are well-posed on $[t, \tau_n]$, establishing uniform integrability requires explicit, path-independent upper bounds to construct the necessary inequalities.

By the Weierstrass Extreme Value Theorem, because these parameters are continuous on a compact set, they attain finite absolute maxima. We define their L^∞ norms as explicit global constants:

$$\bar{\mu} = \sup_{\tau \in [0,T]} |\mu(\tau)|, \quad \bar{\sigma} = \sup_{\tau \in [0,T]} \sigma(\tau), \quad \bar{\Phi} = \sup_{\tau \in [0,T]} |F_\tau \tilde{\rho} \exp(-\rho(T_{\text{fund}} - \tau))|, \tag{58}$$

each finite. The formulation for $\bar{\Phi}$ is robust up to T_{fund} because the exchange funding rate F_τ is constrained by structural caps, and the exponential envelope is continuous. The liquidity parameters $A(\tau)$ and $k(\tau)$ are analogously bounded on $[0, T]$, ensuring that downstream terms such as the maximum jump intensity $A \exp(-k\delta_{\min})$ are capped by finite constants.

Given the inventory capacity $|q_\tau| \leq Q$, each deterministic term of Equation (57) can be bounded from below. The drift integrand satisfies $\mu q_{\tau-} \geq -|\mu||q_{\tau-}| \geq -\bar{\mu} Q$, and integrating against $d\tau$, which preserves pointwise inequalities, gives at worst $-\bar{\mu} Q(\tau_n - t)$. The funding integrand is bounded the same way, at worst $-\bar{\Phi} Q(\tau_n - t)$. Since $\tau_n \leq T$, replacing $(\tau_n - t)$ by the full horizon $(T - t)$ only enlarges both penalties and so preserves the lower bound.

The counting processes N are non-decreasing and bounded by the total number of jumps on the horizon $N_T^{a/b}$. The cumulative cash flow from the spreads is bounded below regardless of

the sign of the floor $\delta_{\min}$. This results in the lower bound:

$$\int_t^{\tau_n} \vartheta_\tau^{a/b} dN_\tau^{a/b} \geq \delta_{\min}(N_{\tau_n-}^{a/b} - N_t^{a/b}) \geq -|\delta_{\min}| N_T^{a/b}.$$

Let $C_{\text{det}} = Q(\bar{\mu}+\bar{\Phi})(T-t)$ represent the combined deterministic worst-case penalty. Because this penalty and the initial wealth $(x+qs)$ are finite scalar constants, their exponential bounds factor entirely outside the expectation. To establish the L^2 bound, we evaluate the expected square of the exponential utility:

$$\begin{aligned}\mathbb{E}_t\left[\exp\left(-2\gamma(X_{\tau_n-} + q_{\tau_n-}S_{\tau_n})\right)\right] \leq \exp\left(-2\gamma(x+qs-C_{\text{det}})\right) \\ \times \mathbb{E}_t\Big[\exp(2\gamma|\delta_{\min}|N_T^a)\,\exp(2\gamma|\delta_{\min}|N_T^b) \\ \times \exp\Big(-2\gamma\int_t^{\tau_n}\sigma q_{\tau-}\,dW_\tau\Big)\Big].\end{aligned}$$

To decouple the distinct sources of stochastic risk, specifically the ask-side jumps, the bid-side jumps, and the Brownian diffusion, we must bound their expectations individually. We achieve this by applying the generalised Hölder inequality. For exponents $p_1, p_2, p_3 > 1$ satisfying $\sum_i p_i^{-1} = 1$, the expectation of a product is bounded by the product of the individual L^{p_i} norms, such that $\mathbb{E}[|Z_1Z_2Z_3|] \leq \prod_i \mathbb{E}[|Z_i|^{p_i}]^{1/p_i}$. Taking $p_1 = p_2 = p_3 = 3$ splits the expectation into three fractional moments, each of which can be bounded in closed form:

$$\begin{aligned}\mathbb{E}_t\left[\exp\left(-2\gamma(X_{\tau_n-} + q_{\tau_n-}S_{\tau_n})\right)\right] \leq \exp\left(-2\gamma(x+qs-C_{\text{det}})\right) \\ \times \mathbb{E}_t\left[\exp(6\gamma|\delta_{\min}|N_T^a)\right]^{\frac{1}{3}}\mathbb{E}_t\left[\exp(6\gamma|\delta_{\min}|N_T^b)\right]^{\frac{1}{3}} \\ \times \mathbb{E}_t\left[\exp\left(-6\gamma\int_t^{\tau_n}\sigma q_{\tau-}dW_\tau\right)\right]^{\frac{1}{3}}.\end{aligned} \tag{59}$$

Because the stochastic intensities of the point processes N^a and N^b are bounded above as a consequence of the admissible control floor, we can define a global maximum intensity $\bar{\lambda} = \sup_{\tau\in[0,T]} A(\tau)\exp(-k(\tau)\delta_{\min})$ (again taking care of the possible time-varying estimates of the, in this case, intensity parameters). The terminal jump counts N_T^a and N_T^b are therefore stochastically dominated by a homogeneous counting process with this constant maximum rate. Taking the conservative maximum over the entire horizon T, we define the integrated bound $\Lambda = T\bar{\lambda}$. The exponential moments of such a dominated counting process are finite for any parameter $\theta \geq 0$:

$$\mathbb{E}_t\left[\exp(\theta N_T^{a/b})\right] \leq \exp\left(\Lambda(\exp(\theta)-1)\right) < \infty. \tag{60}$$

Evaluating this bound at $\theta = 6\gamma|\delta_{\min}| \geq 0$ confirms that the first two expectations in the Hölder expansion are finite and independent of the localisation index n. Stating the estimate using absolute magnitudes ensures the verification remains mathematically robust for any general $\delta_{\min}$, fully decoupling the proof from specific operational calibration choices.

For the Brownian term, let $M_\tau = -6\gamma\int_t^\tau \sigma q_{s-}\,dW_s$. Because the absolute inventory $|q| \leq Q$ and the volatility $\sigma \leq \bar{\sigma}$ are uniformly bounded, the quadratic variation $\langle M\rangle_\tau$ is bounded by

the deterministic constant $K = 36\gamma^2\bar{\sigma}^2Q^2(T-t)$. Because the quadratic variation is deterministically bounded, the stochastic exponential $\exp(M_\tau - \frac{1}{2}\langle M\rangle_\tau)$ is a true martingale of mean one. Consequently, we can bound the expectation:

$$\begin{aligned}\mathbb{E}_t[\exp(M_{\tau_n})] &= \mathbb{E}_t\left[\exp\left(M_{\tau_n} - \tfrac{1}{2}\langle M\rangle_{\tau_n}\right)\exp\left(\tfrac{1}{2}\langle M\rangle_{\tau_n}\right)\right] \\ &\leq \mathbb{E}_t\left[\exp\left(M_{\tau_n} - \tfrac{1}{2}\langle M\rangle_{\tau_n}\right)\right]\exp(K/2) = \exp\left(18\gamma^2\bar{\sigma}^2Q^2(T-t)\right).\end{aligned} \tag{61}$$

Taking the fractional exponent yields a finite deterministic upper bound:

$$\mathbb{E}_t\left[\exp\left(-6\gamma\int_t^{\tau_n}\sigma q_{s-}dW_s\right)\right]^{\frac{1}{3}} \leq \exp\left(6\gamma^2\bar{\sigma}^2Q^2(T-t)\right) < \infty.$$

The finite bounds confirm the sequence $v(\tau_n, X^\vartheta_{\tau_n-}, S_{\tau_n}, q^\vartheta_{\tau_n-})$ is uniformly integrable. By the Vitali Convergence Theorem, which upgrades almost sure convergence to convergence in expectation for uniformly integrable sequences, we can legally pass the limit inside the expectation. Furthermore, because a point process with a bounded intensity jumps at any fixed time T with probability zero, the left limits coincide almost surely with the terminal values, establishing:

$$\lim_{n\to\infty}\mathbb{E}_t\left[v(\tau_n, X^\vartheta_{\tau_n-}, S_{\tau_n}, q^\vartheta_{\tau_n-})\right] = \mathbb{E}_t\left[v(T, X^\vartheta_T, S_T, q^\vartheta_T)\right]. \tag{62}$$

Because we established via the Itô expansion that the integrated expected drift of the value function is non-positive under any arbitrary admissible control policy $\vartheta \in \mathcal{A}$, substituting this limit yields the fundamental supermartingale inequality:

$$\mathbb{E}_t\left[v(T, X^\vartheta_T, S_T, q^\vartheta_T)\right] \leq v(t,x,s,q). \tag{63}$$

Under the standard liquidation penalty terminal condition of Proposition 1, $\omega_q(T) = \exp(-k\nu|q|)$, the fractional component of the ansatz collapses to $\omega_q(T)^{-\frac{\gamma}{k}} = \exp(\gamma\nu|q|)$. The value function at time T precisely recovers the terminal utility of Equation (7):

$$v(T,X,S,q) = -\exp\left(-\gamma(X+qS)\right)\exp(\gamma\nu|q|) = -\exp\left(-\gamma(X+qS-\nu|q|)\right). \tag{64}$$

Substituting this terminal utility back into the fundamental inequality establishes the upper bound for any arbitrary strategy $\vartheta \in \mathcal{A}$:

$$\mathbb{E}_t\left[-\exp\left(-\gamma(X^\vartheta_T + q^\vartheta_T S_T - \nu|q^\vartheta_T|)\right)\right] \leq v(t,x,s,q). \tag{65}$$

However, when the controls are taken equal to the optimal Markovian quotes ($\vartheta = \delta^*$), the Hamiltonian is uniquely maximised. The generator drift becomes exactly zero, upgrading the supermartingale to a true martingale, and equality holds:

$$\mathbb{E}_t\left[-\exp\left(-\gamma(X^{\delta^*}_T + q^{\delta^*}_T S_T - \nu|q^{\delta^*}_T|)\right)\right] = \mathbb{E}_t\left[v(T, X^{\delta^*}_T, S_T, q^{\delta^*}_T)\right] = v(t,x,s,q). \tag{66}$$

Evaluating the supremum over all admissible quoting strategies combines the upper bound

inequality with this exact equality:

$$\sup_{\vartheta\in\mathcal{A}} \mathbb{E}_t \left[-\exp\left(-\gamma(X_T^\vartheta + q_T^\vartheta S_T - \nu|q_T^\vartheta|)\right)\right] = v(t,x,s,q). \tag{67}$$

The candidate function v of Equation (13) therefore coincides with the true value function, and the controls of Equation (15) attain the supremum. This discharges the promise of Theorem 1: ω_q is not merely the unknown of a trial form, but the exact solution from which the optimal controls are derived. ■

D.2.6 Proof of Theorem 2: Asymptotic Optimal Quotes

Let us consider the modified symmetric matrix $M + 2\eta I$. Because the state-dependent penalties $D_q = \alpha q^2 - \beta q$ are confined to the main diagonal, and the constant liquidity jump intensities $-\eta$ symmetrically populate the immediate off-diagonals, the matrix is real and symmetric. It is therefore diagonalisable, and its smallest eigenvalue λ is characterised by the Rayleigh quotient:

$$\lambda = \inf_{x\in\mathbb{R}^{2Q+1}\setminus\{0\}} \frac{x'(M+2\eta I)x}{x'x},$$

and the associated eigenvectors $x \neq 0$ attain this minimum.

Remark on Indefiniteness: At this stage, we note that for sufficiently large $|\beta|$, the diagonal entries D_q may become negative. Consequently, the shifted matrix $M + 2\eta I$ may cease to be positive definite. This does not compromise our analysis: the fact that $M + 2\eta I$ is a real, symmetric matrix is sufficient for diagonalisability and the validity of the Rayleigh quotient characterisation. The smallest eigenvalue λ remains well-defined (though potentially negative), and the Perron-Frobenius-style positivity argument that follows depends only on the off-diagonal signs ($-\eta < 0$).

To evaluate this quadratic form, we first expand the raw matrix multiplication, isolating the diagonal terms, the cross-terms from the jump intensities, and the $2\eta I$ shift:

$$x'(M+2\eta I)x = \sum_{q=-Q}^{Q} (\alpha q^2 - \beta q)x_q^2 \;-\; 2\eta \sum_{q=-Q}^{Q-1} x_q x_{q+1} \;+\; 2\eta \sum_{q=-Q}^{Q} x_q^2.$$

By grouping the adjacent states within the final shift term, we provide the exact components needed to complete the square ($-2x_q x_{q+1} + x_q^2 + x_{q+1}^2 = (x_{q+1} - x_q)^2$), cleanly factoring the expression into:

$$\sum_{q=-Q}^{Q} (\alpha q^2 - \beta q)x_q^2 + \eta \sum_{q=-Q}^{Q-1} (x_{q+1} - x_q)^2 + \eta x_Q^2 + \eta x_{-Q}^2.$$

If x is an eigenvector associated with λ, then applying the absolute value $|x|$ component-wise

yields:

$$\begin{aligned}\lambda &\leq \frac{|x|'(M+2\eta I)|x|}{|x|'|x|} \\ &= \frac{1}{\|x\|^2}\left[\sum_{q=-Q}^{Q}(\alpha q^2-\beta q)|x_q|^2+\eta\sum_{q=-Q}^{Q-1}(|x_{q+1}|-|x_q|)^2+\eta|x_Q|^2+\eta|x_{-Q}|^2\right] \\ &\leq \frac{1}{\|x\|^2}\left[\sum_{q=-Q}^{Q}(\alpha q^2-\beta q)x_q^2+\eta\sum_{q=-Q}^{Q-1}(x_{q+1}-x_q)^2+\eta x_Q^2+\eta x_{-Q}^2\right] \\ &= \frac{x'(M+2\eta I)x}{x'x}=\lambda.\end{aligned}$$

This proves that $|x|$ is also an eigenvector. The inequality holds as a strict equality because the diagonal components $(\alpha q^2-\beta q)|x_q|^2$ are algebraically identical to $(\alpha q^2-\beta q)x_q^2$, and the off-diagonal jump penalties satisfy $(|x_{q+1}|-|x_q|)^2 \leq (x_{q+1}-x_q)^2$. Consequently, to maintain equality, x_q and x_{q+1} must share the same sign (i.e., $x_q x_{q+1} \geq 0$).

Now, suppose $x \geq 0$ is an eigenvector of the shifted matrix $M+2\eta I$ associated with λ. If there exists any interior inventory state q ($|q| < Q$) such that its component $x_q = 0$, we can extract the q-th row of the tridiagonal eigenvalue equation $(M+2\eta I)x = \lambda x$. Substituting $x_q = 0$ eliminates the diagonal term, yielding:

$$0 = \lambda x_q = (\alpha q^2-\beta q+2\eta)x_q-\eta(x_{q+1}+x_{q-1}) = -\eta(x_{q+1}+x_{q-1}) \leq 0.$$

Because we established $x \geq 0$ and the liquidity parameter $\eta > 0$, the sum of the non-negative adjacent components multiplied by a strict negative implies both neighbours must be zero ($x_{q+1} = 0$ and $x_{q-1} = 0$). By immediate induction across the entire tridiagonal inventory grid, this cascading effect forces every component to zero ($x = 0$), which contradicts the definition of an eigenvector ($x \neq 0$).

At the upper boundary, if $x_Q = 0$, then $0 = \lambda x_Q = (\alpha Q^2-\beta Q+2\eta)x_Q-\eta x_{Q-1} = -\eta x_{Q-1} \leq 0$, hence $x_{Q-1} = 0$, leading to the same contradiction. The symmetric argument holds at the lower boundary if $x_{-Q} = 0$. This proves that any non-negative eigenvector must be strictly positive ($x > 0$).

Furthermore, if the eigenvalue λ were not simple, its eigenspace would have a dimension of at least two. We could therefore construct two eigenvectors x and y associated with λ such that $|x|'y = 0$. Because $|x| > 0$, y must contain both positive and negative coordinates. Since we established $y_q y_{q+1} \geq 0$, there must exist some state q where $y_q = 0$. However, this contradicts our preceding point, since $|y| \geq 0$ must also be an eigenvector associated with λ and therefore cannot have coordinates equal to zero. Thus, the eigenspace associated with λ is spanned by a unique, strictly positive vector $f^0 > 0$. We scale this eigenvector such that its $\mathbb{R}^{2Q+1}$ Euclidean norm is normalised to unity ($\|f^0\|_2 = 1$).

Under the theorem's assumption that $\Phi(t,q) \to 0$, the system enters the autonomous regime. The solution to the system state is expressed via the matrix exponential, $\omega(t) = \exp(-M(T-t))\omega(T)$, as it is for any constant-coefficient linear system. Because this time-invariant matrix M is real and symmetric, the spectral theorem further supplies a real orthonormal eigenbasis

$(g_i)_{i=0}^{2Q}$, over which we expand that exponential. We index the basis by the increasingly ordered eigenvalues of M itself, which are those of $M + 2\eta I$ shifted uniformly by -2η. The shift leaves every eigenvector untouched, so the simplicity of the smallest eigenvalue established above carries over and the spectrum satisfies $\lambda_0 < \lambda_1 \leq \cdots \leq \lambda_{2Q}$ (with higher-order eigenvalues repeated if necessary). We naturally assign the principal eigenvector to the smallest eigenvalue, setting $g_0 = f^0$.

For the generic terminal horizon $T \leq T_{\text{fund}}$, let $\omega(T)$ be any strictly positive terminal condition vector; the liquidation boundary, with components $\exp(-k\nu|q|)$, and the epoch continuation boundary f^0 of Section 4.7 are both instances. Expanding the matrix exponential over the eigenbasis yields:

$$\omega_q(t) = \sum_{i=0}^{2Q} \exp(-\lambda_i(T-t))\langle g_i, \omega(T)\rangle (g_i)_q, \quad \forall q \in \{-Q, \ldots, Q\}.$$

As the horizon expands ($(T-t) \to \infty$), the term associated with the smallest eigenvalue λ_0 strictly dominates the summation. The state variables asymptotically behave as:

$$\omega_q(t) \sim \exp(-\lambda_0(T-t))\langle f^0, \omega(T)\rangle f^0_q, \quad \forall q \in \{-Q, \ldots, Q\}.$$

Let $C = \langle f^0, \omega(T)\rangle$ denote the projection of the terminal condition onto the principal eigenvector. Because f^0 and $\omega(T)$ are strictly positive vectors, this inner product evaluates to a strictly positive scalar constant ($C > 0$). Because C is a global dot product over the entire state space, it is independent of the specific inventory level q.

When substituting the asymptotic proportionality $\omega_q(t) \sim \exp(-\lambda_0(T-t)) \cdot C \cdot f^0_q$ into the optimal quote expressions derived via the First-Order Conditions, we evaluate the ratio of adjacent states ($\omega_q/\omega_{q\pm1}$). Both the temporal exponential components and the constant scalar C appear identically in the numerator and denominator, cancelling within the natural logarithms. This yields the time-invariant asymptotic limits:

$$\lim_{(T-t)\to\infty} \delta_t^{b*}(q) = \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right) + \frac{1}{k}\ln\left(\frac{f^0_q}{f^0_{q+1}}\right), \quad q \neq Q,$$

$$\lim_{(T-t)\to\infty} \delta_t^{a*}(q) = \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right) + \frac{1}{k}\ln\left(\frac{f^0_q}{f^0_{q-1}}\right), \quad q \neq -Q.$$

Finally, the characterisation of f^0 stated in Theorem 2 is obtained by writing the Rayleigh quotient associated with the smallest eigenvalue of $M + 2\eta I$:

$$f^0 \in \operatorname*{argmin}_{f\in\mathbb{R}^{2Q+1}, \|f\|_2=1} f'(M + 2\eta I)f,$$

which is algebraically equivalent to the expanded sum form provided in the Theorem. ■

By summing the asymptotic optimal bid and ask half-spreads, we define the total asymptotic

bid-ask spread quoted by the market maker, $\psi^*_\infty(q)$:

$$\psi^*_\infty(q) = \lim_{(T-t)\to\infty} (\delta_t^{a*}(q) + \delta_t^{b*}(q)) = -\frac{1}{k}\ln\left(\frac{f^0_{q+1}f^0_{q-1}}{(f^0_q)^2}\right) + \frac{2}{\gamma}\ln\left(1+\frac{\gamma}{k}\right), \quad |q| \neq Q.$$

D.2.7 Proof of Proposition 3: Continuous Gaussian Approximation

We seek the continuous function $u \in L^2(\mathbb{R})$ (representing our approximated eigenvector) that minimises the functional:

$$\mathcal{J}[u] = \int_{\mathbb{R}} \left[\eta(u'(x))^2 + (\alpha x^2 - \beta x)u(x)^2\right] dx, \tag{68}$$

subject to the normalisation constraint $\|u\|_{L^2} = 1$, which amounts to $\int_{\mathbb{R}} u(x)^2 dx = 1$.

To solve this constrained optimisation problem, we introduce a Lagrange multiplier λ and define the Lagrangian:

$$L(x, u, u') = \eta(u')^2 + (\alpha x^2 - \beta x)u^2 - \lambda u^2. \tag{69}$$

By the fundamental lemma of the calculus of variations, the minimiser $u(x)$ must satisfy the Euler-Lagrange equation:

$$\frac{\partial L}{\partial u} - \frac{d}{dx}\left(\frac{\partial L}{\partial u'}\right) = 0. \tag{70}$$

Calculating the respective partial derivatives yields:

$$\begin{aligned} \frac{\partial L}{\partial u} &= 2(\alpha x^2 - \beta x)u - 2\lambda u, \\ \frac{\partial L}{\partial u'} = 2\eta u' \implies \frac{d}{dx}\left(\frac{\partial L}{\partial u'}\right) &= 2\eta u''. \end{aligned}$$

Substituting these derivatives back into the Euler-Lagrange equation provides:

$$2(\alpha x^2 - \beta x)u - 2\lambda u - 2\eta u'' = 0. \tag{71}$$

Dividing the entire equation by 2 and rearranging the terms yields the classical Sturm-Liouville eigenvalue problem:

$$-\eta u''(x) + (\alpha x^2 - \beta x)u(x) = \lambda u(x). \tag{72}$$

To identify the Lagrange multiplier λ with the eigenvalue of this differential operator, we multiply the equation by $u(x)$ and integrate over $\mathbb{R}$:

$$\int_{\mathbb{R}} \left(-\eta u''(x)u(x) + (\alpha x^2 - \beta x)u(x)^2\right) dx = \lambda \int_{\mathbb{R}} u(x)^2 dx. \tag{73}$$

Because the finiteness of the functional requires both $u \in L^2(\mathbb{R})$ and $u' \in L^2(\mathbb{R})$, implying $u \in H^1(\mathbb{R})$ (where $H^1(\mathbb{R}) = W^{1,2}(\mathbb{R})$ denotes the first-order Sobolev space, distinct from the weighted Hilbert space H introduced subsequently), the product $u\,u'$ belongs to $L^1(\mathbb{R})$ by the Cauchy-Schwarz inequality. An integrable function cannot remain bounded away from zero along its entire tail, so there exist sequences $x_n \to +\infty$ and $y_n \to -\infty$ along which $u(x_n)u'(x_n) \to 0$

and $u(y_n)u'(y_n) \to 0$. Applying integration by parts on $[y_n, x_n]$ and passing to the limit, in which every remaining integral converges absolutely (the potential integral is controlled by the norm of the weighted space, and $u''u = \eta^{-1}(\alpha x^2 - \beta x - \lambda)u^2$ is integrable by the eigenvalue equation itself), the boundary terms vanish and $\int -\eta u'' u \, dx = \int \eta (u')^2 \, dx$. Substituting this back, we recover our original functional $\mathcal{J}[u]$ on the left side, and the normalisation constraint ($\|u\|_{L^2}^2 = 1$) on the right side:

$$\int_{\mathbb{R}} \left[\eta (u'(x))^2 + (\alpha x^2 - \beta x) u(x)^2\right] dx = \lambda(1) \implies \mathcal{J}[u] = \lambda. \tag{74}$$

The Lagrange multiplier λ is therefore the eigenvalue of the differential operator $\mathcal{L} = -\eta \frac{d^2}{dx^2} + (\alpha x^2 - \beta x)$, and the minimiser $u(x)$ its corresponding eigenfunction. Minimising $\mathcal{J}[u]$ subject to the normalisation constraint is thus the same problem as finding the ground state of $\mathcal{L}$, whose eigenvalue λ_0 is the smallest in the discrete spectrum.

To solve this problem and guarantee existence, we rely on the Lax-Milgram theorem to establish a weak solution. This requires the associated bilinear form to be strictly coercive. In the foundational GLT framework, the potential is symmetric ($V(x) = \alpha x^2 \geq 0$, as $\alpha = \frac{k}{2}\gamma\sigma^2 > 0$), naturally guaranteeing coercivity. However, our potential $V(x) = \alpha x^2 - \beta x$ takes negative values between 0 and $\frac{\beta}{\alpha}$, on whichever side of the origin the sign of β places that interval, threatening the coercivity of the bilinear form.

To restore coercivity without changing the function space, we rewrite the potential by completing the square:

$$\begin{aligned} V(x) &= \alpha \left(x^2 - \frac{\beta}{\alpha} x \right) \\ &= \alpha \left(x^2 - \frac{\beta}{\alpha} x + \left(\frac{\beta}{2\alpha} \right)^2 - \left(\frac{\beta}{2\alpha} \right)^2 \right) \\ &= \alpha \left(x - \frac{\beta}{2\alpha} \right)^2 - \frac{\beta^2}{4\alpha}. \end{aligned} \tag{75}$$

Let us define the spatial translation $x^* = \frac{\beta}{2\alpha}$ and the constant $B = \frac{\beta^2}{4\alpha}$. Substituting this restructured potential back into the Euler-Lagrange equation yields the spectrally shifted system:

$$-\eta u''(x) + \alpha (x - x^*)^2 u(x) = (\lambda + B) u(x). \tag{76}$$

Defining $\tilde{\lambda} = \lambda + B$, this becomes the standard eigenvalue problem for the shifted operator. The shift does not alter the problem: adding a scalar constant B to a differential operator moves its entire spectrum by B and leaves the eigenfunctions where they were, and it is the eigenfunctions that set the quotes. Once the principal eigenvalue $\tilde{\lambda}_0$ of the shifted system is found, the original eigenvalue is recovered via $\lambda_0 = \tilde{\lambda}_0 - B$.

To verify that the Lax-Milgram conditions are satisfied for this shifted operator, we follow the approach of GLT and introduce the shifted weighted Hilbert space:

$$H = \left\{ u \in L^1_{\text{loc}}(\mathbb{R}) \;\middle|\; x \mapsto (x - x^*) u(x) \in L^2(\mathbb{R}) \text{ and } u' \in L^2(\mathbb{R}) \right\}, \tag{77}$$

equipped with the norm:

$$\|u\|_H = \sqrt{\int_{\mathbb{R}} (\alpha(x - x^*)^2 u(x)^2 + \eta u'(x)^2)\, dx}. \tag{78}$$

Because the potential function governing our differential operator has been spatially translated from $V(x) = \alpha x^2$ to $V(x) = \alpha(x-x^*)^2$, the domain of the operator must be identically translated. The condition $x \mapsto (x - x^*)u(x) \in L^2(\mathbb{R})$ explicitly guarantees that the shifted potential energy integral remains finite, ensuring the norm $\|u\|_H$ is well-defined.

To prove existence via the Lax-Milgram theorem, we must define the associated bilinear form $a : H \times H \to \mathbb{R}$ and demonstrate that it is bounded and strictly coercive. The bilinear form is defined as:

$$a(u, v) = \int_{\mathbb{R}} \left(\eta u'(x)v'(x) + \alpha(x - x^*)^2 u(x)v(x)\right) dx. \tag{79}$$

A bilinear form is strictly coercive if there exists a strictly positive constant $c > 0$ such that $a(u, u) \geq c\|u\|_H^2$ for all $u \in H$. Evaluating our bilinear form at $u = v$ yields:

$$a(u, u) = \int_{\mathbb{R}} \left(\eta(u'(x))^2 + \alpha(x - x^*)^2 u(x)^2\right) dx. \tag{80}$$

By construction of the H-norm, the bilinear form evaluated at $u = v$ is the norm squared, $\|u\|_H^2$, so coercivity holds with the constant $c = 1$. Boundedness, the second Lax-Milgram hypothesis, holds for the same reason: a is the inner product that induces $\|\cdot\|_H$, so $|a(u, v)| \leq \|u\|_H\|v\|_H$ by Cauchy-Schwarz. Both conditions are therefore met, guaranteeing the existence of a unique weak solution under the Lax-Milgram theorem.

Step 1: $H \subset L^2(\mathbb{R})$ with continuous injection.
The space H carries the requirement $u' \in L^2(\mathbb{R})$ in its definition, which hands us H^1 regularity on the exterior tails at once; on the interior interval it is recovered below through the Fundamental Theorem of Calculus. That regularity is what admits the Cauchy-Schwarz inequality and the Sobolev embedding $H^1(\mathbb{R}) \hookrightarrow C^0(\mathbb{R})$, and it leaves every function of H both continuous and square-integrable, which is the continuous injection into $L^2(\mathbb{R})$ we are after.

Let us consider $u \in H$ and a strictly positive constant $\epsilon > 0$. We partition the real line into a bounded interval centred at x^* and its unbounded complement. On the unbounded complement, the distance satisfies $|x-x^*| \geq \epsilon$, implying $1 \leq \frac{1}{\epsilon^2}(x-x^*)^2$. Multiplying by $u(x)^2$ and integrating yields:

$$\int_{\mathbb{R}\setminus[x^*-\epsilon,x^*+\epsilon]} u(x)^2 dx \leq \frac{1}{\epsilon^2} \int_{\mathbb{R}\setminus[x^*-\epsilon,x^*+\epsilon]} (x - x^*)^2 u(x)^2\, dx \leq \frac{1}{\alpha\epsilon^2}\|u\|_H^2 < +\infty.$$

Hence, because $u' \in L^2(\mathbb{R})$, we have $u \in H^1(\mathbb{R}\setminus[x^*-\epsilon, x^*+\epsilon])$ with a constant C_ϵ independent of u such that $\|u\|_{H^1(\mathbb{R}\setminus[x^*-\epsilon,x^*+\epsilon])} \leq C_\epsilon\|u\|_H$. In particular, u is continuous on $\mathbb{R} \setminus \{x^*\}$.

To prove continuity and boundedness across the shifted center x^*, let us set $\epsilon = 1$. By the Fundamental Theorem of Calculus, for any $x \in (x^*, x^* + 1)$:

$$u(x) = u(x^* + 1) - \int_x^{x^*+1} u'(t)dt.$$

Taking absolute values of the Fundamental Theorem of Calculus representation and applying the triangle inequality yields:

$$|u(x)| \leq |u(x^*+1)| + \left| \int_x^{x^*+1} u'(t)\, dt \right|.$$

Writing the integrand as $1 \cdot u'(t)$ and applying the Cauchy-Schwarz inequality $\left|\int fg\right| \leq \|f\|_{L^2}\|g\|_{L^2}$ with $f = 1$ and $g = u'$ gives:

$$\left| \int_x^{x^*+1} u'(t)\, dt \right| \leq \sqrt{\int_x^{x^*+1} 1^2\, dt} \cdot \sqrt{\int_x^{x^*+1} (u'(t))^2\, dt} = \sqrt{x^*+1-x}\, \|u'\|_{L^2(x,\, x^*+1)}.$$

Combining both terms yields:

$$|u(x)| \leq |u(x^*+1)| + \sqrt{x^*+1-x}\, \|u'\|_{L^2(x^*, x^*+1)}.$$

Because the injection of $H^1(x^*+1, +\infty)$ into $C([x^*+1, +\infty))$ is continuous (by Sobolev embedding), there exists a constant C independent of u such that:

$$|u(x^*+1)| \leq \sup_{x \geq x^*+1} |u(x)| \leq C\|u\|_{H^1(x^*+1, +\infty)}.$$

Since we established that $\|u\|_{H^1(x^*+1,+\infty)} \leq C_\epsilon \|u\|_H$, chaining these two inequalities yields a constant C' such that:

$$|u(x^*+1)| \leq C'\|u\|_H.$$

Substituting both bounds back into the Cauchy-Schwarz inequality, and using $\sqrt{x^*+1-x} \leq 1$ on (x^*, x^*+1), we obtain for every $x \in (x^*, x^*+1)$:

$$|u(x)| \leq \underbrace{|u(x^*+1)|}_{\leq C'\|u\|_H} + \underbrace{\sqrt{x^*+1-x}}_{\leq 1} \cdot \underbrace{\|u'\|_{L^2(x^*, x^*+1)}}_{\leq \|u'\|_{L^2(\mathbb{R})} \leq \frac{1}{\sqrt{\eta}}\|u\|_H} \leq \left(C' + \frac{1}{\sqrt{\eta}}\right) \|u\|_H.$$

Taking the supremum over (x^*, x^*+1) yields a constant C'' such that:

$$\|u\|_{L^\infty(x^*, x^*+1)} \leq C''\|u\|_H.$$

Applying symmetric logic to the lower interval (x^*-1, x^*), using the anchor point x^*-1 instead of x^*+1, yields the identical bound $\|u\|_{L^\infty(x^*-1, x^*)} \leq C''\|u\|_H$.

By combining the L^∞ bounds established on the interior interval $[x^*-1, x^*+1]$ with the square-integrable bounds derived for the exterior tails, we bound the total mass of the function. This allows us to define a global constant K such that $\|u\|_{L^2(\mathbb{R})} \leq K\|u\|_H$. A direct consequence of this global bound is that our custom space H is continuously embedded into $H^1(\mathbb{R})$ and $C^0(\mathbb{R})$, thereby completing the proof of continuous injection.

Step 2: The injection $H \hookrightarrow L^2(\mathbb{R})$ is compact.

On the whole real line a differential operator can carry continuous spectrum, because mass escapes to infinity, and a continuous spectrum would leave no single principal eigenvector to

isolate, which is what the time-invariant asymptotic quotes of Regime I are built on. Compactness of the injection $H \hookrightarrow L^2(\mathbb{R})$ makes the solution operator L of Step 3 compact and forces its spectrum to be discrete. That is the precondition for Steps 4 and 5, which together establish the existence and uniqueness of the principal eigenfunction.

Let us consider a sequence $(u_n)_n$ of functions in H with $\sup_n \|u_n\|_H < +\infty$. Because $H \subset H^1(\mathbb{R})$, for all $m \in \mathbb{N}^*$, we can extract from $(u_n)_n$ a sequence that converges in $L^2((x^* - m, x^* + m))$. Using a diagonal extraction, there exists a subsequence of $(u_n)_n$ and a function $u \in L^2_{\text{loc}}(\mathbb{R})$ such that $u_n(x) \to u(x)$ for almost every $x \in \mathbb{R}$ and $u_n \to u$ in the $L^2_{\text{loc}}(\mathbb{R})$ sense. The inclusion $H \subset H^1(\mathbb{R})$ established in Step 1, combined with the Rellich-Kondrachov theorem, guarantees that the embedding $H^1((x^* - m, x^* + m)) \hookrightarrow L^2((x^* - m, x^* + m))$ is compact for each fixed m, justifying the local subsequence extraction. Now, by Fatou's lemma:

$$\int_{\mathbb{R}} (x - x^*)^2 u(x)^2 dx \leq \liminf_{n\to\infty} \int_{\mathbb{R}} (x - x^*)^2 u_n(x)^2 dx \leq \frac{\sup_n \|u_n\|_H^2}{\alpha}.$$

Fatou's lemma controls the weighted term of the H-norm alone, so the derivative term needs its own argument. The derivatives $(u_n')_n$ are bounded in $L^2(\mathbb{R})$, since $\eta\|u_n'\|_{L^2}^2 \leq \|u_n\|_H^2$, and therefore converge weakly in $L^2(\mathbb{R})$ along a further subsequence; the weak limit is u', because $u_n \to u$ in the sense of distributions and distributional differentiation is continuous there. Weak lower semicontinuity of the L^2 norm then gives $\int_{\mathbb{R}} \eta(u'(x))^2 dx \leq \liminf_{n\to\infty} \int_{\mathbb{R}} \eta(u_n'(x))^2 dx$, finite by assumption. The two bounds together place u in H.

Splitting the integral inside and outside the bounded interval $[x^* - m, x^* + m]$ yields the exact decomposition, valid for all $m \in \mathbb{N}^*$:

$$\int_{\mathbb{R}} |u(x) - u_n(x)|^2 dx = \int_{x^*-m}^{x^*+m} |u(x) - u_n(x)|^2 dx + \int_{\mathbb{R}\setminus[x^*-m,x^*+m]} |u(x) - u_n(x)|^2 dx.$$

Outside the interval $[x^*-m, x^*+m]$, the distance satisfies $|x-x^*| \geq m$, which implies $1 \leq \frac{(x-x^*)^2}{m^2}$. Multiplying by $|u - u_n|^2$ and integrating over the exterior yields:

$$\int_{\mathbb{R}} |u(x)-u_n(x)|^2 dx \leq \int_{x^*-m}^{x^*+m} |u(x)-u_n(x)|^2 dx + \frac{1}{m^2} \int_{\mathbb{R}\setminus[x^*-m,x^*+m]} (x-x^*)^2 |u(x)-u_n(x)|^2 dx.$$

To bound the remaining weighted tail integral, we expand the integrand using the pointwise algebraic inequality $(a - b)^2 \leq 2a^2 + 2b^2$ and apply the H-norm bound to each term:

$$\begin{aligned}\int_{\mathbb{R}} (x - x^*)^2 |u(x) - u_n(x)|^2 dx &\leq 2 \int_{\mathbb{R}} (x - x^*)^2 u(x)^2 \, dx + 2 \int_{\mathbb{R}} (x - x^*)^2 u_n(x)^2 \, dx \\ &\leq \frac{2}{\alpha} \left(\|u\|_H^2 + \sup_n \|u_n\|_H^2 \right).\end{aligned}$$

Both terms are finite: $\sup_n \|u_n\|_H^2 < +\infty$ by our initial assumption, and $\|u\|_H^2 < +\infty$ since $u \in H$ was established above. We therefore define the finite constant:

$$C := \frac{2}{\alpha} \left(\|u\|_H^2 + \sup_n \|u_n\|_H^2 \right) < +\infty,$$

which is independent of both n and m, yielding:

$$\int_{\mathbb{R}} |u(x) - u_n(x)|^2 dx \leq \int_{x^*-m}^{x^*+m} |u(x) - u_n(x)|^2 dx + \frac{C}{m^2}.$$

Taking the limit supremum as $n \to \infty$, the first integral vanishes due to our localised subsequence extraction, yielding:

$$\limsup_{n\to\infty} \int_{\mathbb{R}} |u(x) - u_n(x)|^2 dx \leq \frac{C}{m^2}.$$

Since $\limsup_{n\to\infty} \|u_n - u\|^2_{L^2(\mathbb{R})} \geq 0$ by definition, and is bounded above by $\frac{C}{m^2}$ for every $m \in \mathbb{N}^*$ with C independent of m, sending $m \to +\infty$ forces the lim sup to equal zero:

$$\limsup_{n\to\infty} \int_{\mathbb{R}} |u(x) - u_n(x)|^2 dx = 0.$$

Hence, the sequence $(u_n)_n$ converges to u in $L^2(\mathbb{R})$.

Step 3: Defining the weak operator L and proving its properties.
We consider the equation $-\eta u''(x) + \alpha(x - x^*)^2 u(x) = f(x)$ for $f \in L^2(\mathbb{R})$ and define $u = Lf$ as the weak solution of this equation. Rather than solving this ODE in the classical sense, which would require pointwise second derivatives, we multiply both sides by a test function $v \in H$ and integrate over $\mathbb{R}$. Applying integration by parts to the second-order term, and using the fact that $u, v \in H \subset C^0(\mathbb{R})$ decay at infinity so that the boundary terms vanish, yields the weak formulation:

$$\forall v \in H, \quad \underbrace{\int_{\mathbb{R}} \left(\alpha(x - x^*)^2 u(x)v(x) + \eta u'(x)v'(x)\right) dx}_{= a(u,v)} = \int_{\mathbb{R}} f(x)v(x)\, dx.$$

The bilinear form $a(\cdot, \cdot)$ was shown to be strictly coercive with constant $c = 1$ prior to Step 1, and the right-hand side defines a continuous linear form on H by Cauchy-Schwarz and the continuous injection $H \hookrightarrow L^2(\mathbb{R})$ established in Step 1:

$$\left| \int_{\mathbb{R}} f(x)v(x)\, dx \right| \leq \|f\|_{L^2} \|v\|_{L^2} \leq K \|f\|_{L^2} \|v\|_H.$$

Hence all Lax-Milgram conditions are satisfied, and there exists a unique weak solution $u \in H$ for every $f \in L^2(\mathbb{R})$. Since $H \hookrightarrow L^2(\mathbb{R})$ continuously, the composition $L^2(\mathbb{R}) \to H \hookrightarrow L^2(\mathbb{R})$ is well-defined, and $L : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ is a well-defined linear operator.

We must now establish that L is compact, positive, and self-adjoint.

Compactness. The strategy is to show that L maps bounded sets in $L^2(\mathbb{R})$ to bounded sets in H, and then invoke the compact embedding $H \hookrightarrow L^2(\mathbb{R})$ from Step 2. Substituting $v = Lf \in H$ into the weak formulation yields:

$$\|Lf\|^2_H = a(Lf, Lf) = \int_{\mathbb{R}} f(x) \cdot Lf(x)\, dx = \langle f, Lf \rangle_{L^2}.$$

Applying Cauchy-Schwarz in $L^2(\mathbb{R})$ to the right-hand side, and then using the continuous injec-

tion bound $\|Lf\|_{L^2} \leq C\|Lf\|_H$ from Step 1:

$$\|Lf\|_H^2 \leq \|f\|_{L^2}\|Lf\|_{L^2} \leq C\|f\|_{L^2}\|Lf\|_H.$$

If $Lf = 0$ the bound holds trivially. Otherwise, dividing both sides by $\|Lf\|_H > 0$ yields:

$$\|Lf\|_H \leq C\|f\|_{L^2}.$$

Hence L maps bounded sets in $L^2(\mathbb{R})$ to bounded sets in H. Since the injection $H \hookrightarrow L^2(\mathbb{R})$ is compact, the composition $L^2(\mathbb{R}) \to H \hookrightarrow L^2(\mathbb{R})$ is compact, and therefore L is a compact operator.

Positivity. L is a positive operator if $\langle f, Lf\rangle_{L^2} \geq 0$ for all $f \in L^2(\mathbb{R})$. This follows immediately from the equality established above:

$$\langle f, Lf\rangle_{L^2} = \|Lf\|_H^2 \geq 0.$$

The equality $\langle f, Lf\rangle_{L^2} = 0$ holds only when $Lf = 0$. Because H contains all smooth functions with compact support and is therefore dense in $L^2(\mathbb{R})$, the condition $\int_{\mathbb{R}} f(x)v(x)dx = 0$ for all test functions v implies $f = 0$, proving the injectivity of L.

Self-adjointness. L is self-adjoint if $\langle f, Lg\rangle_{L^2} = \langle g, Lf\rangle_{L^2}$ for all $f, g \in L^2(\mathbb{R})$. To verify this, we use the fact that $Lf \in H$ is itself a valid test function. Substituting $u = Lf$ and $v = Lg$ into the weak formulation yields:

$$\langle f, Lg\rangle_{L^2} = \int_{\mathbb{R}} f \cdot Lg\, dx = \int_{\mathbb{R}} \big(\alpha(x - x^*)^2 Lf(x)Lg(x) + \eta(Lf)'(x)(Lg)'(x)\big)\, dx.$$

The integrand on the right is symmetric in f and g by commutativity of pointwise multiplication. Reversing the substitution, with $u = Lg$ and $v = Lf$, gives:

$$\int_{\mathbb{R}} \big(\alpha(x - x^*)^2 Lg(x)Lf(x) + \eta(Lg)'(x)(Lf)'(x)\big)\, dx = \langle g, Lf\rangle_{L^2}.$$

Hence $\langle f, Lg\rangle_{L^2} = \langle g, Lf\rangle_{L^2}$, and L is self-adjoint.

Since L is a compact, positive, self-adjoint operator on $L^2(\mathbb{R})$, the spectral theorem for compact self-adjoint operators guarantees that its spectrum consists of a discrete sequence of non-negative eigenvalues accumulating only at zero. The largest eigenvalue of L, denoted $\frac{1}{\tilde{\lambda}_0}$, corresponds to the smallest eigenvalue $\tilde{\lambda}_0$ of the original shifted differential operator, since L acts as its inverse. Using the spectral decomposition of L and classical results on Rayleigh ratios, the eigenfunctions f corresponding to the largest eigenvalue $\frac{1}{\tilde{\lambda}_0}$ of L satisfy:

$$\tilde{\lambda}_0 = \frac{\|f\|_H^2}{\|f\|_{L^2}^2} = \inf_{g \in H\setminus\{0\}} \frac{\|g\|_H^2}{\|g\|_{L^2}^2}.$$

This is the Rayleigh quotient of the differential operator: for an eigenfunction f with $Lf = \frac{1}{\tilde{\lambda}_0} f$, the weak formulation gives $\|f\|_H^2 = a(f, f) = \tilde{\lambda}_0 \|f\|_{L^2}^2$, so minimising the quotient over H selects exactly the eigenfunctions associated with the largest eigenvalue of L. Hence, our continuous

approximation problem reduces to proving that the largest eigenvalue of L is simple, and that we can uniquely construct its corresponding positive eigenfunction. This construction and the proof of uniqueness are carried out in Steps 4 and 5.

Step 4: Any positive eigenfunction corresponds to the largest eigenvalue of L. Since $H \subset H^1(\mathbb{R})$ was established in Step 1, any $f \in H$ possesses a weak derivative $f' \in L^2(\mathbb{R})$. This membership in $H^1(\mathbb{R})$ is precisely what legitimises the Sobolev chain rule, which gives $(|f|)' = \operatorname{sgn}(f) \cdot f'$ almost everywhere. Without H^1 regularity, the weak derivative of $|f|$ would not be well-defined. Using this, both norms are unchanged by taking absolute values:

$$\||f|\|_H^2 = \int_{\mathbb{R}} \alpha(x-x^*)^2 \underbrace{|f(x)|^2}_{=f(x)^2} + \eta \underbrace{(|f|'(x))^2}_{=(\operatorname{sgn}(f))^2(f'(x))^2=(f'(x))^2} dx = \|f\|_H^2,$$

and $\||f|\|_{L^2}^2 = \int_{\mathbb{R}} |f|^2\,dx = \int_{\mathbb{R}} f^2\,dx = \|f\|_{L^2}^2$, since $|f|^2 = f^2$ pointwise. Hence the Rayleigh ratios coincide:

$$\frac{\||f|\|_H^2}{\||f|\|_{L^2}^2} = \frac{\|f\|_H^2}{\|f\|_{L^2}^2}.$$

Therefore, if f is an eigenfunction of L corresponding to the largest eigenvalue $\frac{1}{\tilde{\lambda}_0}$, then $|f|$ is also an eigenfunction of L corresponding to $\frac{1}{\tilde{\lambda}_0}$.

Now suppose $\tilde{f}$ is an eigenfunction of L corresponding to a different eigenvalue $\frac{1}{\tilde{\lambda}} \neq \frac{1}{\tilde{\lambda}_0}$. Since $|f|$ is itself a principal eigenfunction of L corresponding to $\frac{1}{\tilde{\lambda}_0}$, and L is self-adjoint (proven in Step 3), eigenfunctions corresponding to distinct eigenvalues are orthogonal in $L^2(\mathbb{R})$:

$$\langle |f|, \tilde{f} \rangle_{L^2} = \int_{\mathbb{R}} |f(x)| \tilde{f}(x)\,dx = 0.$$

Since $|f| \geq 0$ is non-trivial, the weight $|f(x)|$ is strictly positive on a set of positive measure. For the integral to vanish, $\tilde{f}$ must therefore take both positive and negative values; a strictly positive function would make the integrand non-negative and non-trivial, giving a strictly positive integral. Therefore $\tilde{f}$ cannot be strictly positive.

Step 5: Constructing the eigenfunction and proving it uniquely spans the eigenspace.
The shifted operator carries a quadratic potential, which is the quantum harmonic oscillator, and the ground state of a parabolic potential is Gaussian. We therefore take the ansatz, with an unknown constant $c > 0$:

$$g(x) = \exp(-c(x-x^*)^2),$$

where c is determined by substituting into the eigenvalue equation. Differentiating g twice yields:

$$\begin{aligned} g'(x) &= -2c(x-x^*)g(x), \\ g''(x) &= \left(-2c + 4c^2(x-x^*)^2\right) g(x). \end{aligned}$$

Substituting these derivatives into the shifted eigenvalue equation, $-\eta g''(x) + \alpha(x-x^*)^2 g(x) = \tilde{\lambda} g(x)$, written with an eigenvalue $\tilde{\lambda}$ not yet identified as the principal one, yields:

$$-\eta\left(-2c + 4c^2(x-x^*)^2\right) g(x) + \alpha(x-x^*)^2 g(x) = \tilde{\lambda} g(x).$$

Grouping the terms by their spatial dependence gives:

$$2c\eta g(x) + \left(\alpha - 4c^2\eta\right)(x - x^*)^2 g(x) = \tilde{\lambda} g(x).$$

For $g(x)$ to be a valid eigenfunction, the differential operator applied to $g(x)$ must yield a constant multiple of $g(x)$. This requires the spatially dependent term to vanish entirely for all x, forcing its coefficient to zero:

$$\alpha - 4c^2\eta = 0 \implies c^2 = \frac{\alpha}{4\eta} \implies c = \frac{1}{2}\sqrt{\frac{\alpha}{\eta}}.$$

Substituting this exact value back into our ansatz yields the candidate function:

$$g(x) = \exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x - x^*)^2\right).$$

Equating the remaining constant term gives $\tilde{\lambda} = 2c\eta = \sqrt{\alpha\eta}$. Thus g is a strictly positive eigenfunction of our differential operator, and Step 4 established that a strictly positive eigenfunction can belong to no eigenvalue other than the principal one, so $\tilde{\lambda} = \tilde{\lambda}_0 = \sqrt{\alpha\eta}$. Because the weak compact operator L defined in Step 3 acts as the inverse of this differential operator, g is also an eigenfunction of L, corresponding to its largest eigenvalue $\frac{1}{\tilde{\lambda}_0} = \frac{1}{\sqrt{\alpha\eta}}$.

Now, to prove uniqueness, if we look for any other eigenfunction $f \in C^\infty(\mathbb{R}) \cap H$ associated with the same eigenvalue $\sqrt{\alpha\eta}$ (any eigenfunction of L being smooth), we can express it in the form $f(x) = g(x)h(x)$. Substituting this product into the eigenvalue equation for the differential operator yields:

$$\begin{aligned} 0 &= -\eta f''(x) + \alpha(x - x^*)^2 f(x) - \sqrt{\alpha\eta} f(x) \\ &= -\eta\left(g''(x)h(x) + 2g'(x)h'(x) + g(x)h''(x)\right) + \alpha(x - x^*)^2 g(x)h(x) - \sqrt{\alpha\eta} g(x)h(x). \end{aligned}$$

Because $g(x)$ exactly solves the differential equation, the terms containing $h(x)$ cancel out, leaving:

$$0 = 2g'(x)h'(x) + g(x)h''(x) = -2(x - x^*)\sqrt{\frac{\alpha}{\eta}} g(x)h'(x) + g(x)h''(x).$$

Since $g(x) > 0$ everywhere, we can divide it out to obtain a first-order ODE for h':

$$h''(x) = 2(x - x^*)\sqrt{\frac{\alpha}{\eta}}\, h'(x).$$

Setting $p(x) = h'(x)$, this becomes the separable equation $p'(x) = 2\sqrt{\frac{\alpha}{\eta}}\,(x - x^*)\,p(x)$. Separating variables and integrating both sides:

$$\int \frac{dp}{p} = \int 2\sqrt{\frac{\alpha}{\eta}}\,(x - x^*)\,dx \implies \ln|p| = \sqrt{\frac{\alpha}{\eta}}(x - x^*)^2 + \text{const}.$$

Exponentiating yields:

$$\exists\, K_1 \in \mathbb{R}, \quad h'(x) = K_1 \exp\left(\sqrt{\frac{\alpha}{\eta}}(x - x^*)^2\right).$$

Integrating a second time, with basepoint x^* chosen so that the exponent vanishes at the lower limit and $K_2 := h(x^*)$ absorbs the constant of integration:

$$\exists\, K_1, K_2 \in \mathbb{R}, \quad h(x) = K_1 \int_{x^*}^{x} \exp\!\left(\sqrt{\frac{\alpha}{\eta}}(t-x^*)^2\right) dt + K_2.$$

Therefore, the general form of the eigenfunction $f = gh$ is:

$$f(x) = K_1\, g(x) \int_{x^*}^{x} \exp\!\left(\sqrt{\frac{\alpha}{\eta}}(t-x^*)^2\right) dt + K_2\, g(x).$$

To determine if this function belongs to our Hilbert space H, we evaluate its behaviour as $x \to +\infty$. Applying a change of variables $s = t - x^*$ to the integral term yields:

$$g(x) \int_{x^*}^{x} \exp\left(\sqrt{\frac{\alpha}{\eta}}(t-x^*)^2\right) dt = \exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2\right) \int_0^{x-x^*} \exp\left(\sqrt{\frac{\alpha}{\eta}}s^2\right) ds.$$

Since the integrand $\exp\left(\sqrt{\frac{\alpha}{\eta}}s^2\right)$ is strictly increasing in s, we can restrict the domain of integration to a sub-interval to obtain a lower bound:

$$\geq \exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2\right) \int_{\frac{x-x^*}{\sqrt{2}}}^{x-x^*} \exp\left(\sqrt{\frac{\alpha}{\eta}}s^2\right) ds.$$

Because the integrand is strictly increasing, we lower-bound the integral by the interval width times the integrand evaluated at the lower limit $s = \frac{x-x^*}{\sqrt{2}}$:

$$\int_{\frac{x-x^*}{\sqrt{2}}}^{x-x^*} \exp\left(\sqrt{\frac{\alpha}{\eta}}s^2\right) ds \geq \left((x-x^*) - \frac{x-x^*}{\sqrt{2}}\right) \cdot \exp\left(\sqrt{\frac{\alpha}{\eta}}\left(\frac{x-x^*}{\sqrt{2}}\right)^2\right).$$

Noting that $\left(\frac{x-x^*}{\sqrt{2}}\right)^2 = \frac{(x-x^*)^2}{2}$, the exponent at the lower limit simplifies to $\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2$, so multiplying through:

$$\int_{\frac{x-x^*}{\sqrt{2}}}^{x-x^*} \exp\left(\sqrt{\frac{\alpha}{\eta}}s^2\right) ds \geq (x-x^*)\left(1-\frac{1}{\sqrt{2}}\right) \exp\left(\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2\right).$$

Substituting this back and combining with the prefactor $\exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2\right)$, the two exponentials cancel exactly:

$$\begin{aligned} g(x) \int_{x^*}^{x} \exp\left(\sqrt{\frac{\alpha}{\eta}}(t-x^*)^2\right) dt &\geq \exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2\right) \cdot (x-x^*)\left(1-\frac{1}{\sqrt{2}}\right) \\ &\quad \cdot \exp\left(\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x-x^*)^2\right) = (x-x^*)\left(1-\frac{1}{\sqrt{2}}\right). \end{aligned}$$

As $x \to +\infty$, the term $(x-x^*)\left(1-\frac{1}{\sqrt{2}}\right) \to +\infty$. This proves that the integral component of $f(x)$ diverges linearly at minimum, and it is therefore not square-integrable.

Hence, for f to remain bounded and belong to $H \subset L^2(\mathbb{R})$, we must have $K_1 = 0$. Thus, $f(x) = K_2 g(x)$, proving that g uniquely spans the eigenspace corresponding to the largest

eigenvalue of L.

Finally, returning to the original unshifted eigenvalue parameter $\lambda = \tilde{\lambda}_0 - B$, we obtain the true principal eigenvalue $\lambda_0 = \sqrt{\alpha\eta} - \frac{\beta^2}{4\alpha}$.

To finalise the proof of Proposition 3, we must satisfy the initial $L^2(\mathbb{R})$ normalisation constraint $\|\tilde{f}^0\|_{L^2} = 1$. Since Step 5 established that $g(x) = \exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}(x - x^*)^2\right)$ uniquely spans the eigenspace, the normalised eigenfunction must take the form $\tilde{f}^0(x) = C \cdot g(x)$ for some constant $C \in \mathbb{R}$. Imposing the normalisation constraint and squaring $g(x)$, noting that $\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}\right) \times 2 = -\sqrt{\frac{\alpha}{\eta}}$, yields:

$$C^2 \int_{-\infty}^{\infty} \exp\left(-\sqrt{\frac{\alpha}{\eta}}(x - x^*)^2\right) dx = 1.$$

Applying the standard Gaussian integral formula $\int_{-\infty}^{\infty} e^{-cs^2}\, ds = \sqrt{\frac{\pi}{c}}$ directly with $c = \sqrt{\frac{\alpha}{\eta}}$, and noting that the shift by x^* leaves the integral unchanged since the domain is all of $\mathbb{R}$:

$$C^2 \sqrt{\frac{\pi}{\sqrt{\frac{\alpha}{\eta}}}} = 1 \implies C^2 \pi^{\frac{1}{2}} \left(\frac{\alpha}{\eta}\right)^{-\frac{1}{4}} = 1,$$

where the second form follows from $\sqrt{\frac{\pi}{\sqrt{\frac{\alpha}{\eta}}}} = \pi^{1/2}\left(\frac{\alpha}{\eta}\right)^{-1/4}$. Isolating C and taking the square root, where the exponent $\frac{1}{8}$ arises from $\sqrt{\left(\frac{\alpha}{\eta}\right)^{1/4}} = \left(\frac{\alpha}{\eta}\right)^{1/8}$:

$$C = \pm \pi^{-\frac{1}{4}} \left(\frac{\alpha}{\eta}\right)^{\frac{1}{8}}.$$

We select the positive root to be consistent with the strictly positive eigenfunction constructed in Step 5. Finally, recalling the definitions $x^* = \frac{\beta}{2\alpha}$ and $\alpha = \frac{1}{2}k\gamma\sigma^2$, $\beta = k\mu$, a direct substitution gives:

$$x^* = \frac{\beta}{2\alpha} = \frac{k\mu}{2 \cdot \frac{1}{2}k\gamma\sigma^2} = \frac{\mu}{\gamma\sigma^2}.$$

Substituting both the normalisation constant and this explicit form of x^* into $\tilde{f}^0(x) = C \cdot g(x)$ provides the exact, normalised continuous eigenfunction:

$$\tilde{f}^0(x) = \pi^{-\frac{1}{4}} \left(\frac{\alpha}{\eta}\right)^{\frac{1}{8}} \exp\left(-\frac{1}{2}\sqrt{\frac{\alpha}{\eta}}\left(x - \frac{\mu}{\gamma\sigma^2}\right)^2\right). \tag{81}$$

This concludes the proof of Proposition 3. ■

D.2.8 Preservation of Prior Results Under Epoch Stitching

The main text established that the asymptotic matching condition $\omega_q(T^{(k)}_{\text{fund}}) \approx f^0_q$ stitches successive epochs together, with the approximation justified by the exponential decay of higher spectral modes over the macro-horizon ΔT_{epoch}. This appendix verifies that substituting this eigenvector boundary condition in place of the liquidation terminal condition $\omega_q(T) = \exp(-k\nu|q|)$

preserves each of the structural guarantees established in the main proofs. We also note that Proposition 1 implicitly depends on the strict positivity of $\omega_q(t)$ throughout the epoch horizon; this is guaranteed by the preservation of Proposition 2 established below, and requires no separate argument.

Preservation of Proposition 2 (Strict Positivity). The contradiction argument in Appendix D.2.4 required the terminal condition to possess a strictly positive global minimum $m > 0$, defined there as $m = \exp(-k\nu Q)$. Under the eigenvector boundary, we instead define $m = \min_{q\in\{-Q,\dots,Q\}} f^0_q$. By Theorem 2 (Appendix D.2.6), the Perron-Frobenius argument guarantees that every component of the principal eigenvector is strictly positive, so this minimum is bounded away from zero. The remainder of the contradiction argument is structurally identical, and the conclusion $\omega_q(t) \geq m \exp(-(C_{\max} - \eta)(T_{\text{fund}} - t)) > 0$ holds uniformly over the restricted epoch horizon $[t, T_{\text{fund}}]$.

Preservation of Theorem 1 (L^2 Uniform Integrability). The Vitali convergence step in Appendix D.2.5 required the fractional component $\omega_q(\tau)^{-\gamma/k}$ to be uniformly bounded from above along the controlled trajectory. Under the liquidation boundary this followed from $\omega_q(T) = \exp(-k\nu|q|)$ being bounded below by $\exp(-k\nu Q) > 0$. Under the eigenvector boundary, the same bound holds with $m = \min_q f^0_q > 0$, where strict positivity of every component of f^0 is again guaranteed by the Perron-Frobenius argument of Theorem 2. Because the inventory grid is finite, $\max_q (f^0_q)^{-2\gamma/k}$ is therefore finite, and the L^2 integrability argument carries through without modification.

Sharpened Spectral Representation (Preservation of Theorem 2). Under the liquidation boundary $\omega_q(T) = \exp(-k\nu|q|)$, the asymptotic quote limits of Theorem 2 required $(T - t) \to \infty$ for the transient spectral modes to decay. The eigenvector boundary condition sharpens this result within the autonomous regime ($\Phi = 0$, i.e. Regime I of Section 4.8). Since f^0 is the principal eigenvector of the autonomous matrix M satisfying $Mf^0 = \lambda_0 f^0$, the matrix exponential solution collapses to a single exact term:

$$\omega_q(t) = \exp(-M(T_{\text{fund}} - t))\, f^0 = \exp(-\lambda_0(T_{\text{fund}} - t))\, f^0_q.$$

The projections of the boundary onto all higher eigenvectors vanish exactly ($\langle g_i, f^0 \rangle = 0$ for $i \neq 0$ by orthonormality), so no transient modes are excited. Substituting into the First-Order Conditions, the temporal decay $\exp(-\lambda_0(T_{\text{fund}} - t))$ cancels in the ratio $\omega_q/\omega_{q\pm1}$, and the asymptotic quote limits hold exactly in this idealised autonomous epoch. In the true epoch, whose own funding layer excites transient modes near T_{fund}, they hold throughout Regime I up to the matched-asymptotics tolerance ϵ that defines u^* (Appendix D.3), with the residual decaying further as the backward distance from the boundary layer grows. Outside this regime, as $t \to T_{\text{fund}}$ and $\Phi \neq 0$, the non-autonomous matrix $M(t)$ invalidates this decomposition, and the exact numerical backward induction of Regime II takes over.

Remark on Execution Reality and the Prevention of Error Compounding. A natural critique of the continuous approximation of Section 4.1.3 is whether the local error, the difference between the accumulated continuous penalty and the exact discrete fee paid at settlement, compounds over successive epochs and eventually decouples the theoretical utility from the physical P&L.

It is therefore critical to distinguish between the role of the continuous approximation in the mathematical control problem and its role in the execution environment. The continuous cash drain is confined to the HJB equation, where it is used solely to compute the state vector $\omega_q(t)$ and generate the optimal quotes. It is never applied to the strategy's actual P&L. In the execution environment, the discrete jump is executed exactly: at $t = T^{(k)}_{\text{fund}}$, the precise fee $q_{T_{\text{fund}}} S_{T_{\text{fund}}} f_{T_{\text{fund}}}$ is deducted from the cash balance X.

Under the CARA utility formulation, this error structurally cannot compound. As established in the First-Order Conditions (Appendix D.2.3), the optimal controls δ^* are entirely independent of the cash state X, depending only on the inventory q and time t. Because the quoting logic is wealth-independent, the absolute cash discrepancy generated during epoch k does not enter the state matrix $M(t)$ for epoch $k+1$. At the settlement instant $(T^{(k)}_{\text{fund}})^+$, the continuous penalty resets to zero and the state vector resets approximately to f^0, as established in the main text. The mathematical control therefore has no memory of the cash discrepancy from the prior epoch, and the approximation error is bounded within a single epoch.

D.2.9 Extension: Fee-Aware Quoting as an Exact Affine Repair

Section 6.1 shows that the fee-blind objective cannot out-quote the maker fee, since its half-spread is bounded by $1/k$ while the fee exceeds it. The classical repair is to charge the fee inside the objective, and in this framework that repair is exact, closed-form, and calibration-free.

Model. Let ε be the maker fee rate on executed notional. A fill at half-spread δ then pays $\delta - \varepsilon(S \pm \delta)$ per unit rather than δ. Freezing the fee leg at the reference price, the same device as Justification III of Appendix D.4, gives a constant per-unit fee $\varphi := \varepsilon S_{\text{ref}}$, about 14 USDT per BTC at the sample's price level; the neglected cross term $\varepsilon\delta$ is about 0.002 USDT per BTC at the median half-spread, some 2% of a tick.[23]

The mapping. *Fee-blind (before).* In the fill brackets of the HJB equation a bid fill pays δ^b per unit, so the exponent of Proposition 1's computation collects to $X + qS + \delta^b$ (Appendix D.2.3, Equation (38)), and symmetrically on the ask; the optimiser prices every fill at its gross half-spread. *Fee-aware (now).* The per-fill cash becomes $\delta - \varphi$, so that same exponent reads $X + qS + \delta^b - \varphi$. Substituting $\tilde{\delta} = \delta - \varphi$ restores the standard bracket exactly. The market intensity at a given real half-spread is unchanged, $\lambda(\delta) = Ae^{-k\delta}$; rewriting it in the shifted variable, $Ae^{-k\delta} = \big(Ae^{-k\varphi}\big)e^{-k\tilde{\delta}}$, simply relabels the base level as $A_\varphi = Ae^{-k\varphi}$. The fee-aware problem *is* therefore the fee-blind problem in $\tilde{\delta}$ at base level A_φ, and the real quote is recovered by adding the fee back,

$$A_\varphi = A\, e^{-k\varphi}, \qquad \delta^*_{\text{fee}}(q) = \delta^*\big(q;\, A_\varphi\big) + \varphi. \tag{82}$$

What this leaves for the closed forms. Because the substitution changes only the base level, every closed form of Section 4 survives verbatim under $A \to A_\varphi$: concretely $\eta \to \eta\, e^{-k\varphi}$ since η is linear in A, which is Equation (9) with the inventory-independent reweighting $e^{-k\varphi}$ applied to both off-diagonals. That reweighting leaves M symmetric, so the Gaussian machinery is undisturbed, and the funding drain, sitting on the diagonal, passes through untouched. No proposition or theorem is re-derived; one solves the identical system at A_φ and adds φ at the end. *What*

[23] Charging the fee on the true executed price instead is also affine, $\delta(1 \mp \varepsilon) - \varepsilon S_{\text{ref}}$, and the identical substitution then runs with $k \to k/(1 \mp \varepsilon)$, a relative change of 2×10^{-4}. Nothing below depends on the choice.

this means for the quotes. Two consequences read off directly. The inventory-independent term becomes $c_1+\varphi$: the model quotes wider by *exactly* the fee, not by a tuned margin. And although the market fills at any real half-spread just as before, the wider quote is reached less often, so the fill-driven relaxation weakens, η falling by $e^{-k\varphi} \approx 0.28$ at the reference k; inventory then relaxes more slowly and the optimal skew per unit of inventory grows. Covering the fee and controlling inventory are competing uses of quote width, priced against each other inside the same closed form.

D.3 The Regime-Transition Boundary

This appendix specifies the boundary separating the two operating regimes introduced in Section 4.8. The main text established the mathematical rationale for the two-regime decomposition and the epoch-stitching boundary condition; what remains is the boundary itself, the matched-asymptotics stopping condition that fixes the operational transition boundary t^* during the backward induction. The numerical and systems implementation of the architecture it defines, the solver and the lookup table with its memory layout, is given in Appendix E.

Let $u = T_{\text{fund}} - t$ denote backward time elapsed from settlement. The operational transition boundary u^* is determined precisely during the backward induction via a matched asymptotics stopping condition. Let ϵ be a tolerance on the order of the minimum tick size of the exchange. The boundary is the outermost backward time at which any quote still deviates from its autonomous counterpart by more than ϵ:

$$u^* = \sup \left\{ u > 0 \;\middle|\; \max_{q \in \{-Q,\dots,Q\}} |\delta_{\text{exact}}(u,q) - \delta_{\text{discrete_auto}}(q)| > \epsilon \right\}, \tag{83}$$

where δ_{exact} denotes the quotes produced by the numerical backward induction and $\delta_{\text{discrete_auto}}$ denotes the exact discrete autonomous quotes derived directly from f^0. Taking the outermost crossing, rather than the innermost backward time at which the deviation first falls below ϵ, is what makes the definition well posed: the stitching condition $\omega(T_{\text{fund}}) = f^0$ forces the two quote sets to agree exactly at $u = 0$, so the innermost form is satisfied trivially at the settlement itself and would return a boundary of zero. The outermost crossing is also the conservative choice, since the lookup table then covers every u at which the quotes differ materially. Comparing against the discrete autonomous quotes rather than the continuous Gaussian approximation prevents artificial triggering from approximation errors at extreme inventory levels $q \to \pm Q$. The regime-transition boundary in forward time is then $t^* = T_{\text{fund}} - u^*$.

The regime-switching architecture is justified only while that window stays thin relative to the epoch, $u^* \ll \Delta T_{\text{epoch}}$. Were u^* to approach ΔT_{epoch}, as could occur under a small calibrated ρ (slow decay) or persistently extreme carry, the matched asymptotics scale separation would break down, the outer Regime I solution would govern only a negligible fraction of the epoch, and the architecture would collapse back to full numerical backward induction over the entire horizon. The logarithmic dependence on carry makes this remote for BTC/USDT at 8-hour funding intervals: even the $F_t = 40$ stress case occupies 4% of the epoch, and the out-of-sample switch leads of Table 22 confirm it ex post.

D.4 Justification of the Approximations Underpinning the Scalar Ansatz

This appendix provides the formal mathematical justification for the three approximations introduced in Section 4, detailing why they are structurally necessary to preserve the tractability of the $\mathcal{O}(N_q)$ ODE system. The reduction they enable is also validated end-to-end numerically: an un-reduced Crank-Nicolson solve of the full HJB PDE over an explicit (S, q) grid, sharing no code with the production ODE path, reproduces the ODE-reduced quotes to 0.003 ticks at the sample's 99th-percentile carry and to 0.033 ticks at a stress carry of $F_t = 40$ USDT per BTC, a forcing level at which the funding drain itself moves the optimal bid by roughly 113 ticks. The scalar ansatz therefore remains essentially exact well beyond the operating envelope, and the approximations quantified below are the only residual error channels.

All three justifications run into the same obstruction, which we state once here and cite thereafter rather than re-derive.

The exponential-intensity obstruction. Suppose the residual risk penalty θ_q is allowed to depend on a continuous state variable z, whether z is the price S or a parameter such as μ or σ, and suppose we seek closure through a polynomial inner ansatz in z. The maximised Hamiltonian under Avellaneda-Stoikov intensities $\lambda(\delta) = Ae^{-k\delta}$ contributes a term proportional to $\exp(k[\theta_{q\pm1}(t, z) - \theta_q(t, z)])$, derived in Appendix D.2.3. Under any polynomial inner ansatz the bracketed difference is a polynomial in z, and it is not the zero polynomial: in each of the three cases below the z-dependent forcing enters the equation with an inventory-dependent coefficient, μq, $\frac{1}{2}\gamma\sigma^2 q^2$ and qSf respectively, so the coefficients of the inner ansatz must themselves depend on q and cannot cancel in the difference. The exponential of a non-constant polynomial Maclaurin-expands into arbitrarily high powers of z. The continuous drift and diffusion terms of the equation are of finite degree, so no term is available to cancel that expansion at any order beyond the cap, and the method of undetermined coefficients fails. The degree of the inner ansatz is irrelevant: even a linear guess yields e^{bz} with $b \neq 0$ and the same infinite chain. What survives is a genuine PDE in (t, z), one per inventory level, requiring finite-difference discretisation in z at per-timestep cost $\mathcal{O}(N_q \cdot N_z)$ against the $\mathcal{O}(N_q)$ of the scalar system.

The obstruction is specific to continuous state variables. Inventory carries a quadratic penalty αq^2 without any such explosion because q is a finite discrete index: within the equation for a given q, q^2 is a numerical constant, and the system is a grid of ODEs rather than a PDE in q. Justifications I to III differ only in what forces θ_q to acquire the continuous argument in the first place, and each ends at this same barrier.

The linear-response coupling of the ML drift is the third such channel, and the two figures quoted in Section 5.7 are measured as follows. Against an exact solve with the drift built into the backward induction rather than entering as a stored response, the deployed quote is off by at most 2.4 ticks at the median operating signal. The error scales with the size of the signal rather than with inventory, so it is largest exactly where the signal is rarest, which is why it is bounded rather than typical. At the level that decides whether it matters, re-running full replay days with the drift baked into every rebuild moves terminal P&L by 0.4% and markouts by under 0.01 basis points.

D.4.1 Justification I: Timescale Separation and Conditionally Deterministic Parameters

Setup and the theoretical vulnerability In the foundational mathematical framework derived in Section 4, the execution engine treats the drift (μ) and the volatility (σ) as conditionally deterministic scalars over the local integration horizon. However, in high-frequency reality, both the microstructural order flow driving μ and the realised volatility of the asset are themselves stochastic processes. Absent further justification, stochastic optimal control would require any parameter whose stochastic variation materially affects the optimal control to be modelled as an independent state variable, expanding the value function to $v(t, X, S, q, \mu, \sigma)$. The purpose of this subsection is first to show that the scalar ansatz of Proposition 1 is structurally incompatible with this expanded state space, and then to establish via signal persistence that the expansion is unnecessary under the receding-horizon execution architecture.

The scalar ansatz is structurally incompatible with stochastic parameters Suppose μ and σ follow generic Itô diffusions, driven by Brownian motions mutually independent and independent of the price Brownian motion W_t. The specific functional form of these dynamics is immaterial to the argument. What matters is that μ and σ are stochastic state variables adapted to a filtration strictly richer than that generated by (W_t, N_t^a, N_t^b) alone.

When μ and σ are stochastic state variables, the true value function depends on them by the Markov property of the controlled system: the optimal expected utility from state $(t, X, S, q, \mu, \sigma)$ genuinely differs across distinct realisations of (μ, σ), because future drift and diffusion of the price process depend on those realisations. The true value function is therefore of the form $v(t, X, S, q, \mu, \sigma)$, and the corresponding HJB equation is a PDE over the full state space.

The scalar ansatz of Proposition 1, $v(t, X, S, q) = -\exp(-\gamma(X + qS))\,\omega_q(t)^{-\gamma/k}$, contains no μ or σ dependence and therefore cannot represent a value function that varies across realisations of these state variables. It is not a candidate solution to the HJB equation on the richer state space. Substituting it produces an inconsistent equation in which the partial derivatives $\partial_\mu v$ and $\partial_\sigma v$ vanish identically while the HJB equation requires them to absorb the generator terms of the parameter diffusions. The failure is structural rather than algebraic: no choice of $\omega_q(t)$ can rescue an ansatz that does not depend on the relevant state variables.

The expanded ansatz inherits the exponential-intensity obstruction To restore a candidate solution, the ansatz must be expanded to make ω_q depend explicitly on μ and σ:

$$v(t, X, S, q, \mu, \sigma) = -\exp(-\gamma(X + qS))\,\omega_q(t, \mu, \sigma)^{-\gamma/k}. \tag{84}$$

Substituting (84) into the HJB equation and following the change of variables of Proposition 1 yields a coupled system of PDEs in (t, μ, σ), one per inventory level. Closure would require a polynomial inner ansatz in (μ, σ), and this is exactly the situation the exponential-intensity obstruction rules out, now on two continuous axes instead of one: the infinite Maclaurin expansion must fail to cancel on each axis independently. The expanded ansatz therefore generates a PDE requiring discretisation on a two-dimensional parameter grid at per-timestep cost $\mathcal{O}(N_q \cdot N_\mu \cdot N_\sigma)$. The structural situation is the one common to all three justifications: the

scalar ansatz fails, a faithful expanded ansatz exists but is computationally out of reach, and a controlled approximation is required to keep the $\mathcal{O}(N_q)$ structure.

The receding-horizon execution architecture The C++ execution engine never integrates the continuous optimal control problem over a macroscopic horizon. It operates under a receding-horizon paradigm: at each micro-horizon boundary t_0, the engine snapshots the current values of μ and σ, treats them as deterministic constants over the interval $[t_0, t_0 + \Delta t]$, and re-solves the $\mathcal{O}(N_q)$ ODE system. At the boundary $t_0 + \Delta t$, fresh values are snapshotted and the process repeats. The micro-horizon Δt is the interval over which the ODE system is solved with parameters held constant before a fresh snapshot is taken, and it is set by the staleness tolerance on the snapshotted parameters and is 100 ms in the deployed engine, independent of the longer LUT regeneration cadence governed by f_t publication events (Appendix E.3).

Under this architecture, justifying the certainty-equivalence approximation reduces to showing that the optimal quotes computed with μ and σ frozen at their time-t_0 values deviate negligibly from the optimal quotes that would be computed if their full stochastic dynamics were embedded in the state space.

Bounding the approximation error via signal persistence The certainty-equivalence approximation is justified empirically rather than through assumed parametric dynamics. In the execution engine, μ and σ are not realisations of analytical diffusion processes but rather estimators updated continuously from observed market data: μ is the inference output of the LightGBM model evaluated on the one-second feature grid, and σ is a rolling realised-volatility estimator computed over a trailing 60-second window on the same one-second grid. Each carries inherent smoothing: the ML inference is regularised against high-frequency noise by the model's leaf structure and feature half-lives, and the rolling volatility estimator changes slowly relative to its window length by construction. The relevant question is therefore not the variance of an underlying stochastic differential equation, but how much these estimators can drift between successive ODE resolves.

Two distinct errors have to be separated here, and only the second is the certainty-equivalence approximation proper. The first is staleness. Both estimators are piecewise constant on their own update grids, one second for μ and for σ, so over a resolve interval of $\Delta t = 100$ ms the frozen value is either exactly the current one or one grid step behind it. Staleness of this kind is bounded by one grid step and is refreshed at every resolve, so it does not accumulate.

The second is the approximation that the certainty-equivalence label properly refers to. Freezing μ and σ discards the generator terms of their own dynamics, $\partial_\mu v$, $\partial_\sigma v$ and their second-order counterparts, and with them the value of *anticipating* a parameter change rather than merely reacting to one. This error is not bounded by how little the estimator moves over Δt, because the control acts over a horizon longer than one resolve. The horizon that matters is the one over which the position taken under the current quotes persists, which is the relaxation time $1/g$ of the autonomous system, $g = \lambda_1 - \lambda_0$, equal to 155 s at the reference calibration. The quote surface itself carries no memory, since it is rebuilt from the current snapshot at every resolve, but the inventory it accumulates does. Over that horizon the parameters do move.

What bounds the error is that the quotes depend on the parameters only through two smooth

functionals. From Equation (21), the parameters enter through the inventory shift $\mu/(\gamma\sigma^2)$ and the spread scaling $\sqrt{(\sigma^2\gamma/2kA)(1+\gamma/k)^{1+k/\gamma}}$, both evaluated at the current snapshot and σ appearing in each. A controller carrying the full parameter dynamics would differ from the frozen one at first order by the expected drift of those functionals over $1/g$, weighted by the relaxation kernel $e^{-g(s-t)}$. That drift is not zero. Neither estimator is a martingale in its own filtration: the trailing volatility estimator replaces its entire window over one relaxation time and reverts toward the prevailing level of variance rather than holding at its current value, and the inference output forecasts a rolling horizon rather than a fixed terminal quantity. What the frozen controller discards is therefore the predictable component of that reversion, the anticipation value proper, and not the reaction to the parameter change itself, which the next resolve supplies. Because both functionals are smooth, the discarded term is first order in the expected parameter drift over $1/g$, while the unpredictable innovation enters only at second order.

Two things follow. Staleness does not accumulate, because every resolve re-anchors both functionals at the current snapshot. The anticipation error is not removed the same way: it recurs in the same direction for as long as the reversion is persistent, and its size is set by how far the two functionals are expected to travel over a single relaxation window.

Conditionally deterministic treatment of A and k The same certainty-equivalence approximation applies to the rolling-window calibrations of the liquidity parameters A and k described in Section 5. Although A and k are time-varying quantities that affect the controlled dynamics through the fill intensities $\lambda^{a/b}(\delta) = A\exp(-k\delta)$, embedding them as continuous state variables in the HJB equation would expand the ansatz to $\omega_q(t, A, k)$ and inherit the same Avellaneda-Stoikov exponential-Hamiltonian obstruction established in the Justification introduction, in higher dimension. The structural failure of the scalar ansatz is therefore identical to that demonstrated above for μ and σ.

For notational clarity, A and k are retained as static symbols throughout the mathematical derivation rather than written as $A(t)$ and $k(t)$, consistent with the treatment of μ and σ above.

Local constancy of the funding f_t A structurally simpler but related locally-constant assumption concerns the funding rate f_t entering the continuous penalty $\Phi(t, q)$ through F_t (the justification for using S_{ref} is in Appendix D.4.3). Unlike μ, σ, A, and k, which require the persistence argument above to justify the certainty-equivalence approximation, f_t is empirically piecewise constant by the structure of the exchange feed: although Binance publishes f_t on a one-second grid, the published value remains exactly constant for approximately 60-second stretches in the overwhelming majority of cases. The funding rate input therefore carries even less approximation error than the four parameters above.

Summary The scalar ansatz of Proposition 1 does not close under stochastic μ, σ, A, or k: it is structurally incapable of representing a value function that depends on these time-varying quantities, and the expanded ansatz that does represent them generates a high-dimensional PDE blocked by the same Avellaneda-Stoikov exponential Hamiltonian obstruction established at the start of this appendix. We circumvent this dimensional trap via the receding-horizon execution

architecture, which snapshots all four rolling-window parameters at each micro-horizon boundary and re-solves the $\mathcal{O}(N_q)$ ODE system with the snapshots held constant.

D.4.2 Justification II: Continuous Approximation of Discrete Settlement

Setup and the change it induces In the absence of funding, the cash process established in Section 4.1.3 evolves purely through the bid and ask point processes:

$$dX_t = (S_t + \delta_t^a)\, dN_t^a - (S_t - \delta_t^b)\, dN_t^b, \qquad t \neq T_{\text{fund}}. \tag{85}$$

Under (85), the multiplicatively separable ansatz of Proposition 1 reduces the HJB equation to a tractable system of N_q linear ODEs for $\omega_q(t)$, and the verification machinery of Section 4.5 applies unchanged.

Cryptocurrency perpetual futures disrupt this structure. The cost of carry is settled at discrete times T_{fund}. Writing $f_{T_{\text{fund}}}$ for the realised funding fee percentage under the convention $f_{T_{\text{fund}}} > 0 \Rightarrow$ longs pay shorts, the true cash process carries a deterministic jump:

$$X_{T_{\text{fund}}^+} = X_{T_{\text{fund}}^-} - q_{T_{\text{fund}}^-}\, S_{T_{\text{fund}}}\, f_{T_{\text{fund}}}. \tag{86}$$

Inventory does not jump ($q_{T_{\text{fund}}^+} = q_{T_{\text{fund}}^-}$), but the value function must satisfy the interior boundary matching condition:

$$v(T_{\text{fund}}^-, X, S, q) = v(T_{\text{fund}}^+, X - qSf, S, q). \tag{87}$$

The object we are solving is no longer a continuous-time HJB equation over $[t, T]$. It is an optimal control problem featuring an exogenous, deterministic state jump. The value function must stitch across this boundary via (87). The question addressed in this section is whether this exogenous jump can be absorbed into the continuous ODE framework of Proposition 1 via approximation, or whether a structurally different ansatz is required.

Effect of introducing the discrete jump Economically, the jump is a deterministic, price-scaled cash drain proportional to inventory at the settlement instant. Over a single funding epoch of duration $\tau_{\text{fund}} = T_{\text{fund}} - t$, the magnitude is bounded in absolute value by $Q \cdot S \cdot |f_{T_{\text{fund}}}|$. On Binance BTC, typical funding rates are $|f_{T_{\text{fund}}}| \sim 10^{-4}$ per 8-hour interval, and at maximum inventory this is on the order of $10^{-4}QS$. Small relative to instantaneous mark-to-market fluctuations ($\sim Q\sigma\sqrt{\tau_{\text{fund}}}$) but systematically one-sided and predictable, the jump cannot be ignored by a market maker who expects to be flat or long through most of the session.

Mathematically, two properties of the jump determine which ansatz forms can absorb it. First, the jump is *linear in* S: the cash shift is $-qSf$ where $S = S_{T_{\text{fund}}}$. Second, it is *bilinear in* q and S, coupling two state variables that the classical ansatz deliberately decouples.

The classical ansatz does not survive the matching condition The classical Avellaneda-Stoikov-GLT ansatz factors out mark-to-market wealth and leaves a residual risk penalty in (t, q) only:

$$v(t, X, S, q) = -\exp(-\gamma(X + qS + \theta_q(t)))\,. \tag{88}$$

This is equivalent to the ω_q form of Proposition 1 under $\theta_q(t) = \frac{1}{k}\ln\omega_q(t)$. We use the θ form here to match the literature and make the (t, q, S) dependence structure transparent.

Substituting (88) into the matching condition (87):

$$-\exp\big(-\gamma(X + qS + \theta_q(T^-_{\text{fund}}))\big) = -\exp\big(-\gamma(X - qSf + qS + \theta_q(T^+_{\text{fund}}))\big)\,,$$
$$\theta_q(T^-_{\text{fund}}) = \theta_q(T^+_{\text{fund}}) - qSf.$$

The left-hand side is, by construction, independent of S. The right-hand side is linear in S. The only reconciliation is to let θ_q carry explicit S-dependence, violating the construction. The classical ansatz is therefore not preserved under the matching condition: the true value function cannot be of the form (88) across the settlement boundary.

This is a structural observation rather than a contradiction in the mathematics. It tells us the true value function has genuine S-dependence at the settlement instant, and any faithful ansatz must accommodate it. The open question is what accommodating it costs.

Alternative ansatz: the coupled certainty-equivalent form To accommodate the S-dependence, we explicitly enlarge the residual risk penalty to depend on both inventory and price, while maintaining the Mark-to-Market wealth factor:

$$v(t, X, S, q) = -\exp(-\gamma(X + qS + \theta_q(t, S)))\,. \tag{89}$$

This *coupled certainty-equivalent ansatz* coincides structurally with the ansatz used by Bergault et al. (2022) for multi-asset optimal execution under multivariate Ornstein–Uhlenbeck price dynamics, where mean-reverting drift likewise forces S into the risk penalty. Substituting into (87) yields

$$\theta_q(T^-_{\text{fund}}, S) = \theta_q(T^+_{\text{fund}}, S) - qSf, \tag{90}$$

which is consistent: the discrete cash drain on the right-hand side is absorbed by the S-dependence of θ_q on the left. The matching condition is a linear-in-S update to the coefficient structure of θ_q.

The PDE generated on the continuous interval. On $t < T_{\text{fund}}$, the S-dependence propagates into the HJB equation. Applying Itô's lemma to (89) gives:

$$\partial_S v = -\gamma\,(q + \partial_S\theta_q)\,v, \tag{91}$$
$$\partial_{SS} v = \big[-\gamma\,\partial_{SS}\theta_q + \gamma^2(q + \partial_S\theta_q)^2\big]\,v, \tag{92}$$

where the isolated q emerges from the partial derivative of the explicitly separated qS mark-to-market term. Substituting into the HJB, dividing through $-\gamma v$, and collecting terms (following the Itô expansion of Section 4.2) yields:

$$0 = \partial_t\theta_q + \mu(q + \partial_S\theta_q) + \tfrac{1}{2}\sigma^2\big[\partial_{SS}\theta_q - \gamma(q + \partial_S\theta_q)^2\big] + \mathcal{H}_{\text{jumps}}(\theta), \tag{93}$$

where $\mathcal{H}_{\text{jumps}}(\theta)$ encapsulates the maximised Hamiltonian components corresponding to the bid and ask point processes. Unlike the original scalar ansatz, where the second derivative evaluates to a simple scalar ($\partial_{SS} v \propto \gamma^2 q^2 v$), coupling S into the unknown risk penalty forces the spatial derivative to survive the division by v. Because the resulting equation retains the second partial derivative with respect to price ($\partial_{SS}\theta_q$) alongside the time derivative ($\partial_t \theta_q$), it permanently forms a system of quasi-linear second-order PDEs in (t, S), one per inventory level $q \in \{-Q, \dots, Q\}$.

Numerically, this corresponds to a coupled system of $N_q = 2Q + 1$ second-order PDEs requiring spatial discretisation on an S-grid of size N_S to reach a stable solution, yielding per-timestep complexity $\mathcal{O}(N_q \cdot N_S)$ under standard finite-difference schemes, a factor of N_S slowdown relative to the tridiagonal $\mathcal{O}(N_q)$ ODE system that the scalar ansatz preserves.

Does a nested polynomial ansatz close the PDE into ODEs? To avoid resolving the full two-dimensional PDE, a standard mathematical technique is to apply a nested structural guess (an inner ansatz) to the residual risk penalty $\theta_q(t, S)$. Because the matching condition (90) is linear in S, and the continuous PDE (93) contains a squared first-derivative term $(\partial_S \theta_q)^2$, the method of undetermined coefficients limits the polynomial closure to a maximum degree of two. This dictates a quadratic inner ansatz:

$$\theta_q(t, S) = a_q(t)\, S^2 + b_q(t)\, S + c_q(t). \tag{94}$$

Under (94), the matching condition forces:

$$a_q(T^-_{\text{fund}}) = a_q(T^+_{\text{fund}}), \quad b_q(T^-_{\text{fund}}) = b_q(T^+_{\text{fund}}) - qf, \quad c_q(T^-_{\text{fund}}) = c_q(T^+_{\text{fund}}), \tag{95}$$

which preserves the polynomial structure. The continuous drift and diffusion terms in (93) also preserve this polynomial degree: $\partial_S \theta_q = 2a_q S + b_q$ is linear, $\partial_{SS}\theta_q = 2a_q$ is constant, and $(\partial_S \theta_q)^2$ generates a quadratic. This is exactly the structural property exploited by Bergault et al. (2022), whose problem is pure trading-rate control with quadratic execution costs $L(v) = v^\top R\, v$ (their cost matrix, with notation adapted to avoid collision with the jump constant η), an ansatz class exactly closed under those specific continuous operations.

Because our intensities are exponential, performing the first-order condition (FOC) maximisation yields a logarithmic optimal quote:

$$\delta^{b*}(t, S) = \tfrac{1}{\gamma} \ln\!\big(1 + \tfrac{\gamma}{k}\big) - \big(\theta_{q+1}(t, S) - \theta_q(t, S)\big). \tag{96}$$

When substituted back into the HJB equation, the maximised bid-side contribution to the PDE for θ_q is proportional to:

$$\exp\!\big(k\big[\theta_{q+1}(t, S) - \theta_q(t, S)\big]\big). \tag{97}$$

This is the algebraic barrier introduced at the start of this appendix. Alternative structural guesses fail in the same way. A logarithmic ansatz such as $\ln(aS^2 + bS + c)$ trades the polynomial explosion for rational-exponential fractions in the continuous derivatives, which close no better. Furthermore, while one could artificially force closure by truncating the infinite series (analogous to the Taylor approximations in the original AS framework), GLT showed that such truncations

diverge from the true optimal control and underprice risk in volatile regimes.

Therefore, under empirical exponential intensities, keeping the stochastic price S coupled inside the risk penalty leaves a genuine, unresolvable two-dimensional PDE. Integrating that grid at the cadence the lookup table has to be rebuilt, and holding the result in cache, is not feasible. This barrier motivates the continuous-drain approximation and the localised S_{ref} substitution developed in this thesis, which together preserve the $\mathcal{O}(N_q)$ ODE structure.

The continuous-drain approximation We replace the discrete settlement jump at T_{fund} by a continuous, deterministic cash drain over the trading epoch. Let ΔT_{epoch} denote the duration between funding settlements (e.g., 8 hours), and let $t_0 = T_{\text{fund}} - \Delta T_{\text{epoch}}$ be the start of the current epoch. Introducing a smoothing parameter $\rho > 0$, its epoch-normalising constant

$$\tilde{\rho} := \frac{\rho}{1 - e^{-\rho\,\Delta T_{\text{epoch}}}}, \qquad \int_{t_0}^{T_{\text{fund}}} \tilde{\rho}\, e^{-\rho(T_{\text{fund}} - s)}\, ds = 1, \tag{98}$$

and the instantaneous funding rate f_t (the exchange-published funding rate carried on the mark-Price feed), the approximated cash dynamics read:

$$dX_t = (S_t + \delta_t^a)\, dN_t^a - (S_t - \delta_t^b)\, dN_t^b - q_t\, S_t\, f_t\, \tilde{\rho}\, \exp(-\rho(T_{\text{fund}} - t))\, dt. \tag{99}$$

Two distinct quantities govern this approximation, and the normalisation separates them exactly. The *magnitude* of the drain is set by the observable funding rate f_t: because the envelope $\tilde{\rho}\, e^{-\rho(T_{\text{fund}} - s)}$ integrates to one over the epoch by (98), the integrated penalty reproduces the discrete charge qSf exactly whenever inventory, price and rate hold constant, and matches it to leading order otherwise, with the residual quantified in the certainty-equivalent analysis below. The *temporal profile* of the drain, i.e. how that fixed total mass is distributed across the epoch, is shaped exclusively by the smoothing parameter ρ.

Selection of ρ The smoothing parameter ρ is a structural hyperparameter, selected once on the 30 pre-analysis days and held constant across all 64 out-of-sample days, never fitted inside the strategy loop. Under the unit-mass normalisation of Equation (98) it carries no magnitude responsibility, so the selection problem is purely one of timing: over what pre-settlement window should the anticipation of the funding charge concentrate.

The choice of an exponential temporal profile $\exp(-\rho\tau)$ over alternative parametric forms is dictated by three structural imperatives:

1. *Microstructural alignment.* The exponential curve is a natural one-parameter functional form for stochastic processes whose intensity grows multiplicatively with proximity to a deterministic boundary. It is the natural envelope for anticipation that concentrates as a deterministic settlement snapshot approaches.

2. *Analytical tractability.* The exponential parameterisation keeps the HJB system tractable. As the time-to-settlement τ grows large, $\exp(-\rho\tau) \to 0$ smoothly, allowing $\Phi(t, q)$ to vanish. This recovers the autonomous regime and the closed-form asymptotic quotes of Theorem 2. Conversely, power-law alternatives diverge as $\tau \to 0$, demanding ad-hoc

regularisation at the boundary, while piecewise functions introduce non-smooth transitions that break the $\mathcal{O}(N_q)$ ODE reduction. The exponential is the natural one-parameter family that is bounded at $\tau = 0$, vanishes smoothly as $\tau \to \infty$, and keeps $\Phi(t, q)$ within the coefficient class for which the ODE reduction of Proposition 1 holds.

3. *Regularisation.* A rigid one-parameter decay curve leaves the specification with no shape parameters to estimate. A higher-dimensional or nonparametric envelope would have to be fitted to the finite settlement sample (282 settlement events over the 94-day window), with the overfitting risk that carries, and the selection below avoids that demand entirely.

The selection is model-consistent rather than externally fitted. Under the normalised drain, ρ determines only how the fixed unit of anticipation is distributed over the pre-settlement window, and the natural distribution rate is the quoting system's own relaxation rate: the spectral gap $\lambda_1 - \lambda_0$ of the autonomous matrix M, which governs how fast any perturbation of the quote surface decays back onto the principal eigenmode f^0. Setting the drain's anticipation rate equal to that gap means the funding correction concentrates over exactly the window within which the quoting system is able to respond: a faster ρ would compress the anticipation into a layer thinner than the quotes' own relaxation, a slower one would spread it over a horizon where anticipation is unnecessary. Evaluated at the reference calibration (A, k, σ, γ, Q from the pre-analysis window), $\rho = \lambda_1 - \lambda_0 = 6.46 \times 10^{-3}$ s^{-1}, an e-folding time of 155 seconds.

Mathematical derivation of the approximation error. By replacing the true, fluctuating cash jump at settlement with a deterministic continuous integral, we introduce a theoretical modelling error. We formally quantify this error by evaluating the certainty equivalent of the stochastic funding shock under the market maker's CARA utility framework, which is internally consistent with the preference structure of the HJB system.

Let $Y = q_{T_{\text{fund}}} S_{T_{\text{fund}}} f_{T_{\text{fund}}}$ denote the stochastic physical cash jump at settlement. Its variance has two channels, and they must be kept apart. Treating the inventory and the price at settlement as conditionally uncorrelated,

$$\operatorname{Var}_t(Y) = f^2 \Big(\mathbb{E}_t[q^2_{T_{\text{fund}}}] \operatorname{Var}_t(S_{T_{\text{fund}}}) + \operatorname{Var}_t(q_{T_{\text{fund}}}) \, \mathbb{E}_t[S_{T_{\text{fund}}}]^2 \Big). \tag{100}$$

The first channel is the diffusion of the price against a held position, the second the dispersion of the terminal inventory itself. Pegging inventory at capacity maximises the first channel but sets the second to zero, so it is not a bound on the sum. We therefore evaluate the two separately: the price channel in closed form below, at $|q_{T_{\text{fund}}}| = Q$ so that $\mathbb{E}_t[q^2]$ is at its maximum, and the inventory channel numerically, since $\operatorname{Var}_t(q_{T_{\text{fund}}})$ is a property of the controlled birth-death dynamics rather than of the price process. We additionally treat the rate as locally constant over the epoch, $f_t \equiv f_{T_{\text{fund}}}$, consistent with the continuous integration of the observable f_t. This is an auxiliary approximation, and it enters only the second-moment term evaluated here. Let $Y_Q = Q S_{T_{\text{fund}}} |f_{T_{\text{fund}}}|$ represent this *terminal funding exposure at capacity.* Its variance is driven by the diffusion of the reference price alone.

For a CARA utility function $U(W) = -\exp(-\gamma W)$, we evaluate the certainty equivalent of the funding exposure in isolation: the deterministic charge $\text{CE}(Y_Q)$ that the market

maker regards as equivalent to facing the stochastic shock satisfies $-\exp(-\gamma\,(-\mathrm{CE}(Y_Q))) = \mathbb{E}_t[-\exp(-\gamma(-Y_Q))]$, hence:

$$\mathrm{CE}(Y_Q) = \frac{1}{\gamma}\ln\left(\mathbb{E}_t[\exp(\gamma Y_Q)]\right) = \frac{1}{\gamma}\ln(M_{Y_Q}(\gamma)), \tag{101}$$

where $M_{Y_Q}(\gamma)$ is the moment generating function of the stochastic exposure, and $\ln(M_{Y_Q}(\gamma))$ is the cumulant generating function (CGF). The shock is evaluated in isolation deliberately: the baseline wealth cannot be factored out of a joint expectation, since both it and Y_Q depend on $S_{T_{\mathrm{fund}}}$. Interaction terms between the shock and the remaining wealth components are of higher order in γ and are subsumed in the truncation analysis below. In cryptocurrency markets, funding shocks are non-Normal and exhibit heavy tails. Applying a second-order Taylor expansion to the CGF, equivalent to the classical Arrow-Pratt approximation, truncates the contributions of higher-order moments (skewness and kurtosis):

$$\ln(M_{Y_Q}(\gamma)) \approx \gamma\mathbb{E}_t[Y_Q] + \frac{\gamma^2}{2}\mathrm{Var}_t(Y_Q) = \gamma m_t + \frac{\gamma^2}{2}\sigma^2_{Y_Q,t}. \tag{102}$$

Substituting back yields the classical Arrow-Pratt form:

$$\mathrm{CE}(Y_Q) \approx m_t + \frac{\gamma}{2}\sigma^2_{Y_Q,t}, \tag{103}$$

the expected charge plus a variance premium.

The continuous drain captures the first-moment term m_t by construction: the normalised envelope integrates to one over the epoch by (98), and f_t is observed in real time, so the integrated penalty reproduces the discrete charge exactly when inventory, price and rate are constant, and matches it to leading order otherwise, with a residual driven by the covariation of the inventory path q_s with the envelope weights. Therefore, the principal mathematical error introduced by the continuous drain approximation is the discarded variance penalty:

$$\text{CE error} = \frac{\gamma}{2}\mathrm{Var}_t(Y), \tag{104}$$

with the two channels of (100) entering additively. Taking the price channel first, we evaluate it in a deliberately conservative unit convention, per whole BTC of inventory at capacity, which overstates the operational lot-scale exposure by a factor $1/V = 200$: risk aversion per unit $\gamma = 2 \times 10^{-5}$, mid-price $S \approx 77{,}500$ USDT, inventory capacity $Q = 10$ units, 8-hour funding fee $|f_{T_{\mathrm{fund}}}| = 10^{-4}$, and an upper-bound annualised volatility $\sigma = 60\%$ (above the realised daily figures of 55.5% pre-analysis and 43.1% out-of-sample reported in Table 6, so the residual is bounded by this evaluation point). Over the epoch duration $\tau_{\mathrm{fund}} = \frac{8}{24\times365}$, the variance yields:

$$\text{CE error (price channel)} \approx \frac{\gamma}{2}(Qf)^2S^2\sigma^2\tau_{\mathrm{fund}} \approx 2.0 \times 10^{-5}\ \text{USDT}. \tag{105}$$

This is the price channel at its maximum, evaluated under maximum inventory $|q| = Q$, the upper estimate of σ, and the representative epoch-end funding magnitude, so the realised price-channel residual sits below it almost everywhere. It is more than six orders of magnitude below the physical risk being managed in the same convention, a 77.5 USDT cash charge at capacity.

The inventory channel is the larger of the two, and it is measured rather than bounded analytically, because $\mathrm{Var}_t(q_{T_{\mathrm{fund}}})$ depends on the controlled dynamics. Simulating the inventory birth-death process under the optimal autonomous policy over full 8-hour epochs, with the terminal-inventory variance cross-checked against an exact generator matrix-exponential computation, the measured variance correction reaches 1.2×10^{-3} USDT per epoch per BTC of inventory at 99th-percentile carry ($|f| = 1.39 \times 10^{-4}$) and 1.5× the calibrated risk aversion, never exceeding 0.005% of per-epoch gross spread revenue (Table 31). It exceeds the price-channel bound by more than an order of magnitude, the two being evaluated at different risk aversions and carry levels, which is why the decomposition (100) matters: a bound built on pegged inventory alone would understate the total by that factor. Even so, the dominant channel remains more than four orders of magnitude below the 77.5 USDT charge, and the continuous drain discards a negligible penalty on both counts. These figures are illustrative of the BTC calibration. The argument carried forward is the relative ordering of the error channels, which is preserved under rescaling of $f_{T_{\mathrm{fund}}}$, σ, and S. This per-epoch CARA-variance penalty is the

Table 31: Monte Carlo verification of the continuous-drain certainty-equivalent error at the reference calibration (5×10^5 paths per cell, full 8-hour epochs, exact discrete charge versus integrated drain). All cash columns are per BTC of inventory (the conservative $1/V = 200\times$ convention). The first two columns are first-moment gaps dominated by Monte Carlo sampling noise around their analytic value of zero. The headline is the variance-correction column, anchored to the exact generator computation of $\mathrm{Var}(q_T)$ (agreement 0.05%).

γ	$\lvert f\rvert$	$\lvert\mathbb{E}C - \mathbb{E}I\rvert$ (USDT/BTC)	$\lvert J_{\mathrm{disc}} - J^{CE}_{\mathrm{cont}}\rvert$ (USDT/BTC)	var. corr. (USDT/BTC)	% epoch P&L
$0.5\gamma_0$	p50	-1.043e-02	1.013e-02	-2.525e-05	0.000%
$0.5\gamma_0$	p99	-4.230e-02	3.739e-02	-4.150e-04	0.001%
$1\gamma_0$	p50	-8.580e-02	8.521e-02	-4.976e-05	0.001%
$1\gamma_0$	p99	-3.478e-01	3.382e-01	-8.178e-04	0.005%
$1.5\gamma_0$	p50	-7.789e-02	7.701e-02	-7.367e-05	0.001%
$1.5\gamma_0$	p99	-3.158e-01	3.012e-01	-1.211e-03	0.005%

natural reference point for the total error budget of the continuous-drain-plus-S_{ref} formulation against the rejected discrete-jump-plus-S_{ref} alternative, quantified in Appendix D.4.3, where it is shown that even after counting this penalty against the drain side, the drain's total per-epoch residual remains several orders of magnitude smaller than that of the jump alternative.

Truncation of higher-order moments. The first moment of the cash impact is matched to leading order by direct integration of the observable funding rate f_t. To justify relying on the variance (second moment) as the dominant certainty-equivalent error term, we examine the cumulant generating function (CGF) of the stochastic exposure. The CGF expands as:

$$\ln(M_{Y_Q}(\gamma)) = \sum_{n=1}^{\infty} \kappa_n \frac{\gamma^n}{n!} \tag{106}$$

where κ_n is the n-th cumulant of Y_Q. Each successive term is scaled by an additional power of $\gamma \approx 2 \times 10^{-5}$. Truncation is licensed when successive terms fall away, that is when $\gamma|\kappa_{n+1}|/\big((n+1)|\kappa_n|\big) \ll 1$, which holds when the exposure Y_Q does not exhibit pathological tail behaviour beyond standard heavy-tailed funding distributions. The ratio of the third-order to second-order

contribution is:

$$\frac{|\kappa_3|\gamma^3/6}{|\kappa_2|\gamma^2/2} = \frac{|\kappa_3|\gamma}{3\kappa_2} \approx \frac{\gamma \cdot \mathbb{E}[(Y_Q - m_t)^3]}{3\,\mathrm{Var}(Y_Q)} \ll 1 \tag{107}$$

under representative BTC parameters, confirming that skewness ($n = 3$) and kurtosis ($n = 4$) contributions are numerically negligible relative to the variance term. This validates the second-order Arrow-Pratt approximation under normal operating conditions.

Why speed still matters: the regime-switching architecture A natural objection is that if the non-autonomous boundary layer is precomputed offline via backward induction (Section 4.8), the computational cost of the solver is irrelevant to execution latency. Why not precompute the exact coupled-ansatz system in Python and have C++ query the result? Two constraints specific to our setting rule this out.

The LUT must be regenerated at input-change cadence. The boundary-layer LUT is the solution of the ODE system parameterised by the instantaneous funding rate f_t, the rolling volatility estimate σ, the reference-price drift μ, and the Avellaneda-Stoikov intensity parameters A and k. The live reference price S_t and inventory q are deliberately not LUT inputs under the scalar ansatz: S_t enters the live quote post-lookup via the localised reference substitution, and q indexes the LUT rather than parameterising it. Any material change in the parameterising inputs invalidates the prior LUT and triggers regeneration, the full mechanics of which are detailed in Section 4.8. Under normal market conditions these inputs evolve on second-to-minute timescales, but during volatile episodes (liquidation cascades, funding spikes) multiple inputs can shift sharply and concurrently. A PDE solve over (t, q, S) takes 0.60 s against the 64 ms of the scalar rebuild (Table 34), an order of magnitude per solve that widens with N_S, in line with the $\mathcal{O}(N_q \cdot N_S)$ versus $\mathcal{O}(N_q)$ complexity established earlier in this appendix. The offline-replay deliverable absorbs that cost, since its rebuilds are debounced to one every 300 s. A live deployment has no such floor, and that is where the constraint binds. With the inputs moving on second-to-minute timescales, a 0.60 s solve occupies most of the interval between consecutive triggers, and when a cascade moves several of them at once it cannot keep up at all, leaving the table stale through exactly the episodes it most urgently has to track. The 64 ms rebuild leaves an order of magnitude of headroom against the same trigger stream.

The LUT must fit in CPU cache. Even granting a fast offline solve, the live engine queries the LUT on the hot path. Under the scalar ansatz the table is indexed by (t, q) and occupies a few MB, which is cache-resident and reachable at deterministic latency. An S-grid multiplies that footprint by N_S and makes the index computation depend on the live price, which defeats prefetching and replaces a deterministic fetch with a variable-latency one. Appendix E.3 gives the memory layout and Table 34 the measured figures. The scalar ansatz is the only form that keeps the query path both deterministic and cache-resident while permitting regeneration at funding rate-update cadence.

Why the continuous formulation is needed at all. Retaining the discrete jump and simply substituting a deterministic reference price $S \approx S_{\text{ref}}$ directly at the settlement boundary

would produce an approximation error driven by the integrated price variance $\sigma^2\tau_{\text{fund}}$, which is unacceptably large over a typical 8-hour funding epoch. The continuous drain formulation is therefore a structural prerequisite within the scalar-ODE framework: it allows the subsequent affine approximation, formally justified in Appendix D.4.3, to operate iteratively over ultra-short micro-horizons Δt. By doing so, it collapses the macroscopic price variance error to a negligible $\mathcal{O}(\sigma\sqrt{\Delta t})$, preserving the scalar ODE architecture, and the sub-microsecond quote path it feeds, without sacrificing the stochastic fidelity of the macro-horizon.

Summary We replace the discrete settlement jump at T_{fund} by the continuous drain (99), with the smoothing parameter ρ set model-consistently to the spectral gap of the autonomous quoting system. The first-moment magnitude of the penalty is matched by construction through the unit-mass normalisation (98) and direct integration of the observable funding rate f_t, and the temporal profile is shaped by ρ.

D.4.3 Justification III: Affine Constraint via Deterministic Reference Price

Setup and the change it induces After adopting the continuous-drain approximation of Appendix D.4.2, the cash dynamics read

$$dX_t = (S_t + \delta_t^a)\, dN_t^a - (S_t - \delta_t^b)\, dN_t^b - q_t\, S_t\, f_t\, \tilde{\rho}\, \exp(-\rho(T_{\text{fund}} - t))\, dt. \tag{108}$$

The drift now carries a cross-term $q_t S_t$ between two state variables. This section examines whether the scalar ansatz of Proposition 1 survives this cross-term, or whether a further approximation is required. The question is independent of Justification II: even granting the continuous drain, (108) is not immediately ODE-reducible in the form of Proposition 1.

Effect of keeping S_t stochastic inside the drain The affine-jump-diffusion framework of Duffie et al. (2000) characterises the class of stochastic models whose conditional characteristic functions admit closed-form exponential-affine representations. The defining condition is that the drift, variance, and jump intensity coefficients are all affine functions of the state vector. While this result is stated in the context of characteristic functions and derivative pricing, it motivates the broader principle that an exponential-affine ansatz collapses a HJB equation to a system of ODEs in the coefficient functions only when the infinitesimal generator acts affinely on exponentials of linear state functionals.

The funding drain in (108) introduces a cross-term $q_t S_t$ in the drift, which is bilinear in the state pair (q, S) and therefore not affine. To see concretely why this destroys the scalar ansatz, observe that the drain term enters the HJB through $\partial_X v$. Under the scalar ansatz of Proposition 1, this produces a contribution proportional to $q_t S_t$ inside the equation that would otherwise govern $\omega_q(t)$. Since S_t is stochastic, this equation can no longer be a deterministic ODE in (t, q) alone. The scalar ansatz does not close, and the formal argument is carried out explicitly in the following subsection.

The scalar ansatz does not close in the presence of $q_t S_t$ Under the scalar ansatz $v(t, X, S, q) = -\exp(-\gamma(X + qS)) \cdot \omega_q(t)^{-\gamma/k}$, the partial derivatives are (as in Proposition

1):

$$\partial_t v = -\tfrac{\gamma}{k}\, v\, \tfrac{\dot\omega_q}{\omega_q}, \quad \partial_X v = -\gamma v, \quad \partial_S v = -\gamma q v, \quad \partial_{SS} v = \gamma^2 q^2 v.$$

Substituting into the HJB with the drain drift $-q_t S_t f_t \tilde\rho\, e^{-\rho(T_{\text{fund}}-t)}$, the cross-term enters through $\partial_X v$:

$$-q_t S_t f_t\, \tilde\rho\, e^{-\rho(T_{\text{fund}}-t)}\, \partial_X v \;=\; \gamma\, q_t S_t f_t\, \tilde\rho\, e^{-\rho(T_{\text{fund}}-t)}\, v.$$

When we divide the full HJB by $-(\gamma/k)\, v/\omega_q$ to isolate $\dot\omega_q$ (following Proposition 1), this contribution becomes

$$-k\, q_t S_t f_t\, \tilde\rho\, e^{-\rho(T_{\text{fund}}-t)}\, \omega_q(t),$$

which carries an explicit S_t factor in the ODE for ω_q. For the scalar ODE structure of Proposition 1 to hold, this term must be deterministic in (t, q) alone. It is not. The scalar ansatz does not close in the presence of stochastic S_t inside the drain. An alternative ansatz also fails, at the obstruction stated at the start of this appendix.

Alternative: deterministic reference price S_{ref} Rather than enlarging the ansatz, we pursue a second approximation that preserves the scalar form of Proposition 1. Within each solve of the receding-horizon execution engine, we freeze the mid-price inside the drain term only:

$$S_t \;\approx\; S_{\text{ref}} := S_{t_0} \qquad \text{for } t \in [t_0, t_0 + \Delta t], \tag{109}$$

where t_0 is the start of the current micro-horizon and Δt is the resolve interval. The substitution is applied *only* inside the drain. Stochastic S_t is retained everywhere else (in the diffusion, in the qS mark-to-market term within the ansatz, in the jump operators for bid and ask fills). Define the deterministic scalar

$$F_t := S_{\text{ref}} \cdot f_t. \tag{110}$$

The cash dynamics used by the execution engine over the micro-horizon become

$$dX_t = (S_t + \delta_t^a)\, dN_t^a - (S_t - \delta_t^b)\, dN_t^b - q_t\, F_t\, \tilde\rho\, e^{-\rho(T_{\text{fund}}-t)}\, dt, \tag{111}$$

with the drain drift now affine in q_t alone. Substituting (111) into the HJB under the scalar ansatz, the offending cross-term of the previous subsection becomes $-k\, q_t\, F_t\, \tilde\rho\, e^{-\rho(T_{\text{fund}}-t)}\, \omega_q(t)$, which supplies exactly the penalty term $\Phi(t, q)\, \omega_q(t)$ of Proposition 1. The scalar ansatz closes, and we recover the $\mathcal{O}(N_q)$ ODE system of Equations (9) to (11).

Distinction from the rejected S_{ref}-at-settlement approach. A natural theoretical counter-proposal is to avoid the continuous drain entirely by embedding the exact discrete funding fee directly into the epoch boundary condition. Note that this is structurally distinct from the matching-condition analysis of Appendix D.4.2: rather than testing whether the scalar ansatz can absorb the jump at the matching boundary between adjacent epochs, we now ask whether the scalar ansatz can survive when the discrete fee is hard-coded into the terminal condition of the autonomous continuation value at T_{fund}. As we will show, the obstruction is the same in spirit, any explicit S-dependence at the boundary propagates backward through the HJB equation and forces θ_q to acquire S-dependence on the entire continuous interval, but the mechanism

by which it manifests is different and worth tracing explicitly.

$$v(T^-_{\text{fund}}, X, S, q) = -\exp\left(-\gamma(X + qS - qSf_{T_{\text{fund}}})\right) \cdot (f^0_q)^{-\frac{\gamma}{k}} \tag{112}$$

Equating this to the standard scalar ansatz evaluated at T^-_{fund} and isolating the state vector yields the required boundary condition for the ODE system:

$$\omega_q(T^-_{\text{fund}}) = \exp\left(-kqSf_{T_{\text{fund}}}\right) \cdot f^0_q \tag{113}$$

The boundary condition explicitly contains the stochastic reference price S. The state vector can therefore no longer be a function of time and inventory alone. It must expand into a two-dimensional function $\theta_q(t, S)$, which destroys the $\mathcal{O}(N_q)$ ODE reduction and returns a quasi-linear PDE.

To rescue the ODE structure under this discrete formulation, one might propose applying the deterministic reference price directly to the boundary condition, approximating $\omega_q(T^-_{\text{fund}}) \approx \exp(-kqS_{\text{ref}}f_{T_{\text{fund}}})f^0_q$. Evaluating this boundary from a current time t requires projecting the reference price over the macroscopic horizon $\tau_{\text{fund}} = T_{\text{fund}} - t$, and the price error enters the quotes only through its product with the funding rate: substituting the boundary into the first-order-condition log-ratios shifts each quote by $f_{T_{\text{fund}}}S_{\text{ref}}$, so the quoting error of the projection is of order $|f_{T_{\text{fund}}}|\,\sigma\sqrt{\tau_{\text{fund}}}$. Under the representative BTC calibration ($|f_{T_{\text{fund}}}| = 10^{-4}$, $\sigma\sqrt{\tau_{\text{fund}}} \approx$ 1,400 USDT per BTC) this is roughly 0.14, on the order of one tick. Under elevated carry ($|f_{T_{\text{fund}}}| \sim 10^{-2}$) it scales linearly to roughly 14, far beyond any tolerable quoting error, and it is precisely the elevated-carry regimes in which the funding correction is most active.

This motivates the continuous drain approximation developed in Appendix D.4.2. By converting the discrete jump into a continuous, exponentially localised cash drain, the model evaluates the same S_{ref} substitution iteratively over the receding-horizon resolve cadence $\Delta t = 100\,\text{ms}$ (the C++ engine's quote interval), so that the projection window shrinks from τ_{fund} to Δt and the per-resolve quoting error from $\mathcal{O}(|f_{T_{\text{fund}}}|\,\sigma\sqrt{\tau_{\text{fund}}})$ to $\mathcal{O}(|f_{T_{\text{fund}}}|\,\sigma\sqrt{\Delta t})$, refreshed at every resolve.

The per-quote ratio of these two errors admits both a mathematical and a practical reading. Under the full-epoch mathematical bound, in which a terminal S_{ref} perturbation is treated as influencing the boundary value uniformly across the epoch, the ratio is $\sqrt{\tau_{\text{fund}}/\Delta t} \approx 537$ at $\tau_{\text{fund}} = 8\,\text{h}$ and $\Delta t = 100\,\text{ms}$. In practice the boundary perturbation does not influence regime-II quotes uniformly: its effect is weighted by the spectral-gap relaxation kernel of the autonomous matrix M, $e^{-g(T_{\text{fund}}-t)}$, with $g = \lambda_1 - \lambda_0$. The principled influence window is therefore $1/g$, the e-folding time of any terminal perturbation ($1/g = 155\,\text{s}$ at the reference BTC calibration). Substituting $1/g$ for τ_{fund} gives the practical per-quote ratio $\sqrt{(1/g)/\Delta t} \approx 39$, still far above unity, and the measured paired comparison on the 30 pre-analysis days lands on the conservative side of it: refreshing both formulations every 100 ms over the 90 pre-settlement layers, the continuous-drain freeze error has $p99 = 1.1$ basis points against 20.9 basis points for the discrete-jump alternative, a realised ratio of about 20. Both readings are independent of the funding rate level, the volatility, and the asset, and because both errors scale linearly in $|f_{T_{\text{fund}}}|$ the ordering is preserved exactly in the high-carry regimes where the correction matters most.

Total comparison and final justification. The per-quote ratio of the previous paragraph compares only the S_{ref} freeze residuals of the two formulations. A complete comparison of the continuous-drain-plus-S_{ref} formulation actually used by the engine against the rejected discrete-jump-plus-S_{ref} alternative must additionally account for the fact that the drain formulation pays the CARA-variance penalty quantified in Appendix D.4.2, whereas the discrete-jump alternative matches the first moment of the cash impact exactly and does not. On the drain side, this penalty is the sum of the two variance channels of Appendix D.4.2, dominated by the inventory channel measured there, and the cumulative S_{ref} freeze contribution adds negligibly on top of it, because each requote draws on a fresh $\Delta t = 100\,\text{ms}$ Brownian window and the per-resolve errors therefore carry only summable short-range autocorrelation (measured lag-one autocorrelation 0.17, against 0.997 for the discrete-jump alternative, whose errors all reference the same future settlement price). On the discrete-jump side, the boundary S_{ref} errors at consecutive requotes are highly correlated because they reference the same future random variable $S_{T_{\text{fund}}}$, so they accumulate coherently rather than averaging down across requotes, and the cumulative error scales with the regime-II window rather than as its square root.[24] Combining both effects means comparing two channels rather than summing them, since the CARA correction is a certainty-equivalent cash penalty per epoch while the freeze residual is a quote displacement per requote. On the freeze channel, the one both formulations share, the decorrelated cumulative residual of the continuous-drain-plus-S_{ref} formulation stays below the coherently-accumulating residual of the discrete-jump-plus-S_{ref} alternative by a wide margin. The margin is set by the measured per-quote ratio of about twenty, amplified by the difference between $\sqrt{N}$ and N accumulation over the requotes of a regime-II window, and it is not overturned by the inventory channel of the CARA correction, which is four orders of magnitude below the charge it approximates. The advantage on that channel is structural and asset-invariant: it follows from the asymmetric scaling of the two freeze residuals in τ_{fund}, Δt, and the spectral-gap relaxation time $1/g$, and both scale linearly in the funding rate, so their ratio survives any rescaling of the funding rate, the volatility, and the price level. The CARA correction is the one channel that does not share that invariance, scaling quadratically in the funding rate from a level four orders of magnitude below the charge. The CARA-variance penalty derived in Appendix D.4.2 is therefore the price paid for an approximation whose freeze residual is smaller, by a structural margin, than that of the only ODE-preserving alternative, and the discrete-jump-plus-S_{ref} formulation is rejected on these grounds.

Error of the S_{ref} substitution. The pathwise error in the drain term is $q_t(S_t - S_{\text{ref}}) f_t\, \tilde{\rho}\, e^{-\rho(T_{\text{fund}}-t)}$. Over the micro-horizon Δt, the standard deviation of $(S_t - S_{\text{ref}})$ is bounded by $\sigma\sqrt{\Delta t}$. The engine re-solves and refreshes S_{ref} at each micro-horizon boundary, so the error does not accumulate across solves. For $\Delta t = 100\,\text{ms}$ (the engine's quote interval) and σ on the order of 60% annualised for BTC, $\sigma\sqrt{\Delta t}/S \sim 3 \times 10^{-5}$, small relative to both S_{ref} itself and to the other approximations already present (the continuous-drain approximation of Appendix D.4.2, the

[24] Over a window of N consecutive requotes the cumulative-error variance scales as N on the drain side, where consecutive freeze errors decorrelate, and at the coherent-correlation extreme as N^2 on the jump side, where consecutive freeze errors share their future Brownian path. The cumulative-error ratio is therefore at least as large as the per-quote ratio reported above, and is typically larger.

CARA certainty-equivalent correction, and calibration uncertainty in A, k).

Summary We freeze $S_t \approx S_{\text{ref}}$ inside the drain term only, over the receding micro-horizon Δt. The modelling error is $\mathcal{O}(\sigma\sqrt{\Delta t})$ pathwise per resolve, non-accumulating across solves. In exchange, the cross-term $q_t S_t$ becomes affine in q, the scalar ansatz of Proposition 1 closes exactly, the ODE system recovers its $\mathcal{O}(N_q)$ form, and the regime-switching architecture operates at the resolve cadence required by funding rate updates.

Taken together with the continuous-drain approximation of Appendix D.4.2, these two approximations form a coherent two-step reduction from the exact discrete-settlement problem to the scalar ODE system used by the execution engine. The approximations address distinct mathematical obstacles (a discrete jump vs. a bilinear drift term) and carry distinct, independently bounded errors (CARA variance correction vs. $\sigma\sqrt{\Delta t}$ micro-horizon error), both small under representative BTC parameters and both dominated by the calibration uncertainty already present in the framework.

E The Simulation Engine

Every reported number is produced by the quoting engine of Section 5.7, and this appendix specifies it: which of the two Regime-I quote forms the engine evaluates, the numerical methods behind the backward induction that fills the boundary-layer lookup table, the memory layout and inter-process transport that deliver that table to the execution engine, the measured latency profile of both, the bracket runs that price the two conservative choices in the fill model, and the synthetic-market generator and scenario grid that drive the same engine in the stress suite of Section 5.9, together with the realised carry distributions that calibrate its scenarios. The mathematics being implemented is in Appendix D, and the regime-transition boundary at which the engine switches between its two quote sources in Appendix D.3.

E.1 Regime-I Closed Form versus the Exact Eigenvector

Proposition 3 (Appendix D.2.7) relaxes the inventory cap to infinity, so its Gaussian envelope is defined over unbounded inventory, and as Section 4.8 notes it is accurate only when that envelope is already negligible at $\pm Q$. Table 32 tests exactly that at the reference calibration. The envelope has standard deviation $\sigma_q = (\alpha/\eta)^{-1/4} = 8.55$ lots against a band reaching only to $Q = 10$, so 22% of its mass sits beyond the inventory cap, describing positions the maker is not permitted to hold. The resulting quote error is 9.0 ticks in the interior half of the band and up to 62.3 ticks near the cap. This is why the production Regime-I evaluator uses the exact discrete eigenvector, at unchanged cost, while the Gaussian form is retained as the analytic device that explains the solution's shape.

Table 32: Validation of the continuum Gaussian asymptotic against the exact discrete principal eigenvector at the reference calibration ($\gamma = 2 \times 10^{-5}$, $\sigma = 4.57$, $A = 0.274$, $k = 0.090$, $Q = 10$). The asymptotic holds when its envelope width σ_q is small against the inventory cap, so that the mass beyond $\pm Q$ is negligible; here it is not, and the quote error is the gap between the two forms in ticks of \$0.10, reported separately for the interior half of the inventory band and for positions near the cap.

Quantity	Value
Gaussian envelope width σ_q (lots)	8.55
Envelope mass beyond $\pm Q$	22%
Max quote error, $\lvert q\rvert \le Q/2$ (ticks)	8.96
Max quote error, $\lvert q\rvert > Q/2$ (ticks)	62.34

E.2 Numerical Methods for the Backward Induction

The backward induction is performed with Radau IIA, an L-stable implicit Runge-Kutta method of order five, run in log-space on the exact LUT grid with the analytic tridiagonal Jacobian supplied rather than differenced. The implicit method is chosen for two properties. It preserves positivity, since the implicit operator is a diagonally dominant M-matrix and the log-space quotes require $\omega > 0$ throughout; and it remains stable under the volatility and intensity spikes a market can deliver. Supplying the Jacobian analytically avoids finite differencing and makes the Newton stages converge robustly. At the production calibration the measured full LUT rebuild (backward solve, boundary search, and quote grid) completes in 64 ms at the median (77 ms at the 99th percentile), while the un-reduced HJB solve over an (S, q) grid that the scalar ansatz avoids takes 0.60 s per solve (Table 34).

E.3 Lookup Table, Memory Layout and Inter-Process Communication

The backward induction runs asynchronously in a parallel Python process, triggered by material change in its inputs rather than on a clock: the funding scalar F_t moving by a fifth, or the rolling volatility σ by a quarter, with the Regime-I eigenvector refreshed on any change in σ, A or k at up to 1 Hz. The published f_t holds flat for multi-second stretches and then steps as the premium-index average updates, so a table computed at the start of a stable interval stays accurate across it. A live deployment would add a periodic regeneration on top of these triggers, forcing a rebuild every five minutes whether or not anything moved, as insurance against missed or duplicated WebSocket messages. That is a live-feed feature with no offline-replay analogue: a recorded feed cannot silently drop a message, so the replay rebuilds on triggers alone.

What the replay does impose is a floor in the opposite direction. A rebuild is admitted only once 300 seconds have passed since the last, so the triggers above fire at most once every five minutes. This is a compute-bounding choice made for the thesis rather than a modelled feature: rebuilding on every trigger across 64 replayed days is expensive, and the boundary layer does not move fast enough to need it. The 300-second value was fixed as an engineering default before the sweep that prices it, so what follows characterises its cost rather than selecting it. Table 33 reruns the funding strategies on the first eight out-of-sample days, 18 to 25 February 2026, at cadences from 5 to 600 seconds. No cadence in the range moves their Sharpe by more than 0.4% or their total P&L by more than 0.2% against the production setting, while the share of

live time spent in Regime II is flat at 2.4% across the range, reaching 2.5% only at the slowest cadence. Sharpe and total P&L are read here, against the rule of Section 6.1, precisely because they compress mean, volatility and fill count into one number and would therefore register a change in any of them; the Regime II share is the more direct diagnostic, being the one quantity the cadence acts on. None of the three orders consistently with cadence, so the sweep shows noise rather than a cost.

The optimal quotes are evaluated on a downsampled temporal grid at the 100 ms quoting cadence over the boundary layer $[t^*, T_{\text{fund}}]$, producing a lookup table of size $\lfloor u^*/\Delta t_{\text{LUT}} \rfloor \times (2Q+1)$. At the production grid the table is a few megabytes and cache-resident.

Table 33: Robustness of the funding strategies to the Regime-II LUT regeneration throttle *lut_min_rebuild_s* (production default 300 s). Strategies 5 and 6 are the funding and hybrid arms, each replayed at every cadence over the same eight out-of-sample days. Total P&L is USDT over those eight days, the markout is the mean per-fill mid move 5 s after execution in basis points, and Regime II frac. is the share of live time the strategy spends inside the boundary layer, the one quantity the throttle acts on directly. The footer gives each strategy's largest deviation from the production cadence.

Cadence (s)	Strat.	Sharpe	Total P&L	5 s markout (bp)	Regime II frac.
5	5	-53.74	-6,569	-0.92	0.024
5	6	-53.30	-6,281	-0.91	0.024
60	5	-53.74	-6,567	-0.93	0.024
60	6	-53.37	-6,276	-0.91	0.024
300	5	-53.69	-6,563	-0.93	0.024
300	6	-53.16	-6,270	-0.91	0.024
600	5	-53.59	-6,564	-0.93	0.025
600	6	-53.20	-6,272	-0.91	0.025
s5: max $\lvert\Delta\text{Sharpe}\rvert$ vs 300 s = 0.102, max $\lvert\Delta\text{markout}\rvert$ = 0.001 bp					
s6: max $\lvert\Delta\text{Sharpe}\rvert$ vs 300 s = 0.213, max $\lvert\Delta\text{markout}\rvert$ = 0.001 bp					

In the production path the LUT is serialised to the engine via ZeroMQ IPC, a messaging library that lets the two processes exchange messages without either waiting for the other, and so without disk I/O penalties; in offline replay it is loaded from pre-computed binaries. In both cases it is deserialised into a contiguous, page-aligned heap allocation in the C++ process. This memory is locked into physical RAM using `mlock()` to prevent OS page-swapping under memory pressure, and exposed to the hot path via an atomic pointer swap ensuring the execution thread always reads a consistent, fully-written table without locking: the writer prepares the replacement table, then flips a single pointer in one indivisible step, so a reader sees either the whole old table or the whole new one and never has to wait.

During execution, live or replayed, the C++ engine operates in Regime I until the session clock crosses t^*, at which point the state machine transitions to Regime II. The full state-machine transition is summarised as follows:

1. **Regime I (C++, $t < t^*$):** The execution engine operates independently of the Python layer, reading the cached asymptotic quotes derived from the exact discrete eigenvector f^0 (refreshed at 1 Hz as the rolling parameters update) at $\mathcal{O}(1)$ latency.

2. **Asynchronous generation (Python):** Triggered on each f_t publication or material σ drift, the Python process performs implicit backward induction over $[t^*, T_{\text{fund}}]$, initialised at

$\omega_q(T_{\text{fund}}) = f_q^0$, and atomically pushes the updated LUT to the C++ heap (in offline replay, the same triggers are evaluated on the recorded feed and the LUTs are pre-computed at the corresponding timestamps).

3. **Regime II (C++, $t \geq t^*$):** The engine transitions to the LUT, interpolating aggressively skewed quotes to shed toxic inventory prior to settlement.

This architecture guarantees that the computationally intensive resolution of the non-autonomous boundary layer never blocks the execution hot path. Table 34 reports the measured latency profile of each stage on the production calibration: the two hot-path reads execute in under four nanoseconds per quote at the median (worst stage 5.0 ns at the 99th percentile), five orders of magnitude inside the quoting cadence, while every expensive stage (eigensolve, backward induction, full LUT rebuild) runs off-path with at least 15× headroom at the median against its trigger cadence. The Python reference rows quantify what the C++ implementation buys: the identical per-quote lookups, on the same lookup tables, cost 0.31 and 1.31 μs in the Python engine, a factor of 80 to 400, which is immaterial for an offline replay but is the difference between fitting inside and blowing through a live sub-microsecond quoting budget. The final row reports the un-reduced HJB solve over an explicit price grid, the computation the scalar ansatz of Proposition 1 removes: at 0.6 s per solve it is 16× the ODE backward solve it replaces, which is the engineering restatement of why the ODE reduction is load-bearing.

Table 34: Measured latency profile of the quoting architecture at the production calibration ($Q = 10$, 100 ms LUT grid). Hot-path stages are per-quote reads in the C++ engine; off-path stages run asynchronously in Python and are bounded by their trigger cadence, not by the quoting loop.

Stage	Where / cadence	p50	p99
Regime-I f^0 read+compute	C++ hot path / per quote	3.7 ns	4.8 ns
Regime-II LUT interpolation	C++ hot path / per quote	3.3 ns	5.0 ns
Regime-I depths lookup	Python reference / per quote	0.31 μs	0.32 μs
Regime-II LUT interpolation	Python reference / per quote	1.31 μs	1.53 μs
Regime-I eigensolve	Python off-path / ≤1 Hz	0.027 ms	0.051 ms
Regime-II ODE backward solve	Python off-path / f_t trigger	37.9 ms	61.9 ms
full LUT rebuild (+ u^* + quote grid)	off-path / f_t trigger	63.6 ms	77.3 ms
Raw HJB PDE backward solve ($N_S = 512$)	avoided by §3 ansatz	602 ms	622 ms

E.4 Fill-Model Brackets

Two variants of the replay engine's queue model bracket the two conservative choices of Section 5.7, each run for the standard GLT strategy on three out-of-sample days against a baseline that reproduces the headline run exactly. Granting every order front-of-queue priority the moment it is placed moves total P&L by under 2%, because the at-level fills that priority buys inherit the same adverse selection as the sweeps they replace, and net profit per fill is unchanged at −0.062 against −0.063. Suspending the cancel and replace rule, so that a resting order holds its queue position until the ideal quote has moved 5, 20 or 50 ticks away, deepens the loss by 0.8%, 5% and 35% respectively. On this sample, queue priority buys little while reaction speed

is expensive, which suggests the 100 ms decision grid rather than the queue assumption is the tighter of the two execution constraints.

E.5 Monte Carlo Stress Suite: Generator and Scenario Grid

Each synthetic path is one compressed funding day, generated on a 100 ms grid over two funding epochs of one hour, and then replayed through the same engine, fill core, fee schedule and settlement accounting as a recorded day. The mid-price follows $dS = \mu_{\text{true}}\,dt + \sigma_{\text{true}}\,dW$, with μ_{true} an Ornstein-Uhlenbeck process of 20 s half-life. The three strategies run here are drift-free, so no synthetic signal is consumed. Taker flow is a thinned Poisson stream at six arrivals per second whose sweep half-spreads are drawn from $\text{Exp}(1/k_{\text{true}})$, so the last-in-queue fill core reproduces $\lambda(\delta) = Ae^{-k\delta}$ on synthetic flow. The book is a static exponential ladder, read only for queue position at post time, since fills come from the trade stream.

What the compression changes. The drain, the regime switch and the discrete payment are functions of the time remaining to settlement rather than of the wall clock, so the quoting mechanics are unchanged by the shortened epoch. The epoch-normalising constant $\tilde{\rho}$ is likewise unchanged, because $e^{-\rho\Delta T_{\text{epoch}}}$ is negligible at either length. What does change is proportion. The pre-settlement boundary layer occupies a much larger share of a one-hour epoch than of an eight-hour one, so the funding correction is active for more of each synthetic day than it would be in the market. Every claim drawn from the suite is accordingly a paired contrast between scenarios on identical seeds, never a level.

Estimation inputs. The volatility feed is a trailing 60 s realised-volatility estimate of the synthetic mid on a one-second grid, the same estimator the replay runs, so the strategies quote off an estimate with a real lag rather than off the generator's truth. The intensity pair is the generator's own: k is passed exactly, while A is the value implied by the arrival rate and the sweep-half-spread law, refreshed on a five-minute grid. No diagnostic here compares the estimated volatility with the generator's true volatility; the accuracy of that estimator is the subject of Appendix B.8.

Scenario grid. Table 35 lists the five scenarios. Each is run for 200 paths, and path seeds depend on the path index alone, so a change of scenario leaves the draws untouched and every contrast is paired. The funding path and the observation noise are drawn from streams decoupled from the price path, so switching funding regime does not disturb the synthetic market itself.

Table 35: The Monte Carlo scenario grid. Funding scenarios are Ornstein-Uhlenbeck in the rate, clipped to the suite's ±75 bp cap.

Scenario	Specification	Purpose
baseline	constant 1 bp per settlement	the sample's ordinary carry
documented carry	mean 5.5 bp, 4 h half-life	BTC's own top percentile
persistent carry	mean ±45 bp, 4 h half-life	sustained one-sided carry
cap pinned	mean 75 bp, tight	carry riding the suite's cap
sign flip	mean 0, 10 min half-life, wide	whipsawing carry

Cross-symbol funding distributions. Table 36 reports realised per-settlement funding rate distributions for ten Binance USDT-margined perpetuals, fetched from the exchange's pub-

lic funding rate history and per-symbol cap endpoints (Binance Futures, 2026), up to 4,000 settlements per symbol (roughly 3.6 years for the 8-hour contracts). It substantiates the claim of Section 5.9 that the stressed carry regimes are realised in practice by thinner contracts.

Table 36: Realised funding rate distributions across ten USDT-margined perpetuals (Binance public API, accessed July 2026); all rates in basis points per settlement. Int. is the modal settlement interval; Cap is the current per-symbol clamp in bp (†caps are adjusted by the exchange over time, so historical rates can exceed the current value). The last two columns give the share of each symbol's settlements with $|f_{T_{\text{fund}}}|$ at or above the documented-carry level (5.5 bp, the stress suite's calibrated scenario) and at or above BTC's own 30 bp clamp.

			$\lvert f\rvert$ per settlement (bp)			settlements (%)	
Symbol	Int. (h)	Cap†	median	P99	max	$\geq$ doc. carry	$\geq$ BTC cap
BTCUSDT	8	30	0.64	4.4	8.8	0.5	0.00
ETHUSDT	8	30	0.68	4.6	10.2	0.5	0.00
SOLUSDT	8	38	0.93	9.7	92.7	2.8	0.18
XRPUSDT	8	38	1.00	5.8	11.0	1.2	0.00
DOGEUSDT	8	49	0.97	5.6	10.3	1.1	0.00
AVAXUSDT	8	38	1.00	6.4	14.4	1.4	0.00
1000PEPEUSDT	8	200	1.00	11.4	23.8	4.4	0.00
WIFUSDT	4	200	0.50	4.6	63.6	0.7	0.05
TRBUSDT	4	200	0.50	2.7	10.7	0.0	0.00
LPTUSDT	4	200	0.64	56.1	200.0	8.5	1.77

Limits of the synthetic market. A multi-level sweep is represented as a single print at its deepest reached price. That is fill-equivalent for a last-in-queue book, but it does not deplete the synthetic ladder, so a burst of flow can fill more resting size than a depleted book would leave. Taker flow is independent and evenly sided, and the touch half-spread is held fixed, so the synthetic book neither clusters nor widens as a real one under strain would. The suite therefore imposes carry on a market that is otherwise well behaved, which is the point: it isolates the funding mechanism rather than compounding it with microstructure stress. No headline result rests on it; those come from the recorded days.

F Code Manifest

All results in this thesis are produced by the programs listed below Configuration values, random seeds, and file-format details live in the code itself, so a reader with the data can reproduce every table and figure.[25]

Data pipeline. `recorder.py` captures the four Binance WebSocket streams on the recording server. `book_reconstruct.py` rebuilds the 100 ms top-20 order book from diffs and snapshots and flags invalid intervals by update-identifier discontinuity. `markprice_ffill.py` aligns and forward-fills the mark price and funding streams. `data_gap_handler.py` centralises the gap and staleness rules. `feature_engineering.py` computes the one-second feature grid,

[25]The GitHub link for the code: https://github.com/rowan-zwennes/master_qf

and `label_split.py` attaches the multi-horizon labels and fixes the pre-analysis and out-of-sample split. `pipeline_utils.py` holds the loaders and split helpers every later stage shares. `validate_book_reconstruct.py`, `validate_book.py` and `validate_data.py` run the per-day correctness battery. `make_data_section_assets.py` and `trade_size_stats.py` generate the data-chapter tables and figure.

Machine learning. `ml_train.py` runs the walk-forward LightGBM training with inner cross-validation, purging, and the weekly Optuna search; `ml_predict.py` emits the out-of-sample α_{ML} streams. `ml_baseline_ridge.py` is the linear benchmark, `compare_models.py` runs the Diebold-Mariano comparison between the two, and `compare_l2_huber.py` the loss-function robustness check. `compute_purge_window.py` sizes the purge gap from feature autocorrelation. `select_horizon.py` performs the pre-analysis horizon selection. `ml_crossfold_stability.py` and `gen_crossfold_table.py` run and tabulate the hyperparameter-transfer experiment. `ml_preanalysis_diagnostics.py` and `pre_analysis_tests.py` produce the stationarity, distribution, and collinearity batteries, which `export_pre_analysis.py` turns into the Appendix A and B tables. `ml_shap.py` computes the SHAP attributions, and `check_directional_accuracy.py` measures directional accuracy against signal strength.

Calibration. `calibrate_intensity.py` and `calibrate_intensity_rolling.py` implement the queue-aware virtual-order estimator of (A, k), pooled and rolling; `select_intensity_window.py`, `validate_intensity_stride.py` and `validate_intensity_exp.py` select and validate its window, stride, and exponential shape. `calibrate_volatility.py` and `validate_sigma.py` build and validate the 60 s volatility feed. `describe_rho.py` documents the selection of the boundary-layer rate ρ. `calibrate_vpin.py` sets the VPIN bucket sizes.

Quote generation. The `hjb_` family implements the mathematics of Section 4: `hjb_principal_eigenvector.py` (exact discrete f^0 and the Gaussian reference), `hjb_riccati_solver.py` and `hjb_thomas_solver.py` (the backward induction and its tridiagonal inner step), `hjb_lut_builder.py` (boundary-layer LUT construction), `hjb_epoch_stitching.py` and `hjb_regime_boundary.py` (the epoch boundary condition and the Regime I/II switch), with `hjb_asymptotic_reference.py` as the closed-form cross-check. `sim_quote_engine.py` assembles these into the per-tick quoting policy. `quantify_ml_interp_error.py` and `analysis_decaying_drift.py` bound the linear-response and horizon-scaling approximations; `validate_gaussian_asymptotic.py`, `validate_cara_approx.py`, `validate_sref_approx.py` and `validate_boundary_layer.py` verify the analytical approximations.

Replay and stress. `run_simulation.py` is the reference replay engine: last-in-queue fill core, exact fee and funding accounting, and the six-strategy ablation harness, with `fill_core_reference.py` as its standalone fill-mechanics twin. `run_simulation_freshbook.py` is the fork carrying the publication-clock arm of Appendix C.1, kept separate so the headline results stay reproducible from an engine that arm never touched. `export_replay_binary.py`

packages replay inputs for the C++ engine and `validate_cpp_parity.py` proves per-fill parity between the two implementations. `stress_engine.py` generates the synthetic Monte Carlo market, `stress_funding_ou.py` drives the carry scenario grid, `stress_q_sweep.py` the inventory-capacity sweep, and `rebuild_funding_table.py` regenerates Table 23 from the stored per-path outcomes. `fetch_altcoin_funding.py` downloads the cross-symbol funding history of Table 36.

Results and statistics. `statistical_tests.py` and `run_statistical_tests.py` implement the block-bootstrap machinery behind every confidence interval. `analysis_pnl.py` (attribution and ablation tables), `analysis_markout.py` (markout distributions and tests), `analysis_fill_timing.py` (the timing-versus-selection decomposition), `analysis_boundary_layer.py` and `analysis_funding_activation.py` (the u^* census and activation statistics), `analysis_funding_inventory.py` and `analysis_settlement_toxicity.py` (settlement positioning and the toxicity premise test), `analysis_money_figure.py`, `analysis_freshbook.py`, `fig_inventory_dist.py`, `fig_shap_boundary.py` and `analysis_shap_boundary.py` produce the results-chapter exhibits. The `analysis_sensitivity` family and `sensitivity_common.py` run the $\pm 50\%$ and fee sweeps, `analysis_latency.py` and `analysis_latency_offpath.py` benchmark the hot path and the off-path solvers, and `gen_results_tables.py` regenerates every ablation table from the run directories.

C++ engine. The `c++/` tree contains the execution engine: `order_book` and `queue_position` (book state and last-in-queue tracking), `fill_simulator` (the fill core mirrored against Python), `glt_asymptotic`, `riccati_lut`, `regime1_reader` and `regime_switch` (Regime I evaluation, LUT interpolation, and the switch logic), `funding_correction`, `reservation_price` and `pnl_tracker` (quote assembly and accounting), `replay_reader` (binary replay input), `latency_bench` (the Table 34 benchmarks), and `main.cpp` tying them together.